

Mathematics

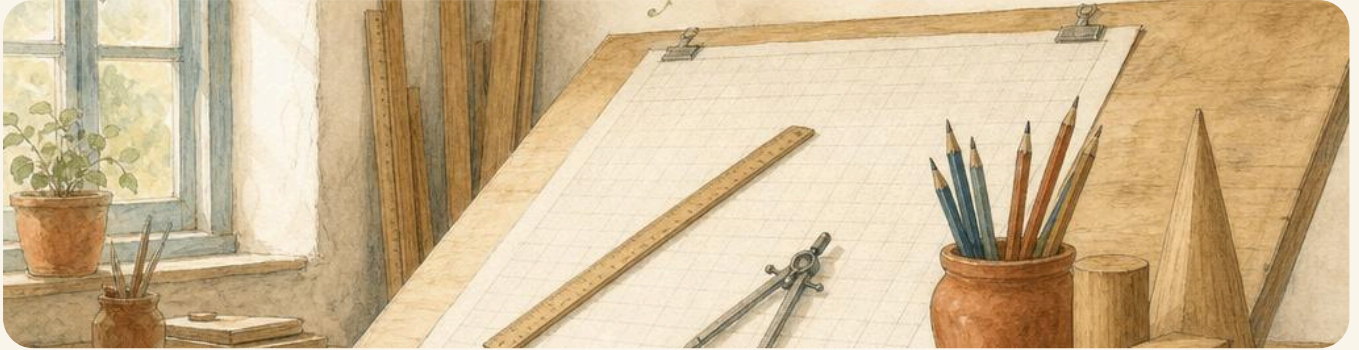
Year 2

Student Manual



ABOUT THIS BOOK

Mathematics · Year 2



The Plan Room: a drafting table, a wooden ruler, a pair of compasses on a squared sheet, a pot of pencils and the solids already cut. Everything in this book is drawn here before it is made next door.

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ABOUT THIS BOOK · CONTINUED

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THE WHOLE OF THIS BOOK IN ONE LINE

Draw it.
Work it out.
Then check it a
different way.

Last year you learned to count something and then count it again to be sure. This year is one step harder, and it is the step that matters.

When you can only get an answer one way, you do not really know whether it is right. When you can get it **two** ways, you do.

What is in this book

Twenty-one units, grouped the way the Cambridge Primary Mathematics framework groups them. Each unit has four parts, and each part ends with your turn to practise.

NUMBER: COUNTING AND PLACE VALUE

1	Numbers to 100 and beyond Counting to 100 and back · Reading and writing numbers · Counting on and back in ones and tens · First, second, third: ordinal numbers
2	Tens and ones What each digit is worth · Partitioning into tens and ones · One more, one less, ten more, ten less · Putting a number on a number line
3	Comparing, ordering and rounding Greater than and less than · Ordering numbers to 100 · Rounding to the nearest ten · Making a sensible estimate
4	Steps, odds and evens Counting in twos, fives and tens · Counting in threes and fours · Odd and even numbers · Sorting numbers by their rule
5	Halves and quarters One half and one quarter · Three quarters, and making a whole · Which shapes are halved, and which are not · Halves and quarters of a set
6	Number pairs and doubles Pairs that make 10 · Pairs that make 20 · Multiples of ten that make 100 · Doubles you know by heart

NUMBER: CALCULATING

7	Adding and taking tens Ten more and ten less • Adding and taking other multiples of ten • Adding a single digit to a two-digit number • Adding four or five small numbers
8	Adding and taking to 100 Adding pairs of two-digit numbers • Making a ten when you add • Finding a small difference • Taking away, and finding the difference
9	Missing numbers and equal sides The equals sign means the same as • Finding the missing number • A shape or a letter for a number you do not know • Making up a number story
10	Rows, columns and times tables Multiplying as repeated addition • An array read two ways • The 2, 5 and 10 times tables • The 3 and 4 times tables
11	Sharing, grouping and what is left Sharing equally between • Grouping into groups of • Division and multiplication together • When something is left over
12	Doubling and halving Doubling two-digit numbers • Doubles of multiples of five • Halving, and how it undoes doubling • Near doubles

GEOMETRY

13	Flat shapes Naming flat shapes · Sides and corners · The same shape, turned round · Sorting flat shapes by a rule
14	Solid shapes Naming solid shapes · Faces, edges and corners · Flat drawings of solid shapes · Making and sorting solids
15	Symmetry A line of symmetry · Symmetry in flat shapes · Completing a symmetrical pattern · Finding symmetry around you
16	Position, turns and right angles Saying where something is · Giving and following directions · Whole, half and quarter turns · A right angle is a quarter turn

MEASURE

17	Money Recognising all coins and notes · Writing amounts of money · Finding totals · Paying, and working out change
18	Length Comparing lengths and heights · Metres and centimetres · Choosing the right unit and the right ruler · Measuring and drawing lines
19	Mass and capacity Heavier and lighter · Kilograms and grams · Full, empty and half full · Litres, and reading a scale
20	Time Seconds, minutes, hours and days · Days of the week, months of the year · O'clock and half past · Measuring how long something takes

HANDLING DATA

21	Sorting and showing what you found Asking a question and collecting the answers · Tally charts and tables · Block graphs and pictograms · Venn and Carroll diagrams
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ALSO IN THIS BOOK

	Welcome, and how this book works the seven panels you will meet on every page
	Before you start the kit you need, and how to set your work out
	Build it and show it six projects along the way, and one big one at the end
	Yard words every word in this book, explained
	Looking back over the year what you can do now that you could not do in September
	For teachers the Cambridge objectives, unit by unit
	Sources and references every link in this book, in one list

Come into the Plan Room

There is a small bright room at the top of the stairs with one tall table in it. The table slopes. On it there is a long wooden ruler, a pair of compasses, a rubber, and a big sheet of paper ruled into squares.

This is where Otto works. Otto is a tawny owl and he wears the same slate-blue coat every day. Before anybody in this book makes anything, Otto draws it here first.

Through the door at the back is the **Making Yard**. There is a workbench, a rack of timber, a pair of brass scales, a money tin, and a slate board on the wall with chalk marks all over it.



The Making Yard through the door: the bench, the scales, the timber rack, the money tin and the tally board.

There are two rules in the Yard, and Bramble the badger will tell you both of them before you have got your coat off.

Nothing gets made until it has been drawn. And nothing leaves the Yard until somebody has checked it a different way.

Who you will be working with

Seven of them, and each one is good at a different part of doing mathematics. You will need all seven.



Otto, Tessa, Hattie, Milo, Bramble, Rook and Pip.

Otto	the tawny owl in the slate-blue coat. He draws it first. Otto turns a question into a picture before he touches a single number.
Tessa	the tortoise in the sage coat. She checks it a different way. Tessa is never satisfied by one method, and she is the reason this book exists.
Hattie	the hedgehog in the dark blue jacket. She estimates first, out loud. Sometimes she is close and sometimes she is wrong, and both of those are useful.
Milo	the mouse in the rust-orange jacket. He asks how do you know. About everything. Including his own answers.
Bramble	the badger in the moss-green waistcoat. He keeps the store. The scales, the timber, the till and the money tin are his, so he asks the questions that have to have a right answer.
Rook	the jackdaw with the charcoal scarf. He lines things into rows and columns, and he keeps whatever is left over. Every array in this book is his.
Pip	the wren. She hops the number line. One hop is one, unless she tells you she is hopping in tens.

None of them will ever tell you an answer. They will tell you how to find one, which is worth more.

The panels you will meet

Each one looks different so you can tell at a glance what a page is asking of you. Here is every single one, doing its real job.

Come into the Plan Room

This panel comes at the start of a unit. It asks you what you already know, before anybody teaches you anything.

In the yard today

Something is happening in the Yard, and it needs some maths.

A panel like this one sets the scene. It is a story, so it is written in handwriting, and there is nothing to answer yet.

✓ **By the end you will**

- know what every panel in this book is for
- know where to look when you are stuck
- know why you are asked to check things twice

**SHARPEN
YOUR
PENCIL**

A quick game to wake your brain up. Count backwards from twenty while your partner claps. Do not stop when they clap faster.

Do you remember?

A panel like this reminds you of something you already learned, just before you need it again.

♣ **Let's work it out**

A white panel like this one is the teaching. It is where something new is explained, with a picture that shows why it is true.

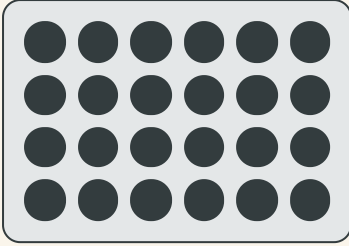
Read it slowly. Read it twice. It is short on purpose.

🔧 **Watch me work one out 1**

How many pegs are in four rows of six?

- 1 Draw it first. Four rows, six in each row.
- 2 Count one row: six. Now count the rows: four.
- 3 Six added four times, or four sixes.
- 4 Count in sixes along the rows: six, twelve, eighteen, twenty-four.

Answer: there are **24** pegs



Four rows of six.

1 A panel with a number is your turn. Write your answer in the space.

↔ **Check it a different way**

You worked it out by counting along the rows. Now try it another way.

Now count the same pegs down the columns instead: six columns of four. If you get twenty-four again, you were right the first time.

How do you know?

Why does counting the columns give the same answer as counting the rows? Tell your partner, using the picture above.

★ Rook's find

A panel like this is something extra that is just interesting. There is nothing to answer. Rook found it and thought you would like it.

→ Take it further

When you have finished, this panel gives you a harder one.

🪵 Yard challenge

This one is meant to be hard. You are allowed to get it wrong several times. That is what it is for.

✂ The material you need

a ruler, a pencil, squared paper and about twenty counters. If you have no counters, dried beans work perfectly.

♂ Find out together

Work with a partner on a panel like this, and find out something neither of you knew when you started.



▣ SCAN TO SEE IT MOVE

Every picture in this book that you could move, all in one place: number lines, arrays, fraction bars, coins and clocks.

mathigon.org/polypad

And at the end of every unit

☐ Check what you can do

- 1 Show what you can do now, on questions you have not seen before.
- 2 These are never the same questions you practised on.

HOW DID IT GO

	confident	almost there	needs more work
say what every panel in this book is for	☐	☐	☐
check an answer a different way	☐	☐	☐

NOW I CAN

- ☐ find my way round a page of this book
- ☐ say why checking twice is worth the trouble

☐ YARD WORDS



method

the way you worked it out



check

work it out another way

The kit, and how to set your work out



A ruler, a pencil, a rubber, squared paper, counters and a handful of coins. That is the whole kit.

✂ The material you need

a sharp pencil, a rubber, a ruler with centimetres on it, squared paper, about twenty counters, and a handful of coins you are allowed to use. A metre stick and a measuring jug live in the classroom, and you will be told when to fetch them.

♣ Setting your work out

Leave a line between one answer and the next. It looks better and, more importantly, you can find your mistake later.

Write one digit in each square of the squared paper. Tens under tens, ones under ones. If the columns do not line up, the answer will not either, and this is the single commonest way to get an addition wrong.

When you draw a number line, use your ruler. A wobbly number line makes the spaces uneven, and the whole point of a number line is that the spaces are even.

BEFORE YOU START · CONTINUED

A number line drawn with a ruler. Every space is the same.



Do you remember?

A pencil is better than a pen for mathematics, because you are going to change your mind, and changing your mind is allowed.

♣ Two habits worth starting today

The first is to **write the question down** before you answer it. Copying it out takes ten seconds and it means you can see, later, exactly what you were being asked.

The second is to **leave your working where it is**. Do not rub out the middle of a calculation once you have the answer. The working is the part that shows how you thought, and your teacher would rather see a crossed out first attempt than a tidy page with nothing behind it.

Both habits feel slow for about a week. Then they stop feeling like anything at all, and you will wonder how anybody works without them.

Set out $24 + 13$ on squared paper, tens under tens.

BEFORE YOU START · CONTINUED

- 1 Sharpen the pencil before you start, not halfway through.
- 2 Write the question, then leave a line.
- 3 Line the digits up in the squares, tens under tens.
- 4 Work it out, and leave the working on the page.
- 5 Check it a different way before you call it finished.

Both numbers split into tens and ones, ready to add.

	Tens	Ones
24	2	4
13	1	3

Do you remember?

A pencil is better than a pen for mathematics, because you are going to change your mind, and changing your mind is allowed.

The Cambridge objectives, and where they live

Each unit opener lists the objective codes it covers, so a teacher can see at a glance what a page is for. The full list is at the back of the book.

For teachers

This Student Book covers the Cambridge Primary Mathematics 0845 Stage 2 framework in twenty-one units, grouped by the framework's own strands: Number, Geometry, Measure and Handling data. Problem solving is not a separate unit because the framework does not treat it as one: the 2Pt codes are delivered inside the four content strands, chiefly through the Check it a different way panels.

Every unit carries four teaching parts, each with a worked example, at least twelve practice items, a reasoning prompt and a second-method panel. Pupils who finish early should be directed to Take it further and then to the Yard challenge rather than to the next unit.

The self-assessment tables use the wording confident, almost there and needs more work rather than colours, so that a pupil who cannot distinguish red from green is not disadvantaged, and so that the judgement is recorded in words your teacher can read at a glance.

The QR codes are optional. Every one of them prints its address as text, and no activity in this book depends on a device. Where a class has tablets, the codes are the fastest route to a manipulative that moves; where it has none, nothing is lost.

✴ Rook's find

The compasses on the cover of this book are not for drawing circles yet. They are Otto's, and he uses them to step off equal distances along a line. You will do that in Unit 18.

UNIT 1

Numbers to 100 and beyond

Otto rules up a hundred squares, and Pip hops every one of them.



Come into the Plan Room

Before anybody teaches you anything, show what you already carry in your head.

Count out loud from one to twenty. Now start at twenty and count back down to one. Which direction was harder?

Now count in tens: ten, twenty, thirty. How far can you get before you have to stop and think?

In the yard today

Otto has pinned up a fresh sheet and ruled it into a hundred squares.

He did it with the long wooden ruler, ten lines down and ten lines across, and it took him most of the morning. Now there are a hundred small squares waiting, and Pip is already standing on the first one.

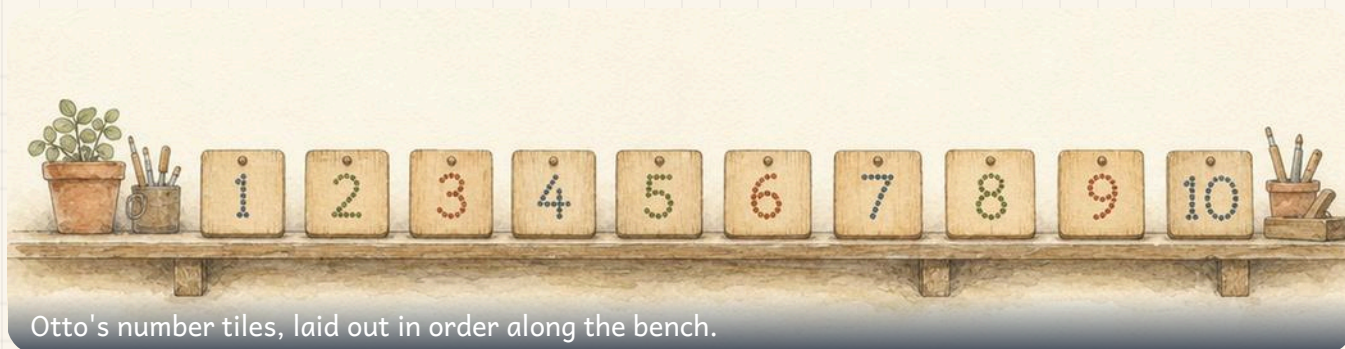
One square, one number, says Otto. All the way to a hundred. And then all the way back, which is the part everybody forgets.

By the end you will

- count to 100 and back again, out loud and on a number line
- read a number and write it in figures and in words
- count on and back in ones and in tens from any number
- use ordinal numbers to say which one came first, second or tenth

Cambridge Primary Mathematics Stage 2: 2Nn1 • 2Nn2 • 2Nn3 • 2Nn11

Counting to 100 and back



SHARPEN YOUR PENCIL

Count round the class in ones. When your teacher claps, turn round and count back the other way. Keep going until somebody laughs.

♣ Counting on, and counting back

Counting **on** means going up: forty-six, forty-seven, forty-eight.

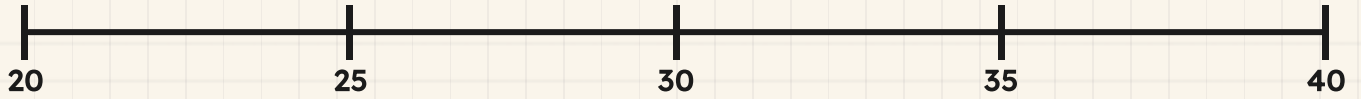
Counting **back** means going down: forty-eight, forty-seven, forty-six.

They use exactly the same words in exactly the same order. The only difference is which way you walk along them.

The hard places are the ones where the tens change. After **twenty-nine** comes **thirty**, not twenty-ten. After **thirty**, going backwards, comes **twenty-nine**.

TOPIC 1.1 · CONTINUED

The stretch where the tens change. Say every number from twenty-eight to thirty-two, out loud.



Do you remember?

A number line always goes up to the right. If you are counting back, you are walking to the left.

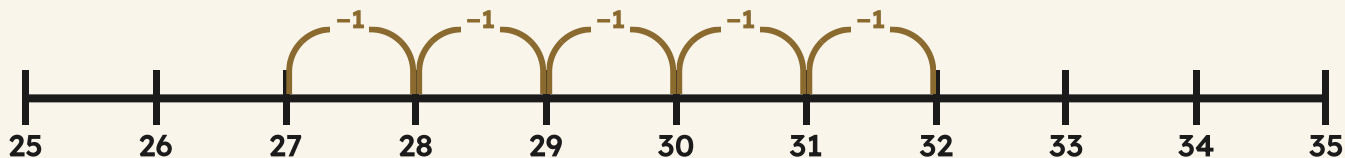
Watch me work one out 1

Count back from thirty-two to twenty-seven.

- 1** Find thirty-two on the number line and put your finger on it.
- 2** Walk one space to the LEFT and say the number: thirty-one.
- 3** Keep walking left, one space at a time: thirty, twenty-nine.
- 4** Careful at the ten. After thirty comes twenty-nine.
- 5** Stop when you reach twenty-seven. Count how many spaces you walked.

Answer: thirty-two, thirty-one, thirty, twenty-nine, twenty-eight, twenty-seven, which is **5** spaces

Pip walks it back with you. Five hops, one number each.



⇌ **Check it a different way**

You worked it out by counting back one at a time. Now try it another way.

You counted back one at a time. Now do it the other way round: start at twenty-seven and count **on** to thirty-two, keeping a tally on your fingers.

If you put up five fingers going up as well as going down, you know the answer is right.

How do you know?

Milo says there are six numbers between twenty-seven and thirty-two, so the answer must be six. Otto says five. Who is right, and what exactly is the other one counting?

Your turn

1 Count out loud from thirty to fifty. Ask a partner to listen.

2 Count back out loud from sixty to forty.

3 Write the next three numbers after sixty-seven.

4 Write the three numbers before ninety.

5 Copy this line and fill in the missing numbers.

41 42 || 44 45 || 47 48 || 50

TOPIC 1.1 • YOUR TURN • CONTINUED

- 6** Copy this line and fill in the missing numbers.

78 || 80 81 || 83 84 || 86 87

- 7** Which number comes straight after thirty-nine? Write it.

- 8** Which number comes straight before seventy? Write it.

- 9** Start at ninety-five. Count on to one hundred and one. Write every number you say.

- 10** Start at one hundred. Count back to ninety-four. Write them all.

TOPIC 1.1 • YOUR TURN, CONTINUED

- 11** Count the acorns in this row by pointing at each one.
Say each number out loud as you point.



Point at the first one and say ONE. Not before you touch it, and not after.

- 12** Count in tens from ten to one hundred. Now start at one hundred and count back in tens.
- 13** A page in this book has a number at the bottom. Find page twenty-nine. What is the number on the next page?

- 14** Count how many chairs are in your classroom. Say your answer to a partner, then count again a different way to check.

★ **Rook's find**

One hundred is a special number because it is the first one that needs three digits. Ninety-nine needs two. Rook says that is why the top row of Otto's grid feels so far away from the bottom one.

→ **Take it further**

Count on in ones from ninety-six all the way past one hundred to one hundred and twelve. The words change shape after a hundred, so say them slowly.

Reading and writing numbers



A numeral is the written sign.

A digit is one of the ten signs 0 to 9.

♣ A numeral, and the digits it is made of

A **numeral** is how you write a number down. A **digit** is one of the ten signs you build numerals out of: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

So **7** is one digit. **47** is two digits. **147** is three digits. The number is the amount; the numeral is the mark on the paper.

Every number also has a **word**. Forty-seven has a hyphen in the middle when you write it in words, and so do all the numbers from twenty-one to ninety-nine except the tens themselves.

ten

twenty

thirty

forty

fifty

sixty

seventy

eighty

ninety

hundred

Do you remember?

Forty has no U in it. Four does. So does fourteen. Otto gets this wrong about once a week.

 Watch me work one out 1

Write the number sixty-three in figures.

- 1** Listen to the first part of the word: sixty. That is six tens.
- 2** Write the six first, because the tens come first.
- 3** Listen to the second part: three. That is three ones.
- 4** Write the three next to the six.

Answer: sixty-three is **63**

Six tens and three ones. The chart proves it.

	Tens	Ones
63	6	3

⇌ **Check it a different way**

You worked it out by listening to the word and writing the tens first. Now try it another way.

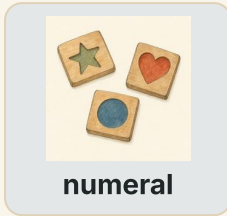
You built it from the word. Now go the other way: cover the word, look at your figures, and read them back out loud.

If your figures read as the word you started with, you wrote it correctly. If you hear sixty-three, you are right. If you hear thirty-six, your digits are the wrong way round.

How do you know?

Thirty-six and sixty-three use the same two digits. Why are they different numbers? Show your answer with the chart above.

Your turn



- 1 Write in figures: forty-two.
- 2 Write in figures: seventy-nine.
- 3 Write in figures: eighty.
- 4 Write in figures: one hundred and six.
- 5 Write 58 in words.



TOPIC 1.2 · YOUR TURN · CONTINUED

6 Write 91 in words.

7 Write 30 in words.

8 Write 115 in words.

9 How many digits are in the numeral 84? How many in 184?

10 Match each numeral to its word.

27 ●

72 ●

17 ●

70 ●

● seventy

● twenty-
seven

● seventeen

● seventy-
two

TOPIC 1.2 · YOUR TURN, CONTINUED



numeral

- 11** Otto wrote 15 when he meant fifty. Write what he should have written, and say what went wrong.
- 12** Write the numeral for the number that has nine tens and no ones.
- 13** Write every numeral from 68 to 74 in a neat row.
- 14** Find a number written in words somewhere in your classroom. Copy it, then write it in figures.



▣ SCAN TO SEE IT MOVE

Drag beads onto a bead string and count them in tens to 100. The string keeps count for you, so you can check what you counted.

mathigon.org/polypad/number

🧠 Yard challenge

Write the numeral that uses the digit 4 twice, has four tens, and is smaller than fifty. Then write it in words. There is only one answer.

Counting on and back in ones and tens



Pip's long number line on the workshop floor, marked in tens.

✿ A hop of ten does not touch the ones

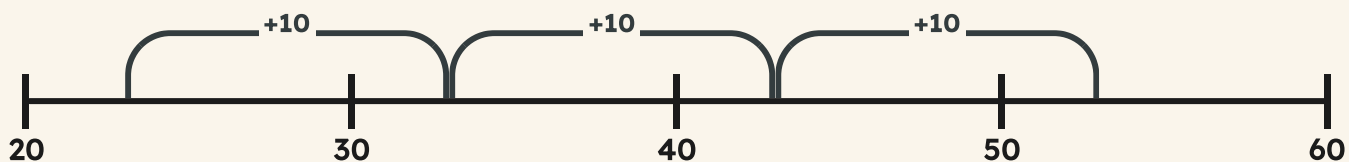
When Pip hops **one**, she lands on the very next number. When she hops **ten**, she flies right over nine numbers and lands on the tenth.

Here is the useful part. When you add ten, the **ones digit does not change at all**. Only the tens digit goes up by one.

Twenty-three, thirty-three, forty-three, fifty-three. The three stays the whole way.

TOPIC 1.3 · CONTINUED

Three hops of ten from twenty-three. Watch the three.



The ones column never moves.

	Tens	Ones
23	2	3
33	3	3
43	4	3
53	5	3

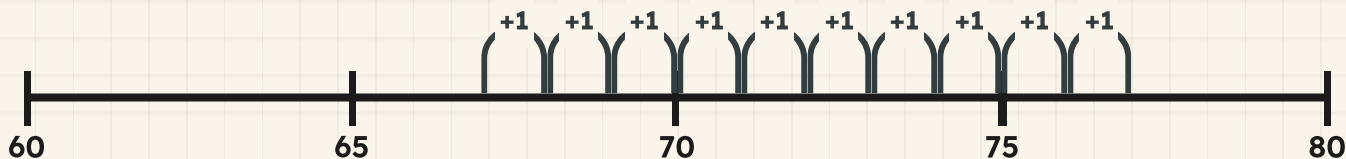
Watch me work one out 1**What is ten more than sixty-seven?**

- 1 The ones digit is seven, and adding ten will not change it.
- 2 So the answer ends in seven. Write the seven down.
- 3 The tens digit is six. Adding ten makes it one more: seven.
- 4 Seven tens and seven ones.

Answer: ten more than sixty-seven is **77****⇌ Check it a different way***You worked it out by looking at the tens and ones digits. Now try it another way.*

You used the digits. Now use the number line instead: start at sixty-seven and count on ten ones, one hop at a time, keeping count on your fingers.

Two completely different methods. If they both say seventy-seven, it is seventy-seven.

Ten single hops. Slower, and it lands in the same place.**How do you know?**

Hattie says adding ten always makes the ones digit bigger. Use the chart on the last page to show her she is wrong.

Your turn



1 Write ten more than 34.

2 Write ten more than 58.

3 Write ten less than 71.

4 Write ten less than 46.

TOPIC 1.3 · YOUR TURN · CONTINUED

5 Write one more than 89.



6 Write one less than 90.

7 Count on in tens from 6. Write the first six numbers you say.

8 Count back in tens from 95. Write every number down to 5.

9 Copy and fill in: 27, 37, 47, and then three more.

10 Copy and fill in: 84, 74, 64, and then three more.

TOPIC 1.3 · YOUR TURN, CONTINUED



- 11** Start at 8 and count on in tens past 100. Write what you say.
- 12** Pip is on 52. She hops ten twice, then one back. Where is she?
- 13** Bramble counts 40, 50, 60, 70. What is he counting on in?
- 14** Write a number between 30 and 40. Now write ten more and ten less than it.

& The material you need

squared paper and a pencil. Draw your own number line from zero to fifty, marking every tenth number, and use your ruler.

🔗 Find out together

Work with a partner. One of you says a two-digit number. The other says ten more. Then swap. Keep going for a whole minute. How many did you manage between you, and which ones made you stop and think?

First, second, third: ordinal numbers

♣ Numbers that say WHICH ONE

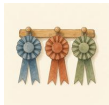
One, two, three tell you **how many**. First, second, third tell you **which one**. Numbers that say which one are called **ordinal** numbers, and they only make sense when things are in an order.

They are written short, with the last two letters of the word: 1st, 2nd, 3rd, 4th. After third they all end in **th** until you reach twenty-first.

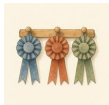
Six pegs on a rail have a first and a sixth. Six pegs in a heap have neither, because a heap has no order.



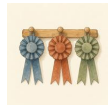
peg
1st



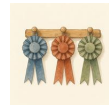
peg
2nd



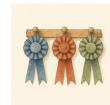
peg
3rd



peg
4th



peg
5th



peg
6th



Three rosettes on three pegs. Only the order makes them first, second and third.

 Watch me work one out 1

Six animals queue at the store. Bramble is third. How many are in front of him?

- 1** Draw the queue as six boxes in a line.
 - 2** Count along and mark the third box. That is Bramble.
 - 3** Now count only the boxes BEFORE his one.
 - 4** First and second come before third.
-

Answer: 2 animals are in front of him

 Check it a different way

You worked it out by counting forwards along the queue. Now try it another way.

You counted forwards to find him. Now count from the OTHER end: if there are six in the queue and Bramble is third, count back from six and see how many are behind him.

Two in front, Bramble, and three behind. Two, add one, add three, makes six. The whole queue is accounted for, so the answer holds together.

How do you know?

Why is the number in front of Bramble one less than his position, and not the same as it? Explain it with a drawing.

Your turn



- 1** Write the first five ordinal numbers in the short way.
- 2** Write the word for 8th.
- 3** Write the short way for the word twelfth.
- 4** Ten pegs hang in a row. Colour the fourth one.

TOPIC 1.4 · YOUR TURN · CONTINUED

5 In the same row, colour the ninth peg a different colour.



number word

6 Seven animals queue. Rook is fifth. How many are behind him?

7 Otto is second in the queue. Who is in front of him?

8 Name the third day of the week, starting on Monday.

9 Name the first month of the year, and the tenth.

10 Line up five books. Say which is first and which is last.

TOPIC 1.4 · YOUR TURN, CONTINUED



number word

- 11** Write your own position in a queue of eight, if you are last.
- 12** Twelve pegs hang in a row. Which peg is halfway along?
- 13** Tessa is fourth from the front of nine animals. What position is she from the back?
- 14** Write the 3rd, 7th and 12th places in words.

→ Take it further

Line up the whole class in height order, shortest first. Now say your own position two ways: from the front, and from the back. Add your two numbers together. Compare your total with somebody else's. Why does everybody get the same total?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1** Write in figures: sixty-one.
- 2** Write 94 in words.
- 3** Write the three numbers that come after 78.
- 4** Write ten more than 45, and ten less than 45.
- 5** Copy this line and fill the gaps: 62, 63, __, 65, __, 67.
- 6** Count back in tens from 88. Write four numbers.
- 7** Eight animals queue. Milo is sixth. How many are behind him?
- 8** Write the short way for the word seventh.
- 9** Which is the larger number, 47 or 74? Say how you know.
- 10** Start at 97 and count on to 104. Write every number.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

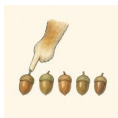
	confident	almost there	needs more work
count to 100 and back again, out loud and on a number line	☐	☐	☐
read a number and write it in figures and in words	☐	☐	☐
count on and back in ones and in tens from any number	☐	☐	☐
use ordinal numbers to say which one came first, second or tenth	☐	☐	☐

UNIT 1 • HOW DID IT GO • CONTINUED

NOW I CAN

- ☐ count to 100 and back again, out loud and on a number line
- ☐ read a number and write it in figures and in words
- ☐ count on and back in ones and in tens from any number
- ☐ use ordinal numbers to say which one came first, second or tenth

▣ YARD WORDS



count

say the numbers in order



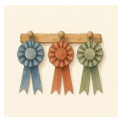
numeral

the written sign for a number



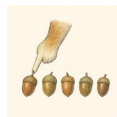
digit

one of the signs 0 to 9



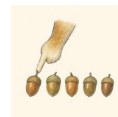
ordinal

a number that says which place



number line

a line with numbers marked in order



hundred square

a square of a hundred numbers

Tens and ones

Every number on the plan comes apart into tens and ones.



Come into the Plan Room

Before anybody shows you a chart, show what you can already do.

Count to one hundred out loud. Now count in tens: ten, twenty, thirty, all the way up. Both counts end in the same place.

Say how many tens are in forty. Say how many ones are in seven. You have just read your first two-digit number the proper way.

☀ In the yard today

Rook has chalked two numbers on the Plan Table, forty-seven and seventy-four, and he says only one of them will fit on the shelf.

They are written with exactly the same two digits. Milo says that means they are the same size. Tessa says look again, because a digit does not just say how many: it says how many **of what**.

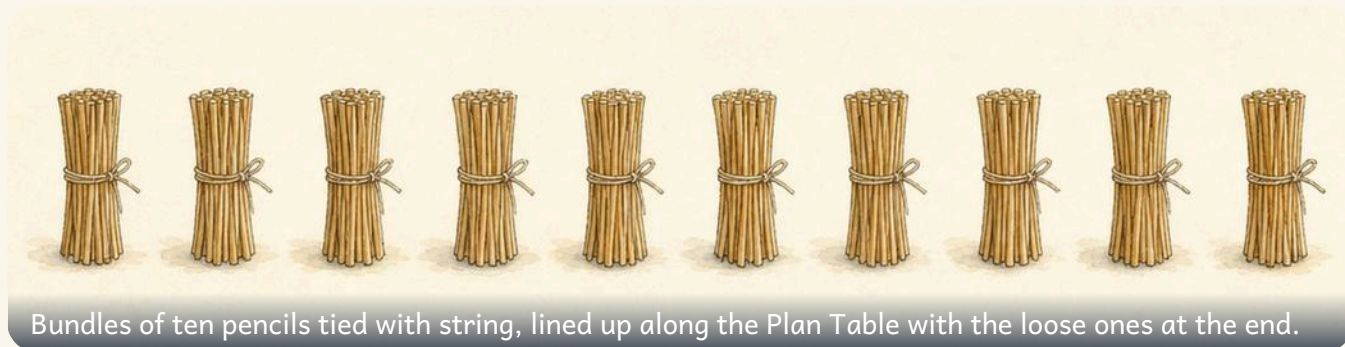
The four in one of them is four tens, says Tessa. In the other one it is four ones. That is the whole difference, and it is a big one.

✓ By the end you will

- say how many tens and how many ones are in a two-digit number
- split a number into its tens and its ones, and put it back together
- find one more, one less, ten more and ten less than any number
- place a two-digit number on a line marked in tens

Cambridge Primary Mathematics Stage 2: 2Nn6 • 2Nn7 • 2Nn10

What each digit is worth



Do you remember?

Count in tens to one hundred. Now count in ones from one to ten. Every two-digit number is made from those two counts.

♣ A digit says how many, and of what

In a two-digit number the digits are not equal. The one on the **left** counts the tens. The one on the **right** counts the ones.

So in forty-seven the four is not worth four. It is standing in the tens column, so it is worth four tens. The seven is standing in the ones column, so it is worth seven ones.

Move the same digits to the other columns and you get a different number altogether. Where a digit stands is what gives it its value.

TOPIC 2.1 · CONTINUED

The same two digits, standing in swapped columns. Read each row aloud before you go on.

	Tens	Ones
47	4	7
74	7	4

$$\boxed{10} + \boxed{6} = \boxed{16}$$

**SHARPEN
YOUR
PENCIL**

Point at a two-digit number on the wall. Say the tens digit, then the ones digit, then the whole number. Five numbers, no mistakes.

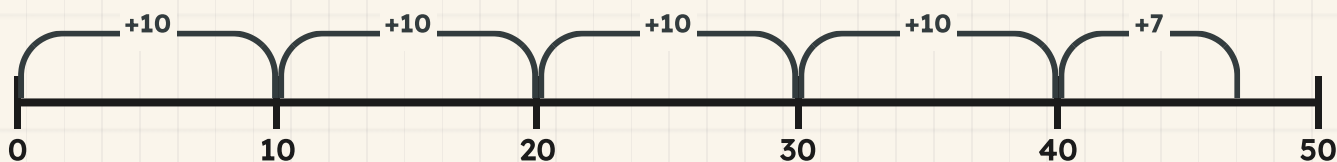
Watch me work one out 1

What is each digit in 47 worth?

- 1 Write the number in a place value chart, tens column first.
- 2 The four is in the tens column, so it counts tens. Four tens is **40**.
- 3 The seven is in the ones column, so it counts ones. That is **7**.
- 4 Put the two values together: **40** and **7**.
- 5 They make **47**, which is the number you started with. The chart has not changed it.

Answer: in **47** the four is worth **40** and the seven is worth **7**

Four hops of ten, then seven single hops. Watch where Pip lands.



⇌ Check it a different way

You worked it out by reading the digits off the place value chart. Now try it another way.

You read the value straight off the chart. Now put the chart away and count it out instead.

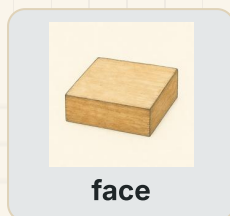
Start at nought on the line above and count on in tens: ten, twenty, thirty, **40**. That is four tens counted, not read. Now count on seven more ones, one hop at a time.

Reading a chart and counting in tens share no steps at all, and they land you in the same place. That is what makes it a real check.

How do you know?

Milo says forty-seven and seventy-four are the same size because they use the same digits. Use the chart to show him what is different, and say which number is larger.

Your turn



- 1 Write how many tens and how many ones are in 46.
- 2 Say what each digit in 72 is worth. Say the tens digit first.
- 3 Write what the 3 is worth in 35.

- 4 Copy these charts and fill in the empty columns.

	Tens	Ones
28		
55		

TOPIC 2.1 · YOUR TURN · CONTINUED

5 Ring the number with six tens.

16

61

6



face

6 Write 34, 58 and 90 in words.

7 Write in figures the number with five tens and two ones.

8 Match each number to what its tens digit is worth.

38 ●

● 90

83 ●

● 30

19 ●

● 10

91 ●

● 80

TOPIC 2.1 · YOUR TURN, CONTINUED



face

- 9** Write the larger number in each pair.

47 74

38 83

- 10** Say how many tens are in 90.

- 11** Draw counters in a chart to show 63.
Say which column holds more counters.

- 12** Build the number with eight tens and one one using tens rods.
The material you need: tens rods and ones cubes

- 13** Explain to a partner why 19 and 91 are not the same.



face

- 14** Order these numbers from smallest to largest.



- 15** Write a two-digit number where both digits are worth the same.

*** Rook's find**

Rook keeps his pencils in bundles of ten because he can read the bundles at a glance. Six bundles and three loose pencils is sixty-three without counting a single pencil. The tens digit is just a count of bundles.

→ Take it further

Take the digits 5 and 8. Make both two-digit numbers you can make from them and put them in a place value chart side by side. Write what the 5 is worth in each one, and say how much bigger one number is than the other.

Partitioning into tens and ones



Six tens rods and three loose cubes, pushed apart on the Plan Table so you can see the two parts.



Partition: split a number into its parts.

♣ Taking a number apart, and putting it back

To **partition** a number is to split it into its parts. A two-digit number splits in one obvious place: the tens go one way and the ones go the other.

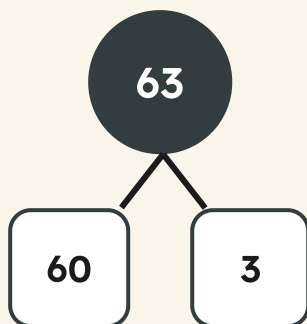
Sixty-three partitions into sixty and three. The sixty is the six tens with nothing left over, and the three is the loose ones. Say it as a sentence: sixty-three is sixty and three.

Partitioning never changes how much you have. Push the parts back together and the number you started with comes back, every time. That is why you can always check your own work.

TOPIC 2.2 · CONTINUED

$$63 = 60 + 3$$

The whole on top, the two parts underneath. Read it downwards, then read it upwards.



TOPIC 2.2 · CONTINUED

- 1 Say the number out loud.
- 2 Write the tens part: the tens digit, then a nought.
- 3 Write the ones part: just the ones digit.
- 4 Add the two parts back together to make sure nothing was lost.

Watch me work one out 1

Partition 63 into tens and ones.

- 1 Say it: sixty-three. The word tells you the tens part is sixty.
- 2 Write the tens part first: six tens is **60**.
- 3 Now the ones part: there are **3** loose ones.
- 4 So the two parts are **60** and **3**.
- 5 Push them back together: **60** and **3** makes **63**, exactly what you started with.

Answer: $63 = 60 + 3$

⇌ Check it a different way

You worked it out by reading the digits and writing a part for each one. Now try it another way.

You partitioned by reading the digits and writing a part for each one. Now prove it with your hands instead of your eyes.

Build the number out of tens rods and loose cubes. Count the rods in tens, then count the cubes in ones, then push the whole lot back into one pile and count it again.

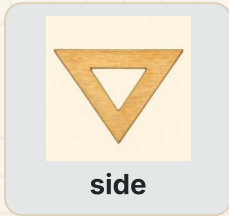
One method reads a written number. The other one builds it and rebuilds it. When the pile you rebuild is the number you started with, your two parts were the right two parts.



How do you know?

Hattie partitioned ninety and wrote ninety and nought. Otto says there is no ones part at all, because **0** ones is nothing. Who is right, and does it matter?

Your turn



1 Partition 47 into tens and ones.

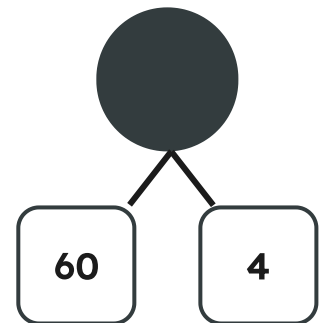
2 Partition 82. Write both parts down.

3 Copy and complete this partition.

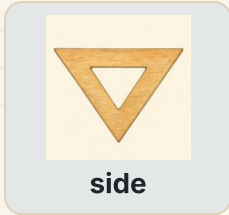
$$35 = \underline{\quad} + \underline{\quad}$$

4 Write the two parts of 59.

5 Add these two parts back together to find the whole.



TOPIC 2.2 · YOUR TURN · CONTINUED



6 Partition 76, then rebuild it by adding your parts.

7 Partition every number in this bank.

24

48

71

95

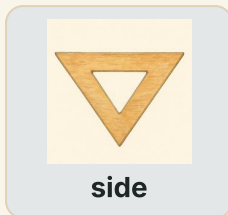
8 Ring the pair of parts that make 48.

forty and eight

four and eight

eighty and four

TOPIC 2.2 · YOUR TURN, CONTINUED



9 Sort these numbers by their tens digit.

Six tens

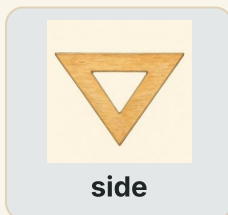
Seven tens

10 Partition 90. Say what happens to the ones part.

11 Draw tens rods and ones cubes to show 34.

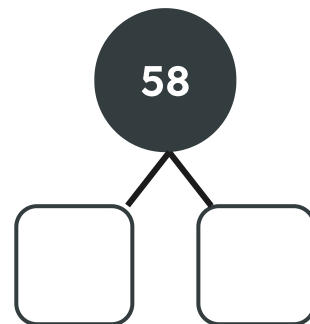
12 Partition 65 and 56. Say what is different about the parts.

13 Write in words the two parts of 27.



14 Find two numbers whose tens part is sixty.

15 Complete this part-whole model.



✂ The material you need

tens rods, loose cubes and a pot for the ones. Build every number you partition, so your hands agree with your pencil before you write anything down.

🔗 Find out together

Work with a partner. One of you calls a two-digit number and the other builds it in rods and cubes as fast as they can. Swap after each go, and play ten rounds. Then talk about it: which numbers were quickest to build, and why were the ones ending in nought so easy?

One more, one less, ten more, ten less



♣ A small step and a big step

One more is the next number along. One less is the number just behind you. On a track you move a single square, and only the ones digit changes.

Ten more is a whole bundle more. You do not count ten squares one at a time: you drop straight down to the number below on a hundred square, or you take one big hop on a number line.

Here is the useful part. Ten more changes the **tens** digit and leaves the ones digit exactly as it was. Fifty-four, sixty-four, seventy-four. The four never moves.

Two big hops of ten from fifty-four. Say each number you land on out loud.



Do you remember?

Adding ten does not touch the ones digit. If your ones digit changed, you took the wrong size hop.

 Watch me work one out 1

Find ten more and ten less than 54.

- 1** Partition fifty-four first, so you can see what you are changing.
- 2** The tens part is **50** and the ones part is **4**.
- 3** Ten more means one more ten. Six tens is sixty, and the **4** ones stay, giving **64**.
- 4** Ten less means one ten fewer. Four tens is forty, and the ones stay again, giving **44**.
- 5** Check the ones digit in all three numbers. It is **4** every time.

Answer: ten more than **54** is **64**, and ten less is **44**

Ten less, the number itself, then ten more. One column is doing all the changing.

	Tens	Ones
44	4	4
54	5	4
64	6	4

⇌ Check it a different way

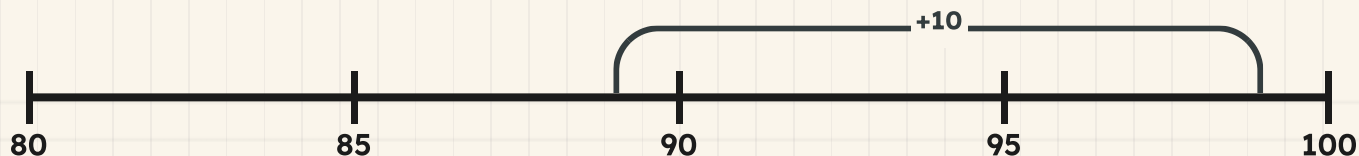
You worked it out by partitioning and changing the tens part. Now try it another way.

You partitioned the number and changed the tens part. Now leave the digits alone and walk it out on the line above.

Put your finger on fifty-four and take one hop of ten forwards. Then go back to fifty-four and take one hop of ten backwards. Read the two numbers you land on.

One method changes a digit. The other moves along a line and never mentions digits at all. Both of them give you a ones digit of **4**, so you were right.

Ten more than eighty-nine. There IS room, and it is ninety-nine.



How do you know?

Bramble says ten more than eighty-nine cannot be worked out, because there is no room for another ten. Use the number line above to show him where he lands, and say what happens to the tens digit.

▣ WORDS FROM THIS PAGE



more



less



jump



tens

Your turn

1 Write one more than 46.

2 Write ten more than 46.

3 Write one less and ten less than 73.

4 Fill in the missing numbers on this track.

70 71 73 74 76 77 79

5 Write ten more than every number in this bank.

17

38

60

85

6 Count on ten from 38. Write the number you land on.

7 Write ten less than 90.

8 Complete these two sentences.

$$54 + 10 = \boxed{}$$

$$54 - 10 = \boxed{}$$

9 Copy this line and mark ten more than 45.

1
50

10 Ring the number that is one less than 60.

61

59

50

11 Start at 27 and count on in tens four times.

12 Say which digit changes when you add ten.

13 Ring
TRUE
or
NOT
TRUE
for
each
line.

One more changes the ones digit.

TRUE / NOT TRUE

Ten more changes the ones digit.

TRUE / NOT TRUE

Ten less takes away one whole ten.

TRUE / NOT TRUE

One less is the number just behind you.

TRUE / NOT TRUE

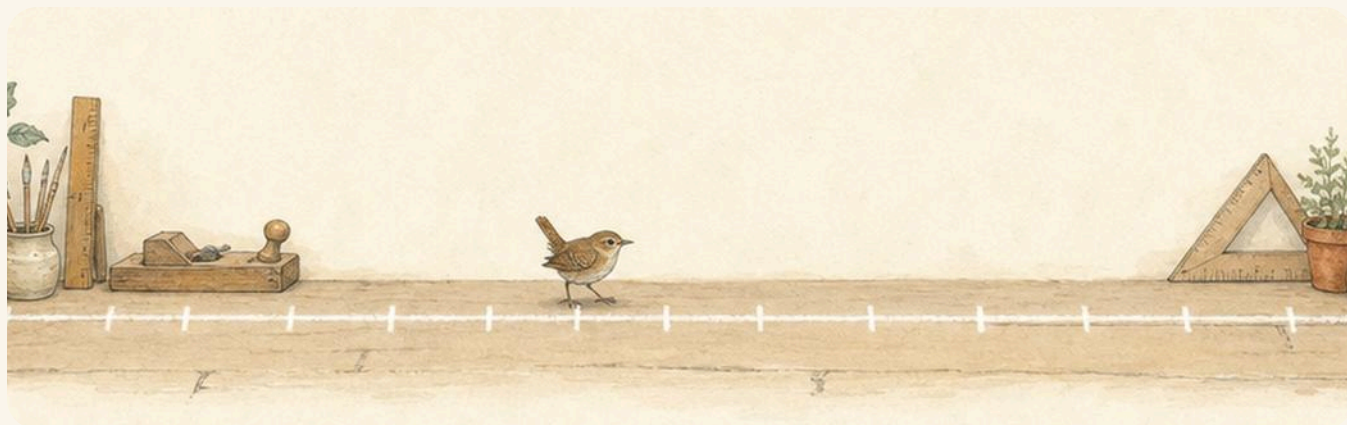
14 Write one more than 79 in words.

15 Play ten more, ten less with a partner. Take turns to call.

🏡 Yard challenge

Start at 6 and keep taking ten more until you cannot go any further without passing one hundred. Write every number you say. Then look at your list: what is the same about every single ones digit, and why did that have to happen?

Putting a number on a number line



A long chalk line down the middle of the yard, marked at every ten, with Pip standing on one of the marks.

♣ Every number has one place on the line

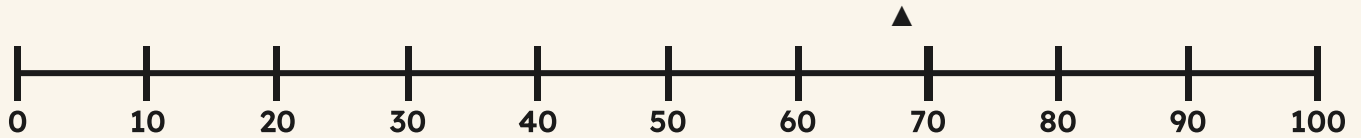
A number line is not a list. It is a road, and every number has its own place on it. The bigger the number, the further along it sits.

Most lines are only labelled at the tens, because labelling all one hundred numbers would leave no room to write. So to place a number you first find the two tens it lies **between**.

Sixty-eight lies between sixty and seventy. Then ask yourself which end it is nearer. Eight ones is most of the way across, so sixty-eight sits close to seventy, not halfway.

TOPIC 2.4 · CONTINUED

The whole road, nought to one hundred, labelled at every ten. One number is marked. Find it.



Do you remember?

Halfway between two tens is the number ending in five. Anything past it is nearer the top ten.

How do you know?

Tessa marked forty-three exactly halfway between forty and fifty. Otto says it belongs much closer to forty. Use the marks on the line to say who is right, and explain how the ones digit told you.

$$68 = 60 + 8$$

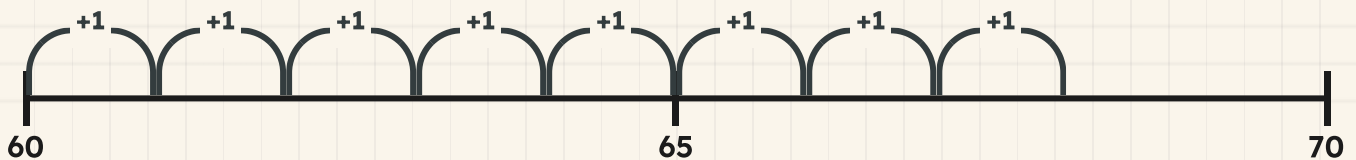
 Watch me work one out 1

Put 68 on a line marked in tens.

- 1 Partition it first, so you know which tens to look at.
- 2 Sixty-eight is **60** and **8**, so it lies between sixty and seventy.
- 3 Find sixty on the line and start there. Now count on **8** ones.
- 4 You stop just before the next label, on **68**.
- 5 Ask which ten it is nearer. It is nearer **70**, because eight ones is more than halfway across.

Answer: 68 goes between **60** and **70**, close to **70**

Start on sixty and count on the ones. The middle mark is there to help you judge.



↔ Check it a different way

You worked it out by counting on in ones from the ten below. Now try it another way.

You counted forwards from the ten below. Now come at it from the other end.

Put your finger on seventy and count **back** in ones: sixty-nine, sixty-eight. Two hops back and you are there, which tells you something the first method never said out loud, that the number is only two away from seventy.

Counting on from below and counting back from above are opposite journeys, and they stop on the same mark. Two hops back is fewer than eight hops forwards, so **70** really is the nearer ten.

**■ SCAN TO SEE IT MOVE**

Build a two-digit number in a place-value chart, then pull the tens and the ones apart again and watch the number change.

www.topmarks.co.uk/place-value/place-value-charts

Your turn

1 Mark 68 on a line marked in tens.

↓
80

2 Write the numbers the empty labels should show.

↓
10

3 Mark 35 on this line.

↓
20

TOPIC 2.4 · YOUR TURN · CONTINUED

4 Say which two tens 47 lies between.

5 Write the nearest ten to 62.

6 Mark 90 and 45 on the same line.



7 Order these numbers from smallest to largest.



8 Fill in the missing numbers on this track.



TOPIC 2.4 · YOUR TURN, CONTINUED

- 9** Mark the number halfway between 40 and 50.

|
40

- 10** Draw a line from 0 to 100 and label every ten.

The material you need: squared paper and a ruler

- 11** Ring the number nearer to 70.

66

74

81

- 12** Estimate where 26 belongs on this line.

|
100

- 13** Write in words the number halfway between 60 and 70.

14 Explain how a number line helps you compare two numbers.

15 Choose a two-digit number and place it on a line in tens.
Say which ten it is nearer, and how you know.

★ **Rook's find**

Rook draws his lines with only the tens labelled, because a line with every number on it is too crowded to read. He says the mark halfway between sixty and seventy, **65**, is the only extra one worth chalking in, and it does half the judging for you.

→ **Take it further**

Chalk a line from nought to one hundred along the yard, marked only at nought, fifty and one hundred. Take turns to stand where a number called by a partner belongs, then check by counting the tens along from the nearest label you can see.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1** Say what the 7 is worth in 71, and what it is worth in 17.
- 2** Write in figures the number with nine tens and four ones.
- 3** Partition 86 into tens and ones.
- 4** Add the parts sixty and nine back together.
- 5** Write ten more than 55.
- 6** Write one less than 40.
- 7** Write ten less than 61, then ten less again.
- 8** Mark 24 on a line labelled at every ten.
- 9** Say which two tens 79 lies between, and which one it is nearer.
- 10** Explain how you would check a partition a completely different way.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
say how many tens and how many ones are in a two-digit number	☐	☐	☐
split a number into its tens and its ones, and put it back together	☐	☐	☐
find one more, one less, ten more and ten less than any number	☐	☐	☐
place a two-digit number on a line marked in tens	☐	☐	☐

UNIT 2 · HOW DID IT GO · CONTINUED

NOW I CAN

- ☐ say how many tens and how many ones are in a two-digit number
- ☐ split a number into its tens and its ones, and put it back together
- ☐ find one more, one less, ten more and ten less than any number
- ☐ place a two-digit number on a line marked in tens

▣ YARD WORDS



tens

how many whole tens a number
has



ones

the ones left after the tens



partition

split a number into parts



place value

what a digit is worth where it
stands



more

a bigger amount



less

a smaller amount

Comparing, ordering and rounding

Two lengths of timber. Which is longer, and how do you know?

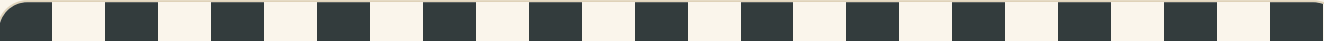


Come into the Plan Room

Before anybody shows you a sign or a rule, show what you can already do.

Count aloud from sixty to seventy. Now count back from seventy to sixty. Say what forty-six is made of, in tens and ones.

Point to the bigger of these two heaps of acorns. You have just compared, and nobody had to teach you how.



☼ In the yard today

Otto has laid two lengths of timber on the Plan Table, and Hattie cannot decide which of them is longer.

Hattie holds them up one at a time and squints. Otto stands them side by side, ends level, against the wall. From across the yard you can see which is longer without measuring anything at all.

Comparing is standing things next to each other, says Otto. You can compare numbers the same way. Line them up, then look at the tens.

✓ **By the end you will**

- compare two numbers and write the $>$ or $<$ sign between them
- put numbers in order from smallest to largest and back again
- round a two-digit number to the nearest ten
- make a sensible estimate, then count to check it

Cambridge Primary Mathematics Stage 2: 2Nn8 • 2Nn9 • 2Nn12 • 2Nn13

Greater than and less than



Do you remember?

Forty-six is four tens and six ones. Say what sixty-four is made of, and say which of those two numbers has more tens.

♣ Two signs, and the hungry end

When two numbers are not the same, one of them is **greater than** the other and one of them is **less than** the other. There is a sign for each.

The sign $>$ means greater than. The sign $<$ means less than. Both signs are wide at one end and pointed at the other, and the **wide end always opens towards the bigger number**. The point always aims at the smaller one.

So you never have to remember which sign is which. Work out which number is bigger, then open the sign towards it.

TOPIC 3.1 · CONTINUED

34

71

80

58

69

8

The same two digits, standing in different columns. The column a digit stands in is what decides its worth.

	Tens	Ones
46	4	6
64	6	4



Two lengths of timber held upright side by side with their ends level. One is clearly the longer.

**SHARPEN
YOUR
PENCIL**

Call out any two-digit number. A partner calls out one that is greater. Take five turns each, then swap and ask for a number that is less instead.

✎ Watch me work one out 1

Write the correct sign between 37 and 52.

- 1** Look at the tens first, because tens are worth more than ones.
- 2** Thirty-seven has **30** in its tens. Fifty-two has **50** in its tens.
- 3** **50** is more than **30**, so fifty-two is the bigger number. The ones do not get a vote.
- 4** Ignore the **7** and the **2** altogether. They cannot rescue thirty-seven.
- 5** Now open the wide end of the sign towards fifty-two.

Answer: 37 < 52

⇌ Check it a different way

You worked it out by comparing the tens digits. Now try it another way.

Put the tens digits away and use the line instead. Find thirty-seven on the number line below, then find fifty-two.

On a number line the numbers get bigger as you walk to the right. Fifty-two sits further right, so fifty-two is the greater number, and the sign opens the same way it did before.

You compared two digits the first time and two positions the second time. When two methods that share no steps agree, you can stop worrying.

The tinted stretch runs from thirty-seven up to fifty-two. Everything to the right is greater.

**How do you know?**

Milo says 80 must be smaller than 8, because 8 is a smaller digit than 80 has room for. Use tens and ones to show him where his thinking goes wrong.

Your turn



1 Write the correct sign in each box. $46 \square 64$ $39 \square 31$

2 Ring the greater number in each pair.

58 or 85

27 or 72

90 or 19

3 Write the correct sign between 27 and 72.

4 Copy these and write the sign in the middle of each. $85 \square 58$
 $19 \square 91$

- 5** Say which is less, 63 or 36. Tell a partner how you know.



half

- 6** Compare 50 and 5. Explain why the digit 5 does not decide it.

- 7** Write a number that is greater than 47 and less than 60.

- 8** Ring
TRUE
or
NOT
TRUE
for
each
line.

The wide end of the sign opens towards the bigger number.

TRUE / NOT TRUE

You compare the ones digits before the tens digits.

TRUE / NOT TRUE

71 is greater than **17**.

TRUE / NOT TRUE

Two numbers with the same tens digit must be equal.

TRUE / NOT TRUE

TOPIC 3.1 · YOUR TURN, CONTINUED



half

9 Write the sign between each pair below. **44** **4068** **86**

10 Write two numbers that both have 6 tens. Ring the greater one.

11 Bramble has 58 pegs. Otto has 62 pegs. Say who has more.

12 Write three numbers that are all less than 30.

13 Mark 46 and 64 on this line. Say which is further right.

!
30



half

14 Write a number that is greater than 89.

15 Explain to a partner what the sign $>$ tells you.
Say greater than out loud every time you use it.

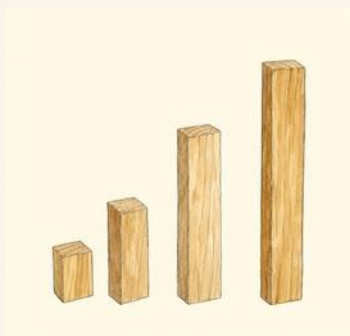
★ Rook's find

Rook keeps his ledger in two columns and never compares the ones first. He says a number with more tens wins before the ones are even read, and the only time the ones matter is when the tens are the same. That is the whole of comparing, in one line of a ledger.

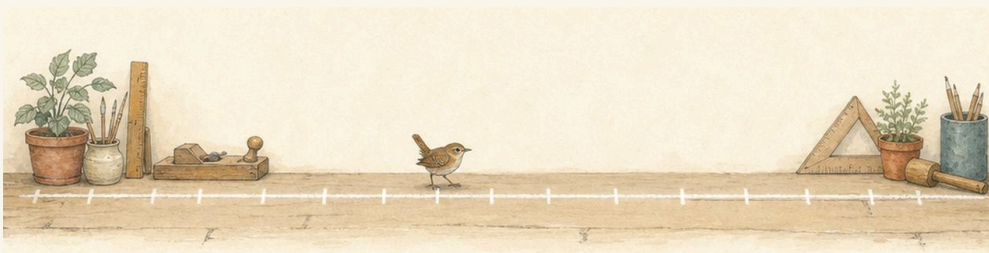
→ Take it further

Take the digits 3 and 8. Make the two different two-digit numbers you can build from them, then write the correct sign between them. Now try it with 4 and 9, and with 2 and 7. Say what is always true about the number that turns out to be greater.

Ordering numbers to 100



Four timber offcuts lined up from shortest to longest.



The long painted number line on the workshop floor, walked from one end to the other.

✿ Ordering is comparing, over and over

To **order** a set of numbers you put them in place, one after another, from smallest to largest or from largest to smallest. You must say which way round you mean, or nobody knows where to start.

Find the smallest number in the whole set and write it first. Cross it off. Now find the smallest of the ones that are left. Keep going until there is nothing left to cross off.

Look at the tens digits to sort them out quickly. Only when two numbers have the same tens digit do you need to look at their ones.

TOPIC 3.2 · CONTINUED

46

19

62

37

Write these four numbers in order, smallest first, one to a box.

--	--	--	--

Find all four numbers on the line. Reading the line from left to right puts them in order for you.



Do you remember?

Smallest to largest goes the same way as counting on. Largest to smallest goes the same way as counting back.

 Watch me work one out 1

Order 46, 19, 62 and 37 from smallest to largest.

- 1** Read the tens digit of each one: four tens, one ten, six tens, three tens.
- 2** The fewest tens is one ten, so **19** goes first.
- 3** Cross it off and look again at the three that are left.
- 4** Three tens is next, then four tens, and six tens is the most.
- 5** So the last one to be written down is **62**, and nothing is left to cross off.

Answer: 19, 37, 46, 62

- 1** Find the smallest number in the set.
- 2** Write it down and cross it off the list.
- 3** Look again at the numbers that are left.
- 4** Keep going until the list is empty.
- 5** Read your order back to check it climbs all the way.

↔ Check it a different way

You worked it out by ordering by looking at the tens digit first. Now try it another way.

Build them instead of reading them. Make four towers of cubes, one tower for each number, and stand the four towers in a row against the wall with their bottoms on the bench.

Now shuffle the towers until they climb like a staircase. Read the numbers off the towers from the shortest end to the tallest end. No digits were compared at all, and the order is the same one.

Digits are quick. Towers are certain. Use the digits to get an order and the towers to prove it.

How do you know?

Hattie ordered 9, 19, 90 and 91 by counting how many letters each number word has. She got the wrong order. Say what she should have looked at instead.

✂ The material you need

a pot of cubes, four small cards and a pencil. Write one number on each card, build a tower of cubes to match each card, then stand the towers in a row and slide the cards to their towers.

Your turn

- 1** Order these from smallest to largest: 27, 72, 17 and 71.



- 2** Order these from largest to smallest: 45, 54, 35 and 53.



- 3** Write the missing numbers on this track.



- 4** Order these from smallest to largest.



TOPIC 3.2 · YOUR TURN · CONTINUED

5 Write the number that comes between 68 and 70.

6 Order these from largest to smallest: 90, 19, 91 and 9.

7 Match each number to the pair of tens it sits between.

34 ●

● 70 and 80

58 ●

● 20 and 30

71 ●

● 30 and 40

26 ●

● 50 and 60

8 Write these numbers in figures: thirty-six, sixty-three, ninety.

TOPIC 3.2 · YOUR TURN, CONTINUED

- 9** Write 28 and 82 in words.
- 10** Order these lengths from shortest to longest: 35 cm, 53 cm, 15 cm.
- 11** Order these from smallest to largest, then read them backwards.
Use 61, 16, 66 and 26.
- 12** Find the smallest number you can make with the digits 4 and 7.
- 13** Order these from smallest to largest: 100, 89, 98 and 9.



14 Write the number that is one less than 70.

15 Tell a partner how you decide which number goes first.

Smaller than fifty

Larger than fifty

▣ WORDS FROM THIS TOPIC



compare



order



greater than



less than

Rounding to the nearest ten

Every number in the tinted half is nearer to fifty. Every number outside it is nearer to forty.



♣ Which ten is it nearest to?

To **round** a number to the nearest ten you swap it for the ten it sits closest to. Forty-seven becomes a ten, and it becomes the ten that is nearest to it, not just any ten.

Every two-digit number sits between two tens. Look at the **ones digit** to decide which ten is nearer. If the ones digit is 1, 2, 3 or 4 the number rounds **down** to the ten below. If it is 6, 7, 8 or 9 it rounds **up** to the ten above.

And if the ones digit is exactly 5, the number sits in the middle, so the rule has to choose. **A five always rounds up.** Everybody uses the same rule, so everybody gets the same answer.

Do you remember?

Rounding does not change how much you really have. It gives you a tidy number to think with, and you keep the real one too.



A wooden number tile with an arrow curving up to a taller peg, showing a number moving to its nearest ten.

**SHARPEN
YOUR
PENCIL**

Say the tens in order: ten, twenty, thirty, all the way to a hundred. Now say them backwards. These are the only numbers a rounded answer is allowed to be.

✎ Watch me work one out 1

Round 47 to the nearest ten.

- 1** Find the two tens it sits between: **40** below it and **50** above it.
- 2** Look at the ones digit. It is **7**.
- 3** **7** is more than five, so the number is past the middle of the stretch.
- 4** Past the middle means nearer the top, so it rounds up to **50**.
- 5** Say it as a sentence: forty-seven to the nearest ten is **50**.

Answer: 47 rounded to the nearest ten is 50

↔ Check it a different way

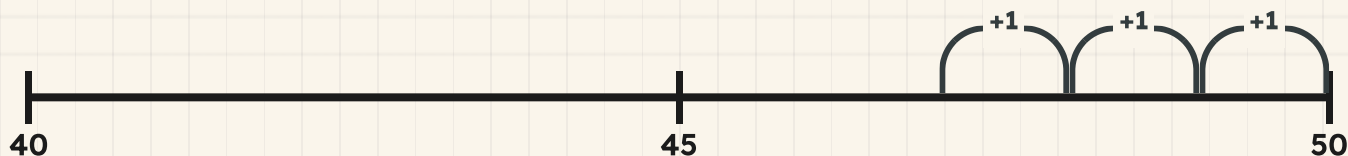
You worked it out by using the rule on the ones digit. Now try it another way.

Forget the rule and count the gaps instead. Put your finger on forty-seven on the line below and hop up in ones to fifty, counting your hops.

Now start on forty-seven again and hop down in ones to forty, counting those hops too. Compare the two counts. The shorter journey is the nearer ten, and it is the same **50** the rule gave you.

The rule is fast and the hops are honest. When a rounding surprises you, count the hops.

Three hops up to fifty. Count the hops down to forty and compare the two journeys.



How do you know?

Otto says **45** could round either way because it is exactly in the middle. Say what the rule does with a five, and why every yard in the world agreeing on one rule is useful.



▣ SCAN TO SEE IT MOVE

A fresh pair of numbers to compare every time you press it, so you never run out of practice at greater than and less than.

www.topmarks.co.uk/ordering-and-sequencing/coconut-ordering

Your turn

1 Round 34 to the nearest ten.

2 Round 47 to the nearest ten.
Use this line to help you.

|
40

3 Round 65 to the nearest ten. Remember what a five does.

4 Round each of these to the nearest ten: 22, 38 and 51.

5 Ring the numbers
that round to
forty.

36

44

46

41

6 Round 85 to the nearest ten.

7 Write two different numbers that both round to 60.

8 Round 19 to the nearest ten. Name the two tens it sits between.

TOPIC 3.3 · YOUR TURN, CONTINUED

9 Round these to the nearest ten: 73, 46 and 95.

10 Draw a line from 20 to 30 and mark 25 on it.

11 Round 8 to the nearest ten.

12 Otto counted 62 nails. Round his count to the nearest ten.

13 Ring
TRUE
or
NOT
TRUE
for
each
line.

A rounded answer is always a ten.	TRUE / NOT TRUE
A number ending in 5 rounds down.	TRUE / NOT TRUE
Rounding changes how much you really have.	TRUE / NOT TRUE
A number ending in 2 rounds down.	TRUE / NOT TRUE

14 Write a number that rounds up and a number that rounds down.

15 Explain the rule for rounding when the ones digit is 5.

Find out together

Work with a partner and a pot of number cards from 1 to 99. Turn one card over each and round it to the nearest ten. Whoever rounded to the larger ten keeps both cards. Play until the pot is empty.

Then look through the cards you kept. How many of them rounded up, and how many rounded down? Say what the ones digits of the ones that rounded up all have in common.

Making a sensible estimate

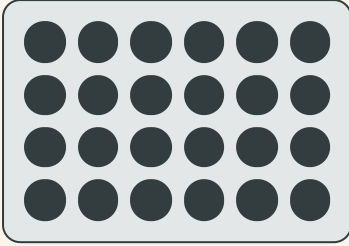
♣ A sensible guess, made on purpose

An **estimate** is a sensible guess about how many there are, made before you count. It is not a wild guess and it is not cheating. It is a real skill, and Rook uses it every day.

To estimate well, find something you already know the size of. Look at ten acorns, then look at the whole heap and ask yourself how many tens like that would fit. If about three would fit, then about thirty is a sensible estimate.

Then **count**, and see how close you were. An estimate tells you what sort of answer to expect, so when your count comes out wildly different you know to look again. That is why an estimate is worth making.

TOPIC 3.4 · CONTINUED



Otto tipped the acorns out on the Plan Table. Look for a few seconds, then look away and estimate.



A jar of acorns on the bench, with nothing written on it at all.

Do you remember?

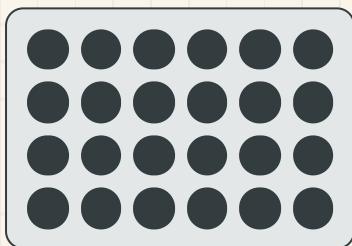
Sensible means it could be true. About thirty acorns in a heap is sensible. About three, or about three hundred, is not.

Watch me work one out 1

Estimate how many acorns are on the Plan Table, then count.

- 1** Do not count yet. Look for a moment and cover them up.
- 2** Look at one row. There are about five or six in a row, and there are about four rows.
- 3** About four rows of about five is about **20**, so about **20** is a sensible estimate.
- 4** Now uncover them and count properly, in rows so you do not lose your place.
- 5** The real count is **24**, and rounding that to the nearest ten gives **20**.

Answer: about 20 acorns, and the count was 24



$$6 + 6 + 6 + 6 = 24$$

Counted in rows, the way you would count them a second time.

⇌ Check it a different way

You worked it out by estimating by eye, in rows. Now try it another way.

Your first estimate was made by eye. Now make one by grouping. Push the acorns into little piles of ten, and see how many full piles of ten you can make.

Count the piles of ten, then count what is left over. That gives you a number without ever counting all of them one at a time, and it lands on the same **24**.

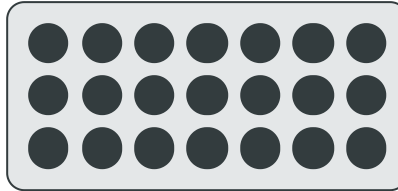
This is the check Cambridge asks for in 2Pt11: estimate, then count, then ask whether the answer is reasonable. If your count is nowhere near your estimate, one of the two is wrong and it is worth finding out which.

How do you know?

Hattie estimated three hundred acorns and then counted **24**. Say what she should have done before guessing, and how the count told her the estimate was not sensible.

Your turn

- 1** Estimate how many counters are here, then count them.



- 2** Estimate the pegs on the bench, then count them in tens.

The material you need: a handful of pegs

- 3** Choose the most sensible estimate for the acorns in a jar.

about five

about thirty

about three hundred

TOPIC 3.4 · YOUR TURN · CONTINUED

4 Estimate how many cubes fill a tray, then fill it and count.

5 Estimate how many steps cross the room, then walk and count.

6 Write a sensible estimate for the pencils in a full pot.

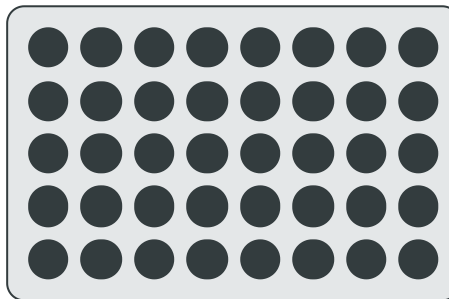
7 Ring the sensible estimate for how tall a Year 2 child is.

12 cm

120 cm

1200 cm

8 Estimate how many acorns, then count them in groups of ten.



TOPIC 3.4 · YOUR TURN, CONTINUED

- 9** Say whether 100 is a sensible estimate for pupils in your class.
- 10** Estimate the length of your table in metres, then measure it.
- 11** Estimate how many books are on one shelf, then count them.
- 12** Round your count from question 11 to the nearest ten.
- 13** Write an estimate that is not sensible, and say why not.
- 14** Estimate the beads in a pot, then count and compare the two.
Say whether your estimate was close, or too big, or too small.

15 Tell a partner why an estimate is useful before you count.

→ **Take it further**

Rook estimates a heap by looking at ten and asking how many tens would fit. Try his method on a heap of pegs, then try it again with a heap twice as big. Say whether the method got harder, and whether your estimates got further from your counts.

 Yard challenge

Fill a jar with acorns and write your estimate on a slip of paper without showing anybody. Ask three friends to do the same. Now count the acorns in tens, round the count to the nearest ten, and order all four estimates from smallest to largest. Whose estimate landed nearest to the rounded count, and can they explain what they looked at first?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1** Write the correct sign between 58 and 85.
- 2** Say which is less, 74 or 47, and how you know.
- 3** Order these from smallest to largest: 30, 13, 31 and 3.
- 4** Order these from largest to smallest: 66, 6, 60 and 16.
- 5** Round 43 to the nearest ten.
- 6** Round 76 to the nearest ten.
- 7** Round 55 to the nearest ten, then say what the rule did.
- 8** Estimate how many pegs fill a tin, then count them in tens.
- 9** Choose a sensible estimate for how wide a door is.
- 10** Explain how you would check a rounding a completely different way.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
compare two numbers and write the $>$ or $<$ sign between them	☐	☐	☐
put numbers in order from smallest to largest and back again	☐	☐	☐
round a two-digit number to the nearest ten	☐	☐	☐
make a sensible estimate, then count to check it	☐	☐	☐

UNIT 3 · HOW DID IT GO · CONTINUED

NOW I CAN

- ☐ compare two numbers and write the $>$ or $<$ sign between them
- ☐ put numbers in order from smallest to largest and back again
- ☐ round a two-digit number to the nearest ten
- ☐ make a sensible estimate, then count to check it

▣ YARD WORDS



compare

look at two things to see which is bigger



greater than

bigger than, written $>$



less than

smaller than, written $<$



order

put in place from smallest to largest



round

change to the nearest ten



estimate

a sensible guess before you count

UNIT 4

Steps, odds and evens

Rook counts the rack in twos, then in fives, then in tens.



Come into the Plan Room

Before anybody teaches you a step, show what your voice already knows.

Count out loud in twos as far as you can go. Now count in tens. Your mouth has done both of these before.

Put ten counters on the table and try to split them into two equal piles. Now try it with nine. One of those jobs works and one of them leaves something over, and that is the whole of Topic 4.3.

In the yard today

Rook has forty-two pegs to count and he refuses to count them one at a time.

He slides them along the Plan Table in twos. Then he sweeps them back and slides them along in fives. Then he does it once more in tens, and he is finished before Milo has reached twenty counting in ones.

Counting in ones is honest work, says Rook, and it is slow work. Take bigger steps and you get the same total sooner, with fewer places to lose your finger.

✓ **By the end you will**

- count in twos, fives and tens, and group things to count them faster
- count on in threes and fours and carry a pattern on
- say whether a number is odd or even, and show why
- sort numbers into groups and say the rule you used

Cambridge Primary Mathematics Stage 2: 2Nn4 • 2Nn5 • 2Nn14 • 2Nn15 • 2Pt8

Counting in twos, fives and tens



Do you remember?

Say the counting numbers from nought to twenty. Now say only every second one. Your voice can already skip.

♣ Bigger steps, same total

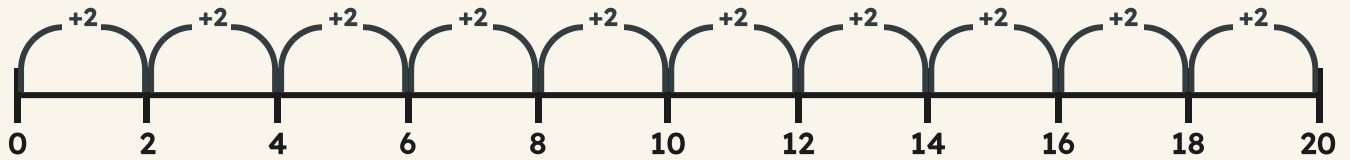
When you count in ones you say every number. When you **skip-count** you say only the numbers you land on, and you get to the total in far fewer goes.

The number you count on each time is the **step**. A step of two, a step of five, a step of ten. The numbers you land on are called **multiples** of your step.

Skip-counting only works if the things are grouped in your step size. Group them first, then count the groups. That is why Rook slides his pegs into pairs before he opens his mouth.

TOPIC 4.1 · CONTINUED

Ten hops of two. Pip says only the numbers she lands on.



Counting in twos, written down as a table.

$$1 \times 2 = 2$$

$$2 \times 2 = 4$$

$$3 \times 2 = 6$$

$$4 \times 2 = 8$$

$$5 \times 2 = 10$$

$$6 \times 2 = 12$$

$$7 \times 2 = 14$$

$$8 \times 2 = 16$$

$$9 \times 2 = 18$$

$$10 \times 2 = 20$$

**SHARPEN
YOUR
PENCIL**

Count round the class in fives. When your teacher claps, the next person counts backwards in fives instead. Keep the step, change the direction.

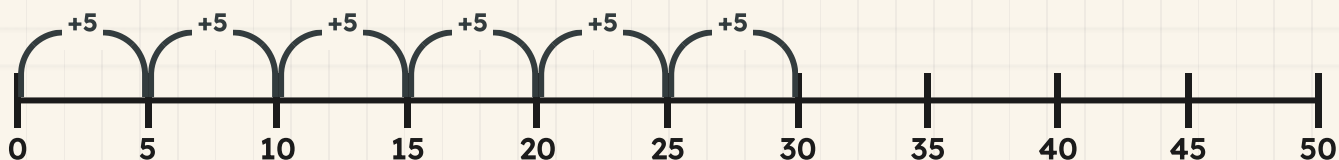
Watch me work one out 1

Count these thirty cubes in fives.

- 1 The cubes are already sitting in piles of five, so use a step of five.
- 2 Start at nought and put your finger on the first pile.
- 3 Hop five for each pile: five, ten, fifteen, twenty.
- 4 Two more piles to go, so two more hops of five.
- 5 6 hops of five land you on **30**.

Answer: there are **30** cubes

Six hops of five. Count the hops, and read where Pip stops.



⇌ Check it a different way

You worked it out by counting on in fives along the number line. Now try it another way.

You counted along the line. Now leave the line alone and go back to the cubes themselves.

Push the piles apart so you can see each one, and count the **piles**: one, two, three, four, five, six. Six piles, and every pile holds five. Now count them in fives with your finger on each pile as you say it.

One method used a number line and no cubes. The other used cubes and no number line. They share no steps at all, and they agree, so the total is right.

How do you know?

Milo counted the same cubes in ones and got a different total. Rook says Milo lost his place. Why is a count in fives harder to lose your place in than a count in ones?

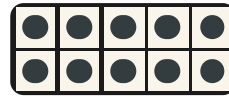
Your turn

- 1 Count in twos from 0 to 20. Write down every number you say.
- 2 Count in fives from 0 to 50. Say each number out loud.
- 3 Count in tens from 0 to 100 with a partner, one number each.
- 4 Write the missing numbers on this counting line.

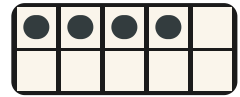


TOPIC 4.1 · YOUR TURN · CONTINUED

- 5 Count these counters in twos.



the first tray



the loose ones

- 6 Count in fives along this line. Ring every number you land on.



- 7 Group these ten socks into pairs, then count the pairs.

Say how many pairs ten socks makes.

- 8 Order these numbers from smallest to largest.



TOPIC 4.1 · YOUR TURN, CONTINUED

9 Count in tens from 30 up to 90. Write each number down.

10 Count back in twos from 20 down to 0.

11 Bramble packs pegs in fives. Count six full packs in fives.

12 Count in twos starting at 1, up to 19. Say what changed.

13 Fill the gaps on this track.

0 1 2 3 : 5 6 7 8 9 : 11 12 13 : 15 16 17 18 19 20

- 14** Count twenty cubes in fives, then count them again in tens.
Say which of the two counts was quicker.

- 15** Tell a partner why counting in tens beats counting in ones.

★ **Rook's find**

Every number you land on counting in tens ends in a nought, all the way from ten to **100**. Rook says a rule you can hear in the last digit is a rule you will never forget, and he is right about that one.

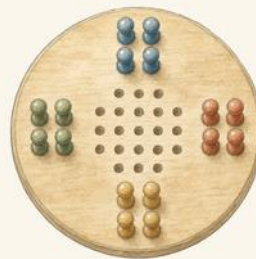
→ **Take it further**

Count in twos from nought to **20**, then count in twos starting at one instead. Both counts take steps of two, and the numbers you say are completely different. Write both lists under each other and say what you notice about the last digits.

Counting in threes and fours



Boards stacked in threes on the Plan Table, with the stacks pushed apart so each three can be seen.



Group: put things together so you can count them in steps.

♣ Steps of three, steps of four

Twos, fives and tens are the friendly steps. Threes and fours are harder, because the numbers you land on do not fall into a tidy pattern of last digits, so you cannot guess them. You have to count them.

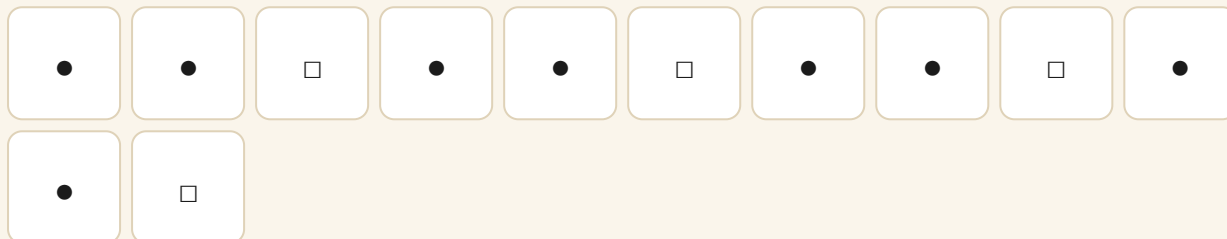
Say them in a rhythm and they stick: three, six, nine, twelve. Four, eight, twelve, sixteen. Twelve turns up in both, because twelve can be built out of threes and out of fours.

A step of three is the same as a repeating pattern of three things. Every time the pattern starts again, you have counted on another three.

TOPIC 4.2 · CONTINUED

- 1 Group the objects in your step size, threes or fours.
- 2 Put your finger on the first group and start at nought.
- 3 Say the new total each time you move to the next group.
- 4 Stop when you run out of groups, and read the total you are on.

A pattern with three cells in it. Every repeat is another three.



Do you remember?

Counting in fours is counting in twos, twice as fast. Two hops of two make one hop of four.

TOPIC 4.2 · CONTINUED

The threes, written out in full.

$$1 \times 3 = 3$$

$$2 \times 3 = 6$$

$$3 \times 3 = 9$$

$$4 \times 3 = 12$$

$$5 \times 3 = 15$$

$$6 \times 3 = 18$$

$$7 \times 3 = 21$$

$$8 \times 3 = 24$$

$$9 \times 3 = 27$$

$$10 \times 3 = 30$$

 **Watch me work one out 1**

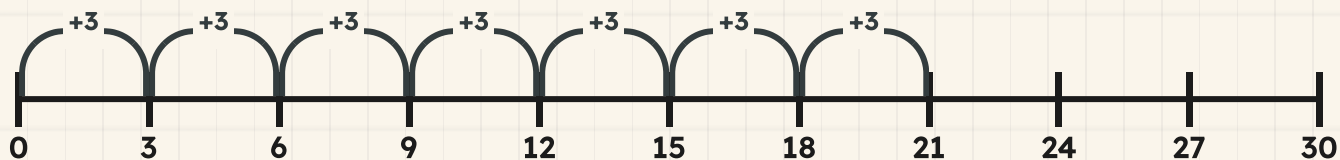
Count seven stacks of three boards.

- 1** Every stack holds three, so the step is three.
- 2** Start at nought with your finger on the first stack.
- 3** Hop three for each stack: three, six, nine, twelve.
- 4** Keep going for the stacks you have not touched yet: fifteen, eighteen.
- 5** The seventh stack takes you to **21**.

Answer: seven threes make **21**

TOPIC 4.2 · CONTINUED · CONTINUED

Seven hops of three. Count the hops, then read where Pip stopped.



⇌ Check it a different way

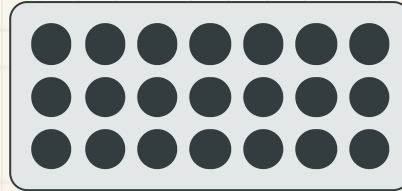
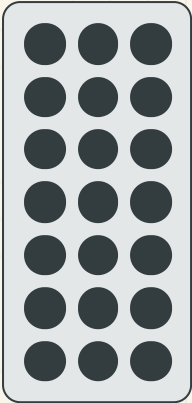
You worked it out by counting on in threes along the number line. Now try it another way.

Build the boards into a rectangle instead: seven rows with three in every row. Now turn the whole rectangle a quarter turn.

Nothing has been added and nothing taken away, but you are now looking at three rows with seven in every row. Count those rows in sevens instead: seven, fourteen, **21**.

You counted in threes the first time and in sevens the second time, and you landed on the same number both times. That is the check Cambridge calls reading an array two ways.

TOPIC 4.2 · CONTINUED · CONTINUED



$$7 + 7 + 7 = 21$$

$$3 + 3 + 3 + 3 + 3 + 3 + 3 = 21$$

The same boards, read down the rows and then across them.

How do you know?

Hattie says twelve can be counted in threes and also counted in fours. Use counters to show she is right, and say why some numbers turn up in both counts.

TOPIC 4.2 · YOUR TURN

**SHARPEN
YOUR
PENCIL**

Count round the class in threes. Anybody who hesitates sits down. Play it again in fours and see whether the class lasts longer.

1 Count in threes from 0 to 30. Write every number you land on.

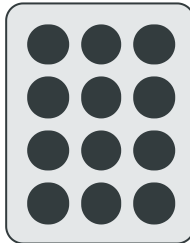
2 Count in fours from 0 to 40. Say them out loud twice.

3 Write the missing numbers on this counting line.



4 Count in fours to find the wheels on five carts.

5 Count these cubes in threes.

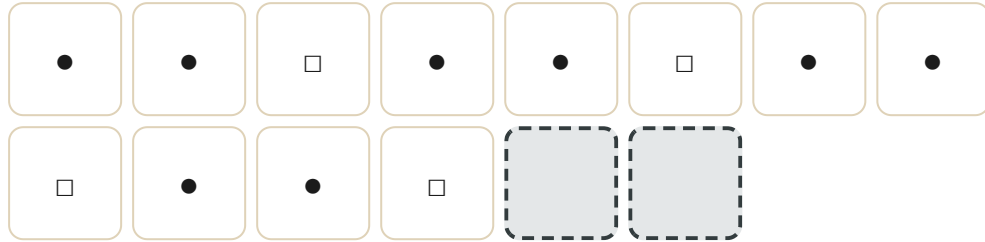


Four rows, three in each row.

TOPIC 4.2 · YOUR TURN · CONTINUED

6 Count in fours to find the legs on six stools.

7 Copy this pattern and carry it on.



8 Ring the number you would not say counting in threes.

9

11

15

18

TOPIC 4.2 · YOUR TURN, CONTINUED

9 Count in threes from 30 back down to 0.

10 Count in fours from 0 to 24. Now count in twos to 24.

11 Fill the gaps in this counting track.

0 1 2 3 4 5 : 7 8 9 10 11 : 13 14 15 16 17 : 19 20

12 Otto stacks boards in threes. Count nine stacks in threes.

13 Count in fours along this line. Write where you stop.



14 Order these numbers from largest to smallest.



15 Explain to a partner why threes are harder to count than tens.

Find out together

Work with a partner and a pot of thirty counters. Take twelve of them and build every rectangle you can: rows of two, rows of three, rows of four, rows of six. Write down which step sizes worked. Now try the same with thirteen counters. How many rectangles can you build this time, and what does that tell you about thirteen?

Yard challenge

Find a number under thirty that you say when counting in threes and also when counting in fours. There is more than one. Then find a number you say in fours but never in threes, and prove it by hopping along a number line from nought to **24**.

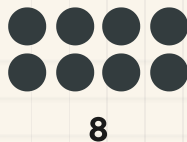
Odd and even numbers

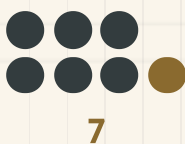
✦ The pairing test

Take a handful of counters and try to put every single one of them into a pair. Do not leave one waiting. If it works, and no counter is standing on its own, your number is **even**.

If one counter is left over with nobody to pair with, your number is **odd**. Trying harder will not help. An odd number cannot be paired up, and that is what odd means.

There is a quick way to tell without any counters at all. Look at the last digit. Nought, two, four, six and eight pair up. One, three, five, seven and nine leave one over. It works for a number of any size, because only the last digit decides.





The ones column holds the last digit, and the last digit is the one that decides.

	Tens	Ones
17	1	7
24	2	4

▣ WORDS FROM THIS PAGE



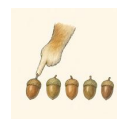
even



odd



pair



count

How do you know?

Milo says fifteen must be even because he can make seven pairs out of it. Lay fifteen counters out in pairs and say what Milo is ignoring.

Watch me work one out 1

Is 17 odd or even?

- 1 Look at the last digit first, because that is the quick test.
- 2 The last digit is seven, and seven does not pair up.
- 3 Now prove it. Put seventeen counters out in pairs.
- 4 8 pairs go down neatly and you run out of counters.
- 5 1 counter is left standing with nobody to pair with.

Answer: 17 is odd



17

Check it a different way

You worked it out by reading the last digit of the number. Now try it another way.

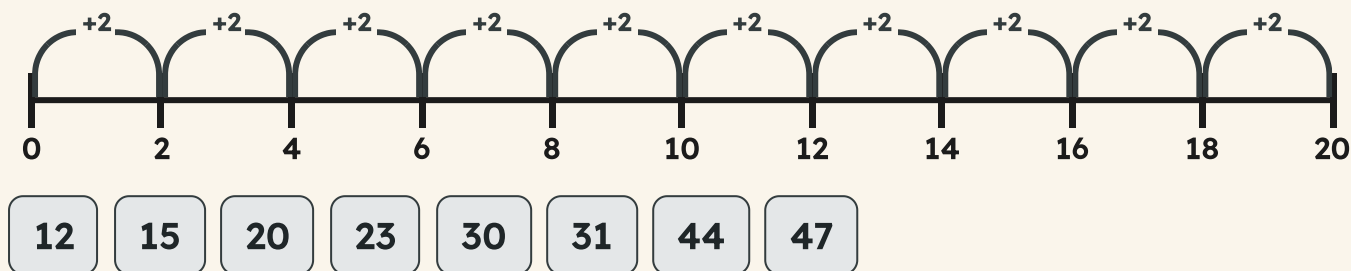
The last digit told you the answer in a second, but it did not show you **why**. So put the counting away and use your hands.

Count out that many counters. Pair them up along the table, two by two, until you cannot make another pair. Look at what is left on the table.

Nothing left over means the number pairs up exactly. One counter stranded means it does not. The digit gave you the answer and the pairing gave you the proof, and the two of them agree.

TOPIC 4.3 · CONTINUED · CONTINUED

Every number Pip lands on counting in twos from nought pairs up exactly. Every number she flies over does not.



✂ The material you need

a pot of counters and a strip of paper. Lay your counters in two straight rows along the strip. If the two rows come out the same length, your number is one that pairs up.



▣ SCAN TO SEE IT MOVE

Watch the counting jumps land on a number line as it counts on in twos, threes, fours and fives.

mathigon.org/polypad/number

Your turn

1 Say whether 14 pairs up exactly. Use counters to show why.

2 Sort these numbers into odd and even.

13

16

21

28

35

40

Odd

Even

3 Write down four numbers between 10 and 20 that pair up exactly.

4 Try to pair up thirteen counters. Say exactly what happens.

Do you remember?

Only the last digit decides. A number as big as ninety-eight is settled by the eight on the end of it.

TOPIC 4.3 · YOUR TURN · CONTINUED

- 5** Ring the odd numbers in this bank.

11

18

25

32

37

46

- 6** Write down the last digit of each number below.

27

54

63

80

95

- 7** Decide whether 26 pairs up exactly. Explain how you decided.

- 8** Ring TRUE or NOT TRUE for each line.

An even number pairs up with nothing left over.

TRUE / NOT TRUE

An odd number always leaves one counter on its own.

TRUE / NOT TRUE

You must count the whole number to know if it is odd.

TRUE / NOT TRUE

The last digit is enough to decide.

TRUE / NOT TRUE

TOPIC 4.3 · YOUR TURN, CONTINUED

9 Count in twos from 0 to **16**. Sort what you say into odd and even.

10 Find every number between 30 and 40 that will not pair up.

11 Pair up eighteen counters. Say what your pairing proves.

12 Sort these house numbers into odd and even.

7

10

19

22

33

48

13 Write an odd number larger than 50.

- 14** Match each number to the words that fit it.

24



29



36



41



● pairs up exactly

● leaves one over

● leaves one over

● pairs up exactly

- 15** Explain how you tell an odd number from an even one.

→ **Take it further**

Add two even numbers together and look at your total. Do it four more times with different pairs. Now add two odd numbers and look at those totals. One of the two rules will surprise you, so test it enough times to be sure of it.

Sorting numbers by their rule

♣ A rule the numbers have to obey

Sorting is putting things into groups, and the group a number goes in is decided by a **rule**. Not by how you feel about the number and not by where there is room. By the rule.

A good rule gives every number one home and only one. Counts in fives. Pairs up exactly. Is bigger than twenty. Each of those is a rule you can test on a number, one number at a time.

The hardest job is the other way round. Somebody hands you two groups already sorted and asks what the rule was. Look at what everything in the first group has in common, then check your idea against the second group.

Counts in fives

Does not count in fives

12

15

20

25

33

40

Do you remember?

Every rule needs testing on a number that fails it. A rule that nothing fails is not sorting anything.

 Watch me work one out 1

Sort 12, 15, 20 and 25 by the rule: counts in fives.

- 1** Take the numbers one at a time and test each one against the rule.
 - 2** Twelve: counting in fives goes ten, fifteen, so twelve is stepped straight over.
 - 3** Fifteen and twenty and twenty-five are all said out loud when you count in fives.
 - 4** So three of the four obey the rule and one of them does not.
 - 5** Put **15, 20** and **25** in the first tray and **12** in the second.
-

Answer: 15, 20 and 25 follow the rule, and 12 does not

⇌ Check it a different way

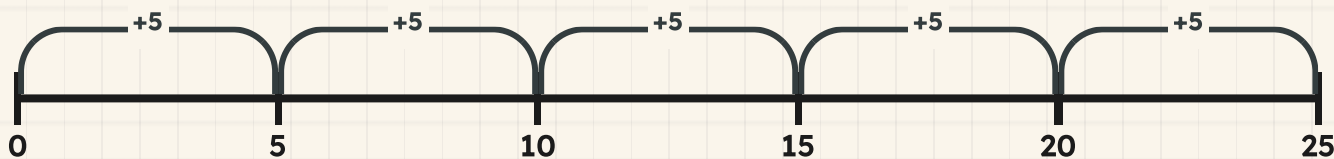
You worked it out by testing each number against the five times table. Now try it another way.

You tested the numbers one at a time against the five times table. Now do it the other way round: forget the numbers and just count.

Send Pip hopping in fives from nought as far as **25** and write down every number she lands on. Then look for your four numbers in her list. Any number in the list obeys the rule. Any number she jumped straight over does not.

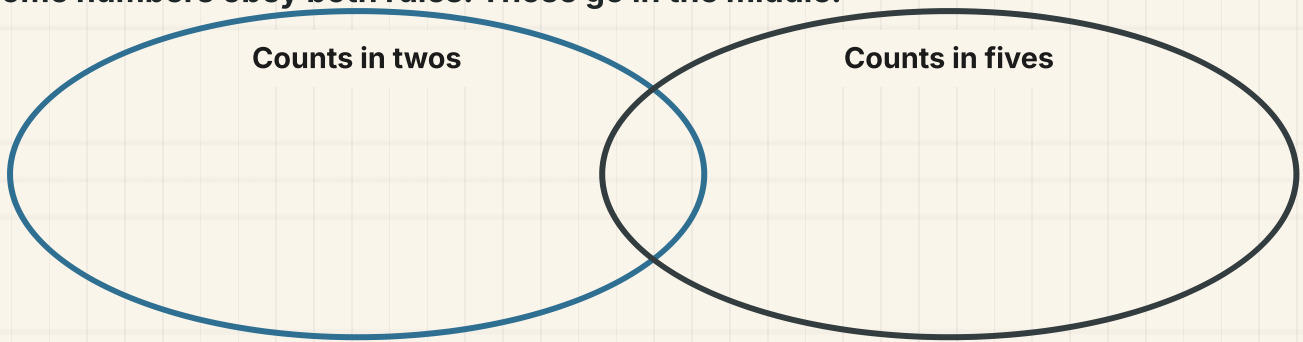
The first method interrogated each number. The second one never looked at your numbers at all until the counting was finished. Same two trays.

Everywhere Pip lands obeys the rule. Everywhere she flies over does not.



TOPIC 4.4 · CONTINUED · CONTINUED

Some numbers obey both rules. Those go in the middle.



How do you know?

Tessa sorted some numbers into two groups and will not say what her rule was. What would you look at first to work it out, and how would you test your idea?

TOPIC 4.4 · YOUR TURN

- 1** Sort these numbers into two trays. Write the rule you used.

6

9

14

21

30

35

My rule

Not my rule

- 2** Ring every number you say when counting in fives.

15

22

30

41

45

58



Numbers already sorted into two groups on the Plan Table, with the rule card turned face down.

TOPIC 4.4 · YOUR TURN · CONTINUED

- 3** Sort these numbers into the table below.
- Two rules at once, so four boxes.**

	Counts in fives	Does not count in fives
Pairs up exactly		
Does not pair up		

- 4** Find the rule for this group: 4, 8, 12, 16, 20.

- 5** Find the rule for this group: 5, 15, 25, 35.

6 Sort these numbers into the two rings.

Counts in tens :

7 Write three more numbers that fit Otto's rule: counts in tens.

8 Choose a rule of your own. Sort ten numbers by it.
Ask a partner to guess your rule.

TOPIC 4.4 · YOUR TURN, CONTINUED

- 9** Sort these numbers into odd and even.

17

24

39

52

Will not pair up

Pairs up exactly

- 10** Ring the number that breaks the rule: counts in threes.

6

12

16

21

- 11** Say which numbers fit both rules: counts in twos, counts in fives.

- 12** Test Milo's rule. He says every number ending in five is even.

13 Write a rule that only numbers which pair up can obey.

14 Sort these numbers by their last digit.

13

20

27

30

43

50

15 Explain how sorting numbers by a rule helps you find one fast.

✂ The material you need

squared paper, a pencil and two paper trays or two hoops. Write your rule on a card and lay it face down between the trays, so a partner has to work out your rule from the sorting alone.

🏠 Yard challenge

Find a rule that puts twenty and thirty in the same group but leaves twenty-five out of it. Then find a different rule that puts all three of them together. Write both rules down and say which one is fussier.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1** Count in twos from 14 up to 30.
- 2** Count in fives from 0 to 45, writing every number down.
- 3** Count in tens from 60 up to 100.
- 4** Count in threes to find how many boards eight stacks of three hold.
- 5** Count in fours to find the wheels on seven carts.
- 6** Say whether 29 pairs up exactly, and prove it with counters.
- 7** Find every number between 42 and 50 that will not pair up.
- 8** Sort 8, 11, 20, 25 and 33 into numbers you say in fives and numbers you do not.
- 9** Otto has sorted 6, 12, 18 and 24 into one group. Work out his rule.
- 10** Explain how you would check a count in fives a completely different way.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
count in twos, fives and tens, and group things to count them faster	☐	☐	☐
count on in threes and fours and carry a pattern on	☐	☐	☐
say whether a number is odd or even, and show why	☐	☐	☐
sort numbers into groups and say the rule you used	☐	☐	☐

UNIT 4 · HOW DID IT GO · CONTINUED

NOW I CAN

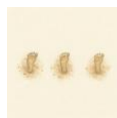
- ☐ count in twos, fives and tens, and group things to count them faster
- ☐ count on in threes and fours and carry a pattern on
- ☐ say whether a number is odd or even, and show why
- ☐ sort numbers into groups and say the rule you used

▣ YARD WORDS



skip-count

count in steps, not in ones



step

how much you count on each time



multiple

a number you land on when counting in steps



odd

a number that will not pair up



even

a number that pairs up exactly



pattern

something that carries on by a rule

The hundred wall

Otto's ruled sheet only goes to a hundred because that is as far as the paper went. Your class is going to build a bigger one, on the wall.

✂ The material you need

a long strip of paper or several sheets joined, a ruler, a pencil, and one colour of crayon for each person in your group.

Build a number line along the wall from zero to one hundred. Mark every tenth number with a long line and a numeral, and every fifth number with a short line and no numeral.

Then, in your own colour, mark on every number you can reach by counting in your own step size. One person counts in twos, one in threes, one in fours, one in fives, one in tens.

When all the colours are on, stand back and look at the numbers that have more than one colour on them.

How do you know?

Which numbers ended up with the most colours on them, and why those ones? Talk about it before you write anything down.

BUILD IT AND SHOW IT · PROJECT 1 · CONTINUED

WHAT TO HAND IN

- 1 Build the line, with tens marked long and fives marked short.
- 2 Mark your own step size in your own colour, all the way to a hundred.
- 3 Write down three numbers that got more than one colour.
- 4 Write one sentence saying why those numbers are special.

HOW IT WILL BE JUDGED

What is judged	Worth	What a good one looks like
Even spacing	30%	every gap is the same width, because you used a ruler and did not guess
Correct counting	30%	your colour lands on every number of your step size, and on no others
Tens and fives marked	20%	long lines and numerals on the tens, short lines on the fives
Your explanation	20%	you say WHY some numbers collected several colours, not just which ones did

Halves and quarters

Tessa cuts the plan sheet in half, then in half again.



Come into the Plan Room

Take a sheet of paper and fold it in half. Open it out and look at the crease. Are the two sides the same?

Now fold it in half again and open it out. How many pieces has the paper got now?

Take eight counters and share them fairly between two people. How many each? That is half of eight, and you just found it without any paper.

In the yard today

Tessa has folded the plan sheet exactly in half, and she is holding it up to the window to check.

The light shows the two halves lying on top of each other. Every edge matches. No paper sticks out anywhere.

Then she folds it again, the same way, and now there are four pieces instead of two, and every one of them is the same.

Fold once for halves, she says. Fold twice for quarters. And if the edges do not meet, you have not halved it, whatever you want to call it.

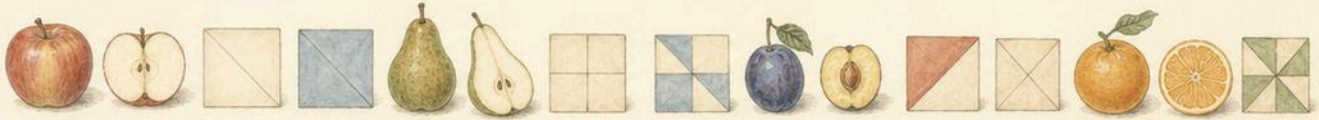
• CONTINUED

✓ **By the end you will**

- write one half and one quarter, and find them on a shape
- know that two halves and four quarters each make one whole
- say whether a shape has been cut into equal parts or not
- find half and a quarter of a small set of objects

Cambridge Primary Mathematics Stage 2: 2Nn16 • 2Nn17 • 2Nn18 • 2Nn19

One half and one quarter



Fruit and paper on the cutting bench, some cut and some whole.

**SHARPEN
YOUR
PENCIL**

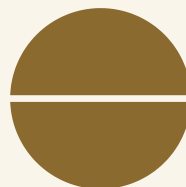
Hold your hands out and hide exactly half of your fingers. Now hide exactly a quarter of them. Which one was impossible, and why?

♣ **Two equal parts, or four equal parts**

When you cut something into **two equal parts**, each part is **one half**. We write it $\frac{1}{2}$.

When you cut something into **four equal parts**, each part is **one quarter**. We write it $\frac{1}{4}$.

The word equal is doing all the work in both of those sentences. Two parts that are not the same size are not halves, however neatly you cut them.



Do you remember?

A quarter is half of a half. Fold once, then fold the same way again, and you have quartered it.

Watch me work one out 1

Find one half of eight counters.

- 1** Half means two equal parts, so you need two equal piles.
- 2** Deal the counters out, one to each pile, and keep going.
- 3** One, one. Two, two. Three, three. Four, four.
- 4** Your hand is empty and both piles are the same size.
- 5** Count ONE pile, because one half is one of the two parts.

Answer: half of eight is **4**



Half of eight counters. Count one pile, not both.

⇌ **Check it a different way**

You worked it out by dealing the counters into two equal piles. Now try it another way.

You dealt them into two piles. Now check by DOUBLING your answer instead.

Double four is eight, which is what you started with, so four really is half of eight. Halving and doubling undo each other, so this check works every single time.

The check.

double **4 = 8**

How do you know?

Milo counted BOTH piles and said half of eight is eight. What did he misunderstand about the word half?

Your turn



- 1** Fold a paper square in half. Draw the crease you made.
- 2** Fold a paper square into quarters. Draw both creases.
- 3** Write the sign for one half.
- 4** Write the sign for one quarter.

TOPIC 5.1 · YOUR TURN · CONTINUED

5 Find half of six counters.



half

6 Find half of ten counters.

7 Find half of fourteen counters.

8 Colour one half of a drawn rectangle.

9 Colour one quarter of a drawn rectangle.

10 Draw a circle and cut it in half with one straight line.

TOPIC 5.1 · YOUR TURN, CONTINUED



half

11 Find half of an apple by drawing where you would cut it.

12 Find a quarter of eight counters. Share them twice.

13 Write how many quarters make one half.

14 Draw two different ways to halve a square.

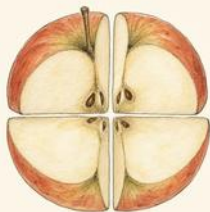
✂ The material you need

paper squares you can really fold, scissors if you are allowed them, about twenty counters, and something round to draw round.

★ Rook's find

Tessa folds rather than measures, and she is not being lazy. A fold cannot be slightly wrong the way a ruler mark can: either the edges meet or they do not, and you can see it from across the room.

Three quarters, and making a whole



Four equal parts. Each one is a quarter.

All four together. That is the whole thing again.

♣ Counting in quarters

Once something is cut into quarters you can count them: one quarter, two quarters, **three quarters**, four quarters.

Four quarters is the **whole** thing, back where you started. Two quarters is the same as one half, because two of four equal parts is half of them.

So $\frac{1}{4}$, then $\frac{1}{2}$, then $\frac{3}{4}$, then the whole. Three quarters is the one people forget, and it is only one quarter short of everything.

TOPIC 5.2 · CONTINUED



- 1 Cut it into four equal parts.
- 2 Count the parts you are keeping.
- 3 Say how many quarters that is.
- 4 Check: do all four parts together make the whole thing?

 Watch me work one out 1

How many quarters make one whole?

- 1** Fold a paper square in half, then in half again the same way.
- 2** Open it out and count the pieces the creases have made.
- 3** There are four, and they are all the same size.
- 4** Put them all back together and you have the square again.

Answer: 4 quarters make one whole

 Check it a different way

You worked it out by folding paper and counting the pieces. Now try it another way.

You counted the folded pieces. Now check with counters instead: take twelve counters and share them into four equal piles.

Each pile has **3**, and four piles of **3** is twelve, which is everything you started with. Paper and counters agree, so four quarters really is the whole thing.

TOPIC 5.2 · CONTINUED · CONTINUED



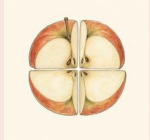
3 each

Twelve shared into four equal piles. One pile is a quarter of twelve.

How do you know?

Why is two quarters the same as one half? Fold a piece of paper to show a partner rather than telling them.

Your turn



quarter

1 Write how many quarters make one whole.

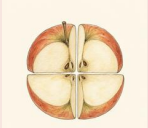
2 Write how many quarters make one half.

3 Write the sign for three quarters.

4 Colour three quarters of a drawn square.

5 Colour two quarters of a drawn square.

TOPIC 5.2 · YOUR TURN · CONTINUED



quarter

6 Write which is bigger, one half or three quarters.

7 Write which is bigger, one quarter or one half.

8 Draw a shape cut into four equal parts. Colour one.

9 Write how many quarters are left if you eat one.

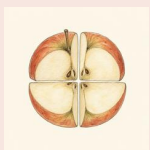
10 Order these from smallest to largest.

one quarter

three quarters

one half

TOPIC 5.2 · YOUR TURN, CONTINUED



quarter

- 11 Find three quarters of eight counters.
- 12 Find three quarters of twelve counters.
- 13 Write two halves as a number of quarters.
- 14 Draw a pizza cut into quarters with one quarter gone.



▣ SCAN TO SEE IT MOVE

Fold and cut fraction bars into halves and quarters with your finger, then put them back together to make a whole.

mathigon.org/polypad/fractions

→ Take it further

Fold a paper square in half three times instead of two. Open it out and count the pieces. What would you call one of them? Keep folding and see how quickly the pieces get too small to fold again.

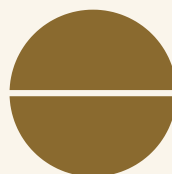
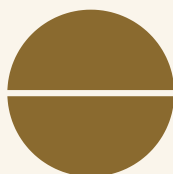
Which shapes are halved, and which are not

✦ If the parts are not equal, they are not halves

This is the most important page in the unit, and it is about one word: **equal**.

A shape cut into two pieces has been **halved** only if the two pieces are exactly the same size. Two pieces of different sizes are just two pieces. Neither of them is a half.

The test is to fold along the cut, or to imagine folding. If one piece would stick out past the other, it was not halved, however straight the line looked.



Do you remember?

Two pieces is not the same as two halves. Halves have to be equal, and you can always test it by folding.

 Watch me work one out 1

Has this square been halved? One piece is clearly wider than the other.

- 1** Look at the two pieces and compare their sizes.
- 2** One is much wider than the other, so they are not the same size.
- 3** Imagine folding along the cut line.
- 4** The wide piece would stick out past the narrow one.
- 5** Halves must be equal, and these are not.

Answer: no, it has been cut into two pieces but it has NOT been halved

⇌ **Check it a different way**

You worked it out by looking at the two pieces and comparing them. Now try it another way.

You judged it by looking. Now check by doing: cut a real paper square along the same line and lay one piece on top of the other.

If one piece covers the other exactly, it was halved. If any paper sticks out, it was not, and now you have proved it with your hands instead of your opinion.

How do you know?

Otto cut a cake into two pieces and gave Milo the smaller one, saying they were halves. What is wrong with what he said, and what should he have said instead?

Your turn



equal

- 1 Write whether two pieces of different sizes can be halves.
- 2 Draw a square that has been halved correctly.
- 3 Draw a square cut into two pieces that are NOT halves.
- 4 Write the test you would use to check a shape was halved.
- 5 Draw a circle halved correctly.

TOPIC 5.3 · YOUR TURN · CONTINUED



equal

6 Draw a rectangle halved along its short way.

7 Draw a rectangle halved along its long way.

8 Write whether both of your rectangles show halves.

9 Draw a shape cut into four pieces that are not quarters.

10 True or not true?
Decide for each line.

Any two pieces are halves.

TRUE / NOT TRUE

Halves must be equal in size.

TRUE / NOT TRUE

You can test halves by folding.

TRUE / NOT TRUE

TOPIC 5.3 · YOUR TURN, CONTINUED



equal

- 11** Sort drawn shapes into halved and not halved.

halved

not halved

- 12** Draw a triangle halved with one straight line.

- 13** Explain why a wonky cut cannot make halves.

- 14** Find something in the room that is exactly half of something else.

♂ Find out together

Work in a pair. Each of you cut a paper square into two pieces, one of you halving it properly and one of you not, without saying which you did. Swap, and let your partner work out which is which by folding. Then swap roles. Could they always tell?

Halves and quarters of a set

♣ Halving a group instead of a shape

You can halve a **group of things** as well as a shape. Half of twelve pegs is not a piece of a peg: it is six whole pegs.

To find half of a set, share it into two equal piles and count one pile.

To find a quarter, share it into four equal piles and count one.

A quarter is half of a half, so you can also halve, and then halve again. Both ways give the same answer, which is a useful thing to know when you want to check.



Half of twelve pegs is 6 pegs. Whole pegs, not pieces.



6 each

Shared into two equal piles.

 Watch me work one out 1

Find one quarter of twelve counters.

- 1** A quarter means four equal piles, so deal into four.
- 2** One to each pile, round and round, until your hand is empty.
- 3** You went round three times, so each pile has three.
- 4** One quarter is ONE of the four piles.

Answer: one quarter of twelve is **3**

 Check it a different way

You worked it out by dealing the counters into four equal piles. Now try it another way.

You dealt into four piles. Now use the other route: a quarter is half of a half, so halve twelve, then halve your answer.

Half of twelve is **6**. Half of **6** is **3**. The same answer by a completely different method, so you can trust it.

Halve, then halve again.

half of **12** = **6** half of **6** = **3**

How do you know?

Why does halving twice give the same answer as sharing into four? Use counters to show a partner.

Your turn



share

- 1 Find half of four counters.
- 2 Find half of sixteen counters.
- 3 Find half of twenty counters.
- 4 Find a quarter of eight counters.

TOPIC 5.4 · YOUR TURN · CONTINUED

5 Find a quarter of sixteen counters.



share

6 Find a quarter of twenty counters.

7 Half of a group is 5. Write how many were in the group.

8 A quarter of a group is 2. Write how many were in the group.

9 Find three quarters of sixteen counters.

10 Share ten counters in half. Write how many in one pile.

TOPIC 5.4 · YOUR TURN, CONTINUED



share

- 11 Find half of eighteen, then halve your answer to check.
- 12 Twelve biscuits are shared in half. How many in each half?
- 13 Write why you cannot find half of seven counters exactly.
- 14 Find half of your own age in years, if it works out whole.

Yard challenge

Find a number under thirty where a quarter of it is a whole number AND half of that quarter is also whole. There is more than one. What do all your answers have in common?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write which is bigger, two quarters or three quarters.
- 2 Colour three quarters of a drawn rectangle.
- 3 Find half of twenty-two counters.
- 4 Find a quarter of twelve counters.
- 5 Write how many halves make one whole.
- 6 Draw a shape cut into two pieces that are not halves, and say why.
- 7 Write the test that proves a shape has really been halved.
- 8 Half of a group is 9. Write how many were in the group.
- 9 Explain why half of eleven counters is not a whole number.
- 10 Order one half, one quarter and one whole from smallest to largest.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

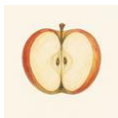
	confident	almost there	needs more work
write one half and one quarter, and find them on a shape	☐	☐	☐
know that two halves and four quarters each make one whole	☐	☐	☐
say whether a shape has been cut into equal parts or not	☐	☐	☐
find half and a quarter of a small set of objects	☐	☐	☐

UNIT 5 • HOW DID IT GO • CONTINUED

NOW I CAN

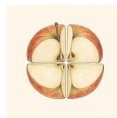
- ☐ write one half and one quarter, and find them on a shape
- ☐ know that two halves and four quarters each make one whole
- ☐ say whether a shape has been cut into equal parts or not
- ☐ find half and a quarter of a small set of objects

▣ YARD WORDS



half

one of two equal parts



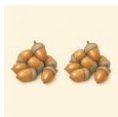
quarter

one of four equal parts



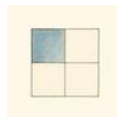
whole

all of it, nothing missing



equal

exactly the same amount



fraction

a part of a whole



share

split into equal parts

Number pairs and doubles

Bramble knows every pair that makes ten without stopping to think.



Come into the Plan Room

Put six counters on a ten frame, filling the top row first. Now look at the empty squares. Do not count them one by one, just look. How many are there?

Say it out loud: six and four make ten. Now say it the other way round: four and six make ten. Both are true, and both are the same pair.

Ask a partner for a number under ten. Say its partner to ten as fast as you can. Swap over, and keep going until neither of you has to stop and think.

In the yard today

Bramble is calling pairs across the yard and Otto cannot get a single one out before her.

Otto is counting on his fingers. Bramble is not counting at all. Otto says seven, and before he has finished saying it Bramble has said three.

You are working them out, she says. I am not working anything out. I just know them. There are only a handful of pairs that make ten, and once they are in your head they never leave.

She taps the board. **And every pair gives you two facts for free. Seven and three make ten, so ten take away three leaves seven. Learn one, get both.**

• CONTINUED

✓ **By the end you will**

- know every pair of numbers that makes 10, without counting
- partition any number to 20 and write the addition and the subtraction
- find pairs of tens that make 100, and write both facts
- double every number to 10, and 15, 20, 25 and 50

Cambridge Primary Mathematics Stage 2: 2Nc1 • 2Nc2 • 2Nc3 • 2Nc5

Pairs that make 10



The reckoning board with counters laid out in pairs, ten to a row.

**SHARPEN
YOUR
PENCIL**

Hold up any number of fingers on one hand. Ask a partner how many more make ten. Then swap hands and swap jobs.

✦ There are not many pairs, so you can know them all

Two numbers that add up to ten are a **pair** that makes ten. People also call it a **number bond**.

Here is the whole list, written in order. Look at how the left column climbs while the right column falls. That pattern is the reason you can trust the list: nothing is missing from it.

Your job is not to work these out. Your job is to **know** them, so that when somebody says six you say four before you have thought about it.

TOPIC 6.1 · CONTINUED

$$0 + 10 = 10$$

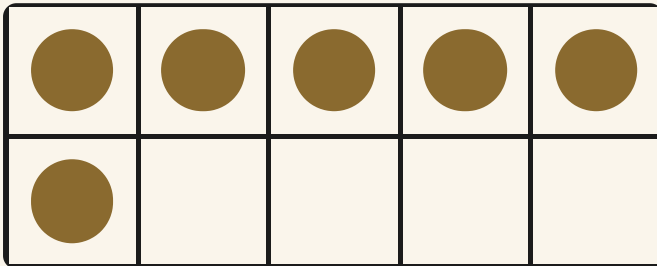
$$1 + 9 = 10$$

$$2 + 8 = 10$$

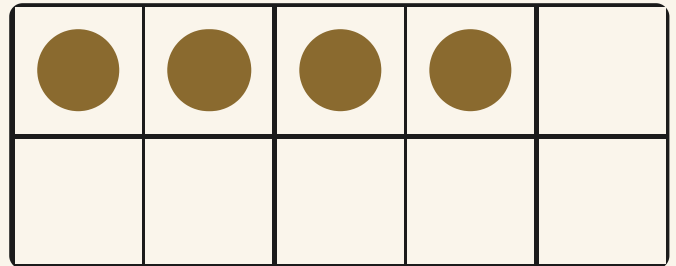
$$3 + 7 = 10$$

$$4 + 6 = 10$$

$$5 + 5 = 10$$



Six counters. Look at the empty squares, do not count them.



The empty squares moved over. The pair is the same pair.

Do you remember?

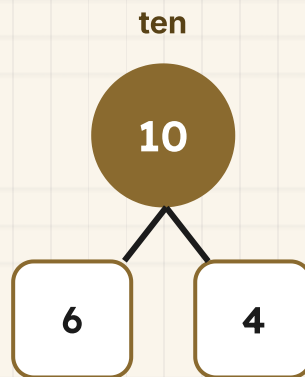
Every pair works both ways round. If you know six and four make ten, you also know four and six make ten. That is one fact, not two.

Watch me work one out 1

You have six. Which number makes ten with it, and what is the subtraction?

- 1** Put the six counters on a ten frame, top row first.
- 2** Do not count the counters again, you already know there are six.
- 3** Look at the squares with nothing in them.
- 4** There is one empty square on the top row and three underneath.
- 5** That is the pair, and now write the taking away as well: ten counters, take the empty ones off, and the six are back.

Answer: six and **4** make ten, and ten take away **4** leaves **6**



$$6 + 4 = 10$$

$$10 - 4 = 6$$

⇔ Check it a different way

You worked it out by counting the empty squares on the ten frame. Now try it another way.

You found the pair by looking at the empty squares on the frame. Now check it with your hand instead, and never look at the frame at all.

Lay out ten counters in a line. Cover **4** of them with your hand and count only the ones still showing. There are **6**. The frame and your hand agree, so the pair is right.

How do you know?

Otto says he has to count the empty squares every time. Bramble says she does not count them at all. What does Bramble know that Otto has not learned yet?

Your turn



pair

1 Write the number that makes ten with seven.

2 Write the number that makes ten with two.

3 Write the number that makes ten with nine.

4 Write the number that makes ten with five.

5 Fill in the missing number in each pair.

Say each one out loud before you write it.

TOPIC 6.1 · YOUR TURN · CONTINUED

$$3 + \square = 10$$

$$8 + \square = 10$$



pair

- 6 Put four counters on a ten frame. Write the empty squares.
- 7 Write the whole list of pairs that make ten, in order.
- 8 Write the addition and the subtraction for eight and ten.
- 9 Order these numbers from smallest to largest. 6 1 9 4
- 10 Say the partner to ten for each number you just ordered.

TOPIC 6.1 · YOUR TURN, CONTINUED



pair

- 11** Match each number to its partner in ten.

3 ●

● 9

1 ●

● 3

7 ●

● 7

- 12** Ask a partner ten pairs. Write how many you knew instantly.

- 13** Explain why nought and ten count as a pair that makes ten.

- 14** Draw a ten frame with three counters. Ring the empty squares.

✂ The material you need

about twenty counters, a ten frame you can put them on, and a partner who will call numbers at you quickly.

* Rook's find

Bramble does not practise pairs by writing them out. She practises them out loud, at speed, with somebody else calling the numbers. A fact you can only find on paper is not yet learned by heart.

Pairs that make 20



Two ten frames side by side. Twenty squares altogether.



A pair. Two numbers that belong together.

✦ Twenty is two full frames, so use the pair you already know

Twenty needs two ten frames. Fill the first one right up, and whatever you are holding goes on the second.

So fourteen is one full frame and four more. The first frame is finished with, and the only question left is a pair to ten on the second frame, which you already know by heart.

That is why the pairs to ten came first. You are not learning a new list here, you are using the old one on a bigger board. And every pair still gives you an addition and a subtraction.

TOPIC 6.2 · CONTINUED

$$10 + 10 = 20$$

$$11 + 9 = 20$$

$$12 + 8 = 20$$

$$13 + 7 = 20$$

$$14 + 6 = 20$$

$$15 + 5 = 20$$

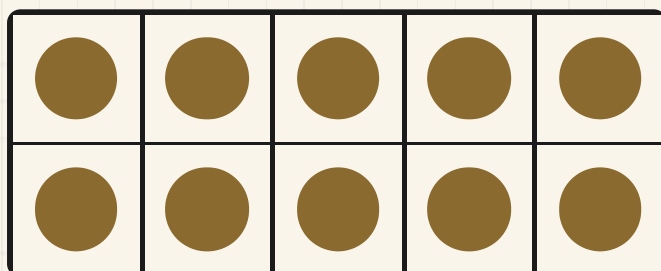
- 1 Split your number into a full ten and the leftovers.
- 2 Ignore the full ten, it is already done.
- 3 Say the pair to ten for the leftovers.
- 4 Write the addition, then write the subtraction that goes with it.

Watch me work one out 1

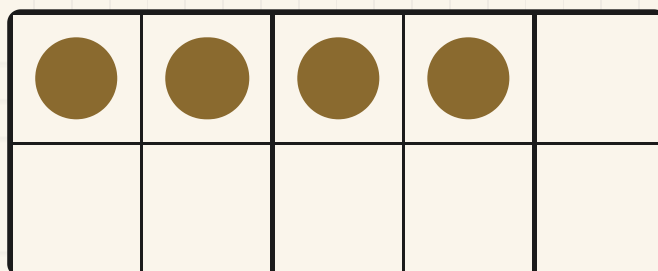
You have fourteen. Which number makes twenty with it?

- 1** Fourteen fills one ten frame and puts four on the next one.
- 2** The full frame is finished with, so stop looking at it.
- 3** Now the question is only about the second frame.
- 4** Four counters are on it, so look at the empty squares.
- 5** That pair is your answer, and the subtraction comes free with it.

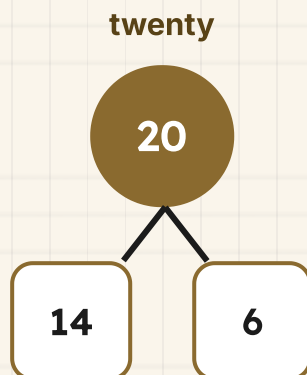
Answer: fourteen and **6** make twenty, and twenty take away **6** leaves **14**



Full. Finished with.



Four on, and the empty squares are the pair.



⇌ **Check it a different way**

*You worked it out by **splitting fourteen into a full ten and four**, then using the **pair to ten**. Now try it another way.*

You used a pair to ten on the second frame. Now check by counting back along a number track, which does not use the pairs at all.

Start on twenty and count back **6** steps, one at a time, touching each number. You land on **14**. Slow, but it agrees with the fast way, so the fast way was right.



How do you know?

Why does knowing six and four make ten also tell you that fourteen and six make twenty? Show a partner with two ten frames.

Your turn



total

- 1 Write the number that makes twenty with fifteen.
- 2 Write the number that makes twenty with eighteen.
- 3 Write the number that makes twenty with eleven.
- 4 Write the number that makes twenty with ten.
- 5 Split seventeen into a full ten and some ones.



total

TOPIC 6.2 · YOUR TURN · CONTINUED

6 Fill in the missing number in each pair.

$$13 + \boxed{} = 20$$

$$16 + \boxed{} = 20$$

7 Write the addition and the subtraction for twelve and twenty.

8 Draw two ten frames. Fill them with nineteen counters.

9 Write how many empty squares your nineteen counters left.

10 True or not true? Decide for each line.

A pair to twenty always uses a pair to ten.

TRUE / NOT TRUE

Twenty needs two ten frames.

TRUE / NOT TRUE

Every pair gives you two facts.

TRUE / NOT TRUE

TOPIC 6.2 · YOUR TURN, CONTINUED



total

11 Write the pair that makes twenty with your own age in years.

12 Order these numbers from largest to smallest.

17

11

14

19

13 Say the partner to twenty for each number you just ordered.

14 Explain how a pair to ten helps you find a pair to twenty.



▣ SCAN TO SEE IT MOVE

Number pairs to 10 and to 20, a new one each time, timed if you want. Good for learning them by heart.

www.topmarks.co.uk/maths-games/hit-the-button

→ Take it further

Bramble asks about pairs that make thirty. Try it with three ten frames. Which part of the job is new, and which part is exactly the same as it was for twenty?

Multiples of ten that make 100

♣ The same short list, counted in tens instead of ones

Multiples of ten are the numbers you land on counting in tens: ten, twenty, thirty, and so on up to a hundred.

A hundred is ten tens. So pairs of tens that make a hundred work exactly like pairs of ones that make ten, because you are pairing up the same ten things. Three and seven make ten, so thirty and seventy make a hundred.

You are not learning a second list. You are saying the list you already know in tens, and every pair still hands you a subtraction as well as an addition.

Every pair of tens that makes a hundred.

$$10 + 90 = 100$$

$$20 + 80 = 100$$

$$30 + 70 = 100$$

$$40 + 60 = 100$$

$$50 + 50 = 100$$

$$60 + 40 = 100$$

$$70 + 30 = 100$$

$$80 + 20 = 100$$

$$90 + 10 = 100$$

$$30 + 30 = 60$$

$$100 - 30 = 70$$

Do you remember?

If you can say three and seven, you can say thirty and seventy. Count the tens, not the ones, and the pair is already in your head.

Watch me work one out 1

You have seventy. Which multiple of ten makes a hundred with it?

- 1** Seventy is seven tens, so count in tens and forget the noughts.
- 2** A hundred is ten tens altogether.
- 3** So the question is really: seven and what make ten?
- 4** You know that pair by heart, so say it.
- 5** Now put the tens back on, and write the subtraction as well.

Answer: seventy and **30** make a hundred, and a hundred take away **30** leaves **70**

The pair, written out.

$$70 + 30 = 100$$

⇌ Check it a different way

You worked it out by using the pair to ten and then putting the tens back on. Now try it another way.

You used the pair to ten and then put the tens back on. Now check by counting on in tens along a number line, which uses no known pair at all.

Start at seventy and hop on in tens until you reach a hundred. Count your hops: eighty, ninety, a hundred, so **3** hops of ten, which is **30**. The two methods agree, so you can trust the fast one.

TOPIC 6.3 · CONTINUED · CONTINUED

Three hops of ten from seventy.



How do you know?

Milo says forty and seventy make a hundred because four and seven look close enough. How would you show him he is wrong without arguing?

Your turn



tens

- 1 Write the multiple of ten that makes a hundred with sixty.
- 2 Write the multiple of ten that makes a hundred with twenty.
- 3 Write the multiple of ten that makes a hundred with ninety.
- 4 Write the multiple of ten that makes a hundred with fifty.
- 5 Write the pair to ten that helped you with sixty.



tens

TOPIC 6.3 · YOUR TURN · CONTINUED

6 Fill in the missing number in each pair.

$40 + \underline{\quad} = 100$

$80 + \underline{\quad} = 100$

7 Write the addition and the subtraction for thirty and a hundred.

8 Write all the multiples of ten from ten up to a hundred.

9 Sort these into pairs that make a hundred and pairs that do not.

makes 100

does not

.....
.....
.....

.....
.....
.....

10 Order these multiples of ten from smallest to largest.

70

20

90

40

TOPIC 6.3 · YOUR TURN, CONTINUED



tens

- 11** Say the partner to a hundred for each multiple you just ordered.
- 12** Explain why ten and ninety make a hundred, using tens.
- 13** Write the pair of tens where both halves are the same.
- 14** Ask a partner five pairs of tens. Write how many were instant.

♣ Find out together

Work in a pair with a hundred counters or a hundred square. One of you covers some whole tens and the other says how many tens are hidden, without counting the ones. Then swap. Does saying the pair to ten first make it quicker every time, or only sometimes?

Doubles you know by heart



Double. The same amount again beside it.

The total. How much there is altogether.

♣ Two of the same, and no working out

To **double** a number you add the same number again. So double four is four and four.

Doubles are worth learning by heart because they turn up everywhere and because they are easy to picture: two equal rows, two equal hands, two equal piles. If you can see them, you can remember them.

Learn every double up to ten first. Then add four more that are useful all the time: double fifteen, double twenty, double twenty-five and double fifty.

TOPIC 6.4 · CONTINUED

The doubles to ten.

double **1** = **2** double **2** = **4** double **3** = **6** double **4** = **8** double **5** = **10**

double **6** = **12** double **7** = **14** double **8** = **16** double **9** = **18**

double **10** = **20**

Do you remember?

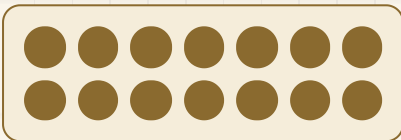
Doubling and halving undo each other. If you can say double seven, you can also say half of fourteen, and that is another fact free.

Watch me work one out 1

Find double seven, and then use it to find half of the answer.

- 1** Double seven means seven and seven altogether.
- 2** Picture two equal rows of seven, one above the other.
- 3** Seven and seven: ten, and four left over.
- 4** Say the whole fact out loud so it starts to stick.
- 5** Now read it backwards to get the halving fact as well.

Answer: double seven is **14**, and half of **14** is **7**



$$7 + 7 = 14$$

Two equal rows of seven. The same double, laid out.

⇔ **Check it a different way**

You worked it out by picturing two equal rows of seven and reading off the total. Now try it another way.

You recalled the double from the two rows. Now check it a different way: use a double you are surer of and step up from there.

Double six is **12**. Seven is one more than six, so double seven must be two more than double six, one extra for each seven. Two more than **12** is **14**. It matches.

One step up.

double **6** = **12** double **7** = **14**

How do you know?

Otto says double nine must be a hard one. Bramble says it is the easiest of the lot. What could she be using to make it easy?

Your turn



double

1 Write double three.

2 Write double eight.

3 Write double ten.

4 Write double nine.

5 Fill in the missing total for each double.

double 4 = _____ double 6 = _____

TOPIC 6.4 · YOUR TURN · CONTINUED



double

6 Draw two equal rows of five counters. Write the total.

7 Write double fifteen.

8 Write double twenty-five.

9 Write double fifty.

10 Order these doubles
from smallest to
largest.

double 5

double 2

double 8

double 4

TOPIC 6.4 · YOUR TURN, CONTINUED



double

- 11** Write half of sixteen by using a double you know.
- 12** Write the double that gives a total of twenty.
- 13** Explain how double six helps you find double seven.
- 14** Ask a partner every double to ten. Write your fastest four.

The four extra doubles.

double **15** = _____ double **20** = _____ double **25** = _____

double **50** = _____

Yard challenge

Find a double whose total is also a pair of tens that makes a hundred with fifty. Then find a number you can double twice, landing on a whole ten both times. What do your answers have in common?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Say the partner to ten for any number a partner calls out.
- 2 Write the pairs that make ten, starting from nine and nothing.
- 3 Write the number that makes twenty with sixteen.
- 4 Write the addition and the subtraction for four and ten.
- 5 Write the multiple of ten that makes a hundred with eighty.
- 6 Explain how a pair to ten gives you a pair of tens to a hundred.
- 7 Write double eight without counting on.
- 8 Write half of twenty by using a double you know.
- 9 Write double twenty and double fifty.
- 10 Explain what learning a pair by heart means, and how it is different from working it out.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
know every pair of numbers that makes 10, without counting	☐	☐	☐
partition any number to 20 and write the addition and the subtraction	☐	☐	☐
find pairs of tens that make 100, and write both facts	☐	☐	☐
double every number to 10, and 15, 20, 25 and 50	☐	☐	☐

UNIT 6 · HOW DID IT GO · CONTINUED

NOW I CAN

- ☐ know every pair of numbers that makes 10, without counting
- ☐ partition any number to 20 and write the addition and the subtraction
- ☐ find pairs of tens that make 100, and write both facts
- ☐ double every number to 10, and 15, 20, 25 and 50

▣ YARD WORDS



pair

two numbers that go together



number bond

a pair that makes a total



total

how much altogether



double

twice as many



add

put together to find the total



subtract

take away to find what is left

Adding and taking tens

A jump of ten lands on the same ones every time. Pip proves it.



Come into the Plan Room

Say the numbers twenty-three, thirty-three, forty-three, fifty-three out loud. What do you hear at the end of every single one of them?

Now start at twelve and count on in tens as far as you can go. Listen to the last part of each number as you say it.

Take ten counters and put them in a line. Add ten more. Could you say how many there are without counting them all again?

☀ In the yard today

Pip the wren has landed on the number line at twenty-four, and Otto has chalked a long arc from her feet to where she is going to land.

She could hop it in ones. She has done that before, and it takes ten little hops, and she nearly always loses count somewhere in the middle.

Today Otto has drawn her one arc instead, a single jump of ten, and written the landing number under it before she has even moved.

You cannot know that, says Pip. **I can,** says Otto. **A jump of ten never touches the ones. Twenty-four has four ones, so wherever you land will have four ones as well. Only the tens change.**

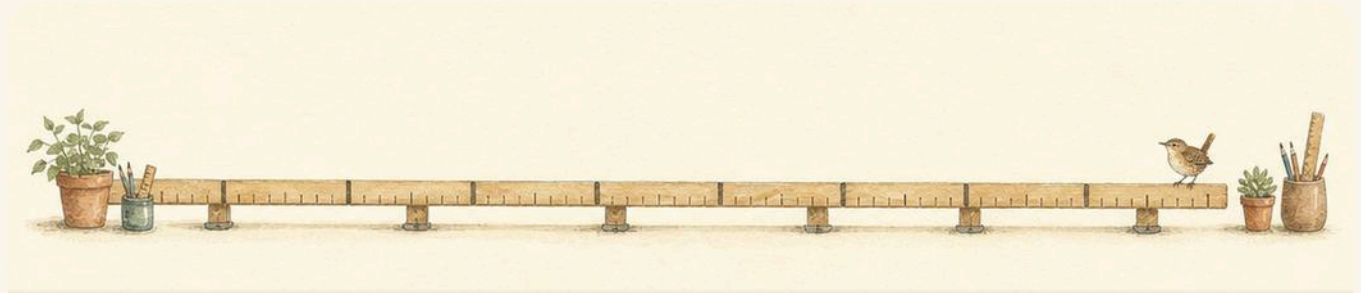
• CONTINUED

✓ **By the end you will**

- add ten and take ten from any two-digit number in your head
- add and take multiples of ten, such as 75 take 30
- add a single digit to a two-digit number, and take one away
- add four or five small numbers, looking for pairs that make ten

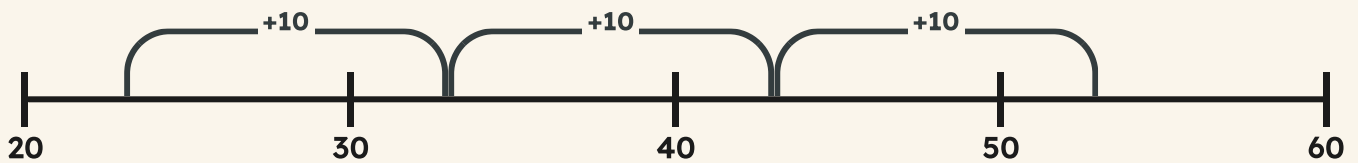
Cambridge Primary Mathematics Stage 2: 2Nc6 • 2Nc11 • 2Nc8

Ten more and ten less



The Reckoning Board, with the long number line chalked along the bench and Pip waiting at one end.

Pip's bench line. Three jumps of ten from twenty-three.



**SHARPEN
YOUR
PENCIL**

Say your own age. Now say the number that is ten more.
Now say the number that is ten less. Which of the two was
harder to find, and why do you think that was?

♣ A jump of ten leaves the ones alone

To add **ten** to a two-digit number you do not have to count. You need **one more ten**, so the tens digit goes up by one and the **ones digit stays exactly where it was**.

Look down the ones column in the chart. Every number in it ends in **3**. The tens column climbs, one step at a time, and the ones column never moves at all.

Taking ten away works the same way backwards. One ten goes, the ones stay. That is why **ten more** and **ten less** are questions you can answer in your head, out loud, straight away.

Four jumps of ten. Watch the ones column refuse to move.

	Tens	Ones
23	2	3
33	3	3
43	4	3
53	5	3

Do you remember?

Ten more changes the tens digit only. If your ones digit has changed, you have added something that was not ten.

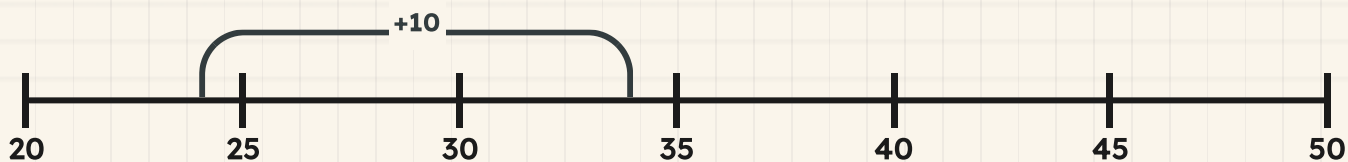
Watch me work one out 1

Pip is standing on twenty-four. She makes one jump of ten. Where does she land?

- 1 Twenty-four is two tens and four ones.
- 2 A jump of ten adds one ten, so now there are three tens.
- 3 The ones are not part of the jump, so there are still four ones.
- 4 Three tens and four ones is the number you want.
- 5 Follow the arc on the line to see her land on the same ones.

Answer: she lands on **34**

One jump of ten from twenty-four.



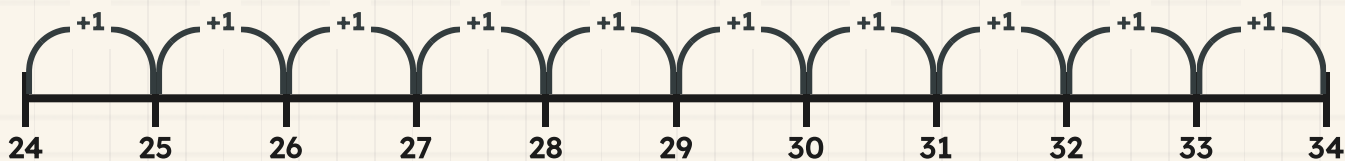
⇌ Check it a different way

You worked it out by changing the tens digit and leaving the ones alone. Now try it another way.

You changed the tens digit and left the ones alone, which took one second. Now do it the slow way and see whether the slow way agrees.

Put your finger on twenty-four and count on ten single hops along the line, one number at a time. Your finger stops on **34**, the same number the rule gave you. Ten little hops and one big jump land in the same place, so the rule is not a trick.

The same jump, counted in ten single hops.



How do you know?

Milo says ten more than thirty-six is thirty-seven, because you add one. What has he added instead of ten, and how would the chart show him he is wrong?

Your turn



jump

- 1 Add ten to twenty-three. Write your answer.
- 2 Add ten to forty-six. Write your answer.
- 3 Take ten from thirty-eight. Write your answer.
- 4 Take ten from seventy-one. Write your answer.
- 5 Write the number that is ten more than fifty-five.



jump

- 6 Write the number that is ten less than sixty-two.

TOPIC 7.1 • YOUR TURN • CONTINUED

- 7** Find the missing jump in this sentence.
the number in the box is the same size as one arc

$$24 + \boxed{} = 34$$

How big was the jump that landed here?

- 8** Copy this chart and fill in the tens and the ones.

Copy the chart. Fill in both columns.

	Tens	Ones
46		
56		

- 9** Add ten to every number in the bank. Write six answers.

12

27

35

48

61

84

- 10** Order these numbers from smallest to largest.

47

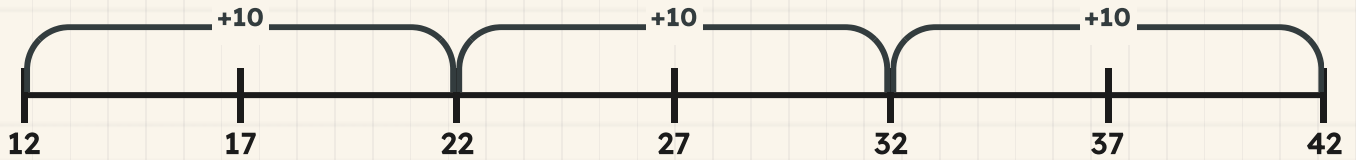
74

17

71

TOPIC 7.1 · YOUR TURN, CONTINUED

The form to copy: ends marked, tens labelled, jumps drawn.



jump

- 11** Draw a number line from twelve to forty-two. Show three jumps of ten.
- 12** Write ten more and ten less than your own house number.

13 Explain why the ones digit never changes when you add ten.

14 Write what stays the same when you take ten from sixty-four.

✂ The material you need

a long number line you can point along, a tens and ones chart with two columns, twenty counters and ten straws you can bundle together into one ten.

★ Rook's find

Otto writes the landing number before Pip has moved, and that is not showing off. He is proving that he understands the number instead of the counting: if you have to hop to find out, the tens have not become real to you yet.

Adding and taking other multiples of ten



One bundle is one ten. Count the bundles, not the straws. One arc is one whole jump of ten along the line.

✦ Thirty is three tens, so it is three jumps

Twenty, thirty, forty and fifty are **multiples of ten**. They are made of whole tens and nothing else, which is why they are so easy to add.

Adding thirty is adding three tens, so it is **3** jumps of ten, one after another. Every one of those jumps leaves the ones alone, so after all **3** of them the ones are still exactly where they started.

Taking thirty away is the same **3** jumps backwards. Count the tens you are adding or taking, make that many jumps, and read the number you have landed on.

TOPIC 7.2 · CONTINUED

Adding ten, then ten, then ten. The ones column still has not moved.

	Tens	Ones
27	2	7
37	3	7
47	4	7
57	5	7

- 1** Look at the number you are adding and count its tens.
- 2** Make that many jumps of ten from where you started.
- 3** Leave the ones digit completely alone.
- 4** Read your landing number, then check the ones still match.

Watch me work one out 1

Seventy-five take thirty. Pip starts at seventy-five and jumps backwards.

- 1** Thirty is three tens, so Pip needs three backward jumps.
- 2** From seventy-five, one jump back is sixty-five.
- 3** A second jump back is fifty-five.
- 4** A third jump back is forty-five, and the jumps are used up.
- 5** Look at the ones: five at the start, five at the end.

Answer: seventy-five take thirty is **45**

Three jumps back from seventy-five.



$$45 = 40 + 5$$

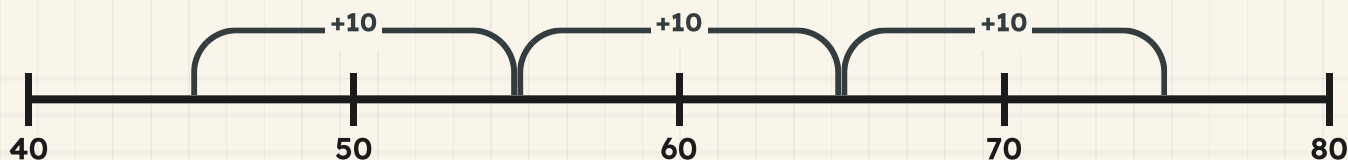
⇌ Check it a different way

You worked it out by making three backward jumps of ten along the line. Now try it another way.

You jumped backwards three times. Now check by going the other way: adding undoes taking away.

Start at your answer, **45**, and jump forwards three tens instead. You land on **75**, which is the number you began the question with. If the journey back brings you home, the journey out was right.

The check. Three jumps forwards, back to where you began.



How do you know?

Tessa says taking thirty is easier than taking three, and she is not joking. What is it about thirty that makes it easier than three for her?

Your turn



tens

- 1 Add twenty to thirty-four. Write your answer.
- 2 Add thirty to forty-five. Write your answer.
- 3 Take twenty from sixty-eight. Write your answer.
- 4 Take thirty from seventy-five. Write your answer.
- 5 Count on in tens from sixteen to fifty-six. Write every number.



tens

- 6 Count back in tens from ninety to fifty. Write every number.

TOPIC 7.2 · YOUR TURN · CONTINUED

7 Find the missing multiple of ten in this box.

$$40 + \boxed{} = 70$$

How many tens went into the box?

8 Write how many jumps of ten each one needs.

add 20

add 40

take 30

take 50

9 Copy this chart and fill in the tens and the ones.

Copy the chart. The ones should surprise nobody.

	Tens	Ones
31		
51		
71		

10 Order these numbers from smallest to largest.

58

85

38

83

TOPIC 7.2 · YOUR TURN, CONTINUED



tens

- 11 Add forty to twenty-seven. Write your answer.
- 12 Take fifty from eighty-three. Write your answer.
- 13 Explain why adding thirty is the same as three jumps of ten.
- 14 Find two multiples of ten that make one hundred altogether.



▣ SCAN TO SEE IT MOVE

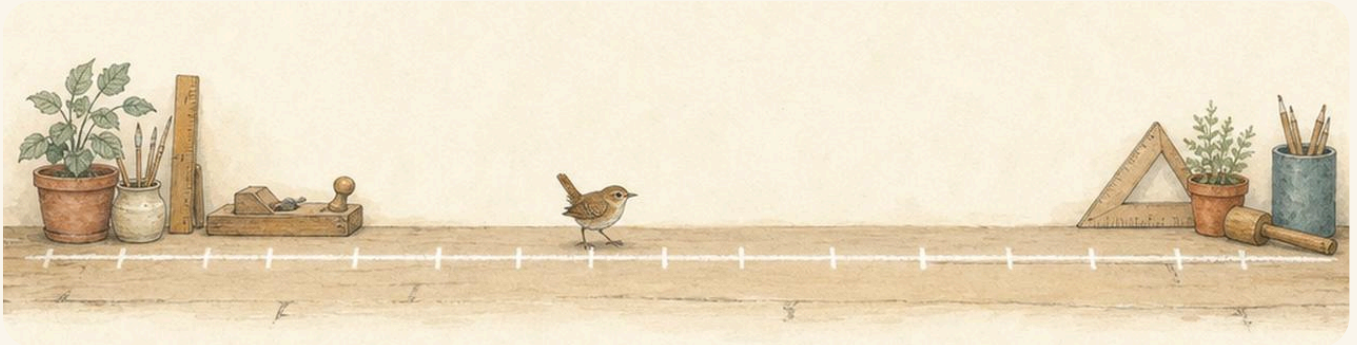
Watch a jump of ten hop along the number line, so you can see why adding ten only changes the tens digit.

mathigon.org/polypad/number

→ Take it further

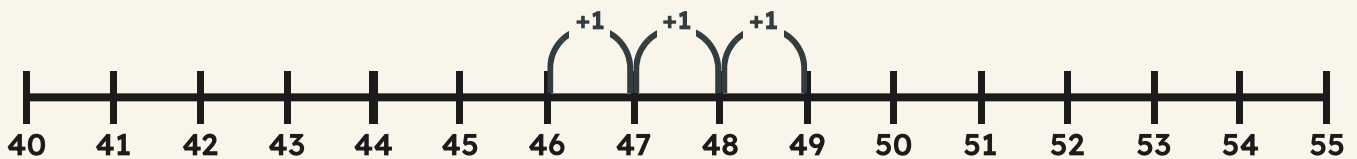
Start at four and add ten again and again, out loud, until you cannot go any further without passing one hundred. Write the list down. Now do the same starting at seven. What is the same about both of your lists, all the way down?

Adding a single digit to a two-digit number



The board's own number line. Every single hop is one number wide, and no hop is ever wider than that.

Three single hops from forty-six. Each hop is one.



♣ This time the ones **DO** change

A single digit is a handful of **ones**, so adding one changes the ones digit. This is the opposite of a jump of ten, and it is worth saying out loud so you do not muddle the two.

Sometimes the ones run out on the way. Thirty-eight and seven is more than forty, so the count crosses a ten. The neat way is to **split the digit**: use **2** to get to forty, then add the **5** you have left.

Taking a single digit away works the same way. Step back to the ten below first, then take away whatever is still owed. Landing on a ten in the middle makes the rest of the journey easy.



Do you remember?

Ten changes the tens. A single digit changes the ones, and it sometimes pushes you over a ten on the way.

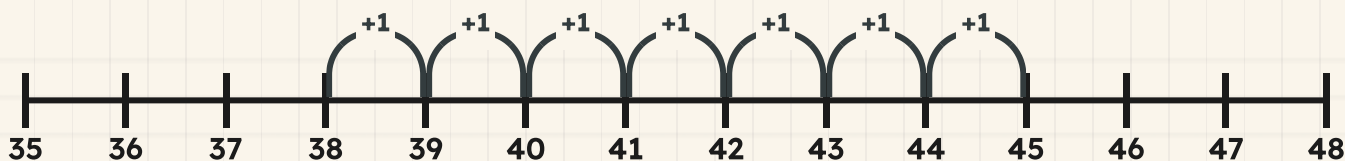
Watch me work one out 1

Thirty-eight add seven. Pip has to cross a ten to do it.

- 1** Count on in ones from thirty-eight, one hop at a time.
- 2** Thirty-nine, forty, forty-one, forty-two.
- 3** Keep going: forty-three, forty-four, forty-five.
- 4** That is seven hops used, and the last one is your answer.
- 5** The ones digit changed, because you added ones and not tens.

Answer: thirty-eight add seven is **45**

Seven single hops from thirty-eight.



⇌ Check it a different way

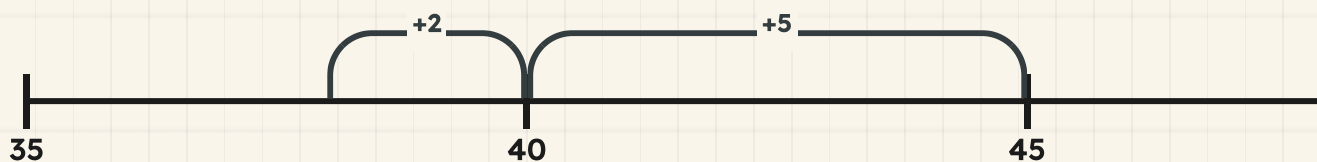
You worked it out by counting on seven single hops one number at a time. Now try it another way.

You counted seven separate hops, and seven hops is seven chances to lose your place. Now split the seven instead.

Thirty-eight needs **2** to reach forty, so hop **2** and land on the ten. You still have **5** of your seven left, so hop that, and you are on **45**. Two hops, the same landing number, and far less to go wrong.

TOPIC 7.3 · CONTINUED · CONTINUED

The same seven, split so you land on forty first.



$$38 + 7 = \boxed{}$$

$$52 - 4 = \boxed{}$$

How do you know?

Otto says fifty-two take four must land below fifty. He is right, and he has not worked it out yet. How can he be sure before he counts a single hop?

Your turn



add

1 Add five to forty-seven. Write your answer.

2 Add six to thirty-eight. Write your answer.

3 Add four to twenty-nine. Write your answer.

4 Take three from fifty-two. Write your answer.

5 Take six from forty-one. Write your answer.



add

TOPIC 7.3 · YOUR TURN · CONTINUED

- 6** Split the seven so you land on forty first.
thirty-six add seven, in two hops instead of seven

- 7** Find the missing single digit in this box.

$$35 + \boxed{} = 42$$

One digit went into the box. Which one?

- 8** Add seven to sixty-five along a number track you draw.

- 9** Write which of these cross a ten as you add.

23 add 4

28 add 5

31 add 6

36 add 8

- 10** Order these numbers from smallest to largest.

52

48

60

44

TOPIC 7.3 · YOUR TURN, CONTINUED



add

- 11** Add nine to fifty-four. Write your answer.
- 12** Take eight from sixty-three. Write your answer.
- 13** Explain why adding six to twenty-eight crosses a ten.
- 14** Find the single digit that takes thirty-six exactly to forty.

15

True
or not
true?
Decide
for
each
line.

Adding ten changes the ones digit.	TRUE / NOT TRUE
Adding six can push you past a ten.	TRUE / NOT TRUE
Landing on a ten first makes the rest easier.	TRUE / NOT TRUE

♣ **Find out together**

Work in a pair with one number line between you. One of you says a two-digit number and a single digit to add. The other adds it in two hops, landing on a ten in the middle, and says both hops out loud. Then swap. Find a starting number where the split is not needed at all, and work out together what all such numbers have in common.

Adding four or five small numbers



Adding is joining. The order you join them in is up to you.

Altogether means the whole amount, once everything has been joined.

✦ Look for the pairs that make ten first

When four or five small numbers are waiting to be added, do not start at the left and plod along. **Look** at all of them first.

You are hunting for **a pair that makes ten**. Seven and three are one. Six and four are another. You already know these pairs by heart from the Ten Wall, and a ten is the easiest thing in the world to add on.

So the plan is: ring the pair that makes ten, add that pair, then add whatever numbers are left over. You may add numbers in any order you like and the total will not change, which is exactly what makes this allowed.

$$7 + 3 = \boxed{}$$

$$6 + 4 = \boxed{}$$

Do you remember?

The total is the same whatever order you add in. So choose the order that gives you tens to work with.

Watch me work one out 1

Add 7, 3, 6 and 4 altogether.

- 1 Read all four numbers before you write anything down.
- 2 Seven and three make ten, so ring that pair.
- 3 Six and four make ten as well, so ring that pair too.
- 4 Now you have two tens and nothing left over.
- 5 Two tens is twenty, and no number was added twice.

Answer: altogether they make **20**

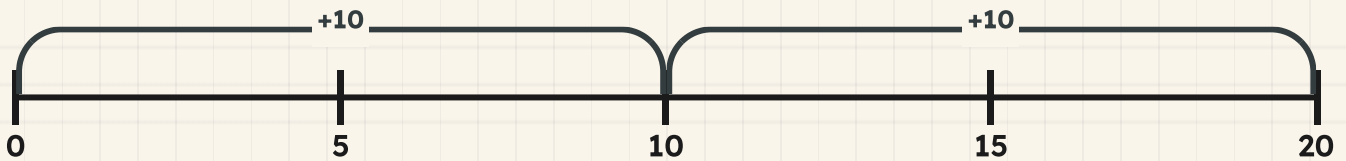
7

3

6

4

Two jumps of ten, because that is what the pairs made.



⇌ Check it a different way

You worked it out by looking for the pairs that make ten and adding those first. Now try it another way.

You hunted for pairs. Now check the plodding way, in the order the numbers were written, and see whether the total dares to be different.

Seven and three is ten. Add six and you have sixteen. Add four and you are on **20**, the same total as before. Adding in a different order gives the same answer every time, so you are free to pick the easy order.

14

15

17

18

20

How do you know?

Milo added **8**, **5**, **2**, **5** and **3** straight along the row and lost his place twice. Which two pairs should he have hunted for first, and what would he have had left to add on?

Your turn



altogether

1 Add four, six, three and seven altogether.

2 Add eight, two, five and five altogether.

3 Add nine, one, four and six altogether.

4 Add three, seven, two and eight altogether.

5 Ring the pair that makes ten, then add the rest.

5

6

5

2



altogether

TOPIC 7.4 · YOUR TURN · CONTINUED

- 6** Match each number to the partner that makes ten.

7 ●

● 4

6 ●

● 2

8 ●

● 3

- 7** Add these five numbers. Hunt for pairs first.

8

5

2

5

3

- 8** Sort these numbers by the partner they need.

needs 10 as a partner

.....

.....

.....

needs no partner

.....

.....

.....

- 9** Add one, nine, two and eight altogether.

- 10** Order these totals from smallest to largest.

10

20

15

25

TOPIC 7.4 · YOUR TURN, CONTINUED



altogether

11 Add six, four, five and five. Then check in another order.

12 Add two, eight, three, seven and four altogether.

13 Explain why hunting for pairs that make ten is quicker.

14 Find three small numbers that make ten altogether.

Yard challenge

Choose five numbers from one to nine, all different, that hide two pairs making ten. Write them in an order that looks as awkward as you can manage, then give them to a partner and time how long the total takes. Can you find a set of five where no pair makes ten at all?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Add ten to fifty-two. Write your answer.
- 2 Take ten from eighty-four. Write your answer.
- 3 Add twenty to sixty-one. Write your answer.
- 4 Take forty from ninety-three. Write your answer.
- 5 Add eight to forty-six. Write your answer.
- 6 Take seven from seventy-two. Write your answer.
- 7 Add two, eight, three and seven altogether. Look for pairs first.
- 8 Write the pair that makes ten from six, five, four and two.
- 9 Explain what happens to the ones digit when you add thirty.
- 10 Order sixty-three, thirty-six, ninety and nineteen from smallest to largest.

Be honest with yourself

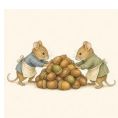
Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
add ten and take ten from any two-digit number in your head	☐	☐	☐
add and take multiples of ten, such as 75 take 30	☐	☐	☐
add a single digit to a two-digit number, and take one away	☐	☐	☐
add four or five small numbers, looking for pairs that make ten	☐	☐	☐

NOW I CAN

- ☐ add ten and take ten from any two-digit number in your head
- ☐ add and take multiples of ten, such as 75 take 30
- ☐ add a single digit to a two-digit number, and take one away
- ☐ add four or five small numbers, looking for pairs that make ten

▣ YARD WORDS



count on

add by counting forwards



count back

subtract by counting backwards



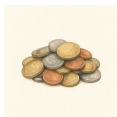
multiple of ten

10, 20, 30 and so on



jump

a step along the number line



altogether

the whole amount when joined



difference

how much more one is than another

The ten-a-day ledger

Bramble writes down what leaves the store every day. He is tired of adding it up. He wants a page he can read at a glance.

✂ The material you need

squared paper, a pencil, a ruler and a rubber.

Invent something your group sells from the Making Yard: pegs, or baskets, or planks. Decide how many you sell each day for five days. Use a two-digit number every day, and make one of them a number you find hard.

Rule a table with a row for each day. Work out the running total after each day, so the last row is the whole week.

↔ Check it a different way

You worked it out by adding the days in order, one at a time. Now try it another way.

Now add your five numbers again in a completely different order, starting from the biggest. You must reach the same weekly total. If you do not, one of the two runs has a mistake in it, and finding which is part of the job.

BUILD IT AND SHOW IT · PROJECT 2 · CONTINUED

WHAT TO HAND IN

- 1 Rule a neat table with one row for each of five days.
- 2 Fill in the day totals and the running total after each day.
- 3 Add the five numbers again in a different order, and show that working.
- 4 Write one sentence about which order was easier, and why.

HOW IT WILL BE JUDGED

What is judged	Worth	What a good one looks like
Correct arithmetic	35%	both runs reach the same total, and that total is right
Two different orders	25%	your second run genuinely uses a different order, not the same one written out again
Clear table	20%	columns line up on the squares, tens under tens
Your explanation	20%	you say which order was easier for you and give a reason

Adding and taking to 100

Two two-digit numbers, one plan, and two ways to be sure.



Come into the Plan Room

Before anybody shows you a method, show what you can already do.

Say what forty-six is made of. Now say what thirty-seven is made of.

Both of them come apart the same way.

Add twenty and thirty in your head. Now add four and three. You have just done most of the work in this unit.

In the yard today

Otto and Tessa have the same two numbers written on the same plan, and they have added them up two different ways.

Otto partitioned both numbers into tens and ones and put the tens together first. Tessa started at the larger number and counted on. Their working looks nothing like each other's, and their total is exactly the same.

That is the point, says Tessa. One method gets you an answer. Two methods tell you the answer is right.

By the end you will

- add two two-digit numbers by partitioning them into tens and ones
- exchange ten ones for one ten, and say why you did it
- find a small difference between two numbers by counting on
- know that adding can be done in any order but taking away cannot

Cambridge Primary Mathematics Stage 2: 2Nc12 • 2Nc13 • 2Nc14 • 2Nc15 • 2Pt6 • 2Pt7

Adding pairs of two-digit numbers



Trays of ten cubes on the Reckoning Board, with the loose ones waiting beside the last tray.

Do you remember?

Forty-six is four tens and six ones. Say what fifty-eight is made of, and what eighty-one is made of.

♣ Two numbers, four parts

Adding two two-digit numbers looks hard until you take both of them apart. Then it is two easy additions instead of one hard one.

Partition each number into its tens and its ones. Now you have four parts, not two.

Add all the tens together. Add all the ones together. Put your two answers back into one number. That is the whole method.

TOPIC 8.1 · CONTINUED

$$46 = 40 + 6$$

$$37 = 30 + 7$$

The two numbers side by side, each digit standing in its own column.

	Tens	Ones
46	4	6
37	3	7

**SHARPEN
YOUR
PENCIL**

Call out a two-digit number to a partner. They say what it is made of, in tens and ones. Swap after five turns, and speed up.

Watch me work one out 1

Add 46 and 37.

- 1** Partition both numbers. Forty-six is forty and six. Thirty-seven is thirty and seven.
- 2** Add all the tens first: forty and thirty make **70**.
- 3** Now add all the ones: six and seven make **13**.
- 4** You have **70** and **13** to put back together.
- 5** **70** and ten more is eighty, and there are **3** ones still to come.

Answer: $46 + 37 = 83$

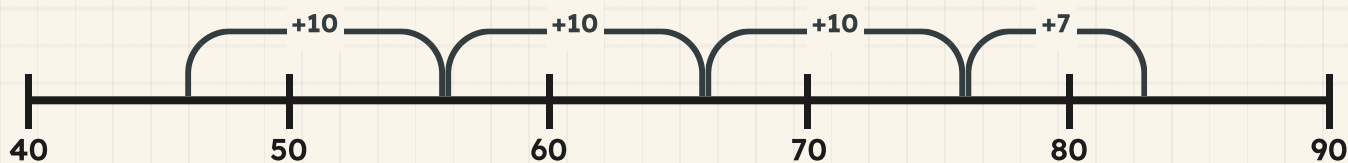
⇌ Check it a different way

You worked it out by partitioning both numbers into tens and ones. Now try it another way.

Leave the tens and ones alone this time. Start at the **larger** number, forty-six, and count on in tens: fifty-six, sixty-six, seventy-six. Three tens is all of thirty. Now count on the seven ones one at a time.

You never partitioned anything, and you landed on the same number. Two different roads, one place. That is what Cambridge means by checking a different way.

Three hops of ten, then one hop of seven. Watch where Pip lands.



How do you know?

Otto put the tens together first. Tessa counted on from the larger number. Why is it safe for two people to use different methods on the same addition?

Your turn



partition

- 1 Add 32 and 45. Partition both numbers before you start.
- 2 Add 54 and 23. Write down the tens total and the ones total.
- 3 Add 46 and 31.
- 4 Work out 27 add 52. Show your partitioning on squared paper.
- 5 Copy these two additions and work each one out.

$35 + 24 = \square$

$43 + 26 = \square$

TOPIC 8.1 · YOUR TURN · CONTINUED



partition

- 6** Work out these two the same way.

$$62 + 25 = \boxed{}$$

$$51 + 38 = \boxed{}$$

- 7** Add 41 and 38. Say your total out loud before you write it.

- 8** Add the two numbers on Otto's plan: 63 and 25.

- 9** Which pair makes the larger total: 34 and 41, or 52 and 26?

TOPIC 8.1 · YOUR TURN, CONTINUED



partition

- 10** Add 15 to 62. Now add 62 to 15 and compare the two totals.
- 11** Bramble stacks 24 boards, then 35 more. How many boards now?
- 12** Add 53 and 44 without writing anything down.
- 13** Choose two numbers between 20 and 40. Add them by partitioning.
- 14** Add 71 and 18. Then add 18 and 71 to check.
Say which order you found easier.



partition

15 Tell a partner how you add two two-digit numbers.

★ **Rook's find**

Adding can be done in any order, so 46 add 37 and 37 add 46 land on the same total every time. Rook says that is a free check, and free checks are the only kind he trusts.

→ **Take it further**

Add three two-digit numbers: 21, 34 and 42. Partition all three, put every ten together, then put every one together. The method does not change just because there is more of it.

Making a ten when you add



Ten loose cubes go in. One tied bundle of ten comes out. Nothing was added and nothing was lost.



Exchange: swap ten ones for one ten.

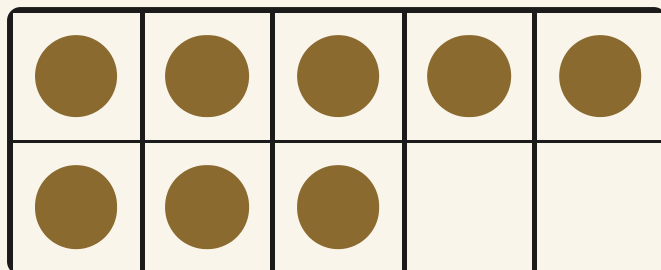
♣ When the ones make more than ten

Sometimes the ones add up to more than ten. Eight ones and six ones make more than a ten frame will hold, so the ones column cannot keep them all.

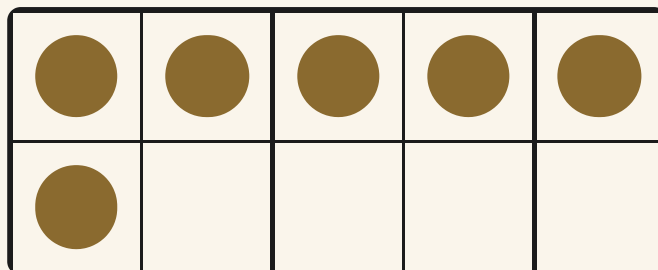
So you **exchange**. Ten of those ones become one ten, and the ten walks over to the tens column. What is left stays behind as ones.

Exchanging never changes how much you have. Ten loose cubes and one bundle of ten are the same amount, tied up differently.

TOPIC 8.2 · CONTINUED



the ones from twenty-eight



the ones from forty-six

Do you remember?

A ten frame is full at ten. If your counters spill out of it, you have a ten to exchange.

$$28 = 20 + 8$$

$$46 = 40 + 6$$

Watch me work one out 1

Add 28 and 46.

- 1 Partition both numbers: twenty and eight, forty and six.
- 2 Add the ones: eight and six make **14**.
- 3 **14** is more than ten, so exchange. It comes apart into **10** and **4**.
- 4 Add the tens: twenty and forty make **60**. The exchanged ten joins them.
- 5 **60** and **10** makes seventy, and **4** ones are still waiting.

Answer: $28 + 46 = 74$

$$14 = 10 + 4$$

How do you know?

Milo says the ones column can hold fourteen. Use a ten frame to show him why it cannot, and say what has to happen instead.

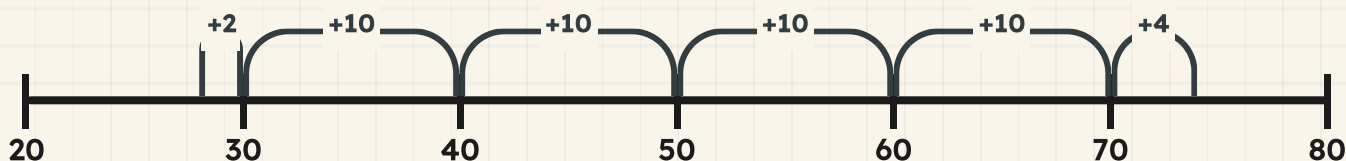
⇌ Check it a different way

You worked it out by adding the ones and exchanging ten of them for one ten. Now try it another way.

Now do it without exchanging anything. Start at twenty-eight and add just enough to reach the next ten: two. That leaves you standing on thirty, which is an easy place to add from.

Add the forty, then the four ones you have not used yet. No exchange, no columns, and the same total. When two methods that share no steps agree, you are right.

Two to reach thirty, then the forty in tens, then the four ones that were left.



■ SCAN TO SEE IT MOVE

Add two two-digit numbers by moving tens and ones around, and see the ten being made when ten ones are exchanged.

mathigon.org/polypad/number

Your turn



exchange

1 Add 27 and 15. Watch for ten ones you can exchange.

2 Add 38 and 24.

3 Add 46 and 27. Write down exactly what you exchanged.

4 Add 55 and 18.

5 Copy these and work them out.

$$29 + 16 = \boxed{}$$

$$47 + 35 = \boxed{}$$

TOPIC 8.2 · YOUR TURN · CONTINUED



exchange

6 Add 34 and 29 by going up to the next ten first.

7 Add 58 and 26.

8 Ring YES where you must exchange ten ones for one ten.

43 add **25**

YES / NO

38 add **27**

YES / NO

56 add **14**

YES / NO

61 add **29**

YES / NO

TOPIC 8.2 · YOUR TURN, CONTINUED



exchange

- 9** Add 49 and 23. Now add 23 and 49. Compare your two totals.
- 10** Draw ten frames to show 26 add 17.
- 11** Test Milo's idea. He says 45 add 15 needs no exchange at all.
- 12** Add 62 and 29.
- 13** Find two two-digit numbers that add up to exactly 100.



exchange

14 Add 36 and 36. Say which digits made the new ten.

Find out together

Play this with a partner. Each of you writes a two-digit number without showing it. Turn them over together and race to add them. Then check each other, one of you partitioning and the other counting on. Play eight rounds. How many of your totals needed an exchange, and how many did not?

Yard challenge

Find a pair of two-digit numbers that makes a total of 100 and needs an exchange. Then find a pair that makes 100 and needs no exchange at all. Both exist. Explain what is different about their ones digits.

Finding a small difference

✿ The difference is the gap

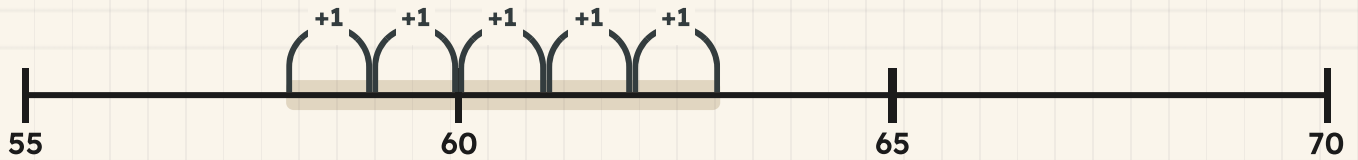
The **difference** between two numbers is how much bigger one is than the other. It is the gap between them, and a gap has a size even when nothing is being taken away.

When the two numbers are close together, do not take one from the other. Stand on the smaller one and **count on** until you reach the bigger one. Count your hops, not the numbers you land on.

Fifty-eight and sixty-three are close. Counting on takes a moment. Taking fifty-eight away takes much longer and goes wrong more often.

TOPIC 8.3 · CONTINUED

The tinted stretch is the gap. Pip hops it in ones and counts the hops.



Two towers side by side. The difference is the bit that sticks up.

How do you know?

Hattie counted the numbers 58, 59, 60, 61, 62, 63 and said the difference is six. Pip counted five hops. Which of them counted the gap, and which counted the numbers?

fifty-eight cubes

sixty-three cubes

 Watch me work one out 1

Find the difference between 58 and 63.

- 1 The numbers are close, so count on rather than take away.
- 2 Put your finger on the smaller number, fifty-eight.
- 3 Hop up in ones: fifty-nine, sixty, sixty-one, sixty-two, sixty-three.
- 4 Count the hops you made, not the numbers you said.
- 5 You stopped on sixty-three after 5 hops.

Answer: the difference between **58** and **63** is **5**

 Check it a different way

You worked it out by counting on from the smaller number. Now try it another way.

You went up. Now go down. Stand on sixty-three and count back in ones until you reach fifty-eight, keeping the count on your fingers.

Then do the check Cambridge asks for in 2Pt7: add your answer to the **smaller** number. If fifty-eight and your answer land you on sixty-three, the gap you found is the real gap.

✂ The material you need

squared paper, a pencil and a pot of counters. Build both numbers as two towers of counters, stand them side by side, and look at the part that sticks up.

Do you remember?

A difference is never a negative number. The gap between two numbers is the same size whichever end you start from.

Your turn



difference

- 1 Find the difference between 58 and 62.
- 2 Find the difference between 45 and 51.
- 3 Count on from 67 to 72. How many hops did you make?
- 4 Find the difference between 89 and 94.
- 5 Copy this line and ring 49 and 55 on it.

1
56

TOPIC 8.3 · YOUR TURN · CONTINUED



difference

- 6** Which gap is smaller: 34 up to 38, or 71 up to 76?
- 7** Hattie is 68 cm tall. Otto is 74 cm tall. Find the difference.
- 8** Count on from 76 to 81 in ones. Write every number you say.
- 9** Match each pair to its difference.
- | | |
|-------------|-----|
| 30 and 32 ● | ● 5 |
| 41 and 45 ● | ● 2 |
| 56 and 59 ● | ● 4 |
| 60 and 65 ● | ● 3 |

TOPIC 8.3 · YOUR TURN, CONTINUED



difference

- 10** Find the difference between 90 and 97.
- 11** Two planks measure 55 cm and 61 cm. Find the difference.
- 12** Ring the pair with a difference of three.
- 38 and 41** **52 and 57** **64 and 70**

13 Bramble sold 43 pegs, then 49 pegs. How many more the second time?

14 Find the difference between 26 and 31 two different ways.



difference

15 Write a pair of numbers with a difference of five.

→ **Take it further**

Hattie says the difference between 30 and 90 is a small difference because both numbers end in nought. Test her by counting on in tens from 30 to 90, and say whether counting on is still the quickest method when the gap gets big.

Taking away, and finding the difference

$$64 - 27 = \boxed{}$$

$$64 - 37 = \boxed{}$$

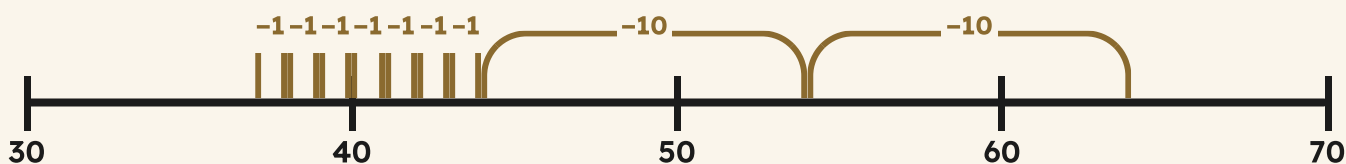
♣ Taking away, and the order it must go in

Taking away is walking backwards. Partition the number you are taking off, then walk back the tens first and the ones after. Landing on a ten on the way makes the rest easy.

Adding can be done in any order. Sixty-four add twenty-seven and twenty-seven add sixty-four give the same total, so you may always start with the larger number.

Taking away cannot. Sixty-four take twenty-seven leaves something on the bench. Twenty-seven take sixty-four leaves you owing Bramble, and Year 2 does not go there. The bigger number goes first.

Two hops of ten back, then seven single hops back. Twenty-seven, walked backwards.



Do you remember?

A jump back of ten does not change the ones digit. Sixty-four, fifty-four, forty-four. The four stays the whole way.

 Watch me work one out 1**Take 27 from 64.**

- 1 Partition the number you are taking off: twenty and seven.
- 2 Count back the two tens from sixty-four: fifty-four, then **44**.
- 3 Now count back the seven ones from **44**.
- 4 Go down to forty first, because forty is an easy place to stand.
- 5 Three more back from forty leaves **37**.

Answer: $64 - 27 = 37$ **Check it a different way**

You worked it out by counting back in tens and then in ones. Now try it another way.

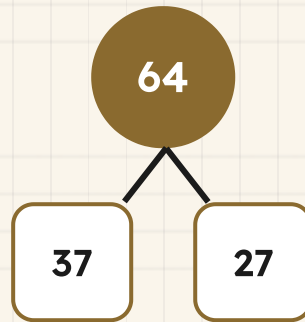
This is the check Cambridge calls 2Pt7, and it is the most useful one you will learn all year. Take your answer and **add it to the number you took away**.

Your answer, add twenty-seven. If that lands you back on **64**, where you started, then nothing went missing on the way down.

Subtracting walks one way. Adding back walks the other. A wrong answer will not survive the return trip.

TOPIC 8.4 · CONTINUED · CONTINUED

What is left, and what you took away, make the number you started with.



How do you know?

Tessa says every take away hides an addition inside it. Use the part-whole model above to explain what she means.

▣ WORDS FROM THIS PAGE



take away



difference



check



method

Your turn



take away

- 1 Take 24 from 57.
- 2 Work out 68 take away 35.
- 3 Take 16 from 43. Then add your answer to 16 to check.
- 4 Work out 72 take away 41.
- 5 Copy these two and work them out.

$$56 - 22 = \boxed{}$$

$$84 - 53 = \boxed{}$$

TOPIC 8.4 · YOUR TURN · CONTINUED



take away

6 Take 30 from 95, then take 4 more away.

7 Is 47 take away 12 the same as 12 take away 47? Explain.

8 Ring
TRUE
or
NOT
TRUE
for
each
line.

Adding can be done in either order.

TRUE / NOT TRUE

Taking away can be done in either order.

TRUE / NOT TRUE

A difference is the gap between two numbers.

TRUE / NOT TRUE

Adding your answer back is a way of checking.

TRUE / NOT TRUE

TOPIC 8.4 · YOUR TURN, CONTINUED



take away

- 9** Take 27 from 60.
- 10** Otto had 84 nails and used 39 of them. How many are left?
- 11** Find the difference between 38 and 63.
- 12** Take 45 from 78. Draw your jumps back on squared paper.
- 13** Take 19 from 51. Now take 51 from 70. Which was harder?



take away

14 Write a take away that leaves exactly 20.

15 Sort these into taking away and finding a difference.

Taking away

.....

.....

.....

Finding a difference

.....

.....

.....

Yard challenge

Bramble's tin held 90 cent this morning and holds 43 cent now. Work out what was spent, then prove it two ways: once by counting back, and once by adding your answer to 43 to see whether you get back to 90.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Add 44 and 25 by partitioning both numbers.
- 2 Add 37 and 48. Say where the new ten came from.
- 3 Find the total of 52 and 29.
- 4 Find the difference between 74 and 79.
- 5 Count on from 85 to 91. How many hops was that?
- 6 Take 33 from 69.
- 7 Take 28 from 50, then add your answer to 28 to check.
- 8 Sixty-three boards, and twenty-nine more arrive. How many altogether?
- 9 Which is larger: 46 add 27, or 81 take away 15?
- 10 Explain how you would check an addition a completely different way.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
add two two-digit numbers by partitioning them into tens and ones	☐	☐	☐
exchange ten ones for one ten, and say why you did it	☐	☐	☐
find a small difference between two numbers by counting on	☐	☐	☐
know that adding can be done in any order but taking away cannot	☐	☐	☐

NOW I CAN

- ☐ add two two-digit numbers by partitioning them into tens and ones
- ☐ exchange ten ones for one ten, and say why you did it
- ☐ find a small difference between two numbers by counting on
- ☐ know that adding can be done in any order but taking away cannot

▣ YARD WORDS



partition

split into tens and ones to work it out



exchange

swap ten ones for one ten



difference

how much more one is than another



take away

remove some to find what is left



method

the way you worked it out



check

work it out again a different way

Missing numbers and equal sides

The equals sign is a balance, not a full stop.



Come into the Plan Room

Hold your arms out like a beam. Put four fingers up on one hand and four on the other. Are you level?

Now put five up on one hand and three on the other. Which arm should dip? Say why.

Take ten counters and split them into two piles any way you like. Say your two numbers out loud. Both of you will have made ten, in different ways, and that is what the equals sign is for.

☀ In the yard today

Tessa has hung the brass scales over the Reckoning Board, and she is loading both pans until the beam sits dead level.

On the left pan she puts a weight of sixteen and a weight of four. On the right pan she puts a weight of seventeen and a weight of three. The beam does not move a hair.

Otto says that cannot be right, because the two sides are different. Tessa points at the beam and says nothing at all.

The equals sign is this beam, she says at last. It does not mean here comes the answer. It means the left side weighs the same as the right side, and you can check it with your eyes.

• CONTINUED

✓ **By the end you will**

- show that 16 add 4 is the same as 17 add 3, and say why
- find the missing number in an addition or a subtraction
- use a box or a shape to stand for a number you do not know yet
- make up a number story to go with a calculation

Cambridge Primary Mathematics Stage 2: 2Nc7 • 2Nc9 • 2Nc10 • 2Pt4

The equals sign means the same as



The brass scales on the Reckoning Board, loaded and level.

SHARPEN YOUR PENCIL

Say a pair of numbers that make ten. Now say a different pair that makes ten. Both pairs are worth the same as each other, and you did not need a sign to know it.

♣ The sign is a beam, not a full stop

The sign = means **is the same amount as**. It does not mean *here comes the answer*.

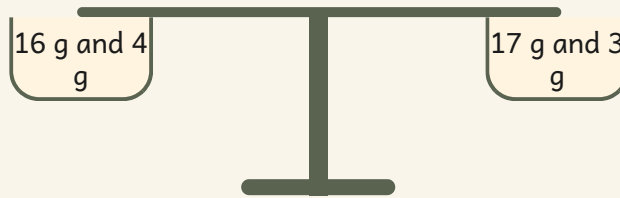
Think of a pair of scales. Whatever sits on the left pan must weigh the same as whatever sits on the right pan, or the beam tips. The equals sign is that beam, written down.

So there can be a sum on both sides. Sixteen add four sits on the left. Seventeen add three sits on the right. Work each side out and you will find the beam is level.

TOPIC 9.1 · CONTINUED

$$16 + 4 = 17 + 3$$

Both sides are worth the same. That is why the sign is allowed to be there.



Do you remember?

If the two sides are not worth the same, you may not write the equals sign. It would be a lie about the beam.

 Watch me work one out 1

Is it true that sixteen add four is the same as seventeen add three?

- 1** The sign asks one question: is the left worth the same as the right?
 - 2** Work out the left side on its own. Sixteen and four more.
 - 3** The left side comes to **20**.
 - 4** Now work out the right side on its own. Seventeen and three more.
 - 5** The right side comes to **20** as well, so the beam is level.
-

Answer: yes, it is true, because both sides are worth **20**

$$16 + 4 = 20$$

$$17 + 3 = 20$$

⇔ Check it a different way

You worked it out by working out both sides separately and comparing the totals. Now try it another way.

You added each side up and compared the two totals. Now check it without adding anything at all.

Look across the sign. Seventeen is **1** more than sixteen, and three is **1** less than four. One has simply crossed over from the four to the sixteen. Nothing was added and nothing was lost, so the pans must still balance. Two methods, one answer.

How do you know?

Milo saw $16 + 4 = 17 + 3$ and wrote 20 in the gap after the sign. What does he think the equals sign means, and what does it really mean?

Your turn



equals

1 Write what the equals sign means, in your own words.

2 Work out both sides.
Write true or not
true.

$$10 + 5 = 12 + 3$$

Check this one.

3 Work out both sides of $9 + 6 = 8 + 7$. Write true or not true.

4 Work out both sides of $11 + 2 = 10 + 4$. Write true or not true.

5 Draw scales with the beam level. Label both pans.

TOPIC 9.1 · YOUR TURN · CONTINUED



equals

- 6** Draw scales with the left pan heavier than the right.
- 7** Write a pair of numbers that make twelve.
- 8** Write a different pair of numbers that make twelve.
- 9** Write your two pairs from tasks 7 and 8 as one sentence.

- 10** True or not true?
Decide for each line.

The equals sign means here comes the answer.

TRUE / NOT TRUE

The equals sign means both sides are worth the same.

TRUE / NOT TRUE

A sum is allowed on both sides of the sign.

TRUE / NOT TRUE

TOPIC 9.1 · YOUR TURN, CONTINUED



equals

- 11** Fill the last pan so both sides balance.

$$14 + 6 = 13 + \underline{\quad}$$

Make the beam level.

- 12** Fill the last pan so both sides balance.

$$8 + 8 = 12 + \underline{\quad}$$

Make the beam level.

- 13** Explain why you may not write $7 + 5 = 13$.

14 Sort these sentences into true and not true.

true

not true

✂ The material you need

a pair of balance scales if the class has them, about twenty counters, cubes you can stack in twos, and a strip of card for covering up a number.

★ Rook's find

Tessa never says the word equals when she is loading the pans. She says the same as, every single time, and she does it on purpose. Children who say the same as out loud stop expecting an answer to come next.

Finding the missing number



The sign that means the same as.



A box where a number is hiding.

♣ A box holds the number you cannot see yet

Sometimes a number sentence has a **box** instead of a number. The box is not empty: there is exactly one number that belongs in it, and your job is to work out which.

Use the balance. The two sides must end up worth the same, so ask yourself how much the side with the box still needs.

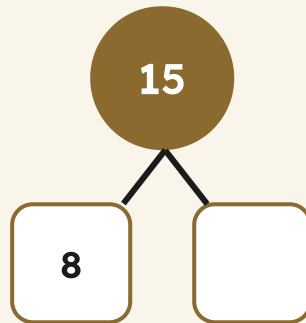
You can **count on** from the number you can see, or you can **take away** from the total. Both roads arrive at the same number, which is how you check yourself.

TOPIC 9.2 · CONTINUED

$$8 + \square = 15$$

Eight, and something more, is fifteen. How much more?

The whole is fifteen. One part is eight.



- 1 Read the sentence and find the box.
- 2 Say what the whole amount has to be.
- 3 Count on from the part you can see, or take it away from the whole.
- 4 Put your number in the box and check both sides balance.

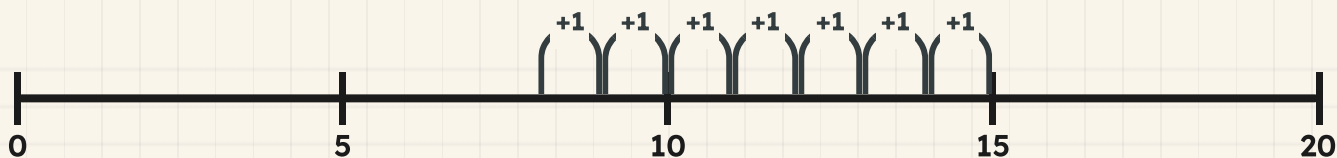
Watch me work one out 1

Find the missing number in eight add box equals fifteen.

- 1 The two sides must be worth the same, and the right side is fifteen.
- 2 So the left side must reach fifteen too, and it starts at eight.
- 3 Put your finger on eight and count on in ones to fifteen.
- 4 Nine, ten, eleven, twelve, thirteen, fourteen, fifteen: that was 7 hops.
- 5 Your finger stopped on **15**, so the box needs **7**.

Answer: the missing number is 7

Counting on from eight until you reach fifteen.



⇔ Check it a different way

You worked it out by counting on from the smaller number to the total. Now try it another way.

You found it by counting on from the smaller number. Now check a completely different way: put your answer back into the sentence and weigh both sides.

The left side becomes eight and **7**, which comes to **15**. The right side is **15**. The beam is level, so **7** is right. Putting your answer back in is the best check there is, because it tests the sentence you were actually given.

$$8 + 7 = 15$$

The answer put back in. Both sides weigh the same.

How do you know?

Otto found the missing number in $8 + ? = 15$ by taking eight away from fifteen. Pip counted on from eight. Why do two such different jobs give the same number?

Your turn



missing
number

- 1** Find the number the box is hiding.

$$6 + \boxed{} = 10$$

Count on, or take away.

- 2** Find the number the box is hiding.

$$7 + \boxed{} = 12$$

Count on, or take away.

- 3** Find the number the box is hiding.

$$20 - \boxed{} = 12$$

Something has been taken.

TOPIC 9.2 · YOUR TURN · CONTINUED

- 4 Find the number the box is hiding. $15 - \square = 9$
Something has been taken.

- 5 Write the missing number. 9 add box equals 14.



missing
number

- 6 Write the missing number. 13 take away box equals 6.

- 7 Write the missing number. Box add 5 equals 11.

- 8 Check task 7 by putting your answer back in.

- 9 Draw a part-whole with a whole of 16 and a part of 9.

- 10 Write the missing part from your drawing in task 9.

TOPIC 9.2 · YOUR TURN, CONTINUED



missing
number

- 11** Find the missing number by counting on.

$$18 + \square = 25$$

Hop along the line.

- 12** Find the missing number by taking away.

$$30 - \square = 24$$

How much went?

13 Write which method you like better, and say why.

14 Cover one number in a sum a partner has done. Swap and solve.



▣ **SCAN TO SEE IT MOVE**

A fresh missing-number sentence every time, so you can practise finding the number that is hidden in the box.

www.topmarks.co.uk/maths-games/mental-maths-train

→ **Take it further**

Write a sentence with a box in it and give it to a partner. Then make one that has no possible answer, such as a box that would have to be less than nothing, and see whether your partner spots that it cannot be done at all.

A shape or a letter for a number you do not know



A number hidden under one pan. The beam still tells you what it weighs.

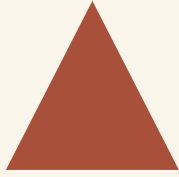
♣ Anything can stand in for the number

The box is not special. You can use a **shape** or a **letter** instead, and it means exactly the same thing: a number you have still to find. That number is called the **unknown**.

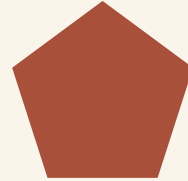
A triangle, a star, the letter **n**: all of them are just a place to stand until you know what belongs there.

One rule matters. In one sentence, the **same shape always stands for the same number**. Two different shapes may stand for two different numbers.

TOPIC 9.3 · CONTINUED



a number



another
number

$$9 + \square = 15$$

Nine add n equals fifteen. The letter n is the unknown.

Do you remember?

A shape or a letter is not a mystery. It is a number wearing a coat, and the sentence tells you how to take the coat off.

 Watch me work one out 1

Nine add n equals fifteen. What number does n stand for?

- 1** The letter n is one number, and the sentence must balance.
- 2** The right side weighs fifteen, so the left side must weigh fifteen.
- 3** Nine is already there, so n is whatever nine still needs.
- 4** Count on from nine: ten, eleven, twelve, thirteen, fourteen, fifteen.
- 5** That was **6** hops, so n stands for **6**.

Answer: n stands for **6**

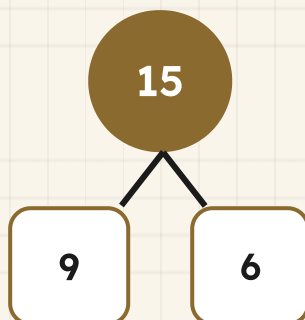
 Check it a different way

You worked it out by counting on from nine until you reached fifteen. Now try it another way.

You counted on from nine. Now use the other route altogether: work with the whole and the parts instead of hopping.

Fifteen is the whole and nine is one part, so take nine away from fifteen and the other part drops out: **6**. Adding and taking away are the same street walked in opposite directions, so the two methods must agree, and they do.

The whole, and both parts, with the coat off.



How do you know?

In one sentence a star stands for 4. Later in the SAME sentence there is another star. What must that one stand for, and why can it not be anything else?

Your turn



unknown

- 1** Write what the letter n stands for. $5 + \square = 13$
5 add n equals 13.

- 2** Write what the triangle stands for. $11 + \square = 20$
11 add a triangle is 20.

- 3** Write what the star stands for. 14 take away star equals 8.

- 4** Write what n stands for. n add 7 equals 16.

- 5** Draw your own shape for an unknown number. Name it.



unknown

TOPIC 9.3 · YOUR TURN · CONTINUED

6 Write a sentence using your shape from task 5.

7 Write whether two of the same shape can hold different numbers.

8 Choose the word that means a number you have not found.

unknown

total

answer

9 Match each sentence to what its shape stands for.

$$4 + n = 9$$

$$2$$

$$n + 3 = 10$$

$$5$$

$$n + 6 = 8$$

$$7$$

10 Ring the sentence where n is the largest.

$$n + 1 = 9$$

$$n + 5 = 9$$

$$n + 8 = 9$$

TOPIC 9.3 · YOUR TURN, CONTINUED



unknown

- 11** Find n , then check by putting it back in. $16 - \square = 9$
16 take away n equals 9.

- 12** Write a sentence where n stands for 10.

- 13** Explain why a shape is easier than writing something is missing.

- 14** Write two shapes that stand for different numbers, and label both.

Find out together

Work in a pair. One of you takes a handful of counters, counts them secretly, and hides some under a cup. Say out loud how many you started with and how many are still showing. Your partner works out what is under the cup and says it before you lift it. Lift the cup and see. Swap over, and do it ten times. How many did each of you get right?

Making up a number story



Level pans. The story and the sentence agree.

The number the story is really asking for.

♣ Every calculation is a story about something

A number sentence is short because it has thrown all the words away. Put the words back and you have a **story**.

A good story needs three things: **who or what**, a **number to start with**, and something that **happens**. Then the missing number is the thing you want to know.

Twelve take away a box equals five could be twelve buns on a tray, some of them eaten, and five left. The box is how many were eaten, and now the calculation is about something real.

TOPIC 9.4 · CONTINUED

$$12 - \square = 5$$

Twelve buns on the tray. Five are left. How many went?

Twelve buns, and five are left on the tray.

$$12 - 5 = \square$$

- 1 Read the sentence and see what it does: adds, or takes away.
- 2 Choose something real to count.
- 3 Say the starting number and what happened to it.
- 4 Ask the question the box is asking, in words.

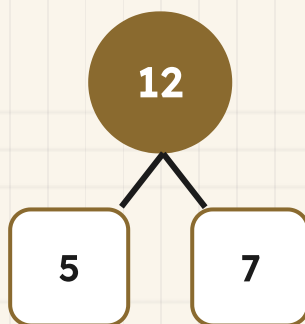
Watch me work one out 1

Make up a story for twelve take away box equals five, and answer it.

- 1** Choose something you can really count. Twelve buns on a tray.
- 2** The sentence takes away, so some of the buns must go.
- 3** Five are still on the tray at the end, so five is what is left.
- 4** The story asks how many buns were eaten, and that is the box.
- 5** Twelve to start, five left, so **7** were eaten.

Answer: Otto ate **7** buns, because twelve take away **7** leaves five

The whole tray, what is left, and what went.



⇌ **Check it a different way**

You worked it out by acting the story out with buns and counting what was left. Now try it another way.

You told the story with real buns and counted what was left. Now leave the buns alone and test the numbers on their own.

Add the five that stayed to the **7** that went: that makes **12**, which is exactly the tray you started with. The story and the sentence weigh the same, so both are right.

$$5 + 7 = 12$$

What stayed and what went, weighed against the whole tray.

How do you know?

Milo made up this story for $12 - \text{box} = 5$: I had twelve marbles and then I had twelve marbles. Why does his story not fit the sentence at all?

Your turn



- 1** Make up a story for 7 add 5.
- 2** Make up a story for 14 take away 6.
- 3** Make up a story about coins for 10 add 5.
- 4** Make up a story about children in a line for 9 add 4.
- 5** Write the question your story in task 4 asks.

TOPIC 9.4 · YOUR TURN · CONTINUED



story

- 6** Make up a story for this sentence. $10 - \square = 4$
Something went. How much?

- 7** Write the answer to your story in task 6.

- 8** Make up a story for this sentence. $13 + \square = 20$
Something arrived. How much?

- 9** Write a sentence to go with this story. Six frogs, two hop away.

- 10** Write a sentence to go with this story. Eight pegs, five more.

TOPIC 9.4 · YOUR TURN, CONTINUED



story

- 11** Tell a partner your best story. Let them write the sentence.
- 12** Make up a story where the missing number is at the start.
- 13** Write why a story needs something you can really count.
- 14** Draw your story from task 12 and label the missing part.

Yard challenge

Make up one story that two different sentences both fit, one adding and one taking away. Write both sentences underneath it. Then explain to a partner why the same story can be written two ways without either of them being wrong.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Say what the equals sign means, without using the word answer.
- 2 Work out both sides of $12 + 5 = 14 + 3$ and say whether it is true.
- 3 Find the missing number in $7 + \text{box} = 16$.
- 4 Find the missing number in $19 - \text{box} = 11$.
- 5 Say what n stands for in $n + 8 = 15$.
- 6 Explain why the same shape must hold the same number all through a sentence.
- 7 Check a missing number by putting your answer back into the sentence.
- 8 Make up a story for $11 - 4$, using something you can count.
- 9 Write a number sentence for this story. Nine cubes, three fall off.
- 10 Draw scales that show $6 + 4$ balancing $7 + 3$.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
show that 16 add 4 is the same as 17 add 3, and say why	☐	☐	☐
find the missing number in an addition or a subtraction	☐	☐	☐
use a box or a shape to stand for a number you do not know yet	☐	☐	☐
make up a number story to go with a calculation	☐	☐	☐

NOW I CAN

- ☐ show that 16 add 4 is the same as 17 add 3, and say why
- ☐ find the missing number in an addition or a subtraction
- ☐ use a box or a shape to stand for a number you do not know yet
- ☐ make up a number story to go with a calculation

▣ YARD WORDS



equals

is the same amount as



balance

the same on both sides



missing number

the number hidden in the box



number sentence

a calculation written with signs



story

a real situation a calculation fits



unknown

a number you have still to find

Rows, columns and times tables

Rook's rack is five across and four down, and he can prove it two ways.

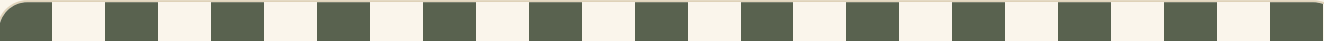


Come into the Plan Room

Take twelve counters. Lay them out in rows, with the same number in every row. How many rows did you make?

Now do it again with a different number in each row. Is there more than one way?

Count in twos to twenty. Now count in fives to fifty. Which one can you already do without thinking?



☀ In the yard today

Rook is standing in front of the new rack with one wing out, counting along the rows.

The rack is five holes across and four holes down, and every hole has a peg in it. Rook has counted the pegs twice this morning and got the same number both times, which has put him in a good mood.

The first time he counted along the rows. The second time he counted down the columns. **Different journeys, he says, same pegs. That is not luck. I did not move a single one of them in between.**

✓ **By the end you will**

- write repeated addition as a multiplication, using the times sign
- read one array as two multiplications, and say why both are right
- know the 2, 5 and 10 times tables, and the division facts with them
- work out the 3 and 4 times table facts from ones you know

Cambridge Primary Mathematics Stage 2: 2Nc16 • 2Nc17 • 2Nc19 • 2Nc22

Multiplying as repeated addition



The rack: five across, four down, every hole full.

SHARPEN YOUR PENCIL

Count in twos round the class. When somebody hesitates, start again from the beginning. See how far the class gets before somebody stops.

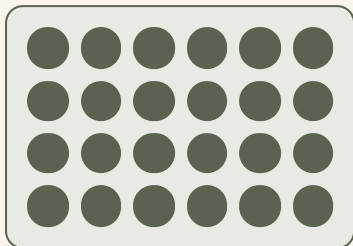
♣ A short way of writing the same thing again and again

Four rows of six pegs means six, and six, and six, and six. Writing that out is slow, so there is a sign for it: $\times$, which means **times**, or **lots of**.

Four times six means four lots of six. The multiplication does not do anything the addition did not do: it just says it in three symbols instead of seven.

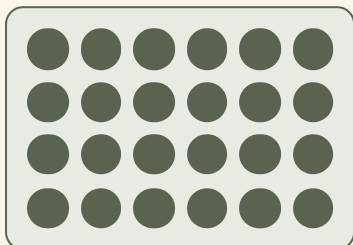
The answer to a multiplication is called the **product**.

TOPIC 10.1 · CONTINUED



$$6 + 6 + 6 + 6 = 24$$

Four rows of six, written as an addition.



$$4 \times 6 = 24$$

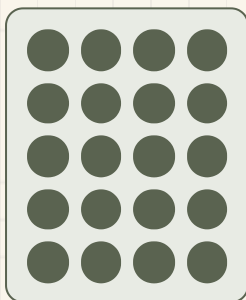
The same picture, written as a multiplication.

Watch me work one out 1

How many pegs are in five rows of four?

- 1** Draw it first: five rows, four pegs in each row.
- 2** Count one row to be sure: four.
- 3** Now count the rows: five.
- 4** Five lots of four, so add four five times: four, eight, twelve, sixteen, twenty.
- 5** Write it the short way as well: five times four.

Answer: there are **20** pegs



$$4 + 4 + 4 + 4 + 4 = 20$$

Five rows of four.

⇌ Check it a different way

You worked it out by adding four five times along the rows. Now try it another way.

You added four five times. Now count the same pegs in a different order: count in FIVES down the columns instead, five, ten, fifteen, twenty.

Four counts of five, instead of five counts of four. Same pegs, same total, and now you have reached twenty by two different routes.

How do you know?

Why is four add four add four add four add four the same as five times four? Point at the picture while you explain.

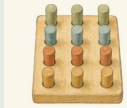
Your turn



multiply

- 1 Write $3 + 3 + 3 + 3$ as a multiplication.
- 2 Write $5 + 5 + 5$ as a multiplication.
- 3 Write $2 + 2 + 2 + 2 + 2$ as a multiplication.
- 4 Write 4×3 as an addition.

5 Write 2×6 as an addition.



multiply

6 Draw three rows of four counters. Write the multiplication.

7 Draw two rows of five counters. Write the multiplication.

8 Work out 3×2 .

9 Work out 4×5 .

10 Work out 2×10 .



multiply

11 Rook fills three rows of five holes. How many pegs?

12 Draw an array for 5×2 and count it.

13 Write what the word product means.

14 Write a multiplication whose product is 12.

✂ The material you need

about thirty counters, and squared paper so your rows line up. An egg box makes an excellent array if you have one.

★ Rook's find

The multiplication sign was invented in 1631, which makes it younger than the plus sign by more than a century. People wrote out the additions in full before that, and Rook says it must have been exhausting.

An array read two ways

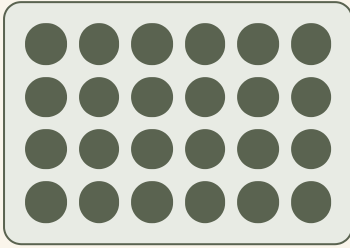
♣ One picture, two multiplications

An **array** is things laid out in rows and columns, with every row the same length. The rack in the yard is an array.

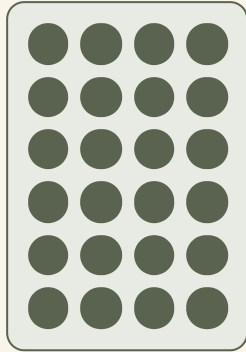
Here is the important part. Read an array along its **rows** and you get one multiplication. Read the SAME array down its **columns** and you get a different multiplication.

But nothing moved. The pegs are exactly where they were. So the two multiplications **MUST** have the same answer, and that is why four sixes and six fours are both twenty-four.

TOPIC 10.2 · CONTINUED



$$6 + 6 + 6 + 6 = 24$$



$$4 + 4 + 4 + 4 + 4 + 4 = 24$$

One array. Read along the rows it is four sixes. Read down the columns it is six fours.
Nobody touched a peg in between.

Do you remember?

Because an array can be read two ways, every multiplication you learn is really TWO facts. Learning your fives gives you a great deal more than the fives.

 Watch me work one out 1

Rook's rack is four rows of six. Show that four sixes and six fours give the same answer.

- 1** Count along the rows: six, twelve, eighteen, twenty-four.
 - 2** So four rows of six is twenty-four.
 - 3** Now turn your head and count down the columns instead.
 - 4** Four, eight, twelve, sixteen, twenty, twenty-four.
 - 5** Six columns of four is also twenty-four, and you moved nothing.
-

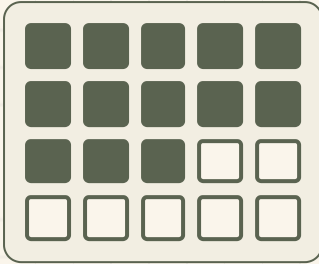
Answer: both are **24**, because they are the same pegs

 Check it a different way

You worked it out by counting along the rows and then down the columns. Now try it another way.

You counted the array twice, in two directions. Now check a third way that does not involve counting at all: turn the paper a quarter turn.

The four rows of six have become six rows of four, and you did not add or remove anything. Turning the page cannot change how many pegs there are, so the two multiplications cannot possibly differ.



A different rack, four by five, with 13 holes used and 7 still empty.

How do you know?

Tessa says learning six fours is a waste of time if you already know four sixes. Is she right? Explain using the array.

Your turn



- 1 Draw an array of three rows of five counters.
- 2 Write the two multiplications your array shows.
- 3 Draw an array of two rows of six.
- 4 Write the two multiplications for that array.

5 Write the partner fact for 5×3 .



array

6 Write the partner fact for 2×8 .

7 Write the partner fact for 10×4 .

8 Rook's rack is 5 across and 4 down. How many holes?

9 Rook fills 13 holes of that rack. How many are empty?

10 Draw an array with 10 counters. Write both multiplications.

TOPIC 10.2 · YOUR TURN, CONTINUED



array

- 11** Match each multiplication to its partner fact.

4×3

5×2

6×4

4×6

3×4

2×5

- 12** Draw an array that can only be read one way.

13 Write why turning an array does not change its total.

14 Find two different arrays that both hold 12 counters.



SCAN TO SEE IT MOVE

Drag counters into rows and columns to build any array, and watch the same array become two different multiplications.

mathigon.org/polypad/number

→ Take it further

Find every array that holds exactly sixteen counters. Draw them all. One of them looks the same whichever way you read it. Which, and why is it special?

The 2, 5 and 10 times tables



A ROW goes across.



A COLUMN goes down.

♣ The three easiest tables, and why

The two, five and ten times tables come first because each one has a pattern you can see, and a pattern is far easier to learn than a list.

Every answer in the **ten** table ends in 0. Every answer in the **five** table ends in 0 or 5. Every answer in the **two** table is an even number.

So before you work one out, you already know what the answer should look like. If you get an odd number in the two times table, you have made a mistake and you do not need anybody to tell you.

The two times table.

TOPIC 10.3 · CONTINUED

$$1 \times 2 = 2$$

$$2 \times 2 = 4$$

$$3 \times 2 = 6$$

$$4 \times 2 = 8$$

$$5 \times 2 = 10$$

$$6 \times 2 = 12$$

$$7 \times 2 = \underline{\hspace{2cm}}$$

$$8 \times 2 = 16$$

$$9 \times 2 = \underline{\hspace{2cm}}$$

$$10 \times 2 = 20$$

The ten times table.

$$1 \times 10 = 10$$

$$2 \times 10 = 20$$

$$3 \times 10 = 30$$

$$4 \times 10 = 40$$

$$5 \times 10 = 50$$

$$6 \times 10 = \underline{\hspace{2cm}}$$

$$7 \times 10 = 70$$

$$8 \times 10 = \underline{\hspace{2cm}}$$

$$9 \times 10 = 90$$

$$10 \times 10 = 100$$

The five times table.

$$1 \times 5 = 5$$

$$2 \times 5 = 10$$

$$3 \times 5 = 15$$

$$4 \times 5 = \underline{\quad}$$

$$5 \times 5 = 25$$

$$6 \times 5 = 30$$

$$7 \times 5 = \underline{\quad}$$

$$8 \times 5 = 40$$

$$9 \times 5 = \underline{\quad}$$

$$10 \times 5 = 50$$

 **Watch me work one out 1**

Work out seven times five.

- 1 You know the answer must end in 0 or 5, because it is a five.
- 2 Count in fives, keeping track on your fingers as you go.
- 3 Five, ten, fifteen, twenty, twenty-five, thirty, thirty-five.
- 4 That was seven fives, because seven fingers are up.
- 5 Thirty-five ends in 5, which is what you expected.

Answer: seven times five is **35**

⇌ **Check it a different way**

You worked it out by counting in fives on your fingers. Now try it another way.

You counted in fives. Now use the partner fact instead: seven times five is the same as five times seven.

Count in sevens five times: seven, fourteen, twenty-one, twenty-eight, thirty-five. Harder counting, same answer, and now you have checked a five with a seven.

How do you know?

Why can no answer in the ten times table end in 7? Explain it without working any of them out.

Your turn



1 Work out 3×2 .

2 Work out 6×2 .

3 Work out 9×2 .

4 Work out 4×5 .

TOPIC 10.3 · YOUR TURN · CONTINUED

5 Work out 7×5 .



multiple

6 Work out 6×10 .

7 Work out 9×10 .

8 Write every answer in the ten times table up to ten tens.

9 Write what every answer in the two times table has in common.

10 Write what every answer in the five times table ends in.

TOPIC 10.3 · YOUR TURN, CONTINUED



multiple

- 11** Ring the numbers that are in the five times table.

15

22

40

37

- 12** Six pegs go in each of five rows. How many pegs?

- 13** Write a division fact that comes from 4×10 .

- 14** Write the two facts that come from 8 and 2.

🔗 Find out together

Work in a pair. One of you says a number under fifty. The other says whether it is in the two times table, the five times table, the ten table, or none of them, and must give a reason from the LAST DIGIT rather than by counting. Twenty turns each. Which numbers were in more than one table?

The 3 and 4 times tables

♣ Building the harder tables out of easier ones

The three and four times tables have no pattern in their last digit, so they take more learning. But you do not have to start from nothing.

The **four** table is the two table **doubled**. Six fours is double six twos. You already know six twos is twelve, so six fours is twenty-four.

The **three** table is the two table with one more lot added on. Six threes is six twos plus another six. Twelve and six makes eighteen.

The four times table.

TOPIC 10.4 · CONTINUED

$$1 \times 4 = 4$$

$$2 \times 4 = 8$$

$$3 \times 4 = 12$$

$$4 \times 4 = 16$$

$$5 \times 4 = 20$$

$$6 \times 4 = \underline{\quad}$$

$$7 \times 4 = 28$$

$$8 \times 4 = \underline{\quad}$$

$$9 \times 4 = \underline{\quad}$$

$$10 \times 4 = 40$$

The two table doubled IS the four table.

double **2** = **4** double **4** = **8** double **6** = **12** double **8** = **16** double **10** = **20**

double **12** = **24**

The three times table.

$$1 \times 3 = 3$$

$$2 \times 3 = 6$$

$$3 \times 3 = 9$$

$$4 \times 3 = 12$$

$$5 \times 3 = 15$$

$$6 \times 3 = \underline{\hspace{2cm}}$$

$$7 \times 3 = \underline{\hspace{2cm}}$$

$$8 \times 3 = 24$$

$$9 \times 3 = \underline{\hspace{2cm}}$$

$$10 \times 3 = 30$$

 Watch me work one out 1

Work out six times four using the two times table.

- ① Start with the two table, which you know: six twos is twelve.
- ② Four is double two, so six fours must be double six twos.
- ③ Double twelve.
- ④ Twenty-four.

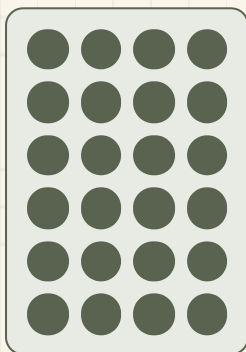
Answer: six times four is **24**

⇌ Check it a different way

You worked it out by doubling the two times table fact. Now try it another way.

You doubled a two-table fact. Now check with the array: draw six rows of four counters and count them.

Four, eight, twelve, sixteen, twenty, twenty-four. The counters do not care which table you used, and they agree with your doubling, so the shortcut is safe to trust next time.



Six rows of four. Count them and see.

How do you know?

Why does doubling a two-table fact give a four-table fact? Draw two arrays side by side to show a partner.

Your turn



product

1 Work out 3×3 .

2 Work out 5×3 .

3 Work out 8×3 .

4 Work out 3×4 .

TOPIC 10.4 · YOUR TURN · CONTINUED

5 Work out 7×4 .



product

6 Work out 9×4 .

7 Use 5×2 to help you work out 5×4 .

8 Use 4×2 to help you work out 4×3 .

9 Write every answer in the three times table up to ten threes.

10 Write every answer in the four times table up to ten fours.

TOPIC 10.4 · YOUR TURN, CONTINUED



product

- 11 Draw an array for 4×3 and write both facts.
- 12 Four rows of nine pegs. How many pegs altogether?
- 13 Write a number that is in both the three and four tables.
- 14 Write a division fact that comes from 8×3 .

Yard challenge

Find every number under fifty that is in BOTH the three times table and the four times table. There are fewer than you expect. Then work out what the smallest one is, and see whether the others are all in its own table.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write $6 + 6 + 6$ as a multiplication.
- 2 Draw an array of four rows of three counters.
- 3 Write both multiplications your array shows.
- 4 Work out 8×5 .
- 5 Work out 7×10 .
- 6 Work out 6×3 .
- 7 Work out 8×4 .
- 8 Write the partner fact for 3×7 .
- 9 Write what every answer in the ten times table ends in.
- 10 Explain why an array can be read two ways and still give one answer.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
write repeated addition as a multiplication, using the times sign	☐	☐	☐
read one array as two multiplications, and say why both are right	☐	☐	☐
know the 2, 5 and 10 times tables, and the division facts with them	☐	☐	☐
work out the 3 and 4 times table facts from ones you know	☐	☐	☐

NOW I CAN

- ☐ write repeated addition as a multiplication, using the times sign
- ☐ read one array as two multiplications, and say why both are right
- ☐ know the 2, 5 and 10 times tables, and the division facts with them
- ☐ work out the 3 and 4 times table facts from ones you know

▣ YARD WORDS



multiply

add the same number again and again



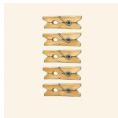
array

rows and columns of the same size



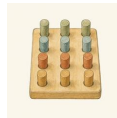
row

things side by side across



column

things one above another



times table

the multiplications for one number



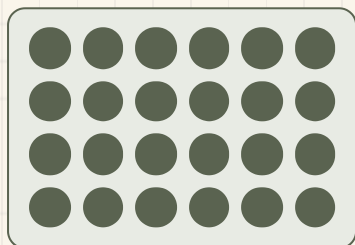
product

the answer to a multiplication

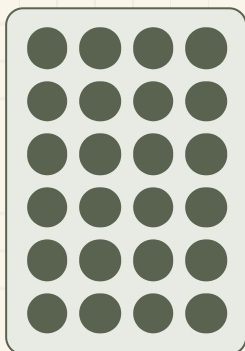
Rook's rack

Rook wants a new rack built for the wall of the yard, and he has opinions about it. It must hold exactly twenty-four pegs, and it must be rows and columns, not a heap.

Find every way to build a rack that holds exactly twenty-four pegs in equal rows and columns. Draw each one on squared paper.



$$6 + 6 + 6 + 6 = 24$$



$$4 + 4 + 4 + 4 + 4 + 4 = 24$$

One of the ways, drawn twice: four rows of six, and six rows of four. Rook counts that as ONE rack turned on its side, not two different racks.

Find out together

Work in a pair. One of you looks for racks with an even number of rows, the other for racks with an odd number. Put your drawings together at the end. Did either of you find one the other missed?

★ Rook's find

Twenty-four is a good number for a rack because it can be built so many ways. Rook says twenty-three is a terrible number for a rack, and when you have finished this project you will know exactly what he means.

WHAT TO HAND IN

- 1 Draw every rack you can find that holds exactly twenty-four pegs.
- 2 Write the multiplication under each drawing.
- 3 Say which two of your racks are really the same rack turned round.
- 4 Try twenty-three pegs. Write down what happens and why.

HOW IT WILL BE JUDGED

What is judged	Worth	What a good one looks like
Complete search	35%	you found all the ways, not just the easy ones
Correct multiplications	25%	the multiplication under each drawing matches the pegs you drew
Turned-round pairs	20%	you spotted which racks are the same one turned
The twenty-three question	20%	you explain why twenty-three will not make a full rack at all

Sharing, grouping and what is left

Twenty-six pegs into groups of four. Rook keeps the two left over.



Come into the Plan Room

Take twelve counters. Share them out fairly between you and one other person. How many did you each get?

Now take the same twelve counters and push them into groups of three. How many groups did you make?

Both times you started with twelve. Did you get the same answer both times? Talk about why not.

☀ In the yard today

Rook has tipped a whole box of pegs onto the bench and he is counting them into trays.

There are twenty-six pegs. There are four trays. Rook is dealing them out one at a time, round and round, the way you deal cards.

When he stops, every tray has the same number in it, and there are two pegs still sitting on the bench.

Those two are not a tray, says Rook, before anybody can suggest it. **They are what is left over. Do not put them anywhere.**

✓ **By the end you will**

- share a set equally between two, three, four or five
- put a set into groups of a given size, using the division sign
- use a multiplication you know to work out a division
- say what is left over, and why it is not another group

Cambridge Primary Mathematics Stage 2: 2Nc18 • 2Nc23 • 2Pt4

Sharing equally between



Four trays on the bench, and two pegs that would not fit.

**SHARPEN
YOUR
PENCIL**

Sit in a circle with twenty counters in the middle. Deal them out round the circle, one each, until you cannot go round again. Who ended up with the most?

♣ Sharing means one for you, one for you

When you **share between**, you know how many **people** there are, and you want to know how many **each** one gets.

The safe way is to deal, exactly like cards. One in this tray, one in that tray, one in that tray, and round again. Nobody can be cheated if you always go round in the same order.

When you cannot go round a whole time more, stop. Whatever is still in your hand is **left over**.



Twelve pegs shared between four trays. Deal them, do not guess them.

 Watch me work one out 1

Share twelve pegs equally between four trays.

- ➊ Put one peg in each tray. That is four pegs gone, eight left.
- ➋ Go round again: one in each tray. Four more gone, four left.
- ➌ Go round a third time. Four gone, none left in your hand.
- ➍ You went round three whole times, so each tray got three.

Answer: each tray gets **3** pegs

 Check it a different way

You worked it out by dealing them out one at a time. Now try it another way.

You dealt them out. Now do it the opposite way: take three pegs out of every tray and count them all back into one heap.

Three, and three, and three, and three. If you get twelve back, three each was right. If you get any other number, one of the trays was wrong.

How do you know?

Milo shared twelve pegs between four trays by putting five in the first tray to save time. What went wrong, and why does dealing one at a time stop it happening?

$$3 + 3 = \boxed{}$$

$$6 + 6 = \boxed{}$$

Your turn



share

- 1** Share ten counters equally between two people. Write how many each.
- 2** Share twelve counters equally between three people.
- 3** Share twenty counters equally between four people.
- 4** Share eighteen counters equally between two people.

- 5** Share fifteen counters equally between five people.



share

- 6** Share sixteen counters equally between four trays.

- 7** Six biscuits are shared between two friends. Draw it.

- 8** Twenty pegs go equally into five trays. How many in each tray?

- 9** Deal fourteen counters between three people. What is left over?

- 10** Share nine counters between two people. Write what happens.

TOPIC 11.1 • YOUR TURN, CONTINUED



share

- 11 Draw four trays. Share eight pegs between them by dealing.
- 12 Bramble shares twenty-four nails between three boxes. Draw it.
- 13 Write a sharing story for twelve counters and four people.
- 14 Share twenty-five counters between five people, then write the sharing as a division sentence.

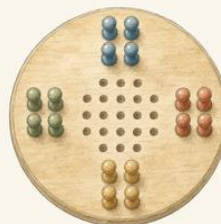
✂ The material you need

about thirty counters and four small trays or four sheets of paper to act as trays. Dried beans and saucers work just as well as anything the school will buy.

★ Rook's find

Dealing is how card games have worked for six hundred years, and it is not a coincidence. Rook says it is the only way of sharing that nobody has ever managed to argue with.

Grouping into groups of



SHARING: three trays, and you found how many EACH.

GROUPING: groups of four, and you found HOW MANY GROUPS.

✿ The same words, a different question

When you **group into groups of**, you know how big each group is, and you want to know **how many groups** you can make.

Sharing asks **how many each**. Grouping asks **how many groups**. They are different questions, and mixing them up is the easiest mistake in this whole unit.

Both are written with the same sign, $\div$. That is unfortunate, and it is why you must always say out loud which question you are answering before you start.



Twelve pegs in groups of four. Count the GROUPS.

Do you remember?

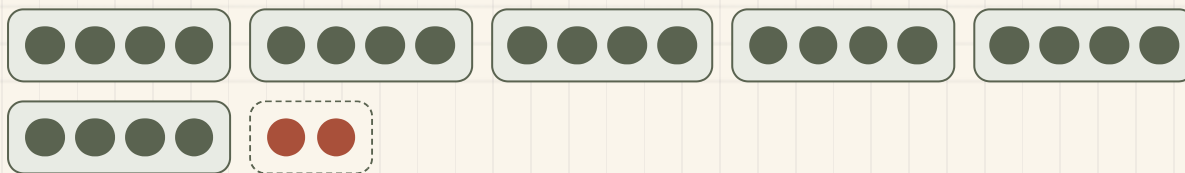
Say it out loud before you touch a counter. Am I finding how many each, or how many groups?

Watch me work one out 1

How many groups of four can you make from twenty-six pegs?

- 1** Push four pegs together. That is one group.
- 2** Push another four together. Two groups.
- 3** Keep going, counting the groups as you make them.
- 4** Stop when you have fewer than four pegs left in your hand.
- 5** Count the groups you made, then look at what is still in your hand.

Answer: 6 groups, with 2 left over



$$26 \div 4 = 6 \text{ remainder } 2$$

Six full groups, and two pegs that will not make a seventh. The dashed group is the leftover, not a group.

⇌ Check it a different way

You worked it out by making the groups and counting them. Now try it another way.

You made the groups and counted them. Now count the other way: six groups of four is four, eight, twelve, sixteen, twenty, twenty-four. Then add the two that were left over.

Twenty-four and two makes twenty-six, which is what you started with. Every peg is accounted for, so the answer holds together.

How do you know?

Why can the leftover never be four or more? Explain it using the picture above.

Your turn

- 1 Make groups of two from twelve counters. How many groups?
- 2 Make groups of five from twenty counters. How many groups?
- 3 Make groups of ten from forty counters. How many groups?
- 4 Make groups of three from eighteen counters. How many groups?
- 5 Make groups of four from sixteen counters. How many groups?
- 6 How many groups of five can you make from twenty-two counters?
- 7 Twenty pegs go into bags of four. How many bags?
- 8 Thirty nails go into boxes of ten. How many boxes?
- 9 Make groups of three from twenty counters. What is left over?
- 10 Make groups of two from fifteen counters. What is left over?

- 11** Write which question each one asks: how many each, or how many groups.

shared between three

in groups of three

- 12** Draw twenty-one counters in groups of five. Ring the leftover.

13 Write a grouping story for twenty-four pegs in groups of six.

14 Bramble has thirty-five nails and boxes that hold eight. How many boxes can he fill?



SCAN TO SEE IT MOVE

Share counters into groups and watch what is left over when they will not share equally.

mathigon.org/polypad/number

→ Take it further

Take twenty-four counters. Find every group size from two to twelve that leaves nothing over. Write your list. Then try twenty-three counters and see how short your list is.

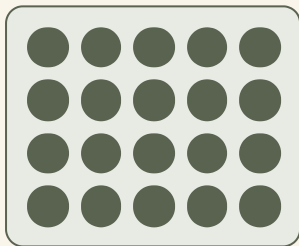
Division and multiplication together

♣ A division you know already

You already know that four fives make twenty. Look at what that tells you for free.

Twenty shared between four is five. Twenty in groups of five is four. You did not have to work either of those out: they were both sitting inside the multiplication you already knew.

Multiplying and dividing **undo** each other. When two operations undo each other, we call them **inverse**.



$$4 \times 5 = 20$$

One array, and it answers three questions at once.

TOPIC 11.3 · CONTINUED

Every one of these is a division as well.

$$1 \times 5 = 5$$

$$2 \times 5 = 10$$

$$3 \times 5 = 15$$

$$4 \times 5 = 20$$

$$5 \times 5 = 25$$

$$6 \times 5 = 30$$

 Watch me work one out 1

Use a times table fact to work out thirty divided by five.

- 1** Ask yourself: how many fives make thirty?
- 2** Count in fives and keep track: five, ten, fifteen, twenty, twenty-five, thirty.
- 3** You said six numbers to reach thirty, so six fives make thirty.
- 4** So thirty divided by five must be six.

Answer: thirty divided by five is **6**

 Check it a different way

You worked it out by counting in fives to thirty. Now try it another way.

You used the times table. Now check it with counters: take thirty counters and actually push them into groups of five.

Count the groups. If there are six groups and nothing left over, the table fact was the right one to reach for.

$$5 + 5 = \boxed{}$$

$$10 + 10 = \boxed{}$$

$$15 + 15 = \boxed{}$$

How do you know?

Hattie says knowing your times tables means you get all the divisions free. Is she right? Use thirty and five to explain.

Your turn

- 1 Two fives make ten. Write the division that goes with it.
- 2 Three fours make twelve. Write the division that goes with it.
- 3 Work out ten divided by two.
- 4 Work out twenty divided by ten.
- 5 Work out eighteen divided by two.

TOPIC 11.3 · YOUR TURN · CONTINUED

6 Work out twenty-five divided by five.

7 Work out sixteen divided by four.

8 Work out twenty-four divided by three.

9 Match each multiplication to its division.

2×5 ●

● $12 \div 3$

4×3 ●

● $20 \div 2$

10×2 ●

● $10 \div 5$

10 Write two divisions that come from four sixes make twenty-four.

TOPIC 11.3 · YOUR TURN, CONTINUED

- 11** Draw an array that shows twenty divided by four.
- 12** Rook says twelve divided by four is three. Draw the array that proves him right.
- 13** Write the four facts that come from three and five.
- 14** Tessa worked out thirty divided by three. Write which times table she should have used.

Find out together

Work in a pair with a set of counters. One of you says a multiplication from the two, five or ten times table. The other must say BOTH divisions that come out of it, and then prove one of them with the counters. Swap after every turn.

When something is left over



Six full trays, and the two pegs Rook refuses to move.

♣ A remainder is not a group

Sometimes a number will not share out or group up exactly. What is still in your hand at the end is called the **remainder**.

The remainder is always **smaller** than the group size. If it were as big as a group, you could make another group out of it, so it would not be left over at all.

Fourteen shared between four is three each, remainder two. Not three and a bit, and not four for some people. Three each, and two left.



3 each, with 2 left over

Fourteen between four. Two of the trays got a fourth peg, so the sharing is not equal yet.



$$14 \div 4 = 3 \text{ remainder } 2$$

Fourteen in groups of four: three full groups, and two left over in the dashed one.

 Watch me work one out 1

Twenty-six pegs go into trays of four. How many full trays, and what is left?

- 1** Count out four pegs into the first tray.
 - 2** Keep filling trays with four each, counting the full ones.
 - 3** Stop when fewer than four pegs are left in your hand.
 - 4** Count the full trays, then count what is in your hand.
 - 5** Check that what is left is smaller than four.
-

Answer: 6 full trays, remainder 2

⇌ Check it a different way

You worked it out by filling the trays and counting them. Now try it another way.

You filled the trays. Now put it back together: six trays of four is twenty-four, and the two left over makes twenty-six.

This is the real test of a remainder answer. Multiply, then add the leftover. If you do not get back to the number you started with, something has been lost.

How do you know?

Rook says a remainder of five is always a mistake when you are grouping in fours. Why must he be right?

Work out six fours, then add two. Do you get twenty-six?

Your turn

- 1 Share thirteen counters between four. Write the remainder.
- 2 Share eleven counters between two. Write the remainder.
- 3 Put seventeen counters into groups of five. Write the remainder.
- 4 Put twenty-three counters into groups of ten.
- 5 Put nineteen counters into groups of three.
- 6 Share twenty-two counters between five people.
- 7 Twenty-nine nails go into boxes of six. How many full boxes?
- 8 Fifteen biscuits are shared between four. What is left over?
- 9 Write a division with no remainder, using groups of five.
- 10 Write a division with a remainder of one, using groups of five.

TOPIC 11.4 · YOUR TURN, CONTINUED

11 True or not true? Decide for each line.	A remainder can be bigger than the group size.	TRUE / NOT TRUE
	A remainder can be zero.	TRUE / NOT TRUE
	Sharing and grouping always give the same answer.	TRUE / NOT TRUE

12 Twenty pegs, groups of six. Draw it and ring the remainder.

13 Check this answer: eighteen in groups of four is four, remainder two. Multiply and add back to test it.

14 Find a number under thirty that leaves remainder three in groups of five.

Yard challenge

Find a number smaller than fifty that leaves a remainder of one whether you group it in twos, in threes, or in fours. There is more than one answer, and finding the smallest is the real puzzle.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Share sixteen counters equally between four. Write how many each.
- 2 Put twenty-one counters into groups of seven. Write how many groups.
- 3 Write the division that goes with five fours make twenty.
- 4 Work out twenty-eight divided by four.
- 5 Twenty-seven pegs go into trays of five. Write the full trays and the remainder.
- 6 Explain the difference between sharing between three and grouping in threes.
- 7 Draw an array that proves eighteen divided by three is six.
- 8 Check this: twenty-two in groups of four is five, remainder two. Show your check.
- 9 Write a number under twenty that leaves no remainder in groups of three.
- 10 Say why a remainder is never as big as the group size.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
share a set equally between two, three, four or five	☐	☐	☐
put a set into groups of a given size, using the division sign	☐	☐	☐
use a multiplication you know to work out a division	☐	☐	☐
say what is left over, and why it is not another group	☐	☐	☐

NOW I CAN

- ☐ share a set equally between two, three, four or five
- ☐ put a set into groups of a given size, using the division sign
- ☐ use a multiplication you know to work out a division
- ☐ say what is left over, and why it is not another group

▣ YARD WORDS



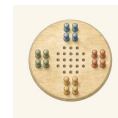
divide

share or group equally



share

give the same amount to each



group

put into sets of the same size



left over

what will not make another group



remainder

the amount left over



inverse

the operation that undoes the other

Doubling and halving

Double it to make two. Halve it to share with one.



Come into the Plan Room

Hold up four fingers on one hand. Now hold up four on the other. How many altogether? That is double four.

Say double one, double two, double three, all the way to double ten, as fast as you can without making a mistake.

Now say half of two, half of four, half of six, up to half of twenty. Which list was faster?

☀ In the yard today

Tessa lays out one row of blocks. Milo lays out a second row exactly the same, right underneath it.

Neither of them has counted anything yet. Milo just copied what Tessa did, block for block, and now there are two identical rows on the bench.

That is doubling, says Tessa. **You do not have to add anything. You just have to do the same thing twice, and then count once.**

Rook, who is watching, points out that you can go the other way as well: cover one row with your wing and you have halved it again.

✓ **By the end you will**

- double a two-digit number by doubling its tens and its ones
- double any multiple of five up to fifty, and halve it back
- halve an even number and check by doubling your answer
- use a double you know to work out a near double

Cambridge Primary Mathematics Stage 2: 2Nc20 • 2Nc21 • 2Pt9

Doubling two-digit numbers



Two identical rows of blocks, one directly under the other.

♣ Double the tens, double the ones

You already know double four is eight. Doubling twenty-four is the same job done twice: once on the tens, once on the ones.

Split twenty-four into twenty and four. Double the twenty to get forty. Double the four to get eight. Put them back together.

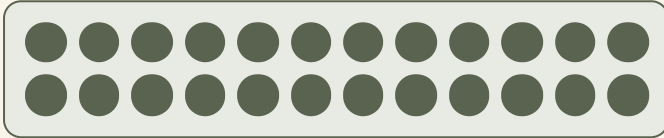
This works because doubling twenty-four means having two lots of everything in it, which means two lots of the tens and two lots of the ones.

TOPIC 12.1 · CONTINUED

$$\mathbf{24} = \mathbf{20} + \mathbf{4}$$

Double each part on its own, then join them.

double **20** = **40** double **4** = **8**



$$\mathbf{12} + \mathbf{12} = \mathbf{24}$$

Two rows of twelve is the same as double twelve. Doubling is just the two times table wearing a different hat.

 Watch me work one out 1**Double twenty-four.**

- 1 Split it into its tens and its ones: **20** and **4**.
- 2 Double the tens. Two lots of twenty is forty.
- 3 Double the ones. Two lots of four is eight.
- 4 Add your two doubles together: forty and eight.

Answer: double twenty-four is **48**

 Check it a different way

You worked it out by doubling the tens and the ones separately. Now try it another way.

You doubled the parts. Now do it as a straight addition instead: write twenty-four, and twenty-four underneath it, tens under tens, and add the whole thing.

Same answer, different route. If the partitioned double and the written addition disagree, one of them has a mistake, and the addition is usually the one to trust least when you are learning.

$$24 + 24 = \boxed{}$$

$$20 + 20 = \boxed{}$$

How do you know?

Otto doubled thirty-six and got sixty-twelve. Explain what he did right, and what he forgot to do next.

Your turn



balance

1 Double 12.

2 Double 21.

3 Double 33.

4 Double 14.

5 Double 41.



balance

6 Double 27.

7 Double 18.

8 Double 44.

9 Write double 13 as an addition, then work it out.

10 Split 26 into tens and ones, then double each part.



balance

- 11 Double 19. Watch the ones carefully.
- 12 Rook has 23 pegs. Bramble brings the same again. How many now?
- 13 Double 30, then double your answer.
- 14 Write two numbers whose doubles are both under fifty.

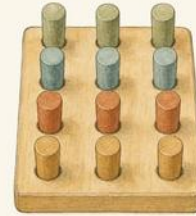
✂ The material you need

squared paper, so your tens sit under your tens when you write a double out as an addition. About forty counters if you would rather build the doubles than write them.

★ Rook's find

Doubling is the oldest shortcut in arithmetic. Rook says the ancient Egyptians multiplied almost everything by doubling over and over again, which means they could multiply large numbers knowing only what you are learning on this page.

Doubles of multiples of five



Two rows the same. That is all doubling is.

And two rows the same is also two times.

♣ Fives double into tens

Multiples of five are the friendliest numbers to double, because two fives make ten. So every double of a five lands on a nice round number.

Double five is ten. Double fifteen is thirty. Double twenty-five is fifty. Double fifty is one hundred.

Learn these by heart. They come up constantly, particularly with money, because coins come in fives, tens, twenties and fifties.

Worth knowing without thinking.

double **5** = **10** double **15** = **30** double **25** = **50** double **50** = **100**

Your fives, with two left for you.

$$1 \times 5 = 5$$

$$2 \times 5 = 10$$

$$3 \times 5 = 15$$

$$4 \times 5 = \underline{\quad\quad}$$

$$5 \times 5 = 25$$

$$6 \times 5 = \underline{\quad\quad}$$

Do you remember?

Two fives make ten. Every double of a multiple of five leans on that one fact.

Watch me work one out 1**Double thirty-five.**

- 1 Split it: thirty and five.
- 2 Double the thirty. Two lots of thirty is sixty.
- 3 Double the five. Two fives make ten.
- 4 Add them: sixty and ten.

Answer: double thirty-five is **70**

⇌ Check it a different way

*You worked it out by **splitting into thirty and five**. Now try it another way.*

You partitioned it. Now check with money, which is what fives are really for: put out two thirty-five cent amounts in coins and count the lot.

Coins cannot lie to you. If you count out seventy cent, double thirty-five is seventy.



Thirty-five cent, one way of making it.

How do you know?

Why is doubling twenty-five easier than doubling twenty-four, even though twenty-five is the bigger number?

Your turn

- 1 Double 5.
- 2 Double 15.
- 3 Double 25.
- 4 Double 45.
- 5 Double 50.
- 6 Double 10, then double 20, then double 40.
- 7 Double 35.
- 8 Double 55.
- 9 Write the doubles of every multiple of ten up to fifty.
- 10 Which is bigger, double 25 or double 24? Say how you know.

TOPIC 12.2 · YOUR TURN, CONTINUED

11 A basket costs 45c. Two baskets cost how much?

12 Double 15c, then double your answer again.

13 Sort these into doubles under fifty and doubles over fifty.

under fifty

over fifty

14 Write a multiple of five whose double is exactly one hundred.



■ SCAN TO SEE IT MOVE

Doubles and halves, a fresh one each time you press it, so you can learn them until they are quick.

www.topmarks.co.uk/maths-games/daily10

→ Take it further

Start at five and keep doubling: five, ten, twenty, forty. How far can you go before you leave the numbers you know? Write your chain down and see how few steps it takes to pass one hundred.

Halving, and how it undoes doubling

♣ Halving walks back the way doubling came

Doubling and halving are **inverse** operations, which means each one undoes the other. If you double a number and then halve your answer, you arrive back exactly where you started.

That is not a coincidence and it is not a trick. It is the most useful thing on this page, because it means **every double you work out comes with its own check built in.**

Halve the tens, halve the ones, put them back together. Forty -six halves into twenty and three.

$$46 = 40 + 6$$

Halve each part, then join them.

$$\text{half of } 40 = 20 \quad \text{half of } 6 = 3$$



Half of eight counters. Deal them into two equal piles and count one pile.

 Watch me work one out 1

Find half of forty-six.

- 1** Split it into tens and ones: forty and six.
 - 2** Halve the forty. Half of four tens is two tens, so twenty.
 - 3** Halve the six. Half of six is three.
 - 4** Join them back together: twenty and three.
-

Answer: half of forty-six is **23**

⇔ Check it a different way

You worked it out by halving the tens and the ones separately. Now try it another way.

You halved it. Now **DOUBLE** your answer, which is the one check that can never let you down here.

Double twenty-three is forty-six. You are back where you started, so the halving was right. If doubling your answer does not return the number you began with, your half is wrong, and you know it immediately.

The check, done.

double **23 = 46**

How do you know?

Why can you not halve twenty-five and get a whole number? What would be left over, and what does that tell you about odd numbers?

Your turn

1 Find half of 10.

2 Find half of 24.

3 Find half of 36.

4 Find half of 48.

5 Find half of 60.

6 Find half of 82.

7 Find half of 100.

8 Find half of 18, then double your answer to check it.

9 Find half of 44, then double your answer to check it.

10 Which of these
can be halved to
a whole number?

16

17

20

25

30

TOPIC 12.3 · YOUR TURN, CONTINUED

11 Twenty-eight pegs are shared between two trays. How many each?

12 Half of a number is 15. Write the number.

13 Double 17, then halve your answer. What do you notice?

14 Write an even number over fifty and find half of it.

Find out together

Work in a pair. One says a number under a hundred, the other doubles it. Then the FIRST person halves the answer, out loud, and must arrive back at the number they said. Play ten rounds. Which numbers made you both slow down?

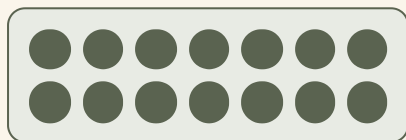
Near doubles

♣ Nearly a double is nearly as easy

Seven add eight looks like a fresh problem. It is not. It is double seven with one more on the end.

Double seven is fourteen, and one more is fifteen. You already knew the hard part before you started.

A sum where the two numbers are next-door neighbours is called a **near double**. Find the double you know, then adjust by one.



$$7 + 8 = 15$$

Double seven. Now add one more counter to one row and you have seven add eight.

- 1 Spot that the two numbers are neighbours.
- 2 Double the smaller one, because you know that answer.
- 3 Add one, because the bigger number has one more in it.
- 4 Say the answer, and notice you never had to count on.

 Watch me work one out 1

Work out fifteen add sixteen using a near double.

- 1** The two numbers are next-door neighbours, so this is a near double.
 - 2** Double the smaller one. Double fifteen is thirty.
 - 3** Sixteen has one more in it than fifteen, so add one.
 - 4** Thirty and one.
-

Answer: fifteen add sixteen is **31**

⇌ Check it a different way

You worked it out by doubling fifteen and adding one. Now try it another way.

You used the near double. Now add them the long way instead: partition both into tens and ones, add the tens, add the ones, and put the answer together.

Ten and ten makes twenty. Five and six makes eleven. Twenty and eleven makes thirty-one. Two routes, one answer, and now you know the shortcut is safe.

$$15 = 10 + 5$$

$$16 = 10 + 6$$

How do you know?

Hattie worked out eight add nine by doubling nine and taking one away. Otto doubled eight and added one. Both got seventeen. Explain why both methods work.

Your turn

- 1 Work out 6 add 7 using a near double.
- 2 Work out 9 add 10 using a near double.
- 3 Work out 12 add 13 using a near double.
- 4 Work out 20 add 21 using a near double.
- 5 Work out 25 add 26 using a near double.
- 6 Work out 4 add 5, then 5 add 6, then 6 add 7.
- 7 Which double helps you with 11 add 12? Write it down.
- 8 Work out 30 add 31.
- 9 Work out 17 add 18.
- 10 Work out 8 add 10 by doubling. Explain which double you used.

TOPIC 12.4 · YOUR TURN, CONTINUED

- 11** Ring the sums that are near doubles.

$7 + 8$

$7 + 12$

$20 + 21$

- 12** Write a near double of twelve, then work it out.
- 13** Explain how double 20 helps you work out 20 add 22.
- 14** Write your own near double, then work it out two ways.

Yard challenge

Every near double gives an odd answer. Test it with five near doubles of your own, then try to explain WHY it always happens. The reason is on the same page as odd and even numbers, back in Unit 4.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Double 32.
- 2 Double 45.
- 3 Find half of 68.
- 4 Find half of 90.
- 5 Work out $14 + 15$ using a near double.
- 6 Double 26, then halve your answer. Write what you notice.
- 7 Two baskets cost 70c altogether. How much is one basket?
- 8 Write an odd number under twenty that cannot be halved to a whole number.
- 9 Explain how doubling and halving undo each other.
- 10 Find half of 54, and show the check you used.

Be honest with yourself

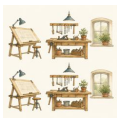
Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
double a two-digit number by doubling its tens and its ones	☐	☐	☐
double any multiple of five up to fifty, and halve it back	☐	☐	☐
halve an even number and check by doubling your answer	☐	☐	☐
use a double you know to work out a near double	☐	☐	☐

NOW I CAN

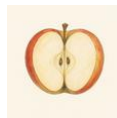
- ☐ double a two-digit number by doubling its tens and its ones
- ☐ double any multiple of five up to fifty, and halve it back
- ☐ halve an even number and check by doubling your answer
- ☐ use a double you know to work out a near double

▣ YARD WORDS



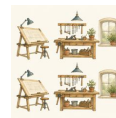
double

twice as many



halve

split into two equal parts



twice

two times as much



near double

a double with one more or one less



undo

go back the other way



check

work it out again a different way

Flat shapes

Otto draws round every shape in the box and counts its corners.



Come into the Plan Room

Look round the room and find something square, something circular and something triangular.

Now count the corners of the page you are reading. How many? And how many sides?

Did the number of corners match the number of sides? Try it on a triangle and see whether that is a coincidence.

☀ In the yard today

Otto is drawing round a wooden hexagon, very slowly, keeping the pencil hard against the edge.

The templates are fanned out on the table beside him: a square, a circle, a triangle, a pentagon, a hexagon. He has drawn round every one of them this morning and written nothing at all.

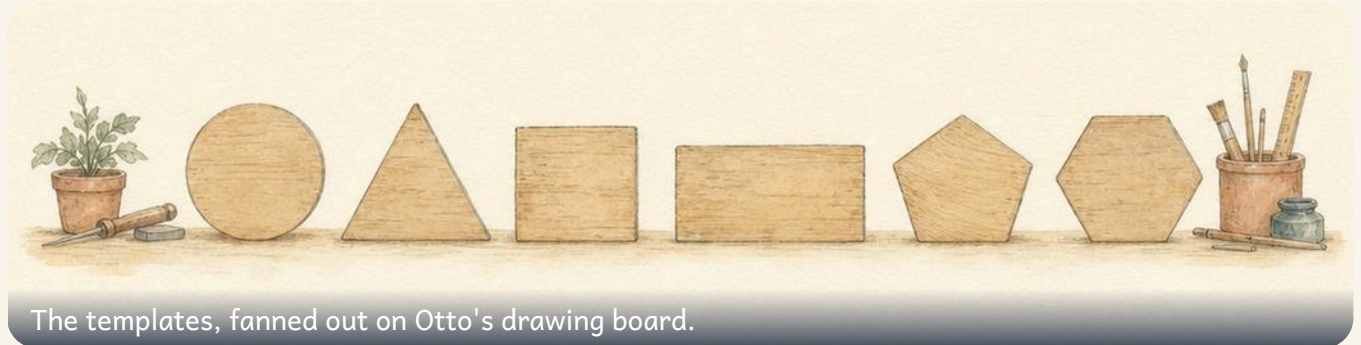
I am counting the corners, he says, when Milo asks. **You cannot count the corners of a shape properly until you have been all the way round it once with a pencil. Then you know exactly how many there are, because you had to turn at every single one.**

✓ **By the end you will**

- name squares, rectangles, circles, triangles, pentagons and hexagons
- count the sides and corners of a shape and describe it
- recognise a shape when it has been turned or stood on a point
- sort shapes by a rule you choose, and say what the rule was

Cambridge Primary Mathematics Stage 2: 2Gs1 • 2Gs4 • 2Pt9

Naming flat shapes



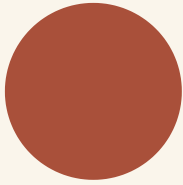
✿ A flat shape has no inside to hold

A **flat** shape is one you can draw on paper. It has a length and a width, and nothing else. You cannot pick a drawn triangle up off the page.

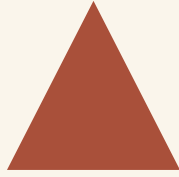
Most flat shapes are made of straight **sides**, and the name usually tells you how many: a pentagon has five, a hexagon has six.

A **circle** is the odd one out. It has no sides and no corners at all, because its edge never stops curving.

TOPIC 13.1 · CONTINUED



circle



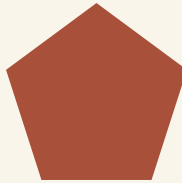
triangle



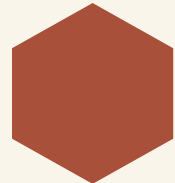
square



rectangle



pentagon



hexagon

Do you remember?

Penta means five and hexa means six. They are Greek, and once you know them the names stop being something to memorise.

 Watch me work one out 1

Which shape has five straight sides?

- 1** Look at each shape and count only the STRAIGHT edges.
- 2** The triangle has three. Too few.
- 3** The square and the rectangle have four each. Still too few.
- 4** The next one along has five, and its name begins with penta.

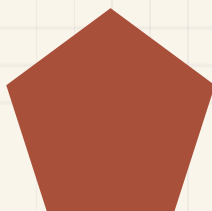
Answer: a pentagon, which has **5** sides

 Check it a different way

You worked it out by counting the straight sides. Now try it another way.

You counted the sides. Now count the **corners** of the same shape instead, going all the way round.

A pentagon has five corners as well as five sides. In any shape made of straight sides, those two numbers always match, so counting corners is a completely separate way to check you named the shape correctly.



**five sides, five
corners**

How do you know?

Why does a circle have no corners? Trace round one with your finger and see whether you ever have to turn.

Your turn



count on

- 1 Name the shape with three straight sides.
- 2 Name the shape with six straight sides.
- 3 Name the shape with no sides and no corners.
- 4 Name the shape with four equal sides.
- 5 Draw a triangle and write how many corners it has.



count on

TOPIC 13.1 · YOUR TURN · CONTINUED

6 Draw a hexagon. Use your ruler for every side.

7 Draw a pentagon and label one of its sides.

8 Find something rectangular in the room. Write what it is.

9 Find something circular in the room. Write what it is.

10 Match each shape to its number of sides.

triangle ●

square ●

pentagon ●

hexagon ●

● six

● three

● five

● four

TOPIC 13.1 • YOUR TURN, CONTINUED



count on

- 11 Draw a shape with five sides that are not all the same length.
- 12 Write the name of a shape with more sides than a square.
- 13 How many sides do a triangle and a hexagon have altogether?
- 14 Draw two different four-sided shapes. Say how they differ.

✂ The material you need

wooden or card shape templates if your classroom has them, a sharp pencil, and a ruler. If you have no templates, draw round the bottom of a cup for a circle and use the corner of this book for a right-angled corner.

★ Rook's find

A hexagon is the shape bees choose for honeycomb, and they are not being decorative. Hexagons fit together with no gaps at all and use less wax than any other shape that does. You will meet that in your tile floor project.

Sides and corners



A SIDE is a straight edge.



A CORNER is where two sides meet.

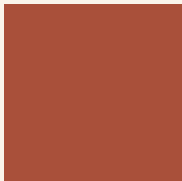
♣ Describing a shape properly

Naming a shape is not the same as describing it. To describe a shape you say how many **sides** it has, how many **corners**, and whether the sides are all the same length.

A square and a rectangle both have four sides and four corners. What tells them apart is that a square's four sides are all **equal**, and a rectangle's are equal only in facing pairs.

So a square is a special kind of rectangle. That surprises most people the first time they hear it.

TOPIC 13.2 · CONTINUED



four equal
sides



two long, two
short

Sides and corners always match. Every time.

Shape	Sides	Corners
triangle	3	3
square	4	4
pentagon	5	5
hexagon	6	6

 Watch me work one out 1

Describe a hexagon to somebody who cannot see it.

- 1** Count the straight sides, going all the way round: six.
- 2** Count the corners the same way: also six.
- 3** Check whether the sides look the same length as each other.
- 4** Say all three facts together, in that order.

Answer: a flat shape with **6** straight sides and **6** corners

 Check it a different way

You worked it out by counting the corners by looking at them. Now try it another way.

You counted round the shape. Now check a completely different way: put one finger on a corner and walk it round, counting only the **URNS** you make.

Every corner makes you turn exactly once, so the number of turns must equal the number of corners. If you turned six times, there are six corners.

How do you know?

Tessa says every shape with straight sides has the same number of corners as sides. Try to draw one that breaks her rule. What happens when you try?

Your turn

- 1 Write how many sides a pentagon has.
- 2 Write how many corners a triangle has.
- 3 Write how many corners a hexagon has.
- 4 Name a shape with four corners and four equal sides.
- 5 Name a shape with four corners but only two long sides.
- 6 Describe a square using sides, corners and equal.

TOPIC 13.2 · YOUR TURN · CONTINUED

7 Describe a triangle in the same three ways.

8 Draw a shape with four sides that is not a square.

9 True or not true? Decide for each line.	A circle has one corner.	TRUE / NOT TRUE
	A square is a kind of rectangle.	TRUE / NOT TRUE
	A hexagon has more sides than a pentagon.	TRUE / NOT TRUE

10 Count the corners of this page and write the number.

TOPIC 13.2 · YOUR TURN, CONTINUED

- 11 How many corners do two triangles have altogether?
- 12 How many sides do three squares have altogether?
- 13 Draw a shape whose sides are all different lengths.
- 14 Write the one thing that tells a square from a rectangle.



▣ SCAN TO SEE IT MOVE

Drag flat shapes together and pull them apart, turn them round, and count their sides and corners as you go.

mathigon.org/polypad/polygons

→ Take it further

Draw a shape with seven sides, then one with eight. Count the corners of each. Now write a sentence that would be true for a shape with a hundred sides, without drawing it.

The same shape, turned round

♣ Turning a shape does not change it

A triangle balanced on one of its points is still a triangle. It has not lost a side or gained a corner. It is just standing differently.

This matters because most people only ever draw a triangle one way up, sitting flat on its base. Then they see one on its point and say it is a different shape. It is not.

The test is always the same: count the sides and count the corners. Those numbers do not change when you turn something round, so they are what a shape really **is**.



a triangle



the same
triangle



still the same



a square



a square,
turned. NOT
a diamond

Watch me work one out 1

Milo says the shape standing on its point is a diamond, not a square. Is he right?

- 1 Count its straight sides. There are four.
- 2 Count its corners. There are also four.
- 3 Check whether all four sides are the same length. They are.
- 4 Four equal sides and four corners is the description of a square.
- 5 Turning something round changes none of those three facts.

Answer: it is a square, with **4** equal sides, just turned round

Check it a different way

You worked it out by counting the sides and corners. Now try it another way.

You checked by counting. Now check by doing: cut a square out of paper, then turn the paper round in your hand.

You did not cut anything off and you did not add anything on. If it was a square before you turned it, it is a square after, and now you have proved it with your hands as well as your eyes.

How do you know?

What DOES change when you turn a shape round, and what does not? Write one thing in each list.

Your turn

- 1 Draw a triangle standing on one of its points.
- 2 Draw a square standing on one of its corners.
- 3 Name this shape whichever way up it is drawn: four equal sides.
- 4 Cut out a paper triangle. Turn it and draw round it twice.
- 5 Write how many sides your turned triangle has.

- 6 Ring the shapes that are really the same shape turned round.

square

square on its corner

rectangle

TOPIC 13.3 · YOUR TURN · CONTINUED

- 7 Draw a rectangle standing on its short side.
- 8 Draw a hexagon, then draw it again turned a little.
- 9 Write whether turning a shape changes its number of corners.
- 10 Find a turned square somewhere in the room or on a floor tile.

TOPIC 13.3 · YOUR TURN, CONTINUED

- 11 Draw the same triangle three times, each one turned differently.
- 12 Explain to a partner why a turned square is still a square.
- 13 Draw a pentagon that is NOT sitting flat on one side.
- 14 Write one thing that stays the same when a shape is turned.

🔗 Find out together

Cut out one paper triangle between you. Take turns to place it on the desk in a way the other person has not used yet, and every time you place it, both of you say the number of sides out loud. Keep going until you run out of positions. Did the number ever change?

Sorting flat shapes by a rule

♣ Say the rule, then sort

You can sort the same pile of shapes lots of different ways. What matters is that you **say the rule first**, and then stick to it.

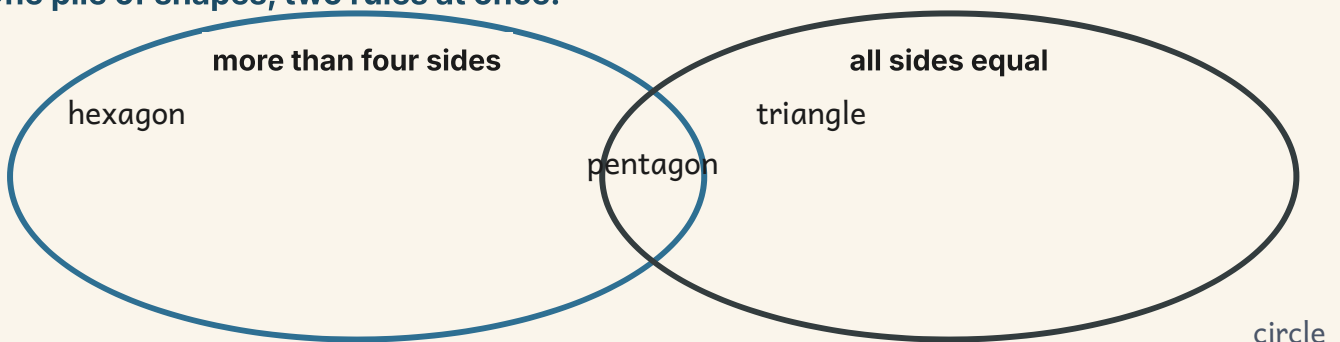
Good rules for flat shapes: has straight sides, has more than four sides, has all its sides equal, has a curved edge.

A rule must be something you can check by looking. **Nice shapes** is not a rule, because nobody else can sort the pile the same way you did.

has straight sides

has a curved edge

One pile of shapes, two rules at once.



 Watch me work one out 1

Sort a square, a circle and a hexagon by the rule has straight sides.

- 1** Say the rule out loud before you touch anything.
- 2** Take the square. Its edges are straight, so it goes in the yes tray.
- 3** Take the circle. Its edge curves the whole way round, so it goes in the no tray.
- 4** Take the hexagon. Six straight edges, so it goes in the yes tray.

Answer: **2** shapes have straight sides, and **1** does not

 Check it a different way

You worked it out by asking the rule about each shape in turn. Now try it another way.

You sorted by asking about each shape in turn. Now check by counting: count how many ended up in each tray, and add the two numbers.

If the total is not the number of shapes you started with, then something was left on the table or something went into both trays, and either one is a mistake.

How do you know?

Why is has straight sides a good rule, and nice shapes a bad one? Your answer should mention somebody else.

Your turn



- 1** Sort these into straight sides and curved.

circle

square

triangle

hexagon

- 2** Write a rule that puts a triangle and a square together.

- 3** Write a rule that separates a triangle from a square.

- 4** Sort four shapes by the rule has more than four sides.

TOPIC 13.4 · YOUR TURN · CONTINUED

- 5 Sort four shapes by the rule has all sides equal.



coin

- 6 Draw two shapes that fit the rule has four corners.
- 7 Draw one shape that does not fit the rule has straight sides.
- 8 Write your own rule, then draw two shapes that fit it.
- 9 Write why nice shapes is not a rule anybody can use.
- 10 Sort the shapes on this page into two trays of your choosing.

TOPIC 13.4 · YOUR TURN, CONTINUED



- 11 Find a rule that fits a square but not a rectangle.
- 12 Fill in a two-rule diagram for straight sides and four corners.
- 13 Write a rule that no shape on this page fits.
- 14 Sort five things in the classroom by a shape rule you invent.

Yard challenge

Find a rule that puts a square and a circle in the same group, and the triangle and hexagon in the other. It exists, and it is not about sides. Think about what you could roll, or fold, or fit through a hole.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Name the shape with five straight sides and five corners.
- 2 Write how many corners a pentagon has.
- 3 Draw a hexagon that is not sitting flat on one of its sides.
- 4 Write what a square and a rectangle have in common.
- 5 Draw a four-sided shape whose sides are all different lengths.
- 6 Write a sorting rule that separates a circle from a hexagon.
- 7 Explain why a turned square is still a square.
- 8 How many sides do a pentagon and a triangle have altogether?
- 9 Describe a rectangle using sides, corners and equal.
- 10 Find a hexagon somewhere outside this book and write where it was.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
name squares, rectangles, circles, triangles, pentagons and hexagons	☐	☐	☐
count the sides and corners of a shape and describe it	☐	☐	☐
recognise a shape when it has been turned or stood on a point	☐	☐	☐
sort shapes by a rule you choose, and say what the rule was	☐	☐	☐

NOW I CAN

- ☐ name squares, rectangles, circles, triangles, pentagons and hexagons
- ☐ count the sides and corners of a shape and describe it
- ☐ recognise a shape when it has been turned or stood on a point
- ☐ sort shapes by a rule you choose, and say what the rule was

▣ YARD WORDS



side

a straight edge of a flat shape



corner

where two sides meet



pentagon

a flat shape with five sides



hexagon

a flat shape with six sides



straight

not curved at all



sort

put into groups by a rule

The tile floor

The Plan Room floor is worn through by the door. Otto has been asked to design a new one, and he has been told to use only two shapes.

✂ The material you need

squared paper, a pencil, a ruler, and two or three colours.

Design a floor for the Plan Room using exactly two different flat shapes. The shapes must fit together with no gaps anywhere, which is the hard part and the whole point.

Draw at least four full repeats of your pattern, so anybody looking at it can see how it carries on.

How do you know?

How do you know your two shapes leave no gaps? Point at the place in your drawing where you can prove it.

WHAT TO HAND IN

- 1 Draw your floor, with at least four full repeats of the pattern.
- 2 Name both your shapes, and write how many sides each one has.
- 3 Mark one place on your drawing where the shapes meet with no gap.
- 4 Say what would go wrong if you used circles.

HOW IT WILL BE JUDGED

What is judged	Worth	What a good one looks like
No gaps	30%	the shapes really do meet with nothing left over
Repeats clearly	25%	four or more full repeats, so the rule is obvious
Correct shape names	25%	both shapes named, with the right side counts
The circle question	20%	you explain why circles leave gaps

Solid shapes

The solids on the bench have faces you can count and edges you can feel.



Come into the Plan Room

Find something in the room shaped like a box. Put one hand flat on the top of it and one hand flat on the bottom.

Now run a finger along the line where the top meets the side. That line is an edge, and you can feel it, which is the point.

How many flat surfaces do you think that box has? Do not count yet. Just decide, and remember what you decided.

☀ In the yard today

Tessa has a wooden cube in one paw and is turning it over and over, saying a number each time it lands.

She has been at it for a while. The other solids are lined up along the bench behind her: a cuboid, a cylinder, a cone, a pyramid, and a wooden ball that keeps trying to leave.

You cannot count the faces of a cube by looking at it, she says. Look at it. You can only ever see three at once. The other three are round the back being just as real as the ones you can see.

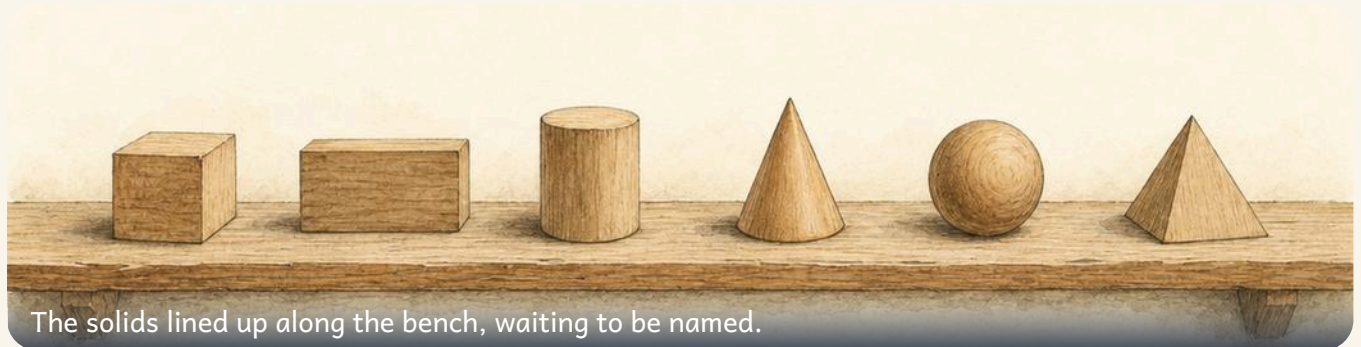
She puts the cube flat on the bench. **So you turn it. Every time a new face touches the bench, that is one. Then you have counted them all, and you have not guessed once.**

✓ **By the end you will**

- name cubes, cuboids, cones, cylinders, spheres and pyramids
- count the faces, edges and corners of a solid shape
- recognise a solid shape from a flat drawing of it
- sort solid shapes by whether they roll, stack or slide

Cambridge Primary Mathematics Stage 2: 2Gs2 • 2Gs4

Naming solid shapes



The solids lined up along the bench, waiting to be named.

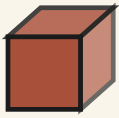
✦ A solid shape is one you can pick up

A **flat** shape can be drawn on paper and left there. A **solid** shape cannot. It has a length, a width and a thickness, and it takes up room in the air.

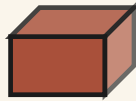
A drawn circle is flat. A ball is solid, and we call it a **sphere**. A drawn square is flat. A box is solid, and we call it a **cuboid**, unless all its faces are squares, in which case it is a **cube**.

The names to learn are cube, cuboid, sphere, cylinder, cone and pyramid. You already own examples of every single one of them.

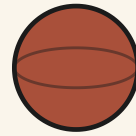
TOPIC 14.1 · CONTINUED



cube



cuboid



sphere



cylinder



cone



pyramid

Do you remember?

A tin of beans is a cylinder. A party hat is a cone. A dice is a cube. Naming solids is mostly remembering things you have already held.

TOPIC 14.1 • CONTINUED



A CYLINDER stands on a flat circle.



A PYRAMID has a flat base and rises to a point.

Watch me work one out 1

Which solid has a flat circle at each end and a curved surface in between?

- 1** Take each solid in turn and feel for a flat circle.
- 2** The sphere is curved everywhere, with no flat part at all.
- 3** The cone has a flat circle at one end, but a point at the other.
- 4** The next one has a flat circle at BOTH ends, joined by a curve.
- 5** That is the shape of a tin, and a tin is a cylinder.

Answer: a cylinder, which has **2** flat faces and **1** curved surface

⇌ Check it a different way

You worked it out by looking for a flat circle at each end. Now try it another way.

You named it by looking for flat circles. Now check a different way, with a pencil: stand the solid on paper and draw round the bottom of it, then turn it over and draw round the other end.

A cylinder leaves you two circles the same size. A cone would leave one circle and one dot. So the paper tells you which solid you had, and you never once had to trust your eyes.

How do you know?

Why is a sphere the only one of the six you cannot stand still on a sloping desk? Say what its surface is doing.

Your turn



data

1 Name the solid with square faces all the way round.

2 Name the solid that rolls in every direction.

3 Name the solid shaped like a tin of beans.

4 Name the solid with a flat base rising to a point.

5 Name the solid shaped like a shoebox.



data

TOPIC 14.1 · YOUR TURN · CONTINUED

6 Find something cuboid in the room. Write what it is.

7 Find something cylindrical in the room. Write what it is.

8 Draw round the flat end of a cylinder. Name the outline.

9 Match each solid to something you would find it in.

The material you need: a cube, a cuboid, a sphere, a cylinder and a cone if your classroom has a set of wooden solids

cube



a football

cylinder



a party hat

sphere



a dice

cone



a tin of soup

10 Sort these solids into rolls and does not roll.

cube

sphere

cylinder

cuboid

TOPIC 14.1 • YOUR TURN, CONTINUED



data

- 11 Name the solid with a curved surface and no flat face.
- 12 Write two solid names that both have flat square faces.
- 13 Hold a cube and say its name aloud to a partner.
- 14 Name the solid you would choose for building a tall tower.

✂ The material you need

a set of wooden or plastic solids if you have one. If you do not, a dice is a cube, a tin is a cylinder, a cereal box is a cuboid, a marble is a sphere, and a rolled paper hat is a cone. Those five are enough for the whole unit.

★ Rook's find

The Great Pyramid at Giza is a square-based pyramid, exactly the shape on the bench, and it stood as the tallest building anywhere on earth for about three thousand eight hundred years. Nobody beat it until people learnt to build with iron.

Faces, edges and corners



A **FACE** is a flat surface you could rest on the bench.

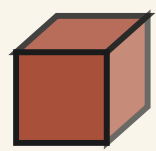
An **EDGE** is the line where two faces meet.

✦ Three things to count, and they are not the same thing

A **face** is a surface of the solid. Rest the solid on the bench and the part touching the bench is one face.

An **edge** is the line where two faces meet. You can run a finger along an edge, and it feels sharp.

A **corner** is the point where edges meet. It is the bit that hurts if you drop the box on your foot. Faces, edges and corners are three separate counts, and a cube gives you three different numbers.



count its
 faces, edges
 and corners

A cube and a cuboid share every number. Only the shape of the faces differs.

Solid	Faces	Edges	Corners
cube	6	12	8
cuboid	6	12	8
pyramid	5	8	5
cylinder	3	2	none
cone	2	1	a point
sphere	1	0	none

**SHARPEN
YOUR
PENCIL**

Put a hand flat on the top of this book, then on the front cover, then on the spine. Each time your hand is flat on a face. Now find a face you have not touched yet.

✎ Watch me work one out 1

How many faces does a cuboid have? Milo says three, because that is all he can see.

- 1** Milo is counting a picture, not a solid. Put a real box down.
- 2** The face on the bench is one, and he could not see that at all.
- 3** Now the top: two. The front and the back: three and four.
- 4** The two ends: five and six. Every face has now touched a hand.
- 5** Milo saw half of them, which is what a drawing always shows you.

Answer: 6 faces, not 3

⇌ Check it a different way

You worked it out by counting the faces by looking at the picture. Now try it another way.

You counted the faces by touching them in the air, which is easy to lose your place in. Now check a completely different way: put the solid on the table, say ONE, then tip it over onto the next face and say TWO.

Keep tipping until a face you have already said comes back down to the table. Only one face touches the table at a time, so this way you cannot count the same face twice or skip one, and it should agree with your first answer.

How do you know?

A cube and a cuboid have exactly the same three numbers. So what IS the difference between them? Look at the faces themselves.

Your turn

- 1 Count the faces of a cube. Write the number.
one, two, three
- 2 Count the edges of a cube. Write the number.
- 3 Count the corners of a cube. Write the number.
- 4 Count the flat faces of a cylinder. Write the number.
- 5 Count the edges of a cone. Write the number.
- 6 Write how many faces a sphere has.
- 7 Run a finger along every edge of a cuboid as you count.

8 Write which has more faces, a cone or a cube.

9 Decide true or not true for each line.

A cube and a cuboid have the same number of edges.

TRUE / NOT TRUE

A sphere has more edges than a cone.

TRUE / NOT TRUE

You can see every face of a cube at once.

TRUE / NOT TRUE

10 Fill in the missing counts for each solid.

Solid	Faces	Edges
cube		
cylinder		
cone		

TOPIC 14.2 · YOUR TURN, CONTINUED

11 Write which two solids have the same number of faces.

12 Order these solids by faces, fewest first and most last.

cube

sphere

cone

cylinder

13 Find a solid whose faces are not all the same shape.

14 Explain to a partner where a corner is on a cuboid.

▣ SCAN TO SEE IT MOVE



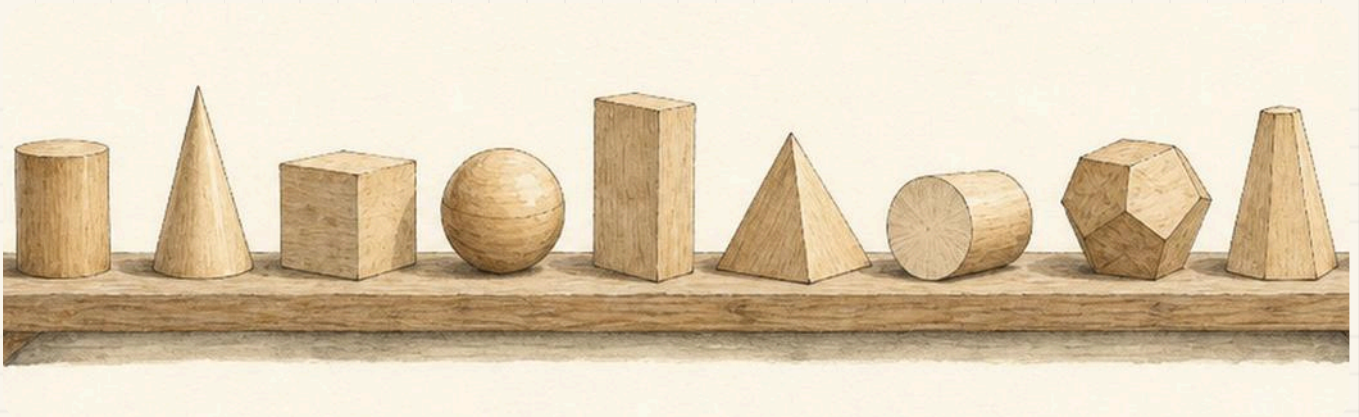
Turn a solid shape round and round with your finger to count every face, edge and corner, which a printed picture cannot let you do.

mathigon.org/polypad/polyhedra

→ Take it further

Add up the faces and the corners of a cube, then take away the edges. Do the same for a cuboid, and then for a pyramid. Write down what you get each time. Something very odd is going on, and a man called Euler noticed it first.

Flat drawings of solid shapes



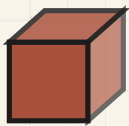
Wooden solids on the bench, and beside them Otto's flat drawings of the same three.

♣ **A drawing of a solid always tells a small lie**

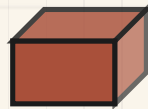
Paper is flat, so a drawing of a cube cannot really be a cube. What an artist does instead is draw the faces you would see and slant them, so your eye believes there is a back to it.

That means some faces are **hidden**. In a drawing of a cube you see three faces at most. The other three are still there on the real solid, and you have to remember them.

It also means the squares come out looking like slanted boxes. They have not changed shape. They are just being seen from the side, the way a door looks narrow when it is half open.



**drawn: you
see three
faces**



**drawn: you
see three
faces**



**drawn: you
see two
faces**

🔗 Watch me work one out 1

Otto has drawn a shape with a flat circle on top and straight sides going down. Which solid did he draw?

- 1 Look at what the drawing shows: a circle, and two straight lines.
- 2 A cone would come to a point, and this does not.
- 3 A cube has no circle anywhere on it, so it is not that.
- 4 A circle at the top with straight sides below is a tin standing up.
- 5 Count what the real solid has: flat faces at each end, one curve.

Answer: a cylinder, with **2** flat faces and **2** edges

↔ Check it a different way

You worked it out by reading the flat drawing on the page. Now try it another way.

You judged it from the drawing, which is a flat thing pretending to be solid. Now check with the real solids and a lamp: hold each one up and look at the shadow it throws on the wall.

Turn each solid until its shadow matches Otto's drawing. Only one of the six will ever give you that shadow, and that is the solid he drew. The shadow is real, so it cannot flatter him.

How do you know?

A drawing of a cube shows three faces. Where are the other faces? Say why the drawing cannot show them.

Your turn

- 1 Draw a cube on squared paper. Slant the top.
- 2 Draw a cylinder standing on its circular end.
- 3 Name the solid each drawing on this page shows.
- 4 Write how many faces you can see in your cube drawing.
- 5 Write how many faces are hidden in your cube drawing.
- 6 Draw a cuboid lying on its longest face.
- 7 Draw round the base of a real cube. Name the outline.

TOPIC 14.3 · YOUR TURN · CONTINUED

- 8** Match each flat drawing to the solid it shows.

a circle with straight sides below ●

a square with slanted lines behind ●

a triangle on a flat base ●

a shaded ring ●



cube



sphere



cylinder



pyramid

- 9** Draw a cone, then draw a pyramid beside it.

- 10** Ring the drawing that shows a solid that rolls.

cube

cylinder

cuboid

TOPIC 14.3 · YOUR TURN, CONTINUED

- 11** Draw a sphere. Shade one side so it looks round.
- 12** Draw the same cuboid twice, standing differently each time.
- 13** Write one thing a flat drawing cannot show you.
- 14** Draw a solid for a partner to name. Do not tell them.

🔗 Find out together

Take one solid between you and put it on the table. Sit on opposite sides and each draw exactly what you can see, without looking at the other drawing. Then put the two drawings side by side. Same solid, same moment, two different pictures: talk about which faces each of you got.

Making and sorting solids



Tessa's cube, held so one face is towards you.

♣ Sort by what a solid DOES

You can sort solids by their names, but a far better rule is what they do when you put them on a sloping book. Some **roll**. Some **slide**. Some just sit there.

And you can sort by what they do in a tower. A solid with a flat face on top will **stack**. A solid with no flat face will not, no matter how carefully you place it.

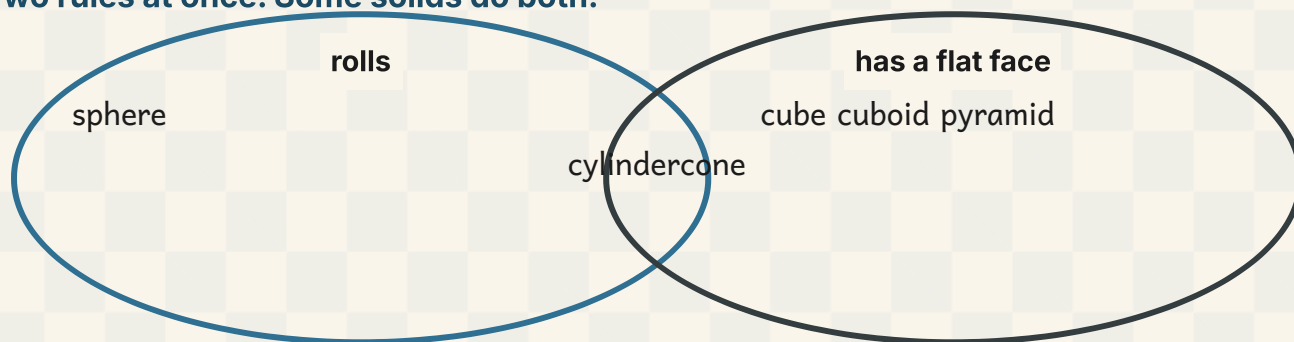
Notice that these rules come from the faces. A curved surface rolls, a flat face stacks. So the sorting is not a new thing to learn: it is what you counted in 14.2, doing something.

TOPIC 14.4 · CONTINUED

rolls

stacks

Two rules at once. Some solids do both.



TOPIC 14.4 · CONTINUED

- 1 Cut out six squares of card, all exactly the same size.
- 2 Lay four of them in a row and tape them edge to edge.
- 3 Curl the row round into a tube and tape the last join.
- 4 Tape one of the two spare squares over the hole at the top.
- 5 Tape the last square over the hole at the bottom. Count what you made.

Watch me work one out 1

Sort a cube, a sphere and a cylinder by the rule it rolls.

- 1 Say the rule out loud before you touch anything.
- 2 Take the cube. Every surface on it is flat, so it will not roll.
- 3 Take the sphere. It is curved all over, so it rolls anywhere.
- 4 Take the cylinder. It has a curve round its middle, so it rolls on its side.
- 5 So look for a curved surface. That is what the rule is really asking.

Answer: 2 of them roll and 1 does not

⇌ Check it a different way

You worked it out by looking for a curved surface on each solid. Now try it another way.

You sorted by looking for a curved surface. Now check by doing it for real: prop one end of a book up on a pencil case and let each solid go at the top.

Whatever reaches the bottom rolled, and whatever stayed put did not. The slope does not care what you predicted, so if it disagrees with your sort, go back and feel for the curve again.

How do you know?

Why does a cone roll round in a circle instead of rolling straight down the slope? Look at its two ends.

Your turn

1 Sort these solids into rolls, stacks and slides.

cube

sphere

cylinder

cone

cuboid

2 Test a cube on a sloping book. Write what happens.

3 Test a sphere on the same slope. Write what happens.

TOPIC 14.4 · YOUR TURN · CONTINUED

4 Write a rule that puts a cube and a cuboid together.

5 Write a rule that separates a sphere from a cube.

6 Build a tower using only the solids that stack.

7 Write why a sphere cannot go partway up your tower.

8 Make a cube from card, or from dough.

The material you need: card, scissors and tape, or a lump of dough

9 Count the faces on the cube you made. Write the number.

10 Fill in a two-rule diagram for rolls and has a flat face.

TOPIC 14.4 · YOUR TURN, CONTINUED

11 Sort five objects at home by whether they roll.

12 Write a rule that fits a cylinder but not a cube.

13 Order these solids by stacking, easiest first and hardest last.

sphere

cube

cone

14 Invent a sorting rule. Sort the solids by it.

Yard challenge

Find a rule that puts the cylinder and the cone on one side, and the sphere, cube, cuboid and pyramid on the other. There is one, and it is not about rolling, because the sphere rolls too. Think about which solids have BOTH a flat face and a curved surface on the same object.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Name the solid with a circle at each end and a curved middle.
- 2 Write how many edges a cuboid has.
- 3 Draw a pyramid and label its flat base.
- 4 Write where an edge is on a solid you can hold.
- 5 Name one solid that stacks and one that rolls.
- 6 Write which has fewer faces, a sphere or a cone.
- 7 Find a cylinder in your kitchen and write what it was.
- 8 Explain why a cube can be stacked but a sphere cannot.
- 9 Name the solid a football is shaped like.
- 10 Count the corners of a cereal box and write the number.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
name cubes, cuboids, cones, cylinders, spheres and pyramids	☐	☐	☐
count the faces, edges and corners of a solid shape	☐	☐	☐
recognise a solid shape from a flat drawing of it	☐	☐	☐
sort solid shapes by whether they roll, stack or slide	☐	☐	☐

NOW I CAN

- ☐ name cubes, cuboids, cones, cylinders, spheres and pyramids
- ☐ count the faces, edges and corners of a solid shape
- ☐ recognise a solid shape from a flat drawing of it
- ☐ sort solid shapes by whether they roll, stack or slide

▣ YARD WORDS



solid

a shape you can hold



face

a flat surface of a solid



edge

where two faces meet



cuboid

a box shape with six rectangle faces



cylinder

a tin shape with two circle faces



pyramid

a shape with a flat base and a point

Symmetry

Fold the plan along the line. If the halves match, it is symmetrical.



Come into the Plan Room

Fold a sheet of paper in half. Open it out and look at the crease. Is one half the same as the other?

Now fold this book in half down the middle of the page. Does the printing on one side land on the printing on the other?

Find something in the room that would fold into two matching halves, and something that certainly would not.

☀ In the yard today

Tessa has folded her plan of the yard gate exactly down the middle, and she is holding it up against the window to see through both halves at once.

The gate on her plan has a post at each side, three bars across, and a lamp on the top. Folded down the middle, every line on the left lands on a line on the right.

That is how I know I drew it properly, she says. I did not measure anything twice. I folded it once. If a line pokes out past the fold, the gate is wrong, and I would rather find out now than after Otto has cut the wood.

Milo folds his own plan and one bar sticks out past the crease by the width of a finger.

✓ **By the end you will**

- fold a shape to find a line of symmetry, and draw the line on
- say how many lines of symmetry a flat shape has
- finish a pattern so that both halves match
- find symmetry in tiles, leaves, buildings and letters

Cambridge Primary Mathematics Stage 2: 2Gs3 • 2Gs4

A line of symmetry



Tessa's gate plan, folded down the middle against the light.

✦ Folding is the test

A shape has **symmetry** when you can fold it so that the two halves land exactly on top of each other. The crease you fold along is called the **line of symmetry**.

Exactly is the important word. The halves must **match** all the way round. If any part pokes out past the fold, even a little, that crease is not a line of symmetry.

A fold straight up and down is called a **vertical** line. A fold straight across is a horizontal one. Both count, and a shape can have several.

TOPIC 15.1 • CONTINUED



FOLD it. The halves must cover each other.



Or stand a mirror on the line and look.

Do you remember?

A crease that cuts a shape into two equal halves is not always a line of symmetry. The halves have to land on each other, not just be the same size.

 Watch me work one out 1

Tessa folds a paper rectangle straight across, top to bottom. Is that crease a line of symmetry?

- 1** Fold along the crease and press it flat.
- 2** Look along the top edge. The two edges lie on top of each other.
- 3** Run a finger all the way round the folded shape.
- 4** Nothing pokes out anywhere, so the halves cover each other.

Answer: yes, that crease is a line of symmetry, because the halves cover each other exactly

 Check it a different way

You worked it out by judging the fold line by eye. Now try it another way.

You judged the crease by eye, looking at it flat on the desk. Now check without folding at all: stand a small mirror upright on the line.

Look into the mirror. If what you see in the glass looks exactly like the half you covered up, the line is a line of symmetry. The mirror does the folding for you, and it never creases your work.



fold it across



fold it down

How do you know?

Otto folds a shape and one corner pokes out past the crease by a whisker. He says that is close enough. Why is he wrong?

Your turn

- 1 Fold a paper square in half so the halves match.
the halves cover each other
- 2 Draw the fold line on your folded square with a ruler.
- 3 Fold a paper circle in half. Draw the crease on.
- 4 Write whether an unmatched crease is a line of symmetry.
- 5 Cut out a paper heart. Fold it once so it matches.

TOPIC 15.1 • YOUR TURN • CONTINUED

6 Test three different creases on a paper rectangle.

7 Draw a shape with exactly one line of symmetry.

8 Stand a mirror on the crease of your folded square.

9 Say
yes
or
no
for
each
line.

A fold that leaves an overhang is a line of symmetry.

YES / NO

A vertical line goes straight up and down.

YES / NO

The two halves must land on each other.

YES / NO

10 Fold a paper triangle. Write whether the halves matched.

TOPIC 15.1 · YOUR TURN, CONTINUED

- 11 Draw a square, then draw one line of symmetry on it.
- 12 Explain what a line of symmetry does to a shape.
- 13 Fold a paper strip in half twice. Count the creases.
- 14 Find a symmetrical object in the room. Write its name.

✂ The material you need

thin paper you may fold and cut, scissors, a ruler, and a small mirror if there is one. Tracing paper helps a great deal: you can trace a half, flip the tracing over and lay it back down. If you have no mirror, a shiny spoon held upright will do.

★ Rook's find

The gate posts of a building, the front of a church, a bee's face and a butterfly's wings are all built round one vertical line. Builders and bees both do it for strength, not for prettiness: matching halves share the load evenly, so nothing pulls to one side.

Symmetry in flat shapes

♣ Try every fold you can

A shape can have more than one line of symmetry, so finding one is not the end of the job. Fold it every way you can think of, and keep count of the folds that work.

A shape with all its sides equal has a line of symmetry through every side. A shape with only its facing sides equal has fewer.

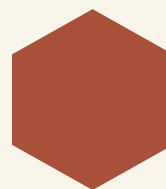
Turning a shape round does not add or remove any lines. The lines turn with the shape, exactly like its sides and corners do.



a square



**the same
square,
turned.
Same lines**



a hexagon

 Watch me work one out 1

How many lines of symmetry has a square?

- 1 Fold it straight across, top edge onto bottom edge. That works.
- 2 Fold it straight down, side onto side. That works too.
- 3 Fold it corner to corner one way. The halves still cover.
- 4 Fold it corner to corner the other way. That works as well.
- 5 Now try any other crease. Something always pokes out.
- 6 So count the folds that worked.

Answer: 4 lines of symmetry, the same as its number of sides

 Watch me work one out 2

Milo draws a line from corner to corner across a rectangle and says it is a line of symmetry. Is he right?

- 1** His line does cut the rectangle into two equal halves, so it looks promising.
- 2** Cut out a paper rectangle and fold it along that corner to corner line.
- 3** Look at the folded shape. One point sticks out past the crease at each end.
- 4** The halves are the same size, but they do not land on each other.
- 5** Now fold straight across, and then straight down. Both of those do cover.
- 6** So count only the folds that covered.

Answer: no: a rectangle has just **2** lines of symmetry, and neither one is a diagonal

⇌ **Check it a different way**

You worked it out by counting the folds that worked. Now try it another way.

You counted the folds that worked. Now check a different way, with a mirror instead of a crease: stand the mirror upright along the diagonal of a drawn rectangle.

Look into the glass. The reflection makes a pointed kite shape, not a rectangle, so the mirror is not rebuilding the shape you started with. Along a real line of symmetry it always does.

Four creases on one rectangle. Only some of them cover.

Fold you might try	Does it cover?
straight across, top to bottom	fold it and see
straight down, side to side	fold it and see
corner to corner one way	fold it and see
corner to corner the other way	fold it and see

How do you know?

Two halves can be the same size and still not match. Fold a rectangle corner to corner and describe what your fingers feel at the ends of the crease.

Your turn

- 1 Write how many lines of symmetry a rectangle has.
- 2 Draw a rectangle and mark both of its lines of symmetry.
- 3 Fold a paper rectangle corner to corner. Write what happened.
- 4 Fold a paper pentagon and count the creases that work.
- 5 Draw a triangle with all three sides the same length.
- 6 Draw a four-sided shape with no line of symmetry at all.
- 7 Order these shapes from fewest lines to most lines.

rectangle

triangle

square

hexagon

TOPIC 15.2 · YOUR TURN · CONTINUED

8	Say true or not true for each line.	A diagonal of a rectangle is a line of symmetry.	TRUE / NOT TRUE
		Turning a shape round removes its lines of symmetry.	TRUE / NOT TRUE
		A shape can have more than one line of symmetry.	TRUE / NOT TRUE

- 9 Ring the shape with the most lines of symmetry.
- rectangle hexagon triangle

- 10 Explain why a diagonal is not a line of symmetry.

TOPIC 15.2 · YOUR TURN, CONTINUED

11 Compare a square and a rectangle by their symmetry lines.

12 Fold a paper circle three different ways. Write what you found.

13 Draw a shape with exactly two lines of symmetry.

14 Match each shape to its number of lines of symmetry.

triangle ●

● 5

square ●

● 3

pentagon ●

● 6

hexagon ●

● 4



■ **SCAN TO SEE IT MOVE**

Draw a line of symmetry and watch the shape fold along it, so you can see whether the two halves really match.

mathigon.org/polypad/polygons

→ **Take it further**

A regular shape has one line of symmetry for every side, so a shape with ten equal sides has ten. Now think about a circle. Fold it any way you like, so long as the crease goes through the middle, and it always covers. Write down how many lines of symmetry a circle has, and see whether you can finish the sentence honestly.

Completing a symmetrical pattern

A blue circular icon with the text "SHARPEN YOUR PENCIL" in white, bold, uppercase letters.

SHARPEN
YOUR
PENCIL

Shade any three squares on the left of a folded sheet. Fold it, hold it to the window, and shade the three squares you can see through the paper. Open it out.

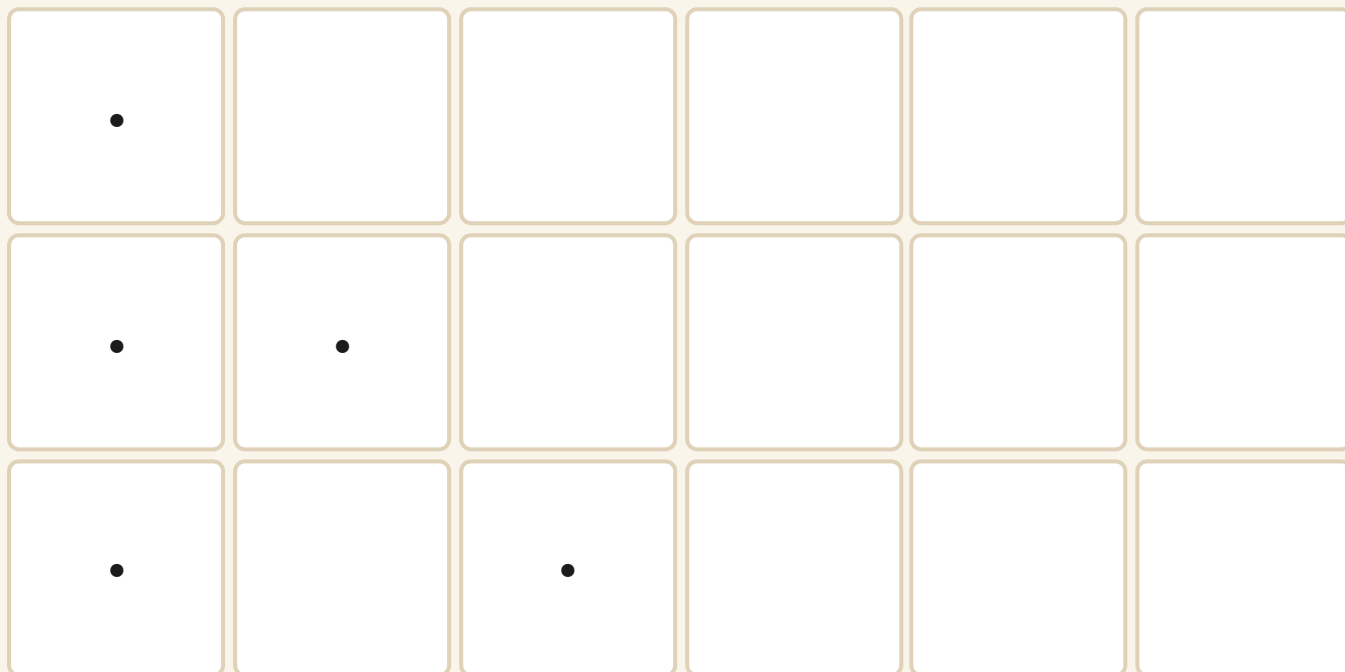
♣ Work one square at a time

To finish a symmetrical pattern, do not try to see the whole thing at once. Take one shaded square on the near side of the line and ask how far from the line it is.

Then count exactly that far on the other side, in the same row, and shade the square you land on. That square is the **mirror image** of the first.

A square touching the line has its partner right beside the line. A square three away has its partner three away. Do them one by one and the pattern finishes itself.

TOPIC 15.3 · CONTINUED



Half a pattern. The line of symmetry runs down the middle.

Tessa's border. Two cells are still missing.



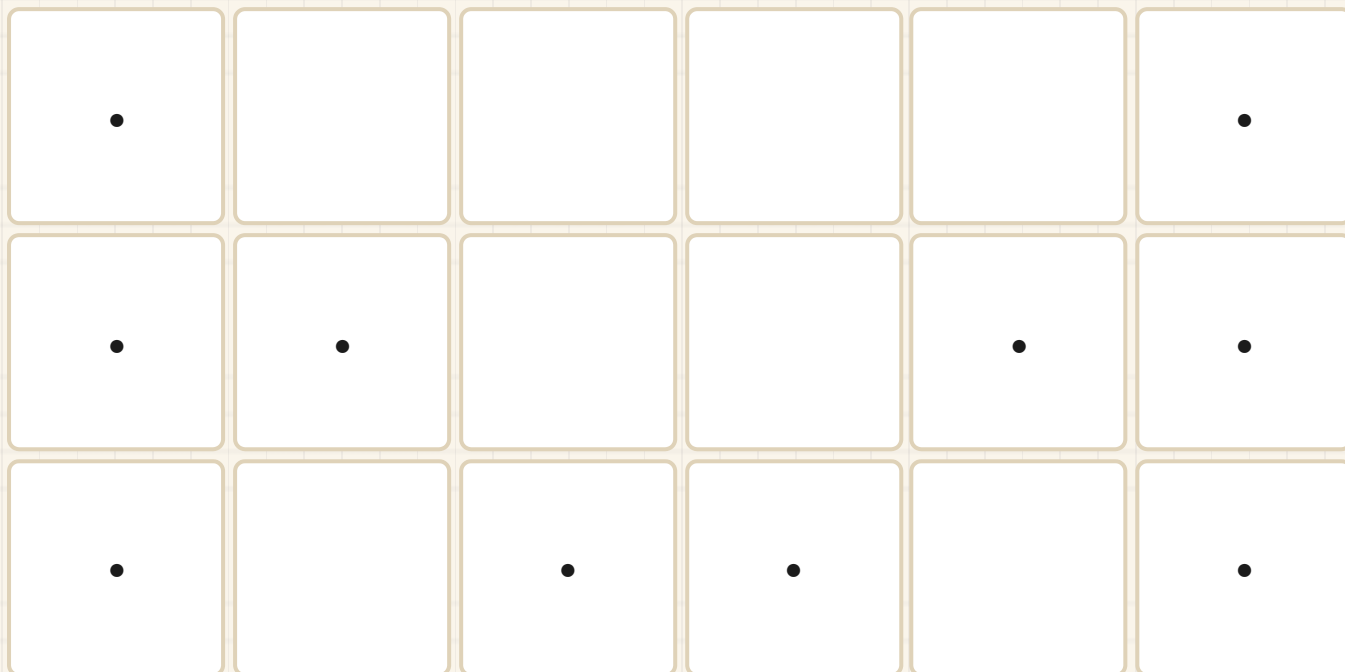
 Watch me work one out 1

Finish this pattern so that both halves match, then say how many squares are shaded altogether.

- 1** Find the shaded square nearest the line. It touches the line.
- 2** Shade the square on the other side that also touches the line, in the same row.
- 3** Take the next shaded square. Count how many squares from the line it sits.
- 4** Count the same number the other way and shade that square.
- 5** Work along every row until no shaded square is left without a partner.
- 6** Count the shaded squares on one side, then double it.

Answer: 5 squares on each side, so **10** shaded altogether

TOPIC 15.3 · CONTINUED · CONTINUED



Both halves. Every shaded square has a partner the same distance from the line.

⇌ Check it a different way

You worked it out by matching one square at a time across the line. Now try it another way.

You matched the squares one at a time, counting across from the line. Now check the whole thing at once: fold the paper along the line of symmetry and hold it up to the window.

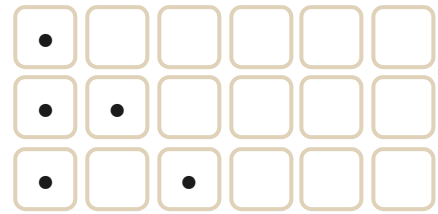
Every shaded square should sit right on top of a shaded square. If you can see a shaded square with white showing through it, that one has no partner and you have found your mistake without counting anything.

How do you know?

How can you tell you have finished a symmetrical pattern, without being told? Say what you would look for.

Your turn

- 1 Complete this pattern so both halves match.



- 2 Shade squares on the right to mirror the left.

- 3 Draw the mirror image of a triangle across a vertical line.

- 4 Finish the tile border so it matches on both sides.

TOPIC 15.3 · YOUR TURN · CONTINUED

- 5 Copy the pattern strip and continue it two more cells.
- 6 Shade four squares, then shade their mirror image.
- 7 Complete a symmetrical pattern using two colours only.
- 8 Count the shaded squares in your finished pattern.
- 9 Make a half pattern for a partner to finish.
- 10 Fold paper, cut a shape, open it out flat.
The material you need: scissors

11 Draw a symmetrical butterfly using one vertical fold line.

12 Shade a pattern that matches across two different lines.

13 Check a partner's pattern by folding along the line.

14 Write how you found the partner of one shaded square.

Find out together

Fold a square of paper in half, then in half again. Cut small shapes out of the folded edges, being careful not to cut right across, then open it out.

Look at what came out. Every single cut you made once has appeared four times, and the whole thing is symmetrical across both creases. Take turns to guess what a cut will look like opened out, then check.

Finding symmetry around you

▣ WORDS FOR THIS UNIT



symmetry

both halves match when folded



mirror image

the same picture the other way round



reflect

flip over a line

✿ Symmetry is everywhere once you look

Tiles, leaves, windows, doors, faces, forks and printed capital letters are nearly all built round a line. Once you start looking for the line, you cannot stop seeing it.

Real things are never perfect. A leaf has a nibbled edge, a face has one eyebrow higher than the other. We say a real thing is symmetrical when it would fold into two halves that very nearly match.

Some things have no line at all. A shoe, a comma and the letter R will not fold into matching halves however you turn them.

TOPIC 15.4 · CONTINUED



would fold into matching halves

would not fold to match

 Watch me work one out 1

How many of the letters A, M, T, H, R and J have a line of symmetry straight down the middle?

- 1** Take A. Fold it down the middle and the two legs meet. It works.
 - 2** Take M. The two humps meet on the fold, so it works too.
 - 3** Take T. The arms are the same length either side. It works.
 - 4** Take H. Both uprights land on each other. It works.
 - 5** Take R. The leg and the loop are only on one side. It fails.
 - 6** Take J. The hook is only on one side, so it fails as well.
 - 7** Count the letters that worked.
-

Answer: 4 of the 6 letters: A, M, T and H

⇔ Check it a different way

You worked it out by imagining the fold and judging by eye. Now try it another way.

You imagined the fold and looked at the letter. Now do it with tracing paper instead of imagination: trace the letter, turn the tracing over, and lay it back down on top of the printed one.

If the flipped tracing sits exactly on the printed letter, the letter has a line of symmetry. If it slides off no matter how you shift it, the letter has none. Your eyes cannot fool you, because the paper does the work.

How do you know?

Milo says every letter of the alphabet must have a line of symmetry somewhere. Find one letter that proves him wrong, and say how you know.

Your turn

- 1 Find a symmetrical leaf outside and draw it.
- 2 Test the letters in your own name for symmetry.
- 3 Write a letter with a line straight across the middle.
- 4 Look at a tiled floor. Find one line of symmetry.
- 5 Draw a symmetrical window from a building near you.
- 6 Sort these into symmetrical and not symmetrical.

a leaf

a shoe

a fork

a comma

TOPIC 15.4 · YOUR TURN · CONTINUED

7 Hold a mirror down the middle of a photograph.

8 Write where you saw symmetry on the way to school.

9 Draw a symmetrical tile of your own design.

10 Say
yes or
no for
each
object.

A shoe would fold into two matching halves.

YES / NO

A butterfly has a line of symmetry down its body.

YES / NO

A real leaf is never quite perfectly symmetrical.

YES / NO

TOPIC 15.4 · YOUR TURN, CONTINUED

- 11 Find an object with two lines of symmetry. Draw it.
- 12 Explain why a real leaf is only nearly symmetrical.
- 13 Draw a wing pattern that matches on both sides.
- 14 Count the symmetrical things on one page of another book.

Yard challenge

Write your own name in capital letters. Now stand a mirror upright at the end of it, so the mirror faces back along the letters, and read what the glass shows you. Some names come back readable and most do not. Work out what a name needs so that it survives the mirror, then invent a word of four letters that does.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Fold a paper hexagon and count the creases that cover.
- 2 Draw a line of symmetry on a capital letter you choose.
- 3 Fold a rectangle two ways that cover and one that does not.
- 4 Shade a symmetrical pattern on a grid of eight squares.
- 5 Name a shape with exactly one line of symmetry.
- 6 Write what a mirror image is, in your own words.
- 7 Draw a symmetrical gate and mark its fold line.
- 8 Explain how folding proves that two halves really match.
- 9 Find something at home with a vertical line of symmetry.
- 10 Write why a circle beats a square for lines of symmetry.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
fold a shape to find a line of symmetry, and draw the line on	☐	☐	☐
say how many lines of symmetry a flat shape has	☐	☐	☐
finish a pattern so that both halves match	☐	☐	☐
find symmetry in tiles, leaves, buildings and letters	☐	☐	☐

NOW I CAN

- ☐ fold a shape to find a line of symmetry, and draw the line on
- ☐ say how many lines of symmetry a flat shape has
- ☐ finish a pattern so that both halves match
- ☐ find symmetry in tiles, leaves, buildings and letters

▣ YARD WORDS



symmetry

both halves match when folded



line of symmetry

the fold line where halves match



mirror image

the same picture the other way round



reflect

flip over a line



match

be exactly the same



vertical

straight up and down

Position, turns and right angles

A quarter turn is a right angle. The corner of the plan sheet proves it.



Come into the Plan Room

Put your pencil down somewhere on your desk. Now tell somebody where it is without pointing.

Stand up and turn yourself a quarter of the way round. What are you facing now that you were not facing before?

Look at the corner of this page. Later in this unit you will find out that the corner and the quarter turn are the same idea.

☀ In the yard today

Otto has chalked a grid of big squares on the yard floor and is standing in one of them, refusing to move.

Milo is at the other end of the yard with the plan sheet, calling out instructions. **Three squares that way**, he shouts, waving.

Which way is that way, says Otto, without turning round. **You are behind me. Your that way and my that way are not the same way.**

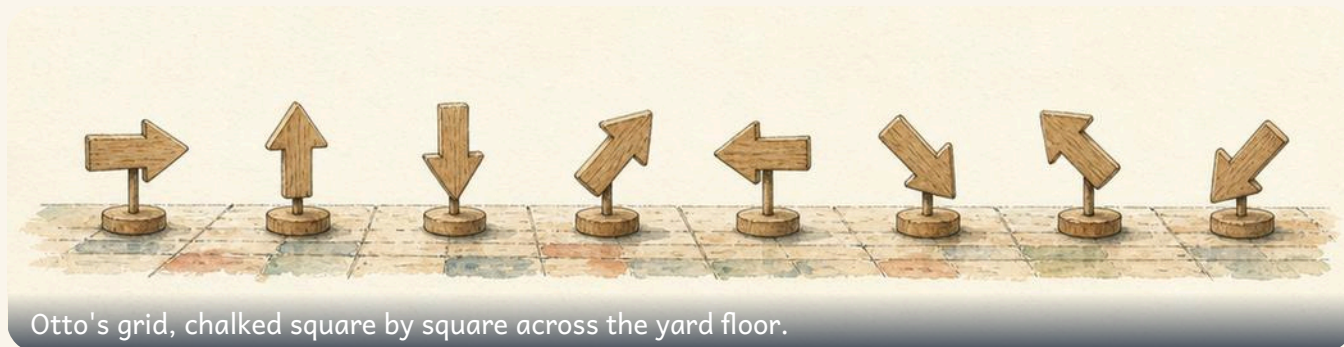
Milo looks at the plan again. **Three squares towards the water butt**, he says, **then two squares towards the gate**. Otto walks it, counting the squares out loud, and arrives exactly where the plan says he should.

✓ **By the end you will**

- say where something is using words such as between and underneath
- give a direction and follow one, on a grid and in the room
- make a whole, half and quarter turn, clockwise and anticlockwise
- recognise a right angle, and find right angles around you

Cambridge Primary Mathematics Stage 2: 2Gp1 • 2Gp2 • 2Gp3

Saying where something is



Otto's grid, chalked square by square across the yard floor.

♣ A position needs something to be near

The **position** of something is where it is. You cannot say where something is on its own. You can only say where it is compared with something else.

So **the peg is there** tells nobody anything, but **the peg is between the two blocks** tells them exactly. The words that do the work are the position words: between, beside, underneath, above, inside, behind, in front of, next to.

On a grid you can be even more exact. You say which column along and which row up, and only one square in the whole grid fits.

TOPIC 16.1 · CONTINUED



position



direction

between

beside

underneath

above

inside

behind

in front of

next to

Do you remember?

Left and right belong to the person, not to the room. Your left is not the left of somebody facing you.

TOPIC 16.1 • CONTINUED



The chalked grid. P is Otto's peg. B is Bramble's bench.

				B
	P			

TOPIC 16.1 · CONTINUED · CONTINUED

Five squares along and four squares up. Every square has its own address, and no two squares share one.

Watch me work one out 1

Say where Otto's peg is, exactly enough for Milo to find it.

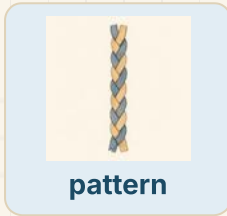
- 1** Point at the peg. Now find something fixed to compare it with.
- 2** Count the columns from the left until you reach the peg's column.
- 3** Count the rows from the top until you reach the peg's row.
- 4** Say the column and the row together, in that order, and stop.

Answer: in column **2**, on row **4**, which is the bottom row

How do you know?

Why is the peg is over there useless, when both of you can see the peg?
Your answer should mention where Milo is standing.

Your turn



- 1 Write where your pencil is, using the word beside.
- 2 Draw a box, then draw a dot inside it.
- 3 Draw a cup with a spoon underneath it.
- 4 Write one sentence about your desk using the word above.
- 5 Name the thing between your book and the edge of the desk.



TOPIC 16.1 · YOUR TURN · CONTINUED

6 Find something behind you. Write what it is.

7 Draw three pegs in a row. Ring the middle one.

The material you need: a ruler

8 Write which column the bench stands in on the grid opposite.

9 Write which row the bench stands on, counting from the top.

10 Sort these into position words and other words.

between

shout

underneath

chalk

beside

count

TOPIC 16.1 · YOUR TURN, CONTINUED



pattern

11 Write where your chair is, using two position words.

12 Order these
from nearest
to you to
furthest from
you.

your pencil

your chair

the window

the door

TOPIC 16.1 · YOUR TURN, CONTINUED · CONTINUED

- 13** Match each picture word to the thing it describes.

between ●

underneath ●

beside ●

inside ●

● in the middle of two

● under something

● next to something

● in something

- 14** Hide a counter. Tell a partner where it is until they find it.

⇌ Check it a different way

You worked it out by counting columns and rows on the plan. Now try it another way.

You worked out the peg's square by counting columns and rows with your finger on the picture. Now check a completely different way: chalk or tape a grid on the floor, stand in the square yourself, and say out loud what you can see beside you and behind you.

If the square is right, the things you name from inside it will match the things drawn next to it on the plan. Standing in a position is a different job from reading one, so it is a real check, not the same check again.

✂ The material you need

squared paper, a ruler, and chalk or masking tape if you are allowed to mark a grid on the floor. A counter or a small peg works for the thing being hidden.

Giving and following directions



A **DIRECTION** is the way something goes.

♣ A direction says which way, and how far

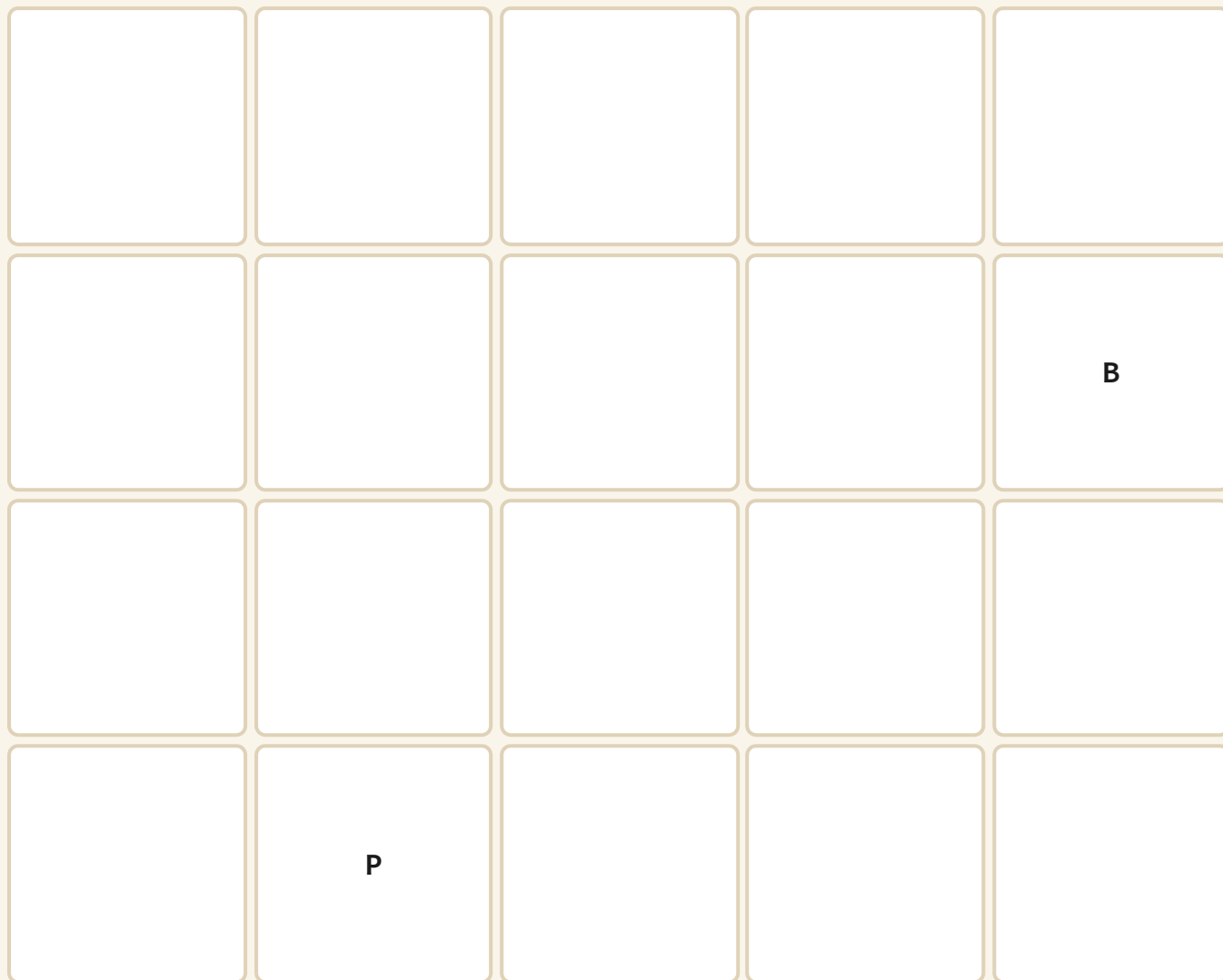
A **direction** is the way something goes. On its own it is not enough. **Go right** could mean one square or twenty, so a useful instruction says which way *and* how far.

A route is just several of those instructions in order. Order matters: two squares right then three up lands you somewhere else than three up then two right, unless you are lucky.

Always name both ends. **From the door to my seat** can be followed by anybody. **Over there** cannot.

- 1 Say where you start.
- 2 Say the first direction, and how many squares.
- 3 Say the next direction, and how many squares.
- 4 Say where you should have arrived, and check that you did.

TOPIC 16.2 · CONTINUED



From P to B. Choose your directions, then count your squares.

 Watch me work one out 1

Write a route from Otto's peg to the bench, using squares.

- 1** Start at P and decide which way the bench lies. It is to the right and higher up.
- 2** Count the columns from P across to B, one square at a time.
- 3** Count the rows from P up to B the same way.
- 4** Say the whole route in order, starting at P and ending at B.

Answer: **3** squares right, then **2** squares up, which is **5** squares walked altogether

⇔ Check it a different way

You worked it out by tracing the route with a finger on the grid. Now try it another way.

You worked the route out by tracing it with a finger on the grid. Now check by **walking** it: chalk the grid on the floor, stand on P, and walk the route counting every square out loud as your foot lands in it.

Your feet cannot skip a square the way a finger can slide over one. If you arrive on B and your count matches the number you wrote, the route is right. If you arrive anywhere else, the walking has found the mistake for you.

How do you know?

Milo says the route back from the bench to the peg needs the same number of squares as the route there. Is he right? Say how you could find out without arguing about it.

Your turn

- 1 Follow this route: two squares right, then one square up.
- 2 Write a route from the classroom door to your own seat.
- 3 Draw an arrow pointing to the right of the page.
- 4 Draw an arrow pointing to the top of the page.
- 5 Write a route from the bench back to the peg.
- 6 Mark where you land after three right, then one down.

TOPIC 16.2 · YOUR TURN · CONTINUED

7
Give a partner three instructions. Watch them follow every one.
one square at a time, please

8
Follow three instructions a partner gives you. Say where you end.

9
True or not true?
Decide for each line.

A direction on its own tells you how far to go.	TRUE / NOT TRUE
Two squares right then one up lands where one up then two right lands.	TRUE / NOT TRUE
A route should name where it starts and where it ends.	TRUE / NOT TRUE

10
Draw your own route, top left corner to bottom right corner.

TOPIC 16.2 · YOUR TURN, CONTINUED

11 Compare the route there with the route back. Write what changed.

12 Write instructions to walk right round your own table once.

13 Sort these into good instructions and useless ones.

says which way and how
far

does not say enough

14 Write a route with four instructions, from your seat to a window.



■ SCAN TO SEE IT MOVE

Give a shape instructions to turn a quarter, a half or a whole turn, clockwise or anticlockwise, and watch it obey.

mathigon.org/polypad/polygons

→ Take it further

Write a route from your seat to the door that has exactly four instructions, then write one for the same journey that has only two. Both should work. Then write a sentence saying why the short one is easier to follow and harder to write.

Whole, half and quarter turns



CLOCKWISE is the way clock hands go.

ANTICLOCKWISE is the other way.

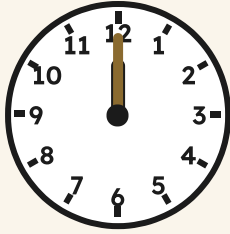
♣ Turning is measured in quarters

A **whole turn** takes you right round and leaves you facing exactly where you began. A **half turn** leaves you facing the opposite way. A **quarter turn** is half of that again.

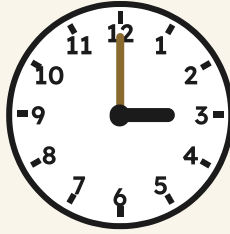
Clock hands go round one way only, and we borrowed their name for it. **Clockwise** is the way they go. **Anticlockwise** is against them.

A turn is not a place, it is an amount. So you must say two things: how much you turn, and which way round.

TOPIC 16.3 · CONTINUED



start, facing twelve



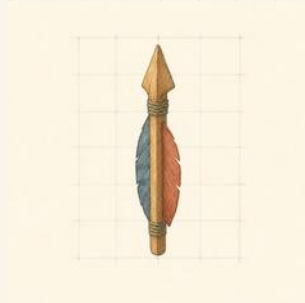
a quarter turn clockwise



a half turn clockwise

Do you remember?

There are **12** hours round a clock face, so a quarter turn carries the hand past **3** of them.



A wooden arrow, shown after a quarter turn.

Watch me work one out 1

A clock hand points at twelve. It makes a quarter turn clockwise. What does it point at now?

- 1** A whole turn goes right round the face, past every hour.
- 2** The face is split into **4** quarters, so share the **12** hours between them.
- 3** Each quarter is worth **3** hours of the face.
- 4** Count that many hours on from twelve, going the way the hands go.

Answer: the **3** on the clock face, because a quarter turn clockwise moves the hand past **3** hours

⇌ Check it a different way

You worked it out by counting hours round the clock face. Now try it another way.

You worked that out on the clock face with your eyes. Now check with your whole body: stand up, choose something fixed to face, such as the door, and say out loud what you are facing.

Make one quarter turn clockwise and name what you face now. Make three more quarter turns the same way, naming each one. If a quarter turn really is a quarter, then **4** of them bring you back to the door, and the last one has to land you exactly where you started.

How do you know?

Tessa turns a half turn clockwise. Otto turns a half turn anticlockwise. They end up facing the same way. Explain why, and then say whether that also happens with quarter turns.

Your turn

1 Stand up. Make a quarter turn clockwise. Say what you face.

2 Stand up. Make a half turn. Say what you face now.

3 Make a whole turn. Write what you face at the end.

4 Turn a pencil a quarter turn anticlockwise on your desk.

5 Draw an arrow, then draw it again after a half turn.

6 Write how many quarter turns make one whole turn.

7 Ring the turn that leaves you facing the opposite way.

a quarter turn

a half turn

a whole turn

TOPIC 16.3 · YOUR TURN · CONTINUED

- 8** Order these turns from smallest to largest.

a half turn

a whole turn

a quarter turn

a three-quarter turn

- 9** Sort these turns into clockwise and anticlockwise.

clockwise

anticlockwise

- 10** Turn this book a half turn. Write what happened to the writing.

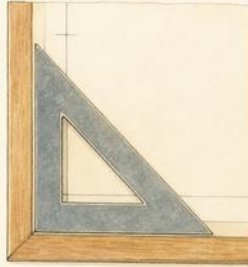
TOPIC 16.3 · YOUR TURN, CONTINUED

- 11** Make two quarter turns clockwise. Name the single turn that equals.
- 12** Write how many half turns make one whole turn.
- 13** Face the door. Make a quarter turn clockwise, then name what you face.
- 14** Draw a clock hand at twelve, then at a quarter turn clockwise.

Find out together

Stand back to back with a partner, both facing something you can name. Take turns to call out a turn, such as a quarter turn anticlockwise, and both make it. Every time you stop, say out loud what you are facing. Keep a count of the quarter turns you have made altogether. Does the count tell you what you should be facing, before you look?

A right angle is a quarter turn



A RIGHT ANGLE is a square corner.

✦ The same thing, standing still and moving

A **right angle** is a square corner. The corner of this page is one. The corner of a door, a window, a tile, a table: all right angles.

Here is the part nobody tells you. A right angle and a **quarter turn** are the same idea. Point your arm along one edge of this page, turn it to point along the other edge, and you have just made a quarter turn. The corner is the amount you turned, drawn and left there.

So the corner is a quarter turn holding still, and a quarter turn is a corner being made. Once you see that, you can measure any corner in the room with the corner of this book.

TOPIC 16.4 · CONTINUED



four square
corners



four square
corners



check each
corner



quarter turn



right angle

Watch me work one out 1

How many right angles are there in the four corners of this page?

- 1 The page is a rectangle, so start by counting its corners.
- 2 A rectangle has **4** corners, one at each turning point.
- 3 Test one corner against the corner of a second book. It fits.
- 4 Slide the test corner round to each of the others in turn.
- 5 Every one fits exactly, so every one is a square corner.

Answer: **4** right angles, one at each corner of the rectangle

Check it a different way

You worked it out by judging the corner by eye. Now try it another way.

Somebody says a corner in this room is a right angle because it **looks** square. Eyes are easily fooled, especially on a big corner or a corner you are looking at from the side.

So check it a different way. Hold the corner of this book right into the corner you are testing, with both edges pressed against it. If you can see daylight along either edge, it is not a right angle. Your book is a measuring tool, and you have had one in your bag all year.

How do you know?

A whole turn is made of **4** quarter turns. So how many right angles do you turn through if you make a whole turn? Explain how you know without turning round.

Your turn

1 Find a right angle in this room. Write what it was.

The material you need: the corner of this book

2 Hold the book corner against your desk corner. Write what fits.

3 Write how many right angles a rectangle has.

4 Draw a right angle using the corner of this book.

5 Ring
the
corners
that
are
right
angles.

a door corner

the point of a pencil

a window corner

6
 Draw a square. Mark one corner with a small square.

7
 Test three corners in the room.
 Answer for each one.

The desk corner is a right angle.	YES / NO
The corner of the door is a right angle.	YES / NO
The corner of the book is a right angle.	YES / NO

8
 Draw a corner smaller than a right angle.

9
 Draw a corner bigger than a right angle.

10
 Write how many right angles you turn through in a half turn.

TOPIC 16.4 · YOUR TURN, CONTINUED

- 11** Sort these corners into right angles and not right angles.

right angle

not a right angle

- 12** Draw a triangle with one right angle. Test it with your book.

- 13** Write why the corner of this book is a good corner tester.

- 14** Count the right angles in one window in this room.

★ **Rook's find**

Bricklayers and carpenters do not trust their eyes either. They carry a set square, which is a triangle with one right angle in it and no other job at all. A wall built without one leans, and a leaning wall does not stay up. The corner of your book is doing exactly the same work for you.

🏡 **Yard challenge**

Fold a sheet of paper in half, then fold it in half again the other way, and open it out. Test the four corners where your folds cross. You have made right angles without a ruler, a set square or a printed page. Write down why folding twice does it, using the words half and quarter.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write where your book is, using the word underneath.
- 2 Name the thing directly in front of you at your desk.
- 3 Write a route from your seat to the classroom bin.
- 4 Follow this route on a grid: one square left, then two squares down.
- 5 Make a three-quarter turn clockwise. Write what you now face.
- 6 Write which turn leaves you facing the way you started.
- 7 Write clockwise or anticlockwise for the way a clock hand goes.
- 8 Find two right angles on the floor of this room. Write where.
- 9 Write how a quarter turn and a right angle are the same idea.
- 10 Test the corner of a book against a corner outdoors. Write what you found.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
say where something is using words such as between and underneath	☐	☐	☐
give a direction and follow one, on a grid and in the room	☐	☐	☐
make a whole, half and quarter turn, clockwise and anticlockwise	☐	☐	☐
recognise a right angle, and find right angles around you	☐	☐	☐

NOW I CAN

- ☐ say where something is using words such as between and underneath
- ☐ give a direction and follow one, on a grid and in the room
- ☐ make a whole, half and quarter turn, clockwise and anticlockwise
- ☐ recognise a right angle, and find right angles around you

▣ YARD WORDS



position

where something is



direction

the way something goes



clockwise

the way clock hands turn



anticlockwise

the way against clock hands



quarter turn

a turn through one right angle



right angle

a square corner

The treasure route

Bramble has hidden something in the yard and he refuses to say where. He will only accept written instructions, and he will follow them exactly, including the mistakes.

Draw a grid of the classroom or the yard on squared paper. Mark a start square and a treasure square.

BUILD IT AND SHOW IT • PROJECT 5 • CONTINUED

Six squares by six is plenty. Mark your start with an S and your treasure with a T.

BUILD IT AND SHOW IT · PROJECT 5 · CONTINUED

Write a route from the start to the treasure using only these instructions: forward one square, quarter turn clockwise, quarter turn anticlockwise. Nothing else is allowed.

Swap with another pair. Follow their route exactly, doing precisely what it says and nothing you think it means.

WHAT TO HAND IN

- 1 Draw your grid and mark the start and the treasure.
- 2 Write your route as a numbered list of instructions.
- 3 Follow another pair's route exactly, and say where you ended up.
- 4 If you did not find their treasure, write down which instruction went wrong.

HOW IT WILL BE JUDGED

What is judged	Worth	What a good one looks like
Route works	35%	somebody else following it exactly reaches the treasure
Only the three allowed moves	25%	no shortcuts, no 'go to the door'
Clear numbering	20%	one instruction per line, in order
Debugging	20%	when a route failed, you found the exact instruction that broke it

Money

Bramble's tin has to balance at the end of the day.



Come into the Plan Room

Put a handful of coins on the table. Name every one of them out loud.

Now find the coin worth the most, and the coin worth the least. Are they the biggest and the smallest to look at?

Try to make exactly twenty cent. How many different ways can you find?

In the yard today

Bramble has tipped the money tin out onto the counter and he is counting it for the second time.

He does this at the end of every single day, and he does it twice. The first time he counts the coins into piles of the same kind. The second time he counts the piles.

Two ways round, he says. If the tin says the same thing twice, the tin is right. If it says two different things, I have made a mistake, and I would rather find it tonight than have it found for me tomorrow.

✓ **By the end you will**

- name every euro coin and note, and say what each is worth
- write an amount of money in euro and cent correctly
- find the total of a handful of coins, and check it another way
- choose coins to pay an amount, and work out the change

Cambridge Primary Mathematics Stage 2: 2Mm1 • 2Mm2 • 2Mm3 • 2Pt5

Recognising all coins and notes



The whole tin, laid out in order along the counter.

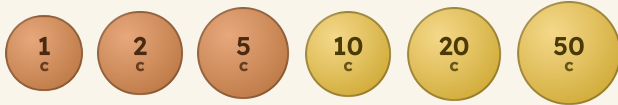
✦ Size does not tell you value

There are eight euro coins: **1c, 2c, 5c, 10c, 20c, 50c, 1 euro and 2 euro**. There are notes as well, and the smallest note is worth five euro.

Here is the thing that catches people out. A 5c coin is **BIGGER** to look at than a 10c coin, but it is worth less. A 50c coin is bigger than a 1 euro coin, and worth half as much.

So you cannot judge a coin by its size. You have to read what is stamped on it, every time.

TOPIC 17.1 · CONTINUED



Six of the eight coins, smallest value first. Look at their sizes.



One euro and two euro. One hundred cent make one euro.

Do you remember?

One hundred cent make one euro. That is the only conversion you need this year, and it is worth knowing without thinking.

Watch me work one out 1

Which is worth more, a 5c coin or a 10c coin?

- 1 Do not look at the size of them. Look at the number stamped on.
- 2 One says 5. That means five cent.
- 3 The other says 10. That means ten cent.
- 4 Ten is more than five, even though the 5c coin looks bigger.

Answer: the 10c coin, which is worth **10c**

Check it a different way

You worked it out by reading the number stamped on each coin. Now try it another way.

You compared the stamped numbers. Now check by swapping: how many 5c coins would you need to be worth the same as one 10c coin?

Two of them. If it takes two 5c coins to match one 10c coin, then the 10c coin must be worth more, and you have proved it without reading either number again.



Two 5c coins are worth the same as one 10c coin.

How do you know?

Milo says the biggest coin must be worth the most. Find two coins in the picture above that show he is wrong.

Your turn

1 Name every euro coin, starting with the one worth least.

2 Write which is worth more, a 20c coin or a 50c coin.

3 Write which is worth more, a 2 euro coin or a 50c coin.

4 Write how many cent make one euro.

5 Find the coin worth least in a handful. Write its value.

6 Draw round a 50c coin and write its value inside.

7 How many 10c coins are worth the same as one 50c coin?

8 How many 20c coins are worth the same as one euro?

9 Order these coins from least valuable to most valuable.

50c

5c

20c

2c

10 Write the value of the smallest euro note.

TOPIC 17.1 • YOUR TURN, CONTINUED

- 11** Find two coins that together are worth 30c. Draw them.
- 12** Find three different coins. Write which is worth most.
- 13** Write two coins that are worth the same as one 20c coin.

- 14** Sort a handful of coins into two trays. Use the headings below.

less than 10c

10c or more

✂ The material you need

a real handful of coins you are allowed to use, or the plastic ones from the classroom. This whole unit is much easier with coins on the desk than without.

★ Rook's find

Every euro coin has the same face on one side wherever it comes from, but the other side is different in every country. Bramble collects the backs. He has nineteen and he is still looking.

Writing amounts of money



Under one euro, write it in cent.

One euro or more, write it with a comma.

✚ Two ways, and when to use each

If an amount is **less than one euro**, write it in cent with a small c after it: **45c**.

If an amount is **one euro or more**, write it with a euro sign and a comma: **€1,45**. In Portugal the comma is the mark between euro and cent.

Never mix them. **€45c** means nothing at all, and neither does **1,45c**. Pick the right form for the size of the amount.



a peg
45c



a basket
€1,45



a nail
80c

Do you remember?

The comma splits euro from cent. Two figures always come after it, because there are always one hundred cent in a euro.

Watch me work one out 1**Write one euro and thirty-five cent correctly.**

- 1 The amount is more than one euro, so use the euro form.
- 2 Write the euro sign, then the number of whole euro: 1.
- 3 Write the comma, which splits euro from cent.
- 4 Write the cent, using two figures: 35.

Answer: €1,35**⇌ Check it a different way**

You worked it out by writing the euro, the comma, then the cent. Now try it another way.

You built it from the words. Now check the other way: count it out in real coins and see whether the coins agree with what you wrote.

One euro coin, one 20c, one 10c and one 5c. If the coins on the desk match the figures on the page, you wrote it correctly.



The same amount, in coins.

How do you know?

Why must there always be two figures after the comma? Think about how many cent there are in one euro.

Your turn

1 Write forty cent the correct way.

2 Write eighty-five cent the correct way.

3 Write two euro and fifty cent the correct way.

4 Write one euro and five cent the correct way.

5 Write €3,20 in words.

6 Write 75c in words.

7 Write €1,99 in words.

8 Which of these is written wrongly?
Write it correctly.

45c

€2,10

€30c

9 Write a price under one euro for something in the store.

10 Write a price over one euro for something in the store.

TOPIC 17.2 · YOUR TURN, CONTINUED

11 Write a price label for a basket costing two euro exactly.

12 Write ninety-nine cent, then write one cent more.

13 Explain why €5c cannot be right.

14 Make three price labels for three things on your desk.



▣ SCAN TO SEE IT MOVE

Count out coins and notes to make any amount, and check your change by putting the coins back.

mathigon.org/polypad/number

→ Take it further

Write every amount from 95c up to €1,05, one for each cent. You will have to change from the cent form to the euro form part way through. Mark the exact place where you change.

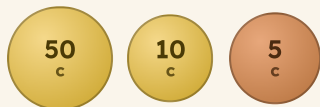
Finding totals

♣ Biggest coin first

To total a handful of coins, always start with the coin worth **most** and work down. Starting with the small ones means holding a big number in your head while you add tiny amounts to it, which is where mistakes come from.

So sort the coins into order first, then add. Sorting is not wasted time: it is most of the work.

Look for pairs that make ten or a hundred as you go. Those are the number bonds you learned in Unit 6, doing a real job.



Sorted biggest first, then added: fifty, sixty, sixty-five.

- 1 Sort the coins with the biggest value first.
- 2 Say the value of the biggest coin out loud.
- 3 Count on the next coin, then the next.
- 4 Say the total, then write it in the correct form.

Watch me work one out 1

Find the total of a 50c, a 20c, a 10c and a 5c coin.

- 1** Sort them biggest first: 50c, 20c, 10c, 5c.
- 2** Start at fifty.
- 3** Count on twenty: seventy.
- 4** Count on ten: eighty.
- 5** Count on five: eighty-five.

Answer: 85c



Eighty-five cent.

↔ Check it a different way

You worked it out by adding from the biggest coin down. Now try it another way.

You added from the biggest down. Now add them in the opposite order, from the smallest up: five, fifteen, thirty-five, eighty-five.

Adding can be done in any order, so the two runs must agree. This is exactly what Bramble does with the tin every evening, and it is why he trusts it.

How do you know?

Why is starting with the biggest coin easier than starting with the smallest, even though both give the same answer?

Your turn

- 1 Total a 20c and a 10c coin.
- 2 Total a 50c and a 20c coin.
- 3 Total two 20c coins and a 5c coin.
- 4 Total a 50c, a 20c and a 20c coin.
- 5 Total three 10c coins and a 2c coin.
- 6 Total a 1 euro coin and a 50c coin.
- 7 Make 45c using exactly three coins. Draw them.
- 8 Make 70c using exactly two coins. Draw them.
- 9 Make 30c two different ways. Draw both.
- 10 A peg costs 25c. Two pegs cost how much?

TOPIC 17.3 · YOUR TURN, CONTINUED

- 11** Buy a 45c peg and a 20c nail. Write the total.
- 12** Buy two things costing 30c and 50c. Write the total.
- 13** Find the fewest coins that make 80c.
- 14** Find two ways to make exactly one euro. Draw both.

♣ Find out together

Work in a pair with about ten coins. One of you closes your eyes and takes three coins. The other says the total. Then swap, and the first person checks the total by adding in a different order. Ten turns each. How many did you disagree on?

Paying, and working out change

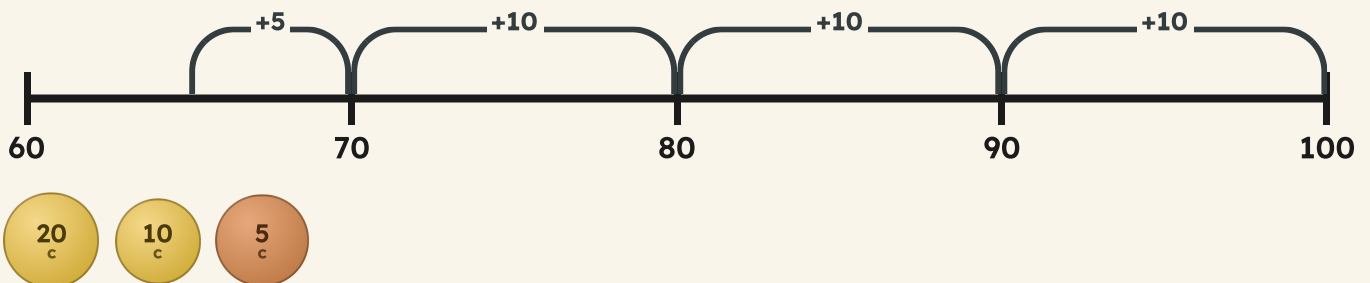
♣ Change is counting UP, not taking away

When something costs 65c and you pay with a one euro coin, the **change** is what you get back.

You could take sixty-five away from one hundred. But there is a much easier way, and it is what real shopkeepers do: count **up** from the price to the coin you were given.

Sixty-five, seventy, eighty, ninety, one hundred. Count how much you added on the way, and that is the change.

From the price up to the coin. Add up the hops.



The change, counted into the customer's paw: 35c.

 Watch me work one out 1

A basket costs 65c. You pay with one euro. What is the change?

- 1** Start at the price: sixty-five.
 - 2** Count up to the next ten: sixty-five to seventy is five.
 - 3** Count up in tens to one hundred: seventy, eighty, ninety, one hundred. That is thirty more.
 - 4** Add up what you counted: five and thirty.
-

Answer: the change is **35c**

⇌ Check it a different way

You worked it out by counting up from the price to the coin. Now try it another way.

The shopkeeper counted UP from the price. Now be the customer and check the other way: add the change you were given back onto the price.

Sixty-five and thirty-five makes one hundred, which is the coin you handed over. That is Cambridge 2Pt7: check a subtraction by adding your answer back. Two people, two directions, and nobody leaves the counter until they agree.

$$65 + 35 = \boxed{}$$

How do you know?

Why is counting up from 65c easier than taking 65 away from 100? Try both and say which one you would rather do standing at a counter with somebody waiting.

Your turn

- 1 Something costs 90c. You pay one euro. Write the change.
- 2 Something costs 80c. You pay one euro. Write the change.
- 3 Something costs 55c. You pay one euro. Write the change.
- 4 Something costs 45c. You pay 50c. Write the change.
- 5 Something costs 15c. You pay 20c. Write the change.
- 6 Something costs 70c. You pay one euro. Draw the change in coins.
- 7 Choose coins to pay exactly 35c. Draw them.
- 8 Choose coins to pay exactly 60c. Draw them.
- 9 A peg is 25c and a nail is 20c. Pay with one euro. Write the change.
- 10 Something costs €1,20. You pay with two euro. Write the change.

TOPIC 17.4 · YOUR TURN, CONTINUED

- 11** Buy two things costing 30c and 45c. Pay one euro. Write the change.
- 12** Check this change by adding back. A 55c basket, one euro paid, 40c change.
- 13** Write a shop story where the change is exactly 25c.
- 14** You have 50c. Write two things you could buy from your own shop.

Yard challenge

Bramble gives 45c change from one euro, but he wants to hand over as FEW coins as possible. Which coins does he choose, and how do you know no smaller number of coins would work?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write which is worth more, a 20c coin or a 10c coin.
- 2 Write two euro and forty cent the correct way.
- 3 Write 65c in words.
- 4 Total a 50c, a 10c and a 5c coin.
- 5 Make 55c using exactly three coins. Draw them.
- 6 Something costs 75c. You pay one euro. Write the change.
- 7 Buy two things costing 20c and 35c. Write the total.
- 8 Write how many 20c coins make one euro.
- 9 Check this: 30c change from 50c on a 25c peg. Show your check.
- 10 Explain why counting up is easier than taking away when giving change.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

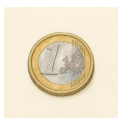
	confident	almost there	needs more work
name every euro coin and note, and say what each is worth	☐	☐	☐
write an amount of money in euro and cent correctly	☐	☐	☐
find the total of a handful of coins, and check it another way	☐	☐	☐
choose coins to pay an amount, and work out the change	☐	☐	☐

UNIT 17 • HOW DID IT GO • CONTINUED

NOW I CAN

- ☐ name every euro coin and note, and say what each is worth
- ☐ write an amount of money in euro and cent correctly
- ☐ find the total of a handful of coins, and check it another way
- ☐ choose coins to pay an amount, and work out the change

▣ YARD WORDS



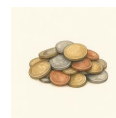
coin

a metal piece of money



note

a paper piece of money



total value

how much it is all worth



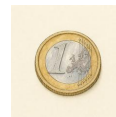
amount

how much money



change

the money you get back



cent

one hundredth of a euro

Length

Cut it twice as long as you need, said nobody good at measuring.



Come into the Plan Room

Stand next to somebody. Who is taller? Now check it with your backs against the wall and no shoes on.

Find something in the room that is about as long as your arm. Hold your arm against it to see how close you were.

Now find something far too long to measure with a ruler. What would you use instead?

☀ In the yard today

Bramble has a plank across two trestles and he is looking at it the way you look at a wasp.

Rook asked for a shelf one metre long. The plank is longer than that, so a piece has to come off. Bramble has measured it three times and he has not picked up the saw yet.

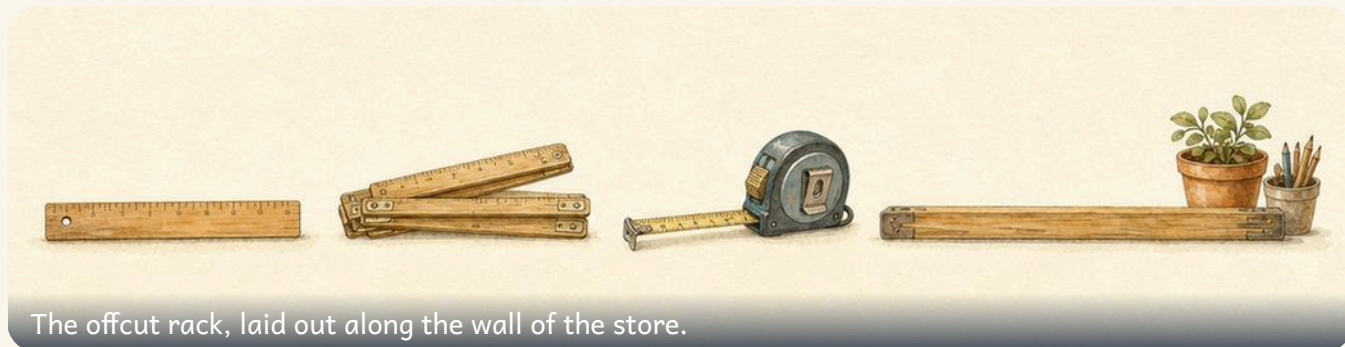
You can always cut more off, he says. You cannot put any back on. So measure it, then measure it again a different way, and only then cut.

✓ **By the end you will**

- compare lengths and put them in order from shortest to longest
- use metres and centimetres, and say which suits what you measure
- choose a ruler, a tape or a metre stick, and say why
- measure a line to the nearest centimetre, and draw one to length

Cambridge Primary Mathematics Stage 2: 2M11 • 2M12 • 2Pt2

Comparing lengths and heights



The offcut rack, laid out along the wall of the store.

✦ Line the ends up first

Length is how long a thing is. **Height** is how tall it is. They are the same idea standing up.

You do not need any numbers at all to compare two lengths. Hold them side by side with one end of each in a line, and look at the other end. The one that sticks out is longer.

The lining up is the whole trick. If one end starts further along than the other, the picture lies to you, and a shorter thing can look like the longer one.

TOPIC 18.1 • CONTINUED

the nail

the peg

the screw

the dowel

Do you remember?

Longer, shorter, taller, higher. Say which two things you are comparing every time, or the word means nothing on its own.

TOPIC 18.1 • CONTINUED

the red ribbon

the blue ribbon

Watch me work one out 1

Which ribbon is longer, the red one or the blue one?

- 1 Put them side by side with their left ends in a line.
 - 2 Look at the right hand end of each one.
 - 3 The red ribbon carries on past the end of the blue one.
 - 4 So the red ribbon is the one that sticks out.
-

Answer: the red ribbon is longer

⇌ Check it a different way

You worked it out by holding the ribbons side by side with their ends in a line. Now try it another way.

You compared them side by side and used no numbers. Now check it a completely different way: measure each ribbon on its own with a ruler in centimetres, and compare the two numbers instead of the two ribbons.

One comes out at **8** centimetres and the other at **5**. The bigger number belongs to the longer ribbon, and it is **3** centimetres longer. Two methods, one answer, so you can trust it.

8 5

How do you know?

Milo lays two pencils on the desk with their ends all over the place, and says the one on the left is longer. Why can he not know that yet?

Your turn

- 1** Order these lengths from shortest to longest.

a broom

a pencil

a metre stick

a nail

- 2** Compare two pencils by lining up one end of each.

- 3** Find something shorter than your book. Draw it.

TOPIC 18.1 • YOUR TURN • CONTINUED

- 4 Find something taller than you in the classroom.
- 5 Write which is longer, your foot or your hand.
- 6 Lay three crayons side by side. Name the longest.
- 7 Order three shoes from smallest to biggest.
- 8 Compare your height with a friend's. Write who is taller.
- 9 Draw two lines, one longer than the other.
- 10 Find the longest stick in the pot. Say how you know.

TOPIC 18.1 • YOUR TURN, CONTINUED

- 11** Sort these into shorter than a ruler and longer than a ruler.

shorter than a ruler

longer than a ruler

- 12** Order four books from thinnest to thickest.

- 13** Compare two ribbons by holding them together, ends in a line.

- 14** Write two things in the room that are the same length.

✂ The material you need

two ribbons or two strips of paper of different lengths, a handful of pencils, and a wall you are allowed to stand against. No ruler yet: this topic is about comparing, not counting.

★ Rook's find

A doorway in Bramble's store is cut to the height of the tallest person who ever asked for a shelf, plus a hand's width. He says a door you have to duck through is a door built by somebody who measured once.

Metres and centimetres



A metre stick. One metre, all in one piece.



A centimetre. About as wide as your fingernail.

♣ One hundred centimetres make one metre

A **centimetre** is small. It is about the width of your little fingernail, and it is the space between two numbered marks on your ruler.

A **metre** is big. It is about as wide as you can stretch your arms. And here is the fact the whole unit stands on: **100 centimetres make 1 metre.**

So to change metres into centimetres, count one hundred for every metre. Two metres is two hundreds of centimetres. Three metres is **300** centimetres, and the number gets big fast.

TOPIC 18.2 · CONTINUED

Ten centimetres of a ruler. Count the spaces, not the marks.



Two units, two short ways of writing them

How we write it	The short way
centimetre	cm
metre	m

Do you remember?

100 centimetres make **1** metre. Learn it until you do not have to think about it, because every length question this year leans on it.

 Watch me work one out 1

How many centimetres are there in 3 metres?

- 1** One metre is **100** centimetres.
- 2** So count one hundred for every metre you have.
- 3** One metre, one hundred. Two metres, two hundred.
- 4** Three metres, three hundred. Stop, because **3** metres is what was asked for.

Answer: **300** centimetres

- 1** Ask yourself which unit the answer has to be in.
- 2** If you are going from metres to centimetres, count in hundreds.
- 3** Count one hundred for every whole metre.
- 4** Add on any spare centimetres at the end.
- 5** Write the number, then write the unit after it.

⇌ Check it a different way

You worked it out by counting one hundred centimetres for every whole metre. Now try it another way.

You counted in hundreds and went from metres down to centimetres. Now go the other way with something real: lay a metre stick along the shelf, end over end, and count the whole metres and the centimetres left over.

A shelf of **1** metre and **20** centimetres flattens into **120** centimetres, because that is one hundred for the metre and the spare centimetres on top. The stick and the counting must agree, and if they do not, one of them was done carelessly.

How do you know?

Which is easier to say out loud, that a corridor is nine metres or that it is nine hundred centimetres? Both are true. Why do we choose one of them?

Your turn

- 1 Write how many centimetres make one metre.
- 2 Write how many centimetres are in two metres.
- 3 Write how many centimetres are in five metres.
- 4 Write one metre and thirty centimetres in centimetres only.
- 5 Write two hundred centimetres in metres.
- 6 Choose metres or centimetres for a pencil's length.

TOPIC 18.2 · YOUR TURN · CONTINUED

7 Measure your desk in centimetres. Write the number.

8 Find something about one metre long. Write its name.

9 Order these lengths from shortest to longest.

1 metre

40 centimetres

2 metres

90 centimetres

10 Write your own height in centimetres.

TOPIC 18.2 · YOUR TURN, CONTINUED

11 Draw a line ten centimetres long. Label it in centimetres.

12 Write which is longer, ninety centimetres or one metre.

13 Measure three things in centimetres. Fill in the table.

What I measured	How many centimetres

14 Explain why we do not measure a corridor in centimetres.



▣ SCAN TO SEE IT MOVE

Measure a line with a ruler you can slide, in centimetres and metres, and see why where you START the ruler matters.

mathigon.org/polypad/geometry

→ Take it further

Write down the length of your hand span in centimetres. Now work out how many hand spans would make one metre, by counting your span again and again along a metre stick. Would your friend need more spans than you, or fewer?

Choosing the right unit and the right ruler



Three tools on the store counter: a ruler, a tape and a metre stick. Each one is right for something.

♣ Big thing, big unit

A **unit** is the size you measure in. Pick a unit that suits the thing in front of you.

Measure a pencil in metres and your answer is a scrap of a metre, which tells nobody anything. Measure the corridor in centimetres and your answer is **800**, which is true, and horrible, and nobody will remember it. Both mistakes are the same mistake.

The tool matters just as much. A ruler is short and stiff, so it is for small straight things. A metre stick is long and stiff, so it is for rooms and walls. A tape bends, so it is for things that curve, such as your waist or a bucket.

What Bramble reaches for, and why

What you are measuring	Tool	Unit
a pencil	a ruler	centimetres
a book	a ruler	centimetres
a corridor	a metre stick	metres
round a bucket	a tape	centimetres



Hold your hands one centimetre apart. Now one metre apart. Do it again with your eyes shut, then check yourself against a ruler and a metre stick.

 Watch me work one out 1

Which unit should you use for the length of the corridor?

- 1** Think how big the corridor is. It is much longer than you.
 - 2** A centimetre is fingernail sized, so you would need heaps of them.
 - 3** A metre is arm span sized, so only a few will do.
 - 4** In metres the answer is **8**, which is a number you can hold in your head and say out loud.
-

Answer: metres

 Check it a different way

You worked it out by choosing the unit from the size of the thing being measured. Now try it another way.

You chose the unit by thinking about the size of the corridor. Now check it by doing the silly thing on purpose: work the same corridor out in centimetres and look at what you get.

800 centimetres. It is exactly the same corridor and the answer is not wrong, but the number is clumsy to say, easy to miscopy and hard to remember. That is how you can tell a poor choice of unit: the number goes strange. Try it the other way too, with a pencil in metres, and the number goes strange in the opposite direction.

How do you know?

Why would a bendy tape be a poor choice for measuring how tall the door is, even though it is long enough?

A pencil is measured in centimetres.	TRUE / NOT TRUE
A playground is measured in centimetres.	TRUE / NOT TRUE
A tape measure can go round a curve.	TRUE / NOT TRUE
A ruler is the best tool for a corridor.	TRUE / NOT TRUE

Your turn

- 1 Choose the best tool to measure a football pitch.
- 2 Choose the best tool to measure a pencil case.
- 3 Choose the best unit for the height of a door.
- 4 Choose the best unit for the length of a finger.
- 5 Write which tool you would use for a curved ribbon.

- 6 Sort these into measure in metres and measure in centimetres.

in metres

in centimetres

TOPIC 18.3 · YOUR TURN · CONTINUED

7 Write why a ruler is no use for measuring the corridor.

8 Match each thing to the best unit for it.

a shoe



metres

a classroom



centimetres

a leaf



metres

a garden
path



centimetres

9 Choose a tool, then measure the width of the door.

10 Write two things you would measure with a metre stick.

TOPIC 18.3 · YOUR TURN, CONTINUED

- 11** Write two things you would measure with a ruler.
- 12** Measure the room with a metre stick. Write it in metres.
- 13** Explain why a tape bends but a metre stick does not.
- 14** Write a length that would sound silly in centimetres.

Find out together

Work in a pair. One of you names something in the room, the other says the tool and the unit and then goes and measures it. Ten things each, and you must swap after every one.

Keep a list. At the end, look for any measurement where you would choose a different unit now that you have seen the number.

Measuring and drawing lines



Find the zero before you find anything else.



End on the zero, then read the other end.

✦ Start at the zero, not at the end of the ruler

Look at your ruler. Before the **1** there is a **0**, and before the **0** there is usually a little strip of plastic with nothing on it at all.

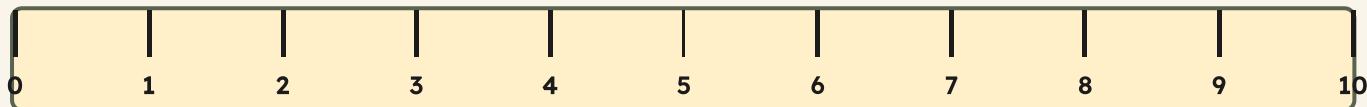
That **0** is where every measurement begins. Put the end of the line exactly on it, hold the ruler still, and read the number at the far end of the line.

Two things go wrong, and they go wrong to nearly everybody. Start at the **1** instead of the **0**, and a line of **7** centimetres reads as **6**: every single length comes out one centimetre short. Start at the very end of the plastic, and your answer is a bit too long instead.

So the first thing you do is not measure. The first thing you do is find the zero.

TOPIC 18.4 · CONTINUED

Put the end of your line on the zero, here at the left.



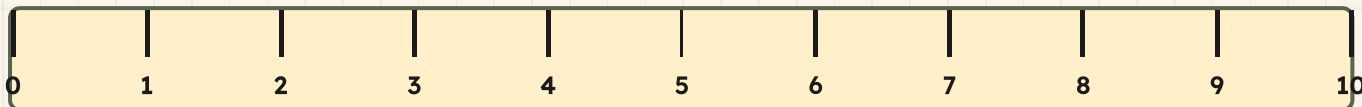
7 cm

Do you remember?

Find the zero. Line the end up on it. Hold the ruler still. Then read the number. In that order, every time.

TOPIC 18.4 · CONTINUED

The ruler Milo used.



Milo's line

Watch me work one out 1

How long is Milo's line, to the nearest centimetre?

- 1** Find the zero on the ruler, before the 1.
- 2** Slide the ruler until the left end of the line sits on the zero.
- 3** Hold it still and look at the right end of the line.
- 4** Read the number the line ends on, then say the unit as well.
- 5** Milo read **6**, because he started at the **1**. Starting at the zero gives one more.

Answer: 7 centimetres

⇌ Check it a different way

You worked it out by measuring with a ruler, with the end of the line on the zero. Now try it another way.

You measured with a ruler, which means you trusted the zero. Now check it with no ruler at all: lay centimetre cubes along the line, end to end, from one end of the line to the other, and count them.

You should count **7** cubes. Cubes cannot start at the wrong mark, because there are no marks, so if the cubes and the ruler disagree then the ruler was in the wrong place. This is the check to use whenever a measurement feels one centimetre out.

How do you know?

Every one of Milo's lines came out exactly one centimetre shorter than everybody else's. What must he have been doing, and how could you prove it?

Your turn

- 1 Measure this line to the nearest centimetre.



- 2 Measure each line.
Write its length in centimetres.



- 3 Draw a line six centimetres long. Start at the zero.

TOPIC 18.4 · YOUR TURN · CONTINUED

- 4 Draw a line nine centimetres long. Check it afterwards.
- 5 Draw a line twelve centimetres long.
- 6 Measure the longest side of your book in centimetres.
- 7 Measure a pencil, then draw a line the same length.
- 8 Write where the ruler must start when you measure.
- 9 Order your drawn lines from shortest to longest.
- 10 Draw a line shorter than five centimetres. Measure it.

TOPIC 18.4 · YOUR TURN, CONTINUED

- 11** Measure these two lines, then check each one with cubes.



- 12** Milo starts at the one mark. Write what that does to his answers.

- 13** Draw a rectangle with sides four and eight centimetres long.



- 14** Measure round your wrist with a strip of paper.

Yard challenge

Bramble needs a shelf one metre long and the plank is longer than that. He has only a ruler that reaches thirty centimetres. Work out how he can mark one metre exactly, using that short ruler and nothing else, and say how he would check the mark before he cuts.

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write which is taller, a door or a chair.
- 2 Order a nail, a broom and a pencil from shortest to longest.
- 3 Write how many centimetres make three metres.
- 4 Write one metre and fifty centimetres in centimetres only.
- 5 Choose the best unit for the length of a swimming pool.
- 6 Choose the best tool for measuring round a bucket.
- 7 Measure the short side of this page in centimetres.
- 8 Draw a line seven centimetres long, starting at the zero.
- 9 Write what goes wrong if you start at the one mark.
- 10 Explain why you would not measure a pencil in metres.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
compare lengths and put them in order from shortest to longest	☐	☐	☐
use metres and centimetres, and say which suits what you measure	☐	☐	☐
choose a ruler, a tape or a metre stick, and say why	☐	☐	☐
measure a line to the nearest centimetre, and draw one to length	☐	☐	☐

UNIT 18 · HOW DID IT GO · CONTINUED

NOW I CAN

- ☐ compare lengths and put them in order from shortest to longest
- ☐ use metres and centimetres, and say which suits what you measure
- ☐ choose a ruler, a tape or a metre stick, and say why
- ☐ measure a line to the nearest centimetre, and draw one to length

▣ YARD WORDS



length

how long something is



height

how tall something is



metre

a unit of length, 100 centimetres



centimetre

a small unit of length



ruler

a tool for measuring length



unit

the size you measure in

Mass and capacity

The scales do not care what you think. They only care what is on them.



Come into the Plan Room

Pick up your reading book in one hand and your pencil case in the other. Which is heavier? Now say how sure you are.

Find the biggest thing on your desk. Is it also the heaviest? Find something small that is heavier than something big.

Fill a cup with water. Pour it into a wide bowl. Does the water look like more, less, or the same amount?

☀ In the yard today

Bramble has two bags on the counter and he will not say which is heavier until the scales have said it first.

One bag is enormous and full of goose feathers. The other is small enough to hold in one paw, and it is full of iron nails.

Everybody guesses the big one, he says. Everybody. And everybody is wrong, because the big one is mostly air.

Your hands can be fooled and your eyes can be fooled. The scales cannot. Put the two bags on and let the beam tell you.

✓ **By the end you will**

- compare masses and order them from lightest to heaviest
- use kilograms and grams, and choose which suits what you weigh
- compare how much two containers hold, and say how you know
- read a measuring jug to the nearest whole or half litre

Cambridge Primary Mathematics Stage 2: 2M11 • 2M12 • 2Pt2

Heavier and lighter



Scales, a sack of flour and marked jugs, along the Measures Store bench.

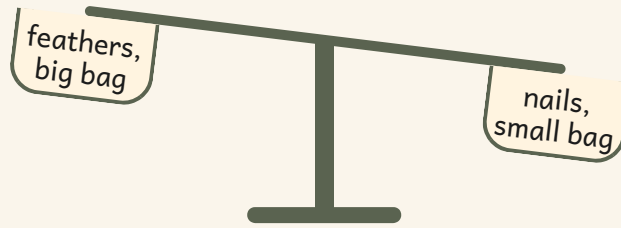
✦ Bigger does not mean heavier

Mass is how heavy something is. You cannot see mass. You can only see how big something is, and big is not the same as heavy.

A huge bag of feathers is **lighter** than a small bag of nails, because feathers are mostly air and nails are solid iron. The bag that looks like more is holding less.

So do not trust your eyes. Use **balance scales**. The pan that goes DOWN holds the heavier thing, every time, and the scales do not care what you guessed.

TOPIC 19.1 • CONTINUED



The beam has tipped towards the small bag of nails, and that settles it.

Do you remember?

The pan that goes down is holding the heavier thing. Level pans mean the two things have the same mass.

 Watch me work one out 1

Which is heavier, the big bag of feathers or the small bag of nails?

- 1** Do not look at how big the bags are. That tells you nothing.
- 2** Put one bag in each pan of the balance scales.
- 3** Let the beam settle, then look at which pan has gone down.
- 4** The pan that went down is holding the heavier bag.

Answer: the small bag of nails, even though it is the smaller bag

 Check it a different way

You worked it out by judging which was heavier by holding one bag in each hand. Now try it another way.

You held one bag in each hand and made a guess. Now check it a completely different way: put the bags on the balance scales and watch the beam.

Your hands can be fooled by how big a thing is. The beam cannot, because it is only ever answering one question: which side pulls down harder? If your hands and the scales disagree, believe the scales.



Level pans. These two things have exactly the same mass.

How do you know?

Milo says the bigger bag must always be the heavier bag. Find two things in the room that prove he is wrong.

Your turn

- 1 Hold a book and a pencil. Write which one feels heavier.
- 2 Find two things on your desk. Write which one is lighter.
- 3 Weigh a rubber against a shoe. Write which pan goes down.
- 4 Order a feather, an apple and a brick from lightest to heaviest.
- 5 Find something heavier than your pencil case. Write its name.
- 6 Find something lighter than your pencil case. Write its name.
- 7 Draw balance scales with the heavier thing in the left pan.
- 8 Guess which is heavier, a full bottle or an empty one.
- 9 Write what level pans tell you about two things.
- 10 Explain why a big box can be lighter than a small box.

TOPIC 19.1 • YOUR TURN, CONTINUED

- 11** Fill one bag with paper and one with pebbles. Write which is heavier.
- 12** Order four school things from lightest to heaviest.
- 13** Write two things in the room that feel about the same weight.
- 14** Sort five things into heavier than a book and lighter than a book.

heavier than a book

lighter than a book

✂ The material you need

balance scales, or two carrier bags hung on the ends of a coat hanger, which work just as well. You also need a handful of small heavy things and a handful of big light things.

★ Rook's find

The heaviest thing Bramble has ever weighed in the store was an anvil, and the lightest was a single goose feather. He needed the same pair of scales for both, but he had to change the weights.

Kilograms and grams



One kilogram. Heavy enough to feel in your arms. Grams. Small units, for weighing small things.

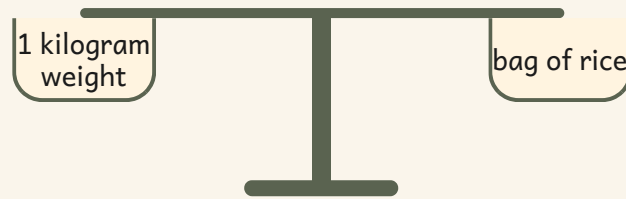
♣ Two units, and choosing the right one

Mass is measured in **grams** and **kilograms**. A gram is tiny: a paperclip is about one gram. A kilogram is a good armful: a bag of sugar is one kilogram.

1000 grams make 1 kilogram. That is the fact this whole topic stands on, so learn it now.

Choose the unit that fits the thing. Weigh a feather in grams, because in kilograms it would be almost nothing. Weigh a child in kilograms, because in grams the number would be enormous.

TOPIC 19.2 · CONTINUED



The pans are level, so the bag of rice has a mass of exactly **1000** grams, which is **1** kilogram.

Do you remember?

1000 grams make **1** kilogram. Half a kilogram is **500** grams, because **500** and **500** make **1000**.

 Watch me work one out 1

A bag of flour is 1500 grams. Is it heavier or lighter than a 1 kilogram weight?

- 1** You cannot compare grams with kilograms until they match.
- 2** Change the kilogram into grams: **1** kilogram is **1000** grams.
- 3** Now both are in grams, so compare them: **1500** and **1000**.
- 4** **1500** is more than **1000**, so the flour has the greater mass.

Answer: the bag of flour is heavier

 Watch me work one out 2

Order the apple, the loaf and the melon from lightest to heaviest.

- 1** Weigh each one and write its mass down before you compare any.
- 2** The bars below are drawn to the masses, so the shortest bar is the smallest mass.
- 3** Read the bars from shortest to longest.
- 4** Write the names in that order, lightest first.

Answer: the apple, then the loaf, then the melon

the loaf

the apple

the melon

⇔ Check it a different way

You worked it out by changing the kilogram into grams and comparing the numbers. Now try it another way.

You changed the kilogram into grams and compared the two numbers. Now check without arithmetic at all: put the flour in one pan and the kilogram weight in the other.

The flour pan goes down. The scales agree with your numbers, and you have proved the same thing twice, once with figures and once with a beam.

How do you know?

Why would you weigh a feather in grams and a bag of potatoes in kilograms? What goes wrong if you swap them over?

Your turn

- 1 Write how many grams make one kilogram.
- 2 Choose grams or kilograms for weighing an apple.
- 3 Choose grams or kilograms for weighing a sack of potatoes.
- 4 Choose grams or kilograms for weighing one feather.
- 5 Choose grams or kilograms for weighing yourself.
- 6 Write which is heavier, 500 grams or 1 kilogram.
- 7 Write which is lighter, 2 kilograms or 900 grams.
- 8 Order these masses in grams from lightest to heaviest. **Write them in the boxes, lightest first.**
- 9 Weigh your reading book. Write its mass in grams.
- 10 Find a food packet at home. Copy the mass from its label.

TOPIC 19.2 · YOUR TURN, CONTINUED

- 11 Write three things with a mass less than one kilogram.
- 12 Write three things with a mass more than one kilogram.
- 13 Add 500 grams and 500 grams. Write the total in kilograms.
- 14 Draw one bag labelled 2 kilograms and one labelled 200 grams.



▣ SCAN TO SEE IT MOVE

Pour water between jugs and watch the level rise up the scale, so you can read a measure that is really moving.

mathigon.org/polypad/number

→ Take it further

Collect ten food packets from a kitchen cupboard and copy the mass off every label. Sort them into less than half a kilogram, about half a kilogram, and more than half a kilogram. Which pile was hardest to decide, and why?

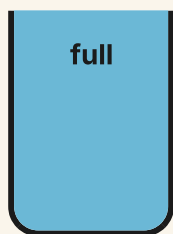
Full, empty and half full

♣ Tall does not mean more

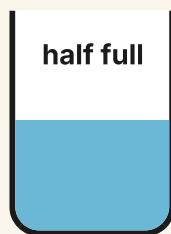
Capacity is how much a container holds. A container can be **full**, **empty**, or somewhere in between. When the water comes exactly halfway up, it is **half full**.

Here is the trap. A **tall thin** jug and a **short wide** jug can hold exactly the same amount. The tall one looks like more, because the water climbs higher, but the wide one is making up for it by being fat instead of tall.

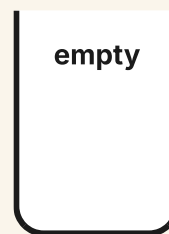
So you cannot judge capacity by height. To be certain, fill one container and pour it into the other. If it fits with nothing left over, they hold the same.



Filled right up to the top.



Exactly halfway up.



Nothing in it at all.

- 1 Look at the two containers and say which you think holds more.
- 2 Fill the first one right to the top with water.
- 3 Pour all of it into the second one.
- 4 If it overflows, the first holds more. If there is room left, the second holds more. If it fits exactly, they hold the same.

 Watch me work one out 1

A tall thin jug and a short wide jug each hold one litre. Which holds more?

- 1** Do not answer from the picture. Height is not capacity.
- 2** Fill the tall thin jug right up to the top.
- 3** Pour every drop of it into the short wide jug.
- 4** It fills the wide jug exactly, with nothing left in the tall one and nothing spilled.

Answer: neither of them, because they hold the same

 Watch me work one out 2

A jug holds 2 litres when it is full. It is half full. How much is in it?

- 1** Half full means the water comes exactly halfway up.
- 2** So you need half of **2** litres.
- 3** Half of **2** is **1**.
- 4** Check it: two halves make the whole jug again.

Answer: **1** litre

⇌ Check it a different way

You worked it out by reading how full each jug looked from the outside. Now try it another way.

You judged which jug held more by looking at how full each one seemed. Now check it a different way: pour each jug in turn into ONE measuring jug of a different shape again, and read the mark.

Both readings land on the same mark, so both jugs hold the same amount. Once every container is poured into the same measure, shape stops mattering.

How do you know?

Sofia pours a full mug into a wide bowl and says she has less water than before. Where has she gone wrong?

Your turn

- 1 Fill a cup to the top. Write whether it is full or empty.
- 2 Draw a jug that is empty.
- 3 Draw a jug that is half full.
- 4 Draw three jugs: full, half full and empty.
- 5 Find a container that holds more than your cup.
- 6 Find a container that holds less than your cup.
- 7 Order three containers from the one holding least to the most.
- 8 Fill a tall thin glass. Pour it into a wide bowl.
- 9 Write which holds more, a mug or a teaspoon.
- 10 Pour water from a full jug until it is half full.

TOPIC 19.3 · YOUR TURN, CONTINUED

- 11 Count how many cups it takes to fill your water bottle.
- 12 Write two containers at home that hold about the same amount.
- 13 Explain why a tall glass may not hold more than a wide bowl.

- 14 Sort five containers into holds more than a mug and holds less.

holds more than a mug

holds less than a mug

🔗 Find out together

Work in a pair at the sink with a tall thin bottle and a short wide tub. Both of you guess which holds more, and write your guesses down before you touch the water. Now fill the tall one and pour it into the wide one. Who was right? Then swap the order and pour the other way to be sure. How many pairs in your class guessed the tall one?

Litres, and reading a scale



Measuring jugs filled to different marks. Read the mark the water reaches.

✿ Read the mark, not the jug

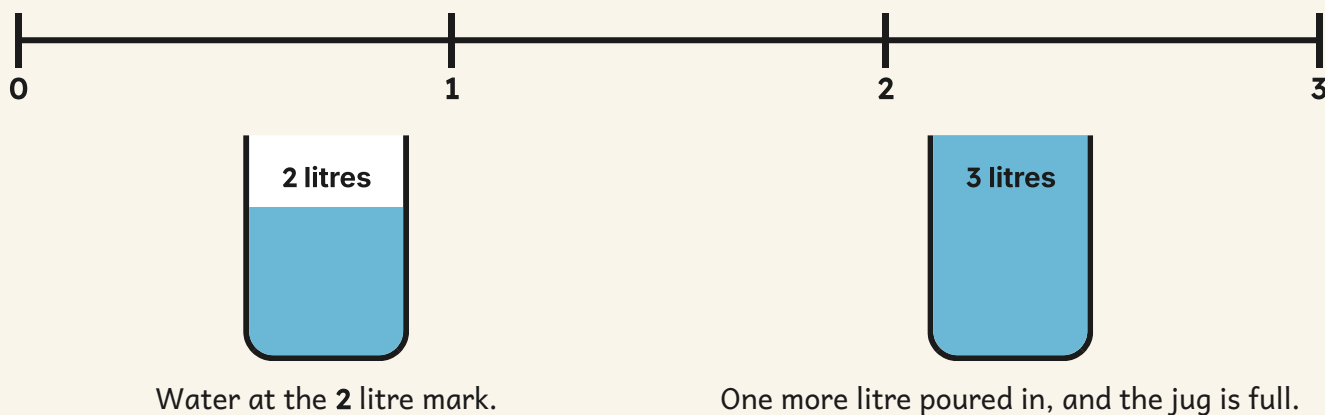
Capacity is measured in **litres**. A big bottle of water holds one litre. A bath holds a great many litres. A teaspoon holds hardly any at all.

A measuring jug has a **scale** up its side: a line of marks with numbers beside them. To read it, put the jug on the table, get your eye level with the water, and read the mark the water reaches.

If the water sits exactly between two marks, it is a half. Between the **1** and the **2** means **1** litre and a half. Do not round it up to make life easy: read what is there.

TOPIC 19.4 · CONTINUED

A jug scale, straightened out. Each mark is one litre.



 Watch me work one out 1

A jug is marked to 3 litres. It holds 2 litres. You pour in 1 litre more. How much is in it now?

- 1 Read the scale first: the water is at the 2 litre mark.
- 2 You are adding 1 litre, so count on from 2.
- 3 2 and 1 makes 3.
- 4 Look at the jug: the water has reached the top mark, which is 3 litres, so the jug is now full.

Answer: 3 litres, and the jug is exactly full

 Check it a different way

You worked it out by reading the mark on the jug scale and counting on. Now try it another way.

You read the mark on the jug scale and counted on. Now check it a different way: pour the water out into a measuring jug of a completely different shape and read that scale instead.

The second jug is wider, so the water sits lower in it, but the number beside the mark is the same. The amount did not change when the shape did, and that is the whole point of using a scale rather than your eyes.

TOPIC 19.4 · CONTINUED · CONTINUED

Fill this in after you have measured, not before.

Container	About how many litres
a mug	
a big water bottle	
a bucket	
a bath	

How do you know?

Two jugs of different shapes both read one litre. How can that be true when one water level is much higher than the other?

Your turn

- 1 Draw a jug marked 0, 1 and 2 litres. Shade one litre.
- 2 Find a drink bottle. Write how many litres it holds.
- 3 Write whether a bath holds more or less than one litre.
- 4 Write whether a teaspoon holds more or less than one litre.
- 5 Draw a measuring jug filled to the two litre mark.
- 6 Draw a measuring jug filled to the half litre mark.
- 7 Pour one litre into a jug. Mark the water level with your finger.
- 8 Order half a litre, 2 litres and 1 litre from least to most.
- 9 Write how many half litres fill a one litre jug.
- 10 Read a jug scale at home. Write the amount you see.

TOPIC 19.4 · YOUR TURN, CONTINUED

11 Estimate how many litres a bucket holds. Then check with a jug.

12 Write two things at home that are measured in litres.

13 Fill a jug to the 1 litre mark. Draw where the water sits.

14 Measure your water bottle in litres. Write what you read.

Yard challenge

Bramble has a jug marked only at 0, 1, 2 and 3 litres, and a customer asks for exactly half a litre. He can pour, tip and share, but he cannot draw a new mark on the jug. How does he measure it, and how would you prove to the customer that it really is half a litre?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write which is heavier, a bag of sand or a bag of feathers the same size.
- 2 Put two rubbers and one pencil on the scales. Write which pan goes down.
- 3 Order a coin, a shoe and a chair from lightest to heaviest.
- 4 Write how many grams are in half a kilogram.
- 5 Choose grams or kilograms for weighing a loaf of bread.
- 6 Write which is heavier, 400 grams or 1 kilogram.
- 7 Draw a jug that is half full and label it.
- 8 Write which holds more, a bucket or a mug.
- 9 Read a jug that is filled to the 2 litre mark. Write what it says.
- 10 Explain why a tall thin jug can hold the same as a short wide jug.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
compare masses and order them from lightest to heaviest	☐	☐	☐
use kilograms and grams, and choose which suits what you weigh	☐	☐	☐
compare how much two containers hold, and say how you know	☐	☐	☐
read a measuring jug to the nearest whole or half litre	☐	☐	☐

NOW I CAN

- ☐ compare masses and order them from lightest to heaviest
- ☐ use kilograms and grams, and choose which suits what you weigh
- ☐ compare how much two containers hold, and say how you know
- ☐ read a measuring jug to the nearest whole or half litre

▣ YARD WORDS



mass

how heavy something is



kilogram

a unit of mass, 1000 grams



gram

a small unit of mass



capacity

how much a container holds



litre

a unit of capacity



scale

the marked line you read a measure from

The shop that has to balance

Bramble is going away for the day and he is leaving the store to you. The tin has to balance when he gets back, and he will check.

✂ The material you need

a handful of coins you are allowed to use, paper price labels, a pencil, and a ruler for the table.

Set up a shop with six things in it. Price each one under one euro, and write a proper price label for each.



a peg
45c



a basket
80c



a nail
25c

Take it in turns to be the shopkeeper. A customer buys two things, pays with a coin that is too big, and the shopkeeper works out the change.

⇌ Check it a different way

You worked it out by counting up from the price to the coin. Now try it another way.

The shopkeeper counts the change UP from the price to the coin. Then the customer counts the change DOWN from the coin to the price. Two people, two directions, and they must agree before the customer leaves.

BUILD IT AND SHOW IT · PROJECT 6 · CONTINUED

WHAT TO HAND IN

- 1 Set up six priced things with written labels.
- 2 Record five sales: what was bought, the total, what was paid, the change.
- 3 Show both ways of working out the change on one of your five sales.
- 4 Add up your takings at the end and check the tin balances.

HOW IT WILL BE JUDGED

What is judged	Worth	What a good one looks like
Correct totals	30%	every total is right for the things bought
Correct change	30%	the change is right, and both methods agree
Money written properly	20%	euro and cent written the correct way round
The tin balances	20%	your takings match what is actually in the tin

Time

The cuckoo clock in the Plan Room is never wrong twice.



Come into the Plan Room

Say the units of time out loud, starting with the shortest one you know. How far along can you get?

Now count slowly to sixty while somebody watches a timer. Did you go too fast or too slow?

Find every clock in the room. How many are there, and do they all say the same thing?

In the yard today

The cuckoo clock in the Plan Room has struck the half hour and Bramble has stopped work to listen to it.

He does this every time. He listens, then he looks at the pocket watch in his apron, and only then does he pick his pencil back up.

A clock can go wrong, he says. A watch can go wrong too. But they do not go wrong in the same direction on the same afternoon, so if the two of them agree I believe them, and if they do not I have found something out.

✓ **By the end you will**

- name the units of time in order, and say how they fit together
- order the days of the week and the months of the year
- read and show o'clock and half past on a clock and on a screen
- time an activity in seconds and in minutes, and compare two

Cambridge Primary Mathematics Stage 2: 2Mt1 • 2Mt2 • 2Mt3 • 2Mt4 • 2Mt5

Seconds, minutes, hours and days



✦ Five facts, and time is mostly done

A **second** is about as long as saying one word carefully. It is the shortest unit you will use this year.

The units go up in steps, and each step is a number worth learning by heart. Sixty **seconds** make one **minute**. Sixty minutes make one **hour**. Twenty-four hours make one **day**.

After the day the units keep going. Seven days make one **week**, and twelve months make one **year**. That is five facts altogether, and they are the whole of time for now.

The five relationships

How many	make	one of these
60	seconds	minute
60	minutes	hour
24	hours	day
7	days	week
12	months	year

Do you remember?

*Sixty seconds make one minute and sixty minutes make one hour. The same number, **60**, does both jobs, which makes it easier to remember than you might expect.*



A second is short. Watch the sand run.



A minute is sixty of them, one after another.

Watch me work one out 1

How many seconds are there in two minutes?

- 1 One minute is sixty seconds. That is the fact you start from.
- 2 Two minutes means sixty seconds, and then sixty seconds again.
- 3 So you need to double sixty.
- 4 Double sixty is one hundred and twenty.

Answer: 120 seconds

⇌ Check it a different way

You worked it out by doubling the sixty seconds in one minute. Now try it another way.

You doubled a number you had learned. Now put the pencil down and check it with a real timer instead of with arithmetic.

Turn a one minute sand timer, count out loud until it empties, then turn it and count again. Keep the same count running through both turns. If your counting is steady you will reach about one hundred and twenty, and the sand has now told you the same thing your doubling did.

How do you know?

Milo says a minute must be longer than an hour, because sixty is a bigger number than the number of hours in his school day. Say what he has muddled up.

Your turn

- 1 Name the units of time in order, starting with the shortest.
- 2 Write how many seconds make one minute.
- 3 Write how many minutes make one hour.
- 4 Write how many hours make one day.
- 5 Write how many days make one week.
- 6 Write how many months make one year.
- 7 Work out how many seconds there are in three minutes.
- 8 Work out how many minutes there are in two hours.
- 9 Work out how many hours there are in two days.
- 10 Work out how many days there are in two weeks.

TOPIC 20.1 · YOUR TURN, CONTINUED

- 11** Order these units from shortest to longest.

hour

second

day

minute

- 12** Choose the sensible unit for brushing your teeth. Say why.
I would measure this in ... because ...

- 13** Choose the sensible unit for one whole school day. Say why.

- 14** Sort these jobs into takes seconds and takes minutes.

clapping once

eating an apple

a sneeze

walking to the yard

takes
seconds

takes
minutes

✂ The material you need

a one minute sand timer if the classroom has one, and a clock or a watch with a second hand. You cannot feel a minute go past by reading about it, so borrow the timer before you start.

✦ Rook's find

Bramble's pocket watch has to be wound every single day or it stops. The cuckoo clock runs on a weight on a chain, and the weight takes eight days to reach the floor. He winds the clock on Sundays so he never has to remember which day it was.

Days of the week, months of the year

♣ Two lists that go round and round

There are seven days in a week and they always come in the same order. When you reach the end of the list you do not stop, you start it again, so the day after the last one is the first one.

There are twelve months in a year and they behave the same way. The month after the last month of the year is the first month of the next year.

Because both lists go round, you can always work out the next one or the one before. You never have to guess.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
1 January	2 February	3 March	4 April			
5 May	6 June	7 July	8 August			
9 September	10 October	11 November	12 December			

Do you remember?

*Seven days make one week and **12** months make one year. Both lists start again the moment they finish, which is why the same days keep coming round.*

 Watch me work one out 1

Which day comes straight after Thursday?

- 1** Find Thursday on the day strip.
 - 2** Put your finger on it and do not move it yet.
 - 3** Now slide your finger one place to the right, because after means further along the list.
 - 4** Read the day your finger has landed on.
-

Answer: Friday

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

⇔ Check it a different way

You worked it out by sliding one place forwards along the day strip. Now try it another way.

You went forwards along the strip. Now go the other way and see whether the strip sends you home again.

Start on Friday, the day you just found, and slide your finger one place to the LEFT. You land on Thursday, which is where you began. Going forwards one day and then backwards one day has to bring you back, so your answer was right.

How do you know?

Sara says there is no day after Sunday, because Sunday is the last one on the strip. Explain what she has forgotten about lists that go round.

Your turn

1 Write the seven days of the week in order, starting with Monday.

2 Write the day that comes straight after Wednesday.

3 Write the day that comes straight before Friday.

4 Write the day that comes two days after Sunday.

5 Fill in the missing days on this strip.

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

TOPIC 20.2 · YOUR TURN · CONTINUED

6 Write the twelve months in order, starting with January.

7 Write the month that comes straight after March.

8 Write the month that comes straight before September.

9 Write the month your birthday falls in.

10 Fill in the missing months on this grid.

1 January	2 February	3 March	4 April
5 May	6 June	7 July	8 August
9 September	10 October	11 November	12 December

TOPIC 20.2 · YOUR TURN, CONTINUED

- 11** Order these days from earliest to latest in the week.

Thursday

Monday

Saturday

Tuesday

- 12** Order these months from earliest to latest in the year.

July

February

November

April

13 Write two things you always do on the same day each week.

14 Write which month comes straight after the last month of the year.

🔗 Find out together

Ask everybody in your group which month their birthday is in, and mark each one on a month grid.

Which month has the most birthdays in it? Is there a month with none at all? Ask another group and see whether the busy month is the same one for them.

O'clock and half past



The cuckoo clock in the Plan Room. Two hands, one short and one long.

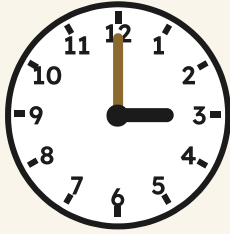
✿ The short hand tells you the hour

A clock face has two hands and they are not the same length. The **short** one is the hour hand. The **long** one is the minute hand.

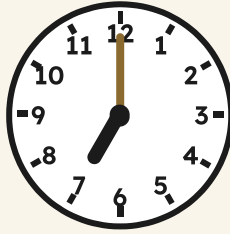
When the long hand points straight up at the twelve, the minute hand has gone all the way round and started again. We say **o'clock**, and the short hand points exactly at the hour.

So to read an o'clock time you only have to look at the short hand and read the number it points at.

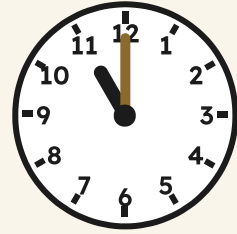
TOPIC 20.3 · CONTINUED



3 o'clock



7 o'clock



11 o'clock

Do you remember?

The long hand is the minute hand and the short hand is the hour hand. Get those two the wrong way round and every time you read will be wrong.

♣ **At half past, the short hand is BETWEEN two numbers**

When the long hand points straight down at the six, half of the hour has gone. We say **half past**.

Now look hard at the short hand, because this is where almost everybody goes wrong. At half past 3 the short hand is NOT pointing at the three any more. Half of the hour has gone, so the short hand has walked half of the way round to the four and it sits BETWEEN them.

That is your rule. If the short hand is between two numbers, the hour is the SMALLER of the two, the one it has not finished yet.



3 o'clock



half past 3



4 o'clock

 Watch me work one out 1

What time is the middle clock above showing?

- 1** Look at the long hand first. It points straight down at the six, so half of the hour has gone.
- 2** That tells you the time is a half past time.
- 3** Now look at the short hand. It is not on a number, it is sitting between the three and the four.
- 4** Take the smaller of the two, because that is the hour the clock has not finished yet.

Answer: half past 3

 Check it a different way

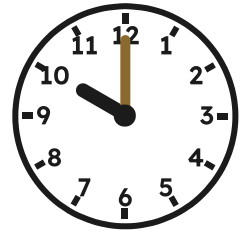
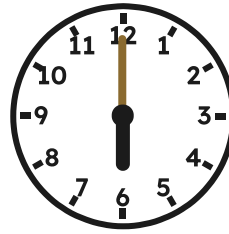
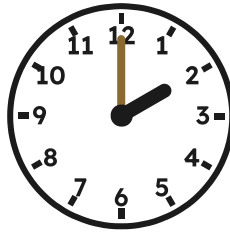
You worked it out by reading the hour hand and the minute hand on the face. Now try it another way.

You read the two hands on the face. Now check a completely different way, by looking at the digital panel on the wall beside the clock.

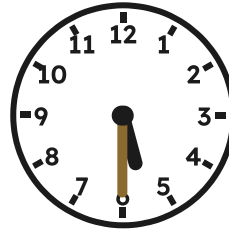
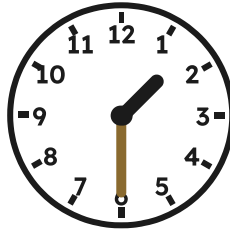
The panel shows **3:30**. The figures before the two dots are the hour and the figures after them are the minutes, and thirty minutes is half of sixty. So the panel says the hour is three and half the hour has gone, which is exactly what the hands said. Two different kinds of clock, one answer.

Your turn

- 1** Write the time shown on each of these clocks.



- 2** Write the time shown on each of these clocks.



- 3** Draw hands on a clock face to show 5 o'clock.

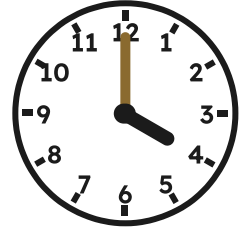
- 4** Draw hands on a clock face to show 12 o'clock.

- 5** Draw hands to show half past 7. Watch the short hand.

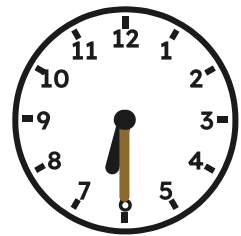
TOPIC 20.3 • YOUR TURN • CONTINUED

6 Draw hands to show half past 11. Watch the short hand.

7 Write this clock time in digital figures.



8 Write this clock time in digital figures.



9 Write which hand is longer, the minute hand or the hour hand.

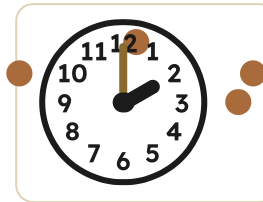
10 Write where the short hand sits at half past 8.

How do you know?

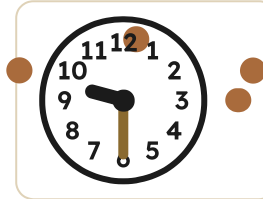
At half past 8 Milo says the short hand points at the nine, because the nine is the number nearest to it. Say what is wrong with his rule, and what the hour really is.

TOPIC 20.3 • YOUR TURN, CONTINUED

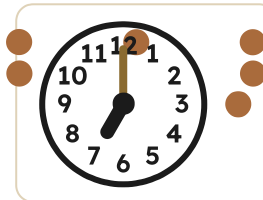
- 11** Join each clock to the time that matches it.



● half past 9



● 7 o'clock



● 2 o'clock

- 12** Write the time half an hour after 2 o'clock.

- 13** Write the time one hour after half past 9.

- 14** Order these times from earliest to latest in the day.

11 o'clock

half past 4

8 o'clock

half past 1



■ **SCAN TO SEE IT MOVE**

Drag the hands of a clock round and watch the digital time change to match, so half past means the same thing on both.

mathigon.org/polypad/clocks

→ **Take it further**

Start at 6 o'clock in the morning and write every time you can read this year, going half an hour at a time, until you reach half past 9 at night. Beside each one draw where the short hand is. How many times have you written, and how many of them have the short hand between two numbers?

Measuring how long something takes

♣ Pick the unit that fits the job

To measure how long something takes you need two things: a way of timing it, and the right unit to say the answer in.

Short jobs go in **seconds**. Hopping ten times, writing your name, tying a shoelace. Longer jobs go in **minutes**. Tidying the desk, eating your lunch, walking to the yard and back.

Saying that lunch took nine hundred seconds is not wrong, but nobody talks like that. Choose the unit that gives you a small, sensible number.

Choosing a unit

TOPIC 20.4 • CONTINUED

The job	The unit that fits	About how many
clapping your hands once	seconds	one
writing your name	seconds	about ten
counting to sixty	minute	one
tidying your desk	minutes	about five

- 1 Decide whether the job is short or long.
- 2 Choose seconds for a short job and minutes for a long one.
- 3 Start the timer at the same moment the job starts.
- 4 Stop it the moment the job is finished.
- 5 Write the number down straight away, with its unit.



The wall clock. Good for hours, useless for seconds.



The same clock, half an hour later.

Watch me work one out 1

Sara skipped for thirty seconds. Milo skipped for one minute. Who skipped for longer, and by how much?

- 1** The two timings are in different units, so you cannot compare them yet. Change one of them.
- 2** One minute is sixty seconds, so Milo skipped for sixty seconds.
- 3** Now both are in seconds: Sara thirty, Milo sixty.
- 4** Sixty is more than thirty, so Milo skipped for longer.
- 5** Take thirty from sixty to find the difference.

Answer: Milo, by **30** seconds

⇌ Check it a different way

You worked it out by counting the seconds out loud as you skipped. Now try it another way.

You counted the seconds out loud and did the arithmetic. Now check it with equipment instead of with your voice, because a voice speeds up when it gets excited.

Skip while somebody turns a one minute sand timer, and stop when the sand runs out. Then do it again with a stopwatch. If the sand and the stopwatch agree with your counting, your count was steady. If they do not, trust the equipment and find out how far out you were.

How do you know?

Two children timed the same job and got different answers. Give two reasons that could happen, without either of them being careless.

Your turn

- 1 Time yourself writing your name. Record the seconds.
- 2 Time yourself hopping ten times. Record the seconds.
- 3 Time a friend tidying the desk. Record the minutes.
- 4 Count aloud to sixty. Ask a friend to time you.
Was I faster or slower than the timer?
- 5 Write which takes longer, eating lunch or clapping once.

TOPIC 20.4 · YOUR TURN · CONTINUED

6

 Write which takes longer, one lesson or one sneeze.

7

 Choose seconds or minutes for tying a shoelace.

8

 Choose seconds or minutes for walking to the yard.

9

 Time two children skipping. Write who took longer.

10

 Fill this table in with three of your own timings.

The job	Seconds or minutes	How many

TOPIC 20.4 · YOUR TURN, CONTINUED

- 11** Order these jobs from shortest to longest.

eating lunch

blinking once

writing your name

tidying the desk

- 12** Write a job that takes about one minute. Then time it.

- 13** Write a job that takes about ten seconds. Then time it.

- 14** Compare two of your timings. Write how many seconds apart.

 Yard challenge

Bramble wants to know how many times he can write his name in one minute, but the only timer in the store runs for ten seconds. Plan how he could still find out, and say how many turns of the ten second timer make the **60** seconds he needs.

 Find out together

Work in a pair. One of you shuts your eyes and says stop when you think one minute has gone past. The other watches a real timer and writes down what it actually said.

Take five turns each. Were you usually too early or too late? Did your guesses get better with practice?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write how many minutes make half an hour.
- 2 Work out how many seconds there are in four minutes.
- 3 Write the day that comes straight after Saturday.
- 4 Write the month that comes straight before June.
- 5 Order these days from earliest to latest in the week: Friday, Tuesday, Sunday.
- 6 Draw hands on a clock face to show 10 o'clock.
- 7 Draw hands on a clock face to show half past 4.
- 8 Write where the short hand sits at half past 6.
- 9 Time yourself hopping twenty times. Record the seconds.
- 10 Explain why you would measure a sneeze in seconds and not in minutes.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
name the units of time in order, and say how they fit together	☐	☐	☐
order the days of the week and the months of the year	☐	☐	☐
read and show o'clock and half past on a clock and on a screen	☐	☐	☐
time an activity in seconds and in minutes, and compare two	☐	☐	☐

NOW I CAN

- ☐ name the units of time in order, and say how they fit together
- ☐ order the days of the week and the months of the year
- ☐ read and show o'clock and half past on a clock and on a screen
- ☐ time an activity in seconds and in minutes, and compare two

▣ YARD WORDS



second

a very short unit of time



minute

sixty seconds



hour

sixty minutes



half past

thirty minutes after the hour



digital

a time shown in numbers



analogue

a time shown by hands

Sorting and showing what you found

Chalk it on the wall so anyone walking past can read it.



Come into the Plan Room

Count how many people in your group are wearing something with a pocket. Count them out loud together.

Now count again, but this time keep a mark on paper for each person as you go.

Which way was easier to check afterwards, and why? That question is the whole of this unit.

☀ In the yard today

Rook is chalking marks on the slate board, in little gates of five, and he will not tell anybody what he is counting until he has finished.

The board is on the wall by the door where everybody walks past. That is deliberate. Rook says a count that nobody can see is a count nobody believes.

Each mark is one thing. Every fifth mark goes across the other four, which makes a gate of five, and a gate of five can be counted at a glance from the other side of the yard.

Anybody walking past should be able to read it, he says. If they cannot, I have wasted the chalk.

✓ **By the end you will**

- ask a question worth asking, then collect the answers in a list
- record what you found in a tally chart and read it back
- show what you found as a block graph or a pictogram
- sort things into a Venn or a Carroll diagram, using not as a rule

Cambridge Primary Mathematics Stage 2: 2Dh1 • 2Dh2 • 2Pt3

Asking a question and collecting the answers



Rook's gates of five, chalked where everybody walks past.

SHARPEN YOUR PENCIL

Ask three people in your class which fruit they like best. Remember all three answers without writing anything down. Now ask three more. How long before you lose one?

✿ A question worth asking has a small number of answers

Before you collect anything, you need a question. A good question for a survey has only a **few possible answers**, and everybody can give one.

Which pet do you have? is a good question. There are only a few common pets, so the answers make neat groups.

What is your favourite thing? is a bad question, because everybody says something different and you end up with twenty groups of one. That tells you nothing at all.

Not every question is worth a chart.

Question	Good or not
Which pet do you have?	good, few answers
How do you come to school?	good, few answers
What is your favourite thing?	not good, too many
What is your name?	not a survey question

 Watch me work one out 1

Rook asked everybody which pet they have. How many did he ask altogether?

- 1 Look at his list of answers, group by group.
- 2 Seven said dog, five said cat, three said fish, two said bird.
- 3 Add them: seven and five is twelve.
- 4 Twelve and three is fifteen. Fifteen and two is seventeen.
- 5 Every person gave exactly one answer, so the total is the number of people.

Answer: he asked **17** people

Rook's answers, written up as a table.

Pet	How many
dog	7
cat	5
fish	3
bird	2

⇔ **Check it a different way**

You worked it out by adding up the groups one at a time. Now try it another way.

You added the groups. Now check a different way: count the people in the room instead, and see whether you get the same number.

If your groups add to seventeen but there are eighteen people, somebody was missed out or somebody gave two answers. Either way, the survey is not finished.

How do you know?

Why must the totals of all the groups add up to the number of people you asked? What would it mean if it did not?

Your turn

- 1 Write a good survey question about journeys to school.
- 2 Write a good survey question about pets.
- 3 Write a question that would give too many different answers.
- 4 Say why your bad question would not make a useful chart.
- 5 List the answers your journey question could get.
- 6 Ask five people your question. Write down every answer.
- 7 Count how many gave each answer. Write the groups.
- 8 Add your groups. Write the total.
- 9 Count the people you asked. Does it match your total?
- 10 Write one thing your survey tells you.

11 Write a survey question with exactly two possible answers.

12 Sort these into good and not good survey questions.

good question

not good

13 Explain why asking somebody's name is not a survey question.

14 Plan a survey of your class. Write the question and the answers.

✂ The material you need

paper for your list, a pencil, and somebody to ask. A clipboard makes you feel official, and Rook says that helps.

★ Rook's find

Rook's gates of five are older than paper. Shepherds cut them into sticks called tally sticks, and the word tally comes from an old word for a cut stick. Nothing about counting in fives is modern.

Tally charts and tables



A gate of five: four marks, and the fifth across them.



The board itself, where anybody can read it.

♣ Why the fifth mark goes across

A **tally** is one mark for each thing you count. You make the mark as you count, not afterwards, and that is what stops you losing your place.

Every fifth mark is drawn **across** the four before it, making a gate. This is not decoration. Five separate marks in a row are slow and easy to miscount; a gate is one shape your eye reads instantly as five.

To total a tally, count the gates in fives and then add the loose marks on the end.

TOPIC 21.2 · CONTINUED

What we found	Tally	How many
dog		7
cat		5
fish		3
bird		2
Altogether		17

Do you remember?

Count the gates in fives first, then add the leftover marks. Never count a tally mark by mark; that is what the gates are for.

 Watch me work one out 1

Read the dog row of the tally chart above.

- 1** Find the dog row and look at its marks.
- 2** Count the complete gates first. There is one gate, which is five.
- 3** Now count the loose marks after it. There are two.
- 4** Five and two.

Answer: 7 people have a dog

 Check it a different way

You worked it out by counting the gates in fives. Now try it another way.

You counted the gates and added the loose marks. Now check by counting every single mark one at a time, touching each one.

Slower, and much easier to get wrong, which is exactly why the gates exist. But if the two ways agree, the row is right.

Both ways, side by side.

Pet	Tally total	Checked by counting
dog	7	7
cat	5	5
fish	3	3
bird	2	2

How do you know?

Why is a gate of five easier to read than five separate marks? Draw both and look at them from across the room.

Your turn

- 1 Draw a tally for seven things.
- 2 Draw a tally for twelve things.
- 3 Draw a tally for five things.
- 4 Write the number this tally shows: two gates and one mark.
- 5 Write the number this tally shows: three gates and four marks.
- 6 Total the cat row of the chart on the last page.
- 7 Total the fish row and the bird row together.
- 8 Which pet had the most? Write it and say how you know.
- 9 Which pet had the fewest? Write it.
- 10 Make a tally chart of eye colours in your group.

TOPIC 21.2 · YOUR TURN, CONTINUED

- 11 Turn your eye colour tally into a table with two columns.
- 12 Add up your whole table. Write the total.
- 13 Count your group. Does it match your table total?
- 14 Explain why you draw a tally mark while counting, not after.



▣ SCAN TO SEE IT MOVE

Sort houses in a street into groups, then change the sorting rule and watch every house move to its new group.

nrich.maths.org/problems/sort-street

→ Take it further

Tally something that is still happening, such as how many people walk past the door in five minutes. You cannot ask them to wait, so the tally has to keep up. Try it once, then try again and see whether you got better.

Block graphs and pictograms

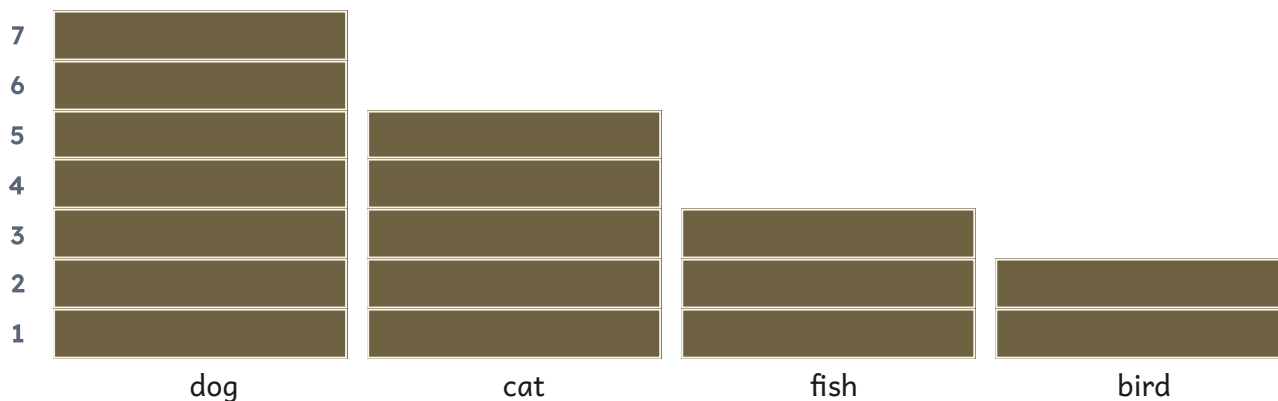
♣ A picture you can read from across the room

A **block graph** stacks one square for each thing counted. The tallest column is the biggest group, and you can see that without reading a single number.

A **pictogram** does the same with small pictures instead of squares. It must always have a **key** saying what one picture stands for, or nobody can read it.

Both are drawings of numbers you already had. They do not add any information: what they add is the ability to see it at a glance.

The same seventeen answers, as a block graph.



And as a pictogram.

dog



cat



fish



bird



one picture means one person

 Watch me work one out 1

How many more people have a dog than a bird?

- 1** Find the dog column and read its height: seven.
- 2** Find the bird column and read its height: two.
- 3** You are being asked how many MORE, which is a difference.
- 4** Count up from two to seven, or take two from seven.

Answer: 5 more people

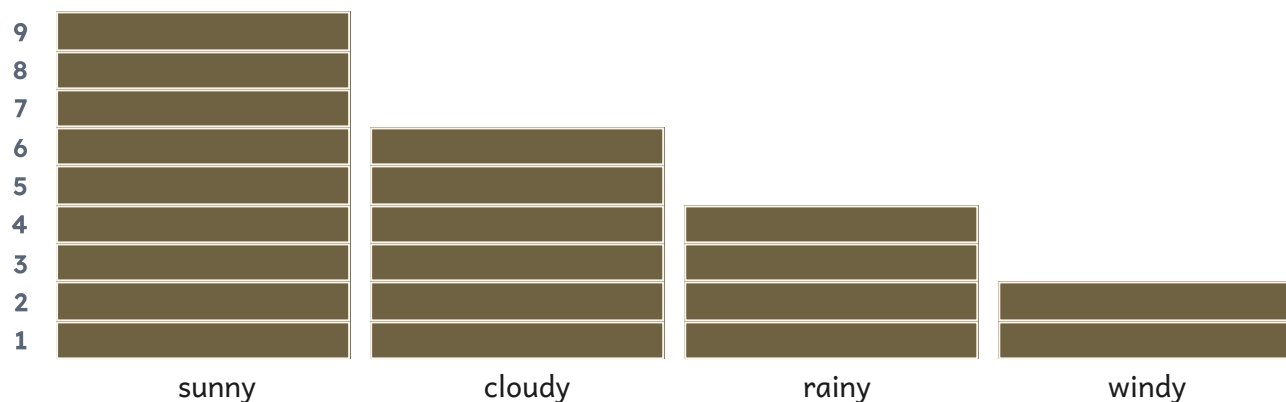
 Check it a different way

You worked it out by reading both heights and subtracting. Now try it another way.

You read the two heights and subtracted. Now check on the graph itself: put a finger on the top of the bird column and count the squares up to the top of the dog column.

Five squares. You never had to subtract anything, and the picture answered the question directly. If the two methods disagree, you misread a column.

A different survey: 21 days of weather, recorded one day at a time.



How do you know?

Rook says you can answer which is biggest from a block graph without knowing any of the numbers. Is he right? Say why.

Your turn

- 1 Read the sunny column. Write how many days.
- 2 Read the rainy column. Write how many days.
- 3 Which weather happened most often? Write it.
- 4 Which weather happened least often? Write it.
- 5 How many more sunny days than windy days?
- 6 How many days were sunny or cloudy altogether?
- 7 Add every column. Write the total number of days.
- 8 Draw a block graph of your eye colour survey.
- 9 Label both parts of your graph so anybody can read it.
- 10 Order the weather columns from fewest days to most days.

TOPIC 21.3 · YOUR TURN, CONTINUED

- 11** Draw a pictogram of your survey. Remember the key.
- 12** Write why a pictogram without a key cannot be read.
- 13** Write one question your graph can answer at a glance.
- 14** Write one question your graph cannot answer.

♣ Find out together

Swap graphs with another pair. Without talking, each of you write down what you think the other pair counted, and which group was biggest. Then compare. If they could not tell what you counted, your labels need work, and that is the most useful thing you will learn today.

Venn and Carroll diagrams

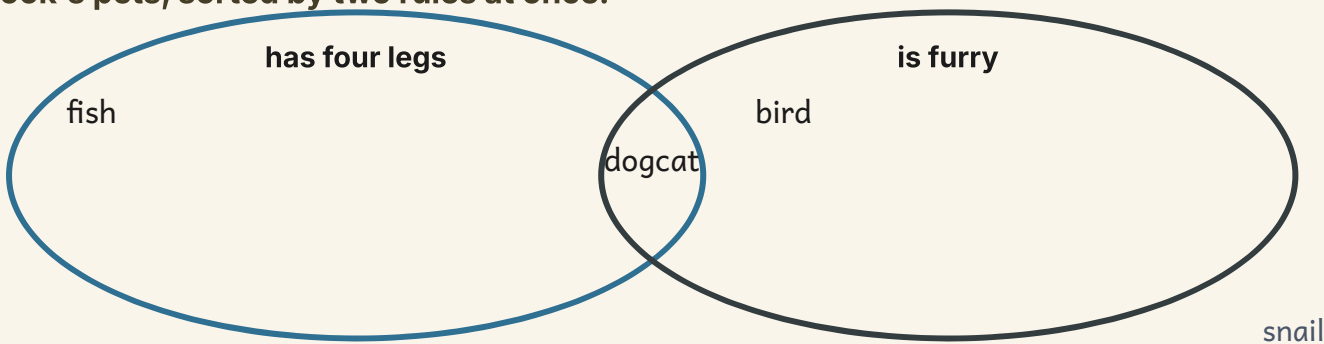
♣ Two rules at once

A **Venn diagram** is rings that overlap. Something that fits both rules goes in the **overlap**, and something that fits neither sits outside both rings.

A **Carroll diagram** does the same job as a table, using yes and no. Every single thing has exactly one box it belongs in, which is why a Carroll diagram is easier to check.

The word **not** is what makes both of them work. Not furry. Not a multiple of five. A rule and its opposite together cover everything.

Rook's pets, sorted by two rules at once.



The same numbers, sorted by two rules in a table.

	under 10	10 or more
even	2 8	12 20
not even	3 9	15 21

Watch me work one out 1**Where does a cat go in the Venn diagram above?**

- 1 Take the first rule: has four legs. A cat does, so it belongs in that ring.
- 2 Take the second rule: is furry. A cat is, so it belongs in that ring too.
- 3 Something that belongs in both rings cannot go in just one.
- 4 It goes in the part where the two rings cross over.

Answer: in the overlap, because it fits both rules

⇌ Check it a different way

You worked it out by testing the cat against each rule in turn. Now try it another way.

You checked the cat against each rule in turn. Now check the whole diagram a different way: count everything in every region, including outside the rings, and compare with how many things you started with. If your regions add up to fewer things than you had, something got left out. If they add up to more, something has been put in two places, and in a Venn diagram nothing may appear twice.

How do you know?

Why can nothing go in two different boxes of a Carroll diagram? Use the word *not* in your answer.

Your turn

- 1 Write where a dog goes in the Venn diagram, and why.
- 2 Write where a fish goes in the Venn diagram, and why.
- 3 Write what goes outside both rings, and why.
- 4 Draw two rings. Sort four animals by furry and not furry.
- 5 Draw a Venn diagram for even numbers and numbers over ten.
- 6 Put 12 in your diagram. Say which region it belongs in.
- 7 Put 7 in your diagram. Say which region it belongs in.
- 8 Draw a Carroll diagram with the rules odd and under twenty.
- 9 Put four numbers into your Carroll diagram.
- 10 Write a number that belongs in the overlap of your Venn diagram.

TOPIC 21.4 · YOUR TURN, CONTINUED

11 True or not true? Decide for each line.	Something can go in two boxes of a Carroll diagram.	TRUE / NOT TRUE
	The overlap of a Venn diagram fits both rules.	TRUE / NOT TRUE
	Nothing can sit outside both rings of a Venn diagram.	TRUE / NOT TRUE

12 Sort the numbers one to ten by odd and multiple of three.

13 Write a pair of rules where nothing fits both.

14 Write a pair of rules where everything fits at least one.

Yard challenge

Find two rules about numbers under twenty where the overlap has exactly one number in it. Then find a pair where the overlap is empty. Which was harder, and why?

Show what you can do

These are new questions. You have not practised these exact ones, which is the whole point of them.

☐ Check what you can do

- 1 Write a good survey question about how pupils travel to school.
- 2 Draw a tally for nine things.
- 3 Write the number shown by four gates and two marks.
- 4 Read a block graph column of six and write what it means.
- 5 Write why a pictogram must always have a key.
- 6 Write where something fitting both rules goes in a Venn diagram.
- 7 Explain why nothing can sit in two boxes of a Carroll diagram.
- 8 Total a survey with groups of eight, five and three.
- 9 Say how you would check that a survey total is right.
- 10 Write one question a block graph answers without reading any numbers.

Be honest with yourself

Tick one box on each line. Nobody is marking this page, and there is nothing wrong with needing more work on something.

	confident	almost there	needs more work
ask a question worth asking, then collect the answers in a list	☐	☐	☐
record what you found in a tally chart and read it back	☐	☐	☐
show what you found as a block graph or a pictogram	☐	☐	☐
sort things into a Venn or a Carroll diagram, using not as a rule	☐	☐	☐

NOW I CAN

- ☐ ask a question worth asking, then collect the answers in a list
- ☐ record what you found in a tally chart and read it back
- ☐ show what you found as a block graph or a pictogram
- ☐ sort things into a Venn or a Carroll diagram, using not as a rule

▣ YARD WORDS



data

the facts you collected



tally

a mark for each one you count



block graph

a graph made of stacked blocks



pictogram

a graph made of pictures



Venn diagram

rings that show what belongs where



Carroll diagram

a table sorting by yes and no

The Making Yard open day

Bramble is opening the yard to the whole school for one afternoon, and he has put your class in charge of it. Everything you have learned this year is needed, and he is not going to tell you which bit goes where.



Otto, Tessa, Rook and Hattie with the finished thing on the bench. It took all of them.

✂ The material you need

squared paper, a ruler, a pencil, coins you are allowed to use, a measuring jug, a metre stick, and whatever your group decides it needs once it has drawn the plan.

BUILD IT AND SHOW IT · THE LAST ONE · CONTINUED

Your group is going to design, build and run **one stall** for the open day. It can be a shop, a game, a measuring challenge or a shape puzzle. What it cannot be is something you have not planned on paper first.

Every stall must do four things, and each one comes from a different part of this book.

- 1 Draw the plan.** On squared paper, with a ruler. Show the sizes of anything you are going to make, in centimetres.
- 2 Work out the numbers.** How many of each thing you need, what it costs, how long a turn takes. Show your working, not just your answers.
- 3 Check every number a different way.** Every total on your stall must have been worked out twice, by two different methods, by two different people.
- 4 Record what happens.** Keep a tally of who visits and what they choose, then turn your tally into a block graph at the end.

What to hand in

- 1 Your plan, drawn on squared paper with the sizes marked.
- 2 A page of working showing every number your stall depends on.
- 3 The same numbers worked out a second way, by somebody else in the group.
- 4 Your tally chart from the afternoon, and the block graph you drew from it.
- 5 One sentence saying what you would do differently next time.

⇔ Check it a different way

You worked it out by whichever method the first person used. Now try it another way.

This is the whole point of the project, and it is worth more marks than anything else. A total that only one person has worked out, one way, is not finished. Swap papers inside your group and check each other by a different method.

HOW IT WILL BE JUDGED

What is judged	Worth	What a good one looks like
Drawn before it was built	20%	there is a real plan on squared paper, with sizes, and the thing you built matches it
Correct arithmetic	25%	every number your stall depends on is right
Checked a different way	25%	every total was worked out twice, by two methods, by two people, and the second method is genuinely different
Data collected and shown	15%	a tally kept properly in fives, turned into a graph anybody can read
Working together	15%	everybody in the group did a part of it, and can explain the part somebody else did

★ **Rook's find**

Bramble has run the open day for eleven years. He says the stalls that go wrong are never the ones with hard sums on them. They are the ones where nobody drew it first.

What you can do now



In September you could count to a hundred if somebody started you off, and you could add two small numbers if you had your fingers free.

Look at what is in your head now.

LOOKING BACK · CONTINUED

- ☐ count to a hundred and beyond, forwards and backwards, from any number
- ☐ say what every digit in a two-digit number is worth
- ☐ compare and order numbers, and round them to the nearest ten
- ☐ know your number pairs to ten and twenty without stopping to think
- ☐ add and take away two-digit numbers, and check by a different method
- ☐ read an array two ways, and know your two, three, four, five and ten times tables
- ☐ share and group, and say honestly what is left over
- ☐ name flat shapes and solid shapes, and describe them by their properties
- ☐ find a line of symmetry by folding, and prove it matches
- ☐ turn a quarter turn, and recognise a right angle when you see one
- ☐ handle money, work out change, and check the change a second way
- ☐ measure length, mass and capacity, choosing the right unit yourself
- ☐ tell the time to o'clock and half past on hands and on figures
- ☐ collect data, tally it, and show it so somebody else can read it

How do you know?

Which one of these was hardest, and what made it click in the end? That answer is worth knowing about yourself.

Every word in this book

In alphabetical order, with the unit that teaches it. If a word here is not the same as the one on your unit card, the unit card is the short version and this is the whole thing.

YARD WORDS · CONTINUED

add To put two or more amounts together to find the total. Its sign is +.

unit 6

altogether The whole amount once everything has been joined together.

unit 7

amount How much money something costs or how much you are holding.

unit 17

analogue A time shown by hands going round a clock face.

unit 20

anticlockwise Turning the opposite way to the hands of a clock.

unit 16

array Things arranged in rows and columns, with every row the same length and every column the same height.

unit 10

balance To be exactly the same on both sides, the way scales hang level when the two pans hold the same amount.

unit 9

block graph A graph made of squares stacked into columns, one square for each thing counted.

unit 21

capacity How much a container holds when it is full.

unit 19

YARD WORDS · CONTINUED

Carroll diagram A table that sorts things by yes and no, so everything has exactly one box it belongs in.

unit 21

cent The small unit of money. One hundred cent make one euro.

unit 17

centimetre A small unit for measuring length. Your finger is about one centimetre wide.

unit 18

change The money you get back when you pay with more than the thing cost.

unit 17

check To work something out a second time by a DIFFERENT method, so that if both ways agree you know the answer is right.

unit 8

clockwise Turning the way the hands of a clock go.

unit 16

coin A piece of metal money. Every coin has an amount stamped on it.

unit 17

column Things one above another, going down.

unit 10

compare To look at two amounts and say which is bigger, which is smaller, or that they are the same.

unit 3

corner The point where two sides of a shape meet.

unit 13

YARD WORDS · CONTINUED

count

unit 1

To say the numbers in order while you touch each thing once, so that the last number you say is how many there are.

count back

unit 7

To subtract by starting at one number and counting backwards.

count on

unit 7

To add by starting at one number and counting forwards.

cuboid

unit 14

A solid box shape with six faces, all of them rectangles.

cylinder

unit 14

A solid tin shape with two flat circular faces and one curved face joining them.

data

unit 21

The facts you collected by asking, counting or measuring.

difference

unit 7

How much more one number is than another. You can find it by counting on from the smaller number to the bigger one.

digit

unit 1

One of the ten signs 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 that numerals are built out of. 47 is one number written with two digits.

digital

unit 20

A time shown as figures, such as 3:30, with no hands at all.

direction

unit 16

The way something is going or facing.

YARD WORDS · CONTINUED

divide

unit 11

To share an amount equally, or to put it into groups of the same size. Its sign is $\div$.

double

unit 6

Twice as much. Double 8 is 16, because 8 and 8 make 16.

edge

unit 14

The line where two faces of a solid shape meet.

equal

unit 5

Exactly the same amount as something else, not just nearly.

equals

unit 9

Is exactly the same amount as. Its sign is $=$, and it means the two sides balance, not that an answer is coming next.

estimate

unit 3

A sensible guess you make on purpose before you count or calculate, so that you can tell afterwards whether your answer makes sense.

even

unit 4

A number that pairs up exactly with nothing left over. 0, 2, 4, 6 and 8 are even, and so is any number ending in one of them.

exchange

unit 8

To swap ten ones for one ten, or one ten for ten ones. Nothing is lost when you exchange: the amount stays the same.

face

unit 14

One flat or curved surface of a solid shape. A cube has six flat faces.

fraction

unit 5

A part of a whole, written with one number above another, such as $\frac{1}{2}$ or $\frac{1}{4}$.

YARD WORDS • CONTINUED

gram

A small unit for measuring mass. A paper clip is about one gram.

unit 19

greater than

Bigger than. It is written with the sign $>$, and the wide end of the sign always faces the bigger number.

unit 3

group

To put an amount into sets of a size you choose, and then count how many sets you made. This is one of the two kinds of dividing.

unit 11

half

One of two equal parts of something. It is written $\frac{1}{2}$, and the two parts must be exactly the same size or neither is a half.

unit 5

half past

Thirty minutes after the hour, when the long hand points straight down.

unit 20

halve

To split an amount into two equal parts. Halving undoes doubling.

unit 12

height

How tall something is from the bottom to the top.

unit 18

hexagon

A flat shape with six straight sides.

unit 13

hour

A unit of time worth sixty minutes.

unit 20

hundred square

A grid of ten rows and ten columns holding the numbers 1 to 100, so you can see the tens going down and the ones going across.

unit 1

YARD WORDS • CONTINUED

inverse

unit 11

The operation that undoes another one. Subtracting is the inverse of adding, and halving is the inverse of doubling.

jump

unit 7

One step along a number line. A jump of ten crosses nine numbers and lands on the tenth.

kilogram

unit 19

A unit for measuring mass, worth 1000 grams. A bag of sugar is about one kilogram.

left over

unit 11

The part of an amount that will not make another full group. It is not a group, however much you want it to be.

length

unit 18

How long something is from one end to the other.

less

unit 2

A smaller amount. Ten less than 25 is 15.

less than

unit 3

Smaller than. It is written with the sign $<$, and the point of the sign always faces the smaller number.

line of symmetry

unit 15

The fold line along which the two halves of a shape match exactly. A shape can have more than one.

litre

unit 19

A unit for measuring capacity. A large bottle of water holds about one litre.

YARD WORDS · CONTINUED

mass

How heavy something is. It is what the scales tell you.

unit 19

match

To be exactly the same, so that one could lie on top of the other with nothing sticking out.

unit 15

method

The way you worked something out. Two people can use different methods and both be right.

unit 8

metre

A unit for measuring length, worth 100 centimetres. A door is about two metres tall.

unit 18

minute

A unit of time worth sixty seconds.

unit 20

mirror image

The same picture the other way round, as a mirror shows it.

unit 15

missing number

The number that has been hidden in a box or behind a shape, which you have to work out from the rest of the sentence.

unit 9

more

A bigger amount. Ten more than 25 is 35.

unit 2

multiple

A number you land on when you count in steps from zero. 15 is a multiple of 5 because 5, 10, 15 lands on it.

unit 4

YARD WORDS · CONTINUED

multiple of ten

unit 7

A number you reach by counting in tens: 10, 20, 30, 40 and so on. All of them end in 0.

multiply

unit 10

To add the same amount again and again a given number of times. Its sign is $\times$.

near double

unit 12

A double with one more or one less, such as 7 and 8. You work it out from a double you already know.

note

unit 17

A piece of paper money, worth more than the coins.

number bond

unit 6

A pair of numbers that make a particular total. 6 and 4 are a number bond for 10.

number line

unit 1

A straight line with numbers marked along it in order, always with the same space between one number and the next.

number sentence

unit 9

A calculation written down with signs, such as $27 + 3 = 30$.

numeral

unit 1

The mark you write on the paper for a number, such as 47. The number is the amount; the numeral is the writing.

odd

unit 4

A number that will not pair up: however you try, one is always left over. 1, 3, 5, 7 and 9 are odd, and so is any number ending in one of them.

YARD WORDS · CONTINUED

ones

unit 2

How many single ones are left after you have taken out all the tens. In 47 there are 7 ones.

order

unit 3

To put things in place one after another by size, saying which end you started from.

ordinal

unit 1

A number that says which one rather than how many, such as first, second or third. Ordinals only work when things are in an order.

pair

unit 6

Two numbers that go together to make a total you want.

partition

unit 2

To split a number into the parts it is made of, such as 47 into 40 and 7, and to be able to put it back together again.

pattern

unit 4

Something that carries on by a rule, so that once you know the rule you can say what comes next.

pentagon

unit 13

A flat shape with five straight sides. The sides do not have to be the same length.

pictogram

unit 21

A graph made of small pictures, where the key tells you how much one picture stands for.

YARD WORDS · CONTINUED

place value

unit 2

What a digit is worth because of where it stands. The 4 in 47 is worth forty, and the 4 in 74 is worth four.

position

unit 16

Where something is, said in words such as between, next to, above or underneath.

product

unit 10

The answer you get when you multiply two numbers together.

pyramid

unit 14

A solid shape with a flat base whose other faces are triangles meeting at a point.

quarter

unit 5

One of four equal parts of something. It is written $\frac{1}{4}$, and two quarters are the same as one half.

quarter turn

unit 16

A turn through one right angle, which is a quarter of the way round.

reflect

unit 15

To flip something over a line, so that it appears the same distance away on the other side.

remainder

unit 11

The amount left over after dividing, when the sharing or the grouping does not come out exactly.

right angle

unit 16

A square corner, the same shape as the corner of this page. It is what you turn through in a quarter turn.

YARD WORDS · CONTINUED

round

unit 3

To change a number to the nearest ten, so 26 rounds to 30 and 24 rounds to 20. A number ending in 5 always rounds up.

row

unit 10

Things side by side, going across.

ruler

unit 18

A straight tool marked in centimetres, used for measuring lengths and for drawing straight lines.

scale

unit 19

The row of marks on a measuring tool that you read the measurement from.

second

unit 20

A very short unit of time. It takes about one second to say one short word.

share

unit 5

To split an amount into equal parts, giving the same to each.

side

unit 13

One straight edge of a flat shape.

skip-count

unit 4

To count in steps bigger than one, such as 2, 4, 6, 8, missing out the numbers in between.

solid

unit 14

A shape you can pick up and hold, which takes up room and has an inside.

sort

unit 13

To put things into groups by a rule, and to be able to say what the rule was.

YARD WORDS · CONTINUED

step

unit 4

How much you count on by each time. Counting 3, 6, 9 has a step of three.

story

unit 9

A real situation that a calculation fits, such as three baskets each holding four pegs.

straight

unit 13

Not curved anywhere along its length.

subtract

unit 6

To take an amount away, or to find the difference between two amounts. Its sign is $-$.

symmetry

unit 15

When a shape can be folded so that the two halves lie exactly on top of each other.

take away

unit 8

To remove some of an amount so that less is left.

tally

unit 21

A mark you make for each one you count, drawn in groups of five so that they are quick to add up.

tens

unit 2

How many whole tens a number has. In 47 there are 4 tens, which is 40.

times table

unit 10

All the multiplications for one number, in order, such as one five, two fives, three fives.

total

unit 6

How much there is altogether when everything is put together.

YARD WORDS · CONTINUED

total value

unit 17

How much a set of coins and notes is worth altogether, which is not the same as how many pieces there are.

twice

unit 12

Two times as much as something.

undo

unit 12

To go back the other way and arrive where you started.

unit

unit 18

The size you choose to measure in, such as centimetres, metres, grams or litres. A measurement without its unit means nothing.

unknown

unit 9

A number you have not found yet. It is often shown by a box or a shape until you work out what it is.

Venn diagram

unit 21

Two or more rings that overlap, so a thing that fits both rules goes in the overlap.

vertical

unit 15

Going straight up and down.

whole

unit 5

All of something, with nothing missing. Two halves make a whole, and so do four quarters.

The Cambridge objectives, unit by unit

 For teachers

This table maps every unit of this Student Book to the Cambridge Primary Mathematics 0845 Stage 2 objective codes it delivers. It is printed so that a teacher can plan against the framework directly, and so that a head of department can audit the coverage without reading the whole book.

The 2Pt problem-solving codes are not given a unit of their own, because the framework does not treat problem solving as separate content. They are delivered throughout, chiefly by the Check it a different way panels, which carry 2Pt6, 2Pt7 and 2Pt11 in every unit, and by the seven projects.

FOR TEACHERS • CONTINUED

Unit	Title	Stage 2 objectives
1	Numbers to 100 and beyondNumber · Plan Table	2Nn1 2Nn2 2Nn3 2Nn11
2	Tens and onesNumber · Plan Table	2Nn6 2Nn7 2Nn10
3	Comparing, ordering and roundingNumber · Plan Table	2Nn8 2Nn9 2Nn12 2Nn13
4	Steps, odds and evensNumber · Plan Table	2Nn4 2Nn5 2Nn14 2Nn15 2Pt8
5	Halves and quartersNumber · Cutting Bench	2Nn16 2Nn17 2Nn18 2Nn19
6	Number pairs and doublesNumber · Reckoning Board	2Nc1 2Nc2 2Nc3 2Nc5
7	Adding and taking tensNumber · Reckoning Board	2Nc6 2Nc11 2Nc8
8	Adding and taking to 100Number · Reckoning Board	2Nc12 2Nc13 2Nc14 2Nc15 2Pt6 2Pt7
9	Missing numbers and equal sidesNumber · Reckoning Board	2Nc7 2Nc9 2Nc10 2Pt4
10	Rows, columns and times tablesNumber · Array Rack	2Nc16 2Nc17 2Nc19 2Nc22
11	Sharing, grouping and what is leftNumber · Array Rack	2Nc18 2Nc23 2Pt4
12	Doubling and halvingNumber · Array Rack	2Nc20 2Nc21 2Pt9

Objective map: geometry, measure and data

Unit	Title	Stage 2 objectives
13	Flat shapesGeometry · Drawing Board	2Gs1 2Gs4 2Pt9
14	Solid shapesGeometry · Drawing Board	2Gs2 2Gs4
15	SymmetryGeometry · Drawing Board	2Gs3 2Gs4
16	Position, turns and right anglesGeometry · Drawing Board	2Gp1 2Gp2 2Gp3
17	MoneyMeasure · Measures Store	2Mm1 2Mm2 2Mm3 2Pt5
18	LengthMeasure · Measures Store	2Ml1 2Ml2 2Pt2
19	Mass and capacityMeasure · Measures Store	2Ml1 2Ml2 2Pt2
20	TimeMeasure · Measures Store	2Mt1 2Mt2 2Mt3 2Mt4 2Mt5
21	Sorting and showing what you foundHandling data · Tally Wall	2Dh1 2Dh2 2Pt3

How the book is built, and why

For teachers

Every unit has four teaching parts. Each part carries a worked example, at least twelve practice items, a reasoning prompt headed How do you know, and a second-method panel headed Check it a different way. Those four are enforced by automated gates at build time rather than by editorial good intentions, so a part cannot ship without them.

Body type is set at 14.5 point and is never reduced to make a page fit. Where content overruns, the page is split. This follows the acceptance criterion agreed after the Year 7 review and applies across the whole Prime Books line.

No answer is printed anywhere in this book except inside a Watch me work one out panel, where printing it is the point. The Check your progress items at the end of each unit are deliberately different from the practice items, so that the check measures learning rather than recall of drilled questions.

The self-assessment tables are worded confident, almost there and needs more work. Colour words are not used, so that a pupil who cannot distinguish red from green is not disadvantaged and so that the record reads as a sentence to a parent.

Pupils who finish a part early should be directed to Take it further, then to the Yard challenge, and then to the unit's QR activity, rather than into the next unit. The projects are designed to absorb a wide range of finishing speeds inside one class.

Every QR code prints its address as readable text. No activity in this book requires a device, and nothing is lost by a class that never scans one.

Where everything came from

All the text, illustrations and diagrams in this book are original to this edition and were made for it. What follows is every web address printed in the book, in one place, so that a teacher can put them on the board without hunting through the pages.

Every QR destination in this book

Where in the book	Kind	Address
Unit 1	INTERACT	mathigon.org/polypad/number
Unit 2	INTERACT	www.topmarks.co.uk/place-value/place-value-charts
Unit 3	GENERATE	www.topmarks.co.uk/ordering-and-sequencing/coconut-ordering
Unit 4	MOVE	mathigon.org/polypad/number
Unit 5	INTERACT	mathigon.org/polypad/fractions
Unit 6	GENERATE	www.topmarks.co.uk/maths-games/hit-the-button
Unit 7	MOVE	mathigon.org/polypad/number
Unit 8	INTERACT	mathigon.org/polypad/number
Unit 9	GENERATE	www.topmarks.co.uk/maths-games/mental-maths-train
Unit 10	INTERACT	mathigon.org/polypad/number
Unit 11	INTERACT	mathigon.org/polypad/number

Every QR destination, continued

Where in the book	Kind	Address
Unit 12	GENERATE	www.topmarks.co.uk/maths-games/daily10
Unit 13	INTERACT	mathigon.org/polypad/polygons
Unit 14	MOVE	mathigon.org/polypad/polyhedra
Unit 15	MOVE	mathigon.org/polypad/polygons
Unit 16	INTERACT	mathigon.org/polypad/polygons
Unit 17	INTERACT	mathigon.org/polypad/number
Unit 18	INTERACT	mathigon.org/polypad/geometry
Unit 19	MOVE	mathigon.org/polypad/number
Unit 20	MOVE	mathigon.org/polypad/clocks
Unit 21	INTERACT	nrich.maths.org/problems/sort-street
How this book works	INTERACT	mathigon.org/polypad
The final project	INTERACT	nrich.maths.org/low-threshold-high-ceiling-resources-primary-teachers

Every address above was checked and answering in July 2026. A code is only printed in this book where the destination does something a printed page cannot: **move**, be **interactive**, or **generate** fresh questions each time.

The curriculum framework

Cambridge Assessment International Education, *Cambridge Primary Mathematics 0845 Curriculum Framework*, Stage 2. Published by the University of Cambridge Local Examinations Syndicate. The objective codes quoted throughout this book (2Nn, 2Nc, 2Gs, 2Gp, 2Mm, 2Ml, 2Mt, 2Dh and 2Pt) are the framework's own codes and are quoted for identification and planning only.

The interactive resources

Mathigon Polypad, mathigon.org/polypad. A free collection of virtual manipulatives, used in this book wherever a printed figure needed to move: number lines, arrays, fraction bars, coins, clocks, polygons and polyhedra. Mathigon is part of Amplify Education, Inc.

NRICH, nrich.maths.org. The primary problem collection of the Millennium Mathematics Project at the University of Cambridge, used for the open-ended problems that extend the projects.

Topmarks, www.topmarks.co.uk. Free browser-based practice games needing no account, used where the value of the link was a fresh set of questions every time.

Typefaces

Inter by Rasmus Andersson. Andika by SIL International. Kalam by Indian Type Foundry. Lexend by Thomas Jockin and Bonnie Shaver-Troup. All are used under the SIL Open Font License 1.1.

A note on the mathematics

Every number, total, array, coin total, measurement and clock face printed in this book is computed by the typesetting engine from the underlying data, not typed by hand. A printed total therefore cannot disagree with the picture beside it. The engine's arithmetic is proved by 306 self-test assertions that run before every build.

PRIME SCHOOL PRESS

Mathematics

Year 2 · Prime School Press · Student Manual

Draw it. Work it out. Then check it a different way.

Last year you learned to count something, and then to count it again to be sure. This book asks for the next thing, which is the thing that makes a mathematician: a method, and then a second, different method that agrees with the first.

Otto draws the plan. Rook lines everything into rows and columns. Bramble keeps the store, the scales and the money tin. And Tessa checks every single thing a different way, which is why this book exists.

INSIDE THIS BOOK

- Twenty-one units covering the full Cambridge Primary Mathematics Stage 2 framework, with the objective codes printed on every unit
- Over a thousand practice items, and not one printed answer to copy
- A worked example, a reasoning prompt and a second method in every part
- Seven Build it and show it projects, with the judging grid printed before the work starts
- QR codes that only appear where the screen can do something the page cannot

Prime School Press · Mathematics

Ages 6–7 · Lower Primary

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